



**GREEN
CLIMATE
FUND**

Meeting of the Board
23 – 26 October 2023
Tbilisi, Georgia
Provisional agenda item 12

GCF/B.37/02/Add.07

2 October 2023

Consideration of funding proposals – Addendum VII

Funding proposal package for FP216

Summary

This addendum contains the following seven parts:

- a) A funding proposal titled "Scaling up climate resilient flood risk management in Bosnia and Herzegovina";
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Secretariat's assessment;
- e) Independent Technical Advisory Panel's assessment;
- f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
- g) Gender documentation.

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Funding Proposal

Project/Programme title:	Scaling up climate resilient flood risk management in Bosnia and Herzegovina
Country(ies):	Bosnia and Herzegovina (B&H)
Accredited Entity:	United Nations Development Programme
Date of first submission:	2021/07/01
Date of current submission	2023/08/01
Version number	[V.010]



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Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

“FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]”

A. PROJECT/PROGRAMME SUMMARY				
A.1. Project or programme	Project	A.2. Public or private sector	Public	
A.3. Request for Proposals (RFP)	Not applicable			
A.4. Result area(s)		GCF contribution	Co-financers' contribution ¹	
	Mitigation total	Enter number %	Enter number %	
	☐ Energy generation and access	Enter number %	Enter number %	
	☐ Low-emission transport	Enter number %	Enter number %	
	☐ Buildings, cities, industries and appliances	Enter number %	Enter number %	
	☐ Forestry and land use	Enter number %	Enter number %	
	Adaptation total	100 %	100 %	
	☒ Most vulnerable people and communities	24.3 %	13 %	
	☒ Health and well-being, and food and water security	24.3 %	12.6 %	
	☒ Infrastructure and built environment	7.7 %	37.7 %	
	☒ Ecosystems and ecosystem services	43.7 %	36.7 %	
A.5. Expected mitigation outcome (Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)	Indicate greenhouse gas (GHG) emission reductions or removals in tCO ₂ eq over total lifespan of the project/programme ²	A.6. Expected adaptation outcome (Core indicator 2: direct and indirect beneficiaries reached)	Total number of direct and indirect beneficiaries : 2.777 million (1.42 Million females; 1.36 million men)	
			799,445 Direct beneficiaries (407,717 females, 391,728 males)	1.977 Million people Indirect beneficiaries (1.01 Million females; 0.97 Million males)
			25% of total population	80% of total rest of population
A.7. Total financing (GCF + co-finance ³)	72.69 Million USD	A.9. Project size	Medium (Upto USD 250 million)	
A.8. Total GCF funding requested	14.4 Million USD			

¹ Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e. GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.

² The total lifespan of the project/programme is defined as the maximum number of years over which the outcomes of the investment are expected to be effective. This is different from the project/programme implementation period.

³ Refer to the Policy of Co-financing of the GCF.

A.10. Financial instrument(s) requested for the GCF funding	☒ Grant <u>14.4 Million USD</u> ☐ Equity <u>Enter number</u> ☐ Loan <u>Enter number</u> ☐ Results-based payment <u>Enter number</u> ☐ Guarantee <u>Enter number</u>		
A.11. Implementation period	6 years	A.12. Total lifespan	20 years
A.13. Expected date of AE internal approval	17 June 2021	A.14. ESS category	B
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☐ No ☒
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☐ No ☒
A.19. Complementarity and coherence	<i>Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1.</i> Yes ☒ No ☐		
A.20. Executing Entity information	UNDP		
A.21. Executive summary (max. 750 words, approximately 1.5 pages)			

1. Climate change in Bosnia and Herzegovina (B&H) has been observed through continuous rise of the mean annual temperature and significant seasonal and spatial changes in precipitation, that have become more pronounced over the last fourteen years⁴. As a result of these changes in climate, years with damaging floods have become very common. The Third National Communication (TNC) shows that the temperature is expected to increase further until the end of the 21st century, and the variability of rainfall regime and extreme precipitation will further intensify under warmer climate conditions.
2. Approximately 26% of the B&H population is highly vulnerable to climate change induced flood risk⁵. The increasing intensity and variability of rainfall driven by climate change have been causing more frequent and intensive floods resulting in significant impacts on people, their property and critical infrastructure. The most important economic sectors at risk include agriculture, water management and hydropower. Between 2006 and 2012 flooding accounted for 62% of damages from all-natural disasters in most of the B&H territory. In 2010, the second largest flood on record resulted in damages of US\$ 200 million or approximately 1% of GDP⁶. In May 2014, Bosnia and Herzegovina experienced its worst flooding in 150 years which affected 25% of its territory and resulted in 26 deaths and US\$2.7 Billion worth of damages (15% of GDP). The most damaging floods have had devastating impacts on the most vulnerable groups including the rural poor, war returnees and displaced persons who generally occupy the cheapest and most risk-exposed land on the floodplain. For example, in several municipalities up to 100% of affected households have been identified as war returnees or displaced persons and are least equipped to cope with and recover from floods. This has led to a deepening of poverty in flood affected areas. The 2014 flood was the most extensive, affecting 5 of the 7 major rivers, compared to previous floods which generally impacted only 1 major river. In the last two decades, B&H experienced 10 major flood events (2001, 2004, 2006, 2009, 2010, 2014, 2017, 2018, 2019, 2020), compared to 8 major flood events in the preceding 75 years (1925 to 2000)⁷. In each of the last four years 2017, 2018, 2019 and 2020, flooding occurred in at least two of the seven major river basins. Socio-economic and climate risk modeling conducted during the feasibility study, demonstrate that by mid-century, due to climate change, 13% more people would be affected by floods leading to 12.5% higher losses of household assets compared to baseline scenario.
3. In order to adapt to the intensifying climate-induced flood risk, B&H needs to move towards a more proactive integrated approach to flood risk management that prioritizes risk reduction, risk prevention and preparedness through enhanced use of climate information, establishment of a national flood forecasting and early warning system and enhancement of adaptive capacity and resilience of at-risk communities.
4. Objective of the project *Scaling up climate resilient flood risk management in Bosnia and Herzegovina (hereinafter referred to as the project)* is to address increasing vulnerability of B&H communities and livelihoods to intensified climate-induced flood-related disasters. The project will establish an integrated and innovative gender-responsive climate-resilient flood risk management (FRM) approach, strengthen institutional, technical and financial capacity to implement long-term FRM strategies, including a combination of structural and non-structural measures and ecosystem-based approaches. The project will strengthen the use of climate information, flood forecasting, early warning and emergency response systems to enhance adaptive capacity and resilience of at-risk communities. The project is aligned with the new UN multi-partner initiative “Early Warnings for All (EW4All)” announced by the UN Secretary-General in March 2022 with the aim to promote universal coverage of the early warning, and will contribute to the implementation of EWS4All Executive Action Plan developed in March 2023.
5. The project will scale-up successful FRM solutions and technologies tested through a UNDP/SCCF project⁸ in Vrbas River Basin, promote and implement technological, institutional and financing innovations in FRM sector, and will leverage considerable public and private investment into resilient FRM. The latter will be achieved through the development of an FRM investment framework supported with incentives, risk transfer products and private sector partnership models. The willingness of the B&H private sector to engage with the intervention is reconfirmed through the materialized co-financing commitments (Annex 13), surveys and consultations carried out during the project design phase (Annexes 2 and 7). The project has two Outcomes, and three Outputs as follows:
 - Outcome 1:** Climate informed gender-responsive impact-based FFEWS and increased generation and use of climate data reduce vulnerability to flood related disasters
 - Output 1:** Fully integrated impact-based Flood Forecasting and EWS facilitates timely preparation and response
 - Outcome 2:** Vulnerable communities in B&H including women, children, disabled and elderly are better protected from and less exposed to climate induced flood risks through a combination of structural and non-structural flood risk reduction solutions
 - Output 2:** Non-structural flood risk reduction measures and nature-based solutions mainstreamed in sectoral policies and plans and effectively contribute to protection of people and livelihoods from climate-induced flood risk

Output 3: Climate-proof flood protection measures scaled-up through new and improved national and local investment frameworks increasing resilience of the most vulnerable groups to climate induced flooding.

6. Socio-economic risk modelling undertaken to assess potential adaptation benefits of the proposed project intervention for baseline and climate change (2050) scenarios for each of the five most vulnerable sub-basins in B&H (Una-Sana, Vrbas, Bosna, in the Sava basin, and Neretva and Trebišnjica in the Adriatic Sea basin)⁹, shows that 799,445 people and 272,811 households are currently at risk from floods (Section 2.3 and Annex E of FS). Under climate change, by 2050 the number of affected people and households will increase to 902,906 people and 306,907 households. The total damage to household assets is expected to increase from US\$ 3.2 Billion (baseline) to US\$ 3.6 Billion (2050), while flood affected agriculture areas and agricultural damages are expected to increase from 12,896 ha and US\$ 3.34 Billion to 15,064 ha and US\$ 3.7 Billion. Currently, 2,825 business sector assets across B&H with a total value of US\$ 3.2 Billion are at risk of flooding and would sustain US\$ 548 Million in asset damages in an extreme flood event¹⁰.
7. The project targets 799,445 direct beneficiaries by the end of the project (25% of the total population). It will prevent loss of life, property, critical infrastructure, and economic assets and will build resilience to flooding in the most vulnerable communities. It will achieve this by:
 - (i) developing institutional capacity to generate climate risk information for long-term strategic climate-resilient flood risk management (FRM);
 - (ii) scaling up successful innovative FRM technologies and implementing a nationwide impact-based Flood Forecasting and Early Warning System (FFEWS), which will result in improved warning times and reduced losses;
 - (iii) empowering local communities to undertake local flood risk reduction and response measures, including community-based early warning systems (CBEWS) that complement the centralized national EWS system.
 - (iv) enhancing the capacity to identify, plan and implement long-term climate resilient flood risk management strategies at the basin and sub-basin scale by introducing combined structural and non-structural methods including ecosystem-based adaptation (EbA) measures;
 - (v) promoting innovative EbA strategies and nature based FRM solutions, which enhance the environmental and ecological protective functions of the river basins.
 - (vi) enhancing long-term flood risk investment planning and enabling environment for climate risk financing to secure long-term sustainability of interventions and adequate operations and maintenance of the enhanced climate information and early warning services;
 - (vii) engaging private sector in climate information services and FRM activities and facilitating an increased use of climate risk information in their operations.

⁴ TNC for Bosnia and Herzegovina. See also Section 2.1 of the FS

⁵ Based on flood socio-economic risk modelling undertaken for this proposal (See Section 2.3, Table 2.6 of FS). In addition, the 2014 flood event impacted 25% of the population of BiH (PDNA)

⁶ Based on up-to-date flood event information recorded by HMLs and WA's in Bosnia. See Sections 2.2.1 and 2.2.2 (and tables 2.4 and 2.5.) of FS.

⁷ Ibid.

⁸ "Technology transfer for climate resilient flood management in Vrbas River basin" (Vrbas project, 2015-2020)

⁹ See Section 2.3 and Annex E of the FS.

¹⁰ The analysis of socio-economic impact of floods in B&H under current and climate change conditions, was carried out using the PFRA map and the Vrbas detailed EUFD flood mapping as the basis. In the Vrbas basin detailed climate change flood hydrological and hydraulic modelling resulted in the increased flood extent throughout the basin. In other basins (apart from Vrbas), flood plains were defined through PFRA (developed in 2010 and 2012) without hydrologic and hydraulic modelling and without flood probability of occurrence. Considering that Vrbas river Basin is located on 12% of the territory and borders with all other basins, it is assumed that the change in flood extent calculated for the Vrbas climate change scenario will be similar in other basins. Hence given the lack of hydraulic and hydrologic modelling for other basins, their CC scenarios were constructed by increasing flood zones defined in PFRA by the same % increase as Vrbas.

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

Climate Change Problem

Geophysical and Developmental Context

8. B&H is a sovereign state with a decentralized political and administrative structure consisting of two Entities: The Federation of Bosnia and Herzegovina (FB&H) and the Republic of Srpska (RS) and Brčko District. B&H is a middle-income country with an estimated 3.531 million inhabitants and total surface area of 51,209.2 km². Agriculture employs 19.1% of the workforce, while industry employs 32.2% and services 48.7% (2017 est.)¹¹. The 1992-1995 war has had a devastating impact on human, social and economic resources of the country, leading to enormous challenges of the post-war reconstruction and economic and social recovery. Above all, the war affected flood risk management capacities of the country through the direct damage and the lack of maintenance of flood control facilities and hydrometric monitoring stations, and minefields around some facilities (TNC, pg. 19). This challenge has been further compounded by the transition towards a market economy requiring structural reforms and improved governance.
9. There are two main river basin systems and seven major rivers in B&H¹² – Sava (Vrba, Una-Sana, Ukrina, Bosna and Drina) and Adriatic Sea Basin (Neretva and Trebisnjica rivers). The Sava is the largest river of the country and forms its northern natural border with Croatia. It drains 76% of the country's territory into the Danube¹³ and then the Black Sea. Sava river sub-basins are at highest risk of mainly fluvial, pluvial and torrential flooding¹⁴. The Una-Sana and Vrba are in the north-western region of B&H. The Bosna river gave its name to the country, and is the longest river fully contained within it, stretching through central Bosnia, from its source near Sarajevo to Sava in the north. The Drina flows through the eastern part of Bosnia, and for the most part it forms a natural border with Serbia. The Neretva, the largest karst river in the Dinaric Alps, is the major river of Herzegovina and the only major river that flows south, into the Adriatic Sea. Trebisnjica river is one of the longest sinking rivers in the world (90.5km of its 187km length flows underground). Both Neretva and Trebisnjica are heavily modified by HPP dams. These Adriatic Sea sub-basins face risk of flooding mainly from groundwater due to their karstic geology.

Observed and Projected Climate Change in B&H

10. All B&H's National Communications to UNFCCC (INC, SNC, TNC) have identified that climate change is affecting B&H and will accelerate during the remainder of the twenty-first century. According to the TNC, **meteorological observations** over the period of 1961–2014 demonstrated continuous rise of the mean annual temperature across the entire territory of B&H. Trends in annual temperatures on all analyzed stations were found to be statistically significant, while the changes are more pronounced in the continental part. The increase in annual air temperature ranges from 0.4 to 1.0°C, while the increase in temperature during the growing season (April– September) reaches 1.2°C. Analysis of air temperature during the period 1961–2014 found that all indices of warm temperature extremes have positive trends, while indices of cold temperature extremes have negative trends. The most significant change in this period was observed in the number of cold days (FD) and the number of warm days (SU). In all the meteorological stations the number of cold days (FD) has a negative trend. In the central mountain areas the number of cold days has decreased by 4 days per 10 years, while in the south of the country the decrease is slightly less and it ranges around 2 days per 10 years. The number of warm days (SU) has a positive trend and it is statistically significant.
11. In the period from 1961 to 2014 most of the B&H territory was characterized by a slight increase in the amount of observed annual rainfall (TNC). Analysis of historical records carried out for the FNC¹⁵ (draft) show that, compared

¹¹ UNDP Human Development Index Report (2018)

¹² Chapter 4 of the FS provides a detailed description of all major sub-basins

¹³ Bosnia and Herzegovina is a member of the International Commission for the Protection of the Danube River (ICPDR).

¹⁴ A pluvial, or surface water flood, is caused when heavy rainfall creates a flood event independent of an overflowing water body. There are two common types of pluvial flooding: 1) Intense rain saturates an urban drainage system. The system becomes overwhelmed and water flows out into streets and nearby structures and 2) Run-off or flowing water from rain falling on hillsides that are unable to absorb the water. In the Balkans and Southern Europe (e.g. Spain). Torrential flooding is the term used to describe a flashflood with significant erosive power and which (in combination with the readily available sediments) carries heavy debris/sediment load. The modelling of torrential flooding requires detailed erosion mapping in order to develop torrent susceptibility and hazard mapping which is used for strategic management of torrent flooding. With regard to hazard modeling and mapping of torrents, because they generally occur on small headwater catchments, the density of the rainfall observation network is important.

¹⁵ The Forth National Communication to UNFCCC is currently in an advanced state of preparation but not yet officially submitted.

to the reference period 1961-1990, there was an increase in annual maximum rainfall in the Sava Baisn of 16.4%, 22.55% and 22.5% for the periods of 1991-2010, 1991-2014 and 1991-2018 respectively (the largest flood was in 2014). In the Adriatic Basin, the annual maximum rainfall increased by 25.4% for all the above periods against the reference period. The records also show that changes in the amount of rainfall are more pronounced by seasons than annually. In the central part it is negative during spring and summer (the most pronounced in the area of Herzegovina – up to 20%), while during autumn the increase in rainfall was observed, especially in the north-western and central parts. The records show that significant floods in the last two decades have occurred in DJF (2010, 2019); MAM (2004, 2018, 2019); JJA (2014, 2019); SON (2014, 2015). Prior to this, floods were recorded in JJA (1975, 1976) and SON (1925, 1934). Hence floods have become more frequent in late winter/early spring and in late spring¹⁶. The results of **observed flow data trend analysis** (FNC) demonstrate strong changes in seasonal regime and extreme events, with the most significant change being the increase in runoff during the winter season.

12. Analyses (TNC, FNC) show that there has also been a reduced amount of snowfall in winter months due to increased daily winter temperatures. Hence, winter precipitation falling as rain instead of being stored as snow, is contributing to an increase in flood risk, which has been a contributing factor in many of the major flood events of the last decade. Furthermore, late spring snowfalls have been observed, which resulted in flooding in spring months when subjected to rapid melting due to large differences in daily temperature from one day to the next. These observed changes in the precipitation regime, including increasing maximum annual rainfall, decrease in winter snow fall, increase in spring snowfall, when combined with temperature increases, resulted in an increased likelihood of floods due to the increased runoff and an increasing intensity and frequency of rain events¹⁷. According to the TNC, the increased intensity and variability of rainfall and the increased share of heavy rains in the total amount of rainfall resulted in an increased risk of flooding especially in the north-eastern part of B&H, where the most disastrous floods in history were recorded during May 2014.¹⁸ All these changes in climate have been observed in the project target basins, but the most flood damages have been caused by the increases in rainfall intensity and frequency.
13. B&H TNC **projections**¹⁹ show that extreme precipitation will intensify under warmer climate conditions. Scenarios RCP8.5 and A2 anticipate an increase in annual precipitation for the time horizon 2011-2040 in relation to the reference period 1971–2000 (+5% for most of the B&H territory). Precipitation is projected to increase in winter, spring and autumn on greater part of B&H during 2011-2040, while summer precipitation is projected to decrease for all time horizons. Scenario A1B, A2 and RCP8.5, downscaled for B&H, also anticipate further increase in variability of the rainfall regime. The draft FNC confirms that projected changes in rainfall are complex with possibly both positive and negative changes relative to the reference period. For all four scenarios (RCP8.5, RCP 2.6, RCP 4.5, RCP6.0) the FNC projections show that possible changes are in the range from -5 to + 5% for the near future time horizon relative to the reference period values.
14. Whilst climate change projections from the TNC and FNC for the periods beyond mid-century show that some annual precipitation anomalies are negative in relation to the reference climate period, all scenarios show a positive change in rainfall amounts for the near future periods considered, which justifies the assumption that within the project climate change horizon (up to 2050), there will be an increase in rainfall amounts (annual and seasonal for all seasons except summer), contributing to increased flood risk. Furthermore, changes of indices of extreme

¹⁶ See Section 2.2.1 of FS

¹⁷ B&H SNC based on SPI analysis of precipitation and drought (Section 3.1.4) states: Six out of eight stations marked the last decade as a decade with the largest number of dry summers, which indicates that the last decade saw an increase in frequency of summer droughts. Nevertheless, if wet summers are taken into consideration, it is evident that five stations marked the last decade (independently or combined with another) as a period with the largest number of wet summers. Hence, in addition to dry summers, the last decade also saw an increase in frequency of wet summers, which contributed to the fact that in six stations the 2001- 2010 decade had the largest number of extreme summers in terms of precipitation quantity.

¹⁸ B&H TNC

¹⁹ Three climate scenarios for the territory of Bosnia and Herzegovina have been used in producing the TNC report), and these have been carried out with two different regional models, nonhydrostatic regional NMMB model and related hydrostatic regional model EBU-POM. Regional model NMMB is a nonhydrostatic atmospheric model, while Regional climate model EBU-POM is fully linked atmospheric and oceanic model (Djurdjevic and Rajkovic, 2008; Djurdjevic and Rajkovic, 2010). The atmospheric component of the model is Eta model and oceanic component is the Princeton Ocean Model (POM). With the NMMB Model the regionalization of climate scenario RCP8.5 was made (Moss et al., 2008), as defined in the fifth report of the Intergovernmental Panel on Climate Change (IPCC – AR5), while the EBU-POM model was used for regionalization of the scenarios A1B and A2 (Nakicenovic and Swart, 2000), as defined in the fourth report of the Intergovernmental Panel on Climate Change (IPCC– AR4). Horizontal resolution of the NMMB model was 8 km, and the resolution of EBU-POM model was 25 km. The selected reference period was the period from 1970 to 2000, while the future climate integrations covered the period from 2011 to 2100. For the boundary conditions in the integration of RCP8.5 scenario, the results of global climate model CMCC-CM were used (Scoccimarro et al. 2011), while for the boundary conditions for the scenarios A1B and A2 the results of the global climate model ECHAM5 were used (Roegner et al, 2003).

precipitation indicate that there might be an increase in the daily accumulations in the days with precipitation greater than 20 mm or greater than the 95th percentile. In some cases, the increase of the total precipitation during the days with extreme precipitation can result in a positive anomaly on significant parts of the territory, with a change of up to several dozen percentages for some seasons compared to the reference period. This situation is consistent with the fact that warmer air can carry a greater amount of water vapor, which, under favorable synoptic situations, primarily through convective processes, can be a source of abundant precipitation.

Climate Change Impacts

Historical flooding in B&H

15. The increased intensity, frequency and variability of rainfall due to climate change observed in the past decades resulted in more frequent, more intensive and more geographically extensive floods. Years with floods have become increasingly common. In the last two decades, Bosnia and Herzegovina were hit by several extreme floods, with 10 recorded major flood events (2001, 2004, 2006, 2009, 2010, 2014, 2017, 2018, 2019, 2020) compared to 8 major flood events in the preceding 75 years (1925 to 2000)²⁰. In April 2004 significant floods affected 48 municipalities (out of a total of 142) in the basins of Una, Vrbas, Bosna and Drina. About 20,000 hectares of agricultural land and 300,000 people were endangered by floods, of which several hundred families were evacuated. In December 2010 over 140,100 hectares of land in FB&H were flooded including approx. 91,360 hectares of agricultural land; 1261 landslides were recorded. In the Republic Srpska about 100,649 hectares of land were flooded, 48 houses were destroyed, 1513 objects were damaged. Moreover, 15 bridges were destroyed, 40 bridges were damaged, and 156 landslides were recorded²¹.
16. In May 2014 extreme floods occurred affecting B&H and the wider region with up to three times the normal May total and half the annual average rainfall falling in a 5 day period²². The country was hit with the most intense precipitation ever recorded since the beginning of the organized hydrometeorological observations, i.e. in the last 120 years. A combination of this prolonged extreme rainfall and snowmelt resulted in the extreme increase of water levels in an exceptionally short period of time, especially on rivers Bosna, Sava and Drina and their tributaries. On 17-18 May 2014, dams were breached on several sites along the Sava River and its main tributaries, causing floods and significant damage to property in the areas of Middle Posavina, Odžaka Posavina and Semberija. Additional damages to the basin were caused due to numerous landslides, some of which have completely transformed the environment. Flooding, erosion, flash floods and landslides marked 2014. After the May floods, further precipitation caused flooding again in July, August and September 2014, in areas already devastated by the May floods. Damages and losses incurred in critical sectors due to 2014 floods were as follows: Agriculture – EUR 187.2 Million; Education – EUR 8.7 Million; Energy – EUR 102 Million; Health – EUR 53 Million; Flood protection infrastructure – EUR 49.2 Million; Housing and household items – EUR 453. 2 Million; Public Services and Facilities – EUR 27.4 Million; Transportation and Communications – EUR 347.7 Million, Water and Sanitation – EUR 7.5 Million²³.
17. In the Adriatic basin in B&H the hydrological regime has been significantly altered with a multifunctional water management system of hydropower plants (HPP) dams. As a result, flooding has been less severe, although annual prolonged groundwater flooding occurs. In December 1999, a great flood wave on the Neretva River caused enormous damage in Mostar, but also in the entire lower course of Neretva. In 2004, when large river overflows were recorded in the Sava River basin in B&H, in the Neretva River basin particularly high-water levels were recorded with sporadic overflows.
18. The events of the last decades show that floods have been escalating in B&H due to strong regional precipitation, which has caused catastrophic floods. The flood risk has also been increasing due to mild winters with scarce snow but heavy and prolonged rainfall, or when there is abundant snow with extreme temperature oscillations. An analysis of the numbers of significant flood events and fatalities from 1925 to the present, shows that of the 40 total life losses recorded since 1925, 28 (70%) occurred since 2010 (see Section 2. 2 of the Feasibility Study). The 2014 event on its own, accounts for 26 deaths.

Flood risk under baseline and climate change conditions

19. Of the two main river basin systems in B&H – Sava (Vrbas, Una-Sana, Ukrina, Bosna and Drina rivers) and Adriatic Sea Basin (Neretva and Trebišnjica rivers) - Sava river basin is at highest risk of flooding, while Adriatic Sea Basin

²⁰ Section 2.2.1 of the FS

²¹ https://www.icpdr.org/main/sites/default/files/nodes/documents/icpdr_flood_report_2010.pdf

²² https://ec.europa.eu/fpi/sites/fpi/files/pdna/pdna_bih_2014_-_final_report_0.pdf

²³ https://ec.europa.eu/fpi/sites/fpi/files/pdna/pdna_bih_2014_-_final_report_0.pdf

faces a lower risk of flooding due to its karstic geology and heavily modified hydrology from hydropower dams. The main flooding sources in B&H are fluvial, pluvial, torrents and groundwater, with high flood risk identified in five sub-basins - the Una (in B&H 9,130 km²), Vrbas (6,386 km²), Bosna (10,457 km²), Neretva and Trebišnjica (10,110 km²).

20. Modelling of the socio-economic risk of flooding for baseline and climate change (2050) scenarios has been completed as part of the feasibility study for each of the five most vulnerable sub-basins (Section 2.3 of FS). In these basins under current conditions 799,445 people (22.5% of total B&H population) and 272,811 households (23.5% of total households) are at risk from floods. Under current conditions the expected damages to household assets is US\$ 3.2 Billion. Under climate change conditions, 902,906 people and 306,907 households will be affected, which is 25.6% of total B&H population or 26.6% of total households in B&H. The total expected damages to household assets under climate change come to US\$ 3.6 Billion. The Bosna River basin is the most vulnerable to flood risk under climate change with approx. 50% of total estimated damages or US\$ 1.824 Billion, followed by the Vrbas river basin (24% of total estimated damages) and the Una river basin (17% of total estimated damages). Current flood risk to agriculture is expected to increase from 12,896 ha and US\$ 3.34 Billion in damages to 15,064 ha and US\$ 3.7 Billion in damages under climate change. Agriculture in the Neretva River basin is expected to be most affected²⁴, with 35% of total modelled agricultural damage. With regard to the vulnerability of the public sector infrastructure (including health, education, judiciary, administration, social welfare, police and defense, cultural and historical facilities), 505 public sector buildings/facilities are exposed to flood risk, with estimated damages of US\$ 31.2 Billion. These estimates do not include economic losses from loss of business continuity, loss of income generation, loss of production, etc. A total of 2,825 business sector assets of US\$ 3.2 Billion in total value are currently at risk of flooding and would sustain US\$ 550 Million in asset damages in case of an extreme flood event, 57% of which would be in the Bosna River basin.
21. Based on the risk assessment under current and future climate change, the five highest risk basins in B&H – Bosna, Neretva, Trebišnjica, Una-Sana, and Vrbas – have been selected as the target basins for project intervention. In addition, Drina River basin (7,283 km² in B&H) where flood risk management studies (focused on optimizing HPP dam operations) have been undertaken by the World Bank, has also been identified as a target basin. The project interventions will complement the WB study. Overall, the project target basins comprise 85% of the total surface area of B&H.
22. It is clear that more frequent and intensive floods in B&H due to climate change will result in significant impacts on people, their property and critical infrastructure. In addition to climatic drivers, land use change such as deforestation is affecting watershed functions - drainage control, erosion processes and landslide susceptibility - that may increase exposure and aggravate consequences of flooding in vulnerable watersheds.

Social and Economic vulnerability

23. Climate-induced floods have a significant impact on the most important economic sectors including agriculture, water management, and hydropower, and limit B&H already low adaptive capacity. As outlined above, changes in annual rainfall distribution and increasing rainfall extremes have been driving increased frequency and intensity of floods and increasing economic damages from floods have been significant, especially in the water, agricultural and housing sectors. The total economic impact of the 2014 floods is estimated to have reached EUR 2.04 billion, or 15% of B&H's GDP in 2014²⁵.
24. **Agriculture** employs approximately 20% of the total work force and 46% of land use in B&H and is the most vulnerable sector to climate change. Agricultural losses due to floods can be substantial depending on the timing of the flood during the growing period (see Section 2.2, Annex E and Section 18.5 of FS). In addition, agricultural land suffers from pollution following floods and must be left uncultivated until rehabilitated. In torrent catchments, the sediment load and debris carried by flash floods to agricultural land result in significant losses and abandonment of productive land. Socio-economic risk modelling shows that in the most vulnerable sub-basins in B&H (Una-Sana, Vrbas, Bosna, in the Sava basin, and Neretva and Trebišnjica in the Adriatic Sea basin), a single extreme flood event would impact 12,896ha of agricultural land and result in \$3.33 Million USD in damages in the agricultural sector. This increases to 15,064ha and \$3.71 Million USD under climate change (2050).

²⁴ Section 2.3 of FS

²⁵ https://ec.europa.eu/fpi/sites/fpi/files/pdna/pdna_bih_2014_-_final_report_0.pdf

25. The **hydropower** potential of B&H is estimated at 8,000 megawatts (MW), with technical potential of 6,800 MW and economic potential of 5,800 MW, placing the country among the leading hydropower production nations in Southern Europe. Large hydropower plants are part of 3 existing power companies in B&H, which are joint stock companies, while small power companies are mainly private limited companies. More frequent and intense rain events lead to intensive runoff and increased peak river flows and could result in reduced power production and damage to hydropower infrastructure. During 2010 and 2014 floods, hydro power plants (HPPs) did not adequately adjust their discharge before the events, and as a result significantly aggravated the flood damage. The Adriatic Sea Basin is particularly heavily modified with many existing and planned hydropower dams and have already experienced an extensive variability in generation due to climate change. While generally designed and licensed to include flood alleviation functions, the HPPs in Adriatic Sea basin require improved management through **forecasting** to minimize climate change impacts on their operations (See Annex C of FS). In the Sava sub-basins in B&H with HPPs, there is a need to use enhanced climate risk information and consider climate induced flood risks in HPPs' operations. While increased flood risk could result in damage to HPPs the type and size of flood events that would result in infrastructural damage to HPPs is different to the flood events being considered for protection of population. Hence HPPs are included in the project in order to optimize their essential dual role in flood alleviation by optimizing their operations for generation and flood alleviation using climate risk information and forecasting. In addition, the risk from a failing dam, although of low probability, would result in significant damages to property and assets and loss of life downstream. In terms of economics, climate change could have a net positive impact in the near period on the level of average annual production for the baseline in the amount of 162.2 GWh, or an increase in average production of 7% per year resulting in an average annual additional income of € 11.1 million²⁶. However, the damages and losses that may result in the downstream area could be significant.
26. B&H has the highest proportion of **forests** and the largest variety of forest species in the Western Balkans. Approximately 20% of forest land is privately owned and 80% is state-owned. Between 1990 and 2005, Bosnia and Herzegovina lost 1.1% of its forest cover, or around 25,000 hectares. However, taking into account the total rate of habitat conversion (defined as change in forest area plus change in woodland area minus net plantation expansion) during 1990-2005, B&H lost 4.4% of its forest and woodland habitat. The increase in average temperature (3.4 to 3.8°C by end of the century) for A2 scenario will result in extremely different forest communities compared to the current communities. The assumed changes in forest ecosystems indicate that drastic changes would occur even in the mildest scenario (TNC). According to the TNC, climate change represents a threat for all four macro-regions in B&H (ecological-vegetation areas). The Dinarides Region, a very important and rich centre of endemic species in Balkans, will be under particular threat from both climate change and human activity such as deforestation and land degradation²⁷ which will result in shifting of water supplies and distribution of agricultural pests and diseases. Deforested and degraded catchments increase the vulnerability to flooding from extreme events and impact on people and the assets in river basins and sub-catchments (See FS Section 10.2). It also increases maintenance costs²⁸ of built infrastructure in such basins. The National Communications (INC, SNC, TNC) have suggested a number of mitigation measures including increase in forest productivity and maintaining or increasing the forest area through afforestation/reforestation and rehabilitation of bare lands²⁹. Mitigation measures are needed to address any climate-induced changes that will impact the structure and density of the forest cover, which will, in turn, greatly exacerbate flood risk in river basins. In terms of the institutional framework, there is a noticeable lack of integration of climate change and FRM considerations into forestry policies and strategies.
27. **Settlements** and development/economic activities in high-risk areas: One of the root causes of increasing vulnerability and damage from floods is increasing exposure. Spatial planning legislation and development control does not take account of flood risk nor do flood zones include flood resilient land use and construction rules. As a result, irreversible adverse land use practices in the floodplain and the building of houses and economic activity (including subsistence agriculture) in the floodplain, are common practices which significantly contribute to

²⁶ NAP project – Economic assessment of HPPs

²⁷ TNC page 16: The annual increment in the forests is relatively low, because so-called economic forests (forests that can be managed on an economic basis) cover only about 13,000 km² (approximately 25% of the BiH territory), and even these forests have low timber reserves (as low as 216 m³/ha with an incremental increase of timber of almost 5.5 m³/ha from half of the potential of the habitat). There are about 9,000 km² (approximately 17%) of low and degraded forests with a very low incremental increase (approximately 1 m³/ha) and with no economic value from the timber production perspective. Based on this increment, about 7,000,000 m³ per year was cut in BiH before the war

²⁸ Steep deforested land are the main sediment source areas for high energy torrential floods which carry a high percentage of debris, soil and mud. Such high energy torrential flood phenomena can damage existing infrastructure, river banks, stone/concrete embankments and increase annual maintenance costs of those objects

²⁹ IPARD 2014-2020 programme of the European Commission focus also points to afforestation and agroforestry, fire prevention and restoration after fire, and improving the resilience and environmental value of forest ecosystems are highlighted.

exposure. Lack of a regulatory framework, as well as limited awareness of the risks and continued uncontrolled adverse practices, will further exacerbate the problem, and will increase the vulnerability and costs from flooding. In addition, due to cheap land, some of the highly exposed flood risk areas have been inhabited by the most vulnerable populations, such as war refugees and displaced persons. Socio-economic risk modelling shows that in the most vulnerable sub-basins in B&H (Una-Sana, Vrbas, Bosna, in the Sava basin, and Neretva and Trebišnjica in the Adriatic Sea basin), 799,445 people and 272,811 households (23% of population) are currently at risk from floods (Section 2.3 and Annex E of FS). Under climate change, by 2050 the number of affected people and households will increase to 902,906 people and 306,907 households (26% of population). The total damage to household assets is expected to increase from US\$ 3.2 Billion (baseline) to US\$ 3.6 Billion (2050).

Institutional, Legislative and Policy Context

28. B&H is a sovereign state with a decentralized political and administrative structure. B&H consists of two Entities and one district: **The Federation of Bosnia and Herzegovina (FB&H)** and the **Republic of Srpska (RS)** and **Brčko District**. The Federation of Bosnia and Herzegovina is sub-divided into 10 Cantons. Decision-making involves the Council of Ministers, the two Entities and Brčko District. The two entities and Brčko District manage environmental issues through laws, regulations and standards. The State of Bosnia and Herzegovina as the central authority has only limited and specific powers with regard to the water sector and environmental protection: The Ministry of Foreign Trade and Economic Relations (MoFTER) has water-related competencies at the level of Bosnia and Herzegovina. Due to the lack of a State-level framework and the constitutional character of B&H and its entities, the current state of affairs is complex and heterogeneous, and the responsibilities for water management rest with the Entities (Ministry for Agriculture, Forestry and Water Management of RS and Ministry for Agriculture, Water Management and Forestry of FB&H). The B&H constitutional and legal framework does not contain specific and clear principles to guide the constitutive elements of the State in their management of shared water resources across entity or district borders. The State-level authorities therefore have no responsibility for regulating these inter-entity relations.
29. The two Entities and the Brčko District have relevant political, administrative and legal jurisdiction in their own territories, with some coordination and cooperation between them. Furthermore, the FB&H is divided into 10 Cantons which have their own authorities (ministries) with responsibilities in the water sector, including adoption of their own relevant laws. This complex administrative structure results in a number of different institutions in charge of water management issues and increases the need for coordination at the B&H level. The legal framework is not unified across the country and there are certain discrepancies in legislation between Entities (FB&H and RS) and even among Cantons within FB&H.

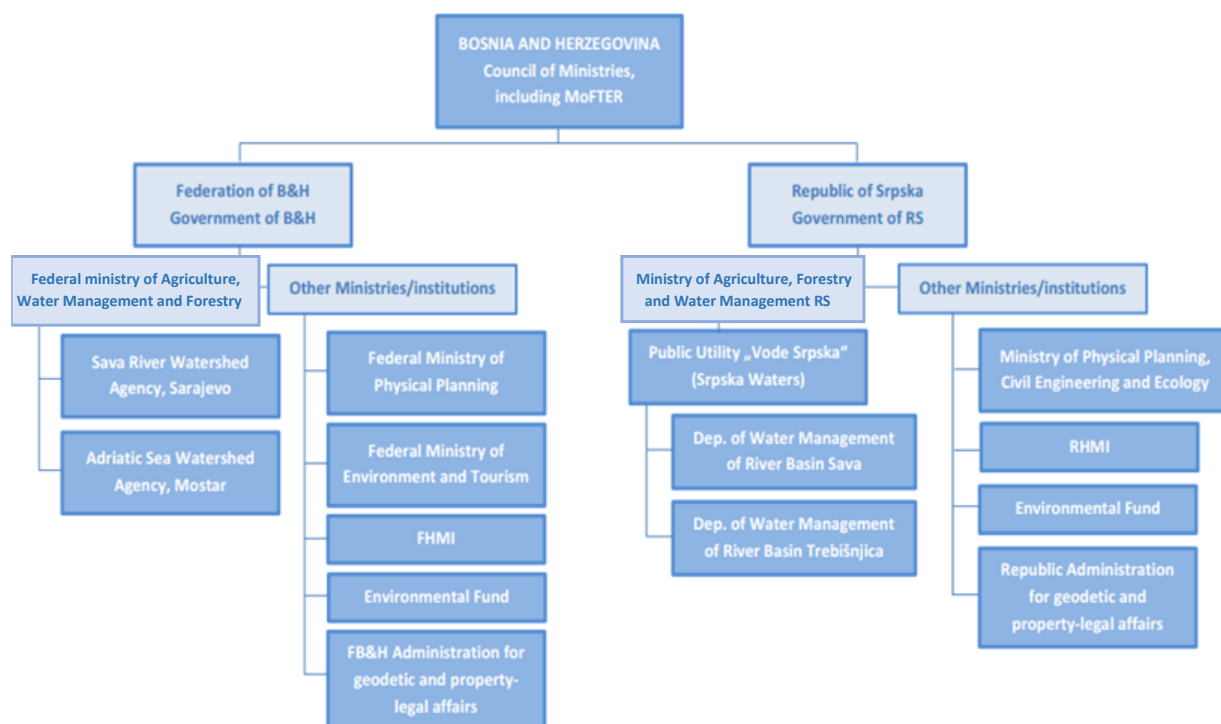


Figure B-1: Institutional framework for water management in B&H

30. The Law on Water of Republika Srpska was aligned with the **EU Floods Directive in 2017** to provide the legal basis for better flood risk management and defines the method of flood risk assessment and management. In FB&H the Draft Amendment of the Water Law was officially submitted to the FB&H Parliament at the end of 2017 and is not yet passed.
31. The **FB&H Environmental Protection Strategy 2022–2032** whose integral part is water management strategy and the **Republika Srpska Strategy for Integral Water Management (2014–2024)** both acknowledge climate change but do not identify concrete adaptation measures to address it. These entity strategies require coordination and harmonization in order to provide an aligned B&H-wide strategy.
32. **B&H's Climate Change Adaptation and Low Emission Development Strategy** (adopted in 2013) is of key importance to defining and planning adaptation action. As one of the worst impacts of climate changes in B&H the Strategy identifies more frequent and severe floods, including more frequent flood-related damage to HPP and damage to agriculture and forest eco-systems caused by changes in rainfall and weather extremes. The lack of reliable data on climate risks has been identified in the Strategy as a critical limiting factor for the adaptation action. The Strategy also identified the need for improvements in infrastructure (both flood protection and water storage infrastructure) and in the mechanisms for better water management. Hydrological modelling and flood mapping were identified as priority research projects. Strengthened hydro-meteorological monitoring, establishment of the early warning system, water information system, natural disaster data bases are identified as priority actions in the Strategy. However, its implementation has been slow mainly due to lack of knowledge and institutional capacity to identify and attract required finances and undertake adaptation measures. The Climate Change Adaptation and Low Emission Development Strategy was updated in 2020 and includes adaptation priorities for 2020 to 2030, with 4 expected outcomes. Outcome 1: Climate change risks, vulnerabilities and opportunities are reliably identified, quantified and effectively communicated, supporting evidence-based policy development; Outcome 2: An effective institutional and regulatory framework supporting adaptation, coupled with delivery capacity, is addressing climate change risks and opportunities across sectors; Outcome 3: Adaptation approaches are mainstreamed into decision-making, implemented and their objectives and planned delivery are understood by an informed society; Outcome 4: A well-resourced Adaptation Strategy being implemented on time, effectively, and delivering the expected outcomes.
33. In 2017, B&H submitted its **Third National Communication (TNC)** to the Conference of the Parties to the UNFCCC, which summarizes information on vulnerabilities to climate change, steps taken to adapt to climate change, measures related to public awareness and education, research and observation and technology transfer. TNC highlights that “due to the increased intensity of rainfall and its greater variability, as well as due to the increased share of heavy rains in the total amount of rainfall, there is the increased risk of flooding especially in the north-eastern part of B&H.” B&H's Climate Change Adaptation and Low Emission Development Strategy is of key importance to the national adaptation process. The 4th National Communication is currently under preparation.
34. The Government has initiated the development of the **National Adaptation Plan**, which should lay out specific activities with timeframes, indicative costs and implementing entities, necessary to combat climate change consequences in the most vulnerable sectors: water management, agriculture, forestry, biodiversity, hydro-energy, human health and tourism. It is anticipated that the National Adaptation Plan will be ready for UNFCCC submission by the end of 2020.
35. Key strategic documents in B&H including water management strategies and rural development strategies point out that climate changes affect their sectoral operations, but these documents do not propose measures to mitigate future risks or adapt to climate change. The water management institutional framework is also fragmented and lacks vertical and horizontal cooperation.
36. At the state level, the **system for civil protection** is headed by the Ministry of Security through the Sector for Protection and Rescue. It leads policy design. However, because of the decentralized nature of the system, the brunt of the responsibility for measures taken to deal with protection and rescue is placed on the entities and Brcko District. The Ministry of Security and the Coordinating Body of B&H (composed of the Council of Ministers, five representatives from the Federation of B&H, five from RS and two from Brcko District) take the leading role only in the event of a large-scale accident with trans-boundary effects or which cannot be resolved at the entity level. Local governments lack capacity and resources to fulfil even basic DRR functions. Civil protection centers are organized on municipal, cantonal and entity levels. There is an emergency warning center being organized on B&H (country) level, but it is still not in operation. Split responsibilities in the past have already led to artificially triggered floods

(e.g., 1999 floods in Neretva river basin were triggered by the erroneous opening of the gates on several dams as a result of the lack of coordination among various electricity agencies).

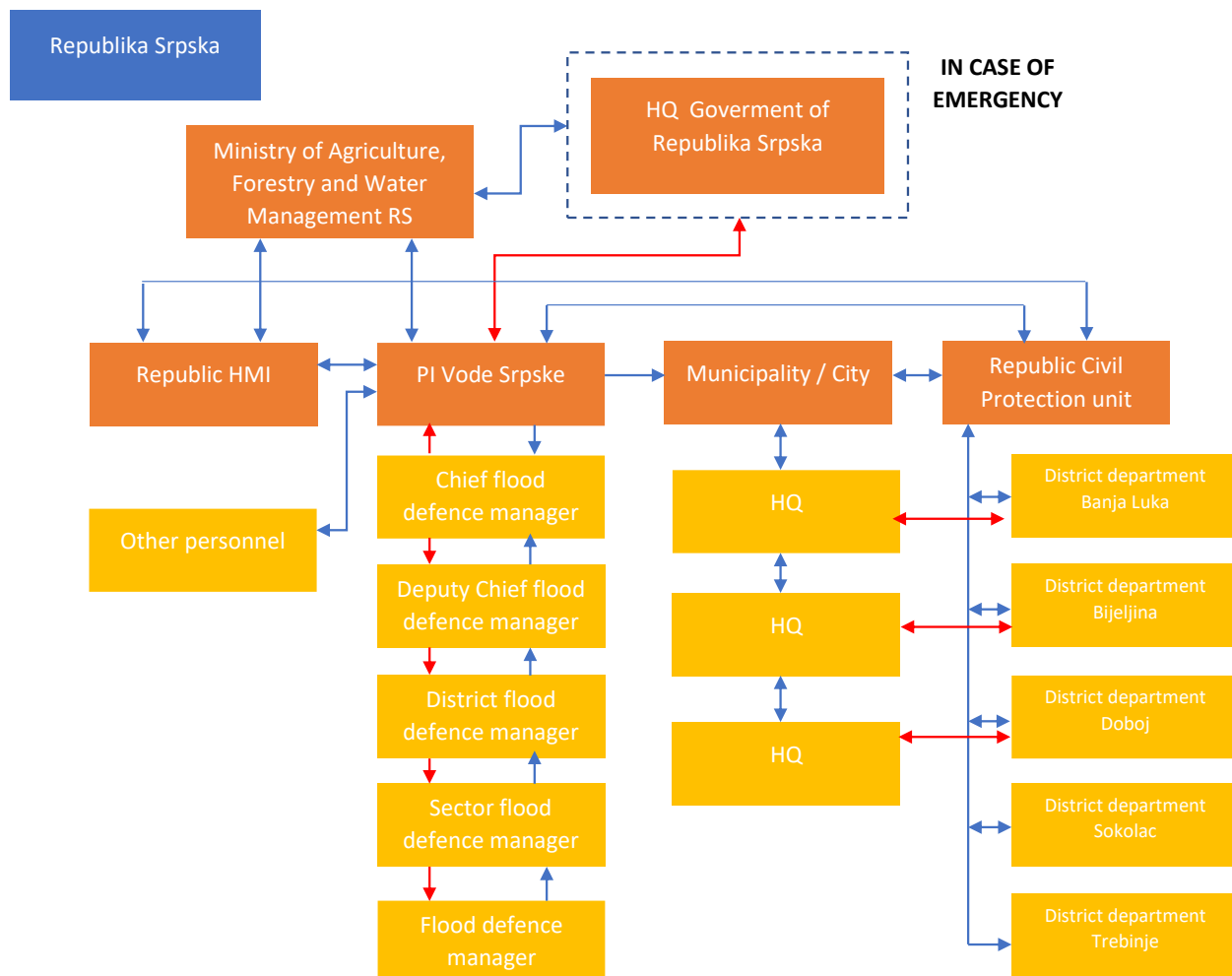


Figure B-2: Institutional arrangement for Civil Protection in Republika Srpska (RS)

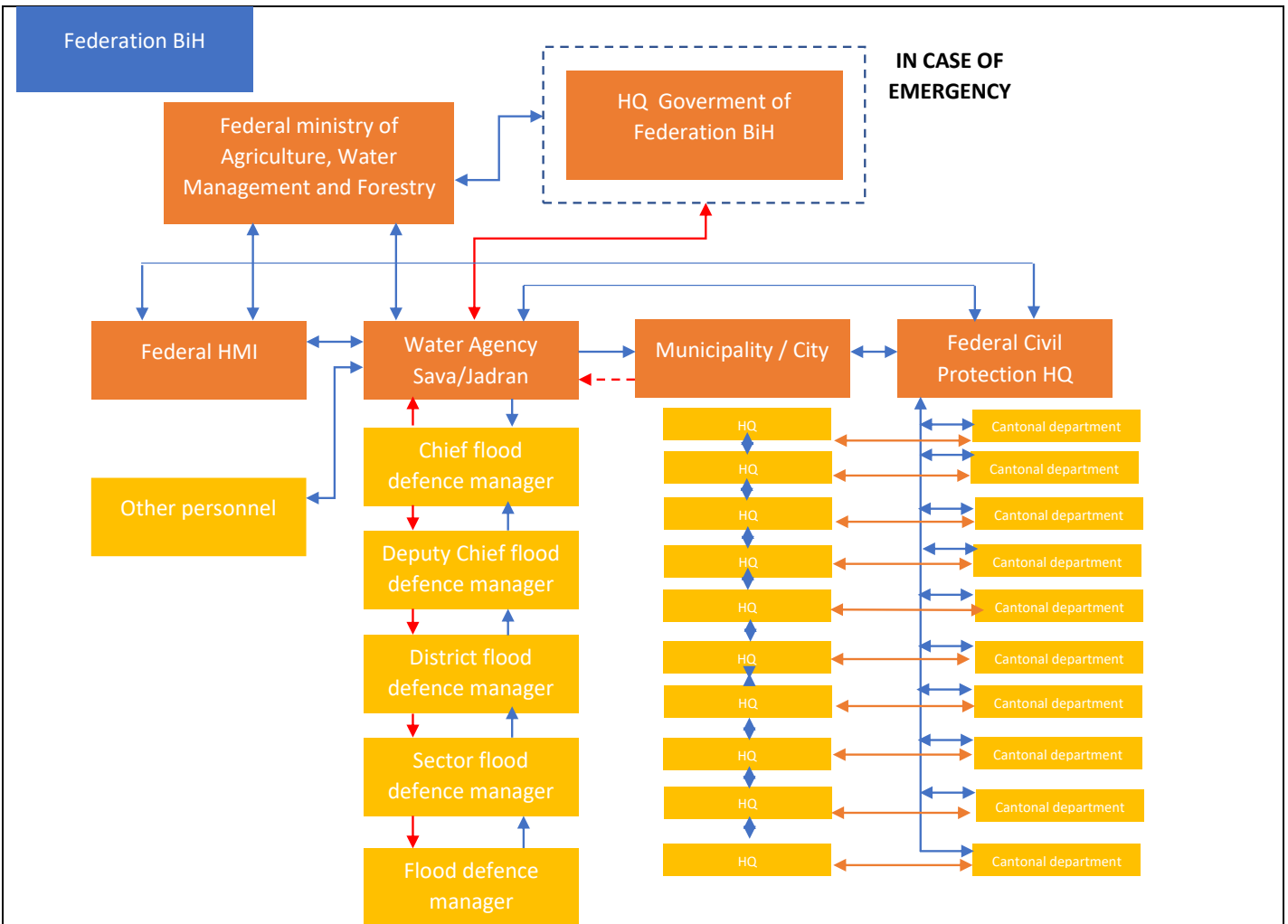


Figure B-3: Institutional arrangement for Civil Protection in Federation of Bosnia and Herzegovina (FB&H)

37. Following the devastating 2014 floods, B&H Council of Ministers adopted in November 2015 an **Action Plan for Flood Protection and River Management** for the period 2014-2017. The Action plan focuses on the following activities: recovery of the infrastructure damaged in 2014 floods, transposing of EU flood directive, establishment of the flood forecasting system, adoption of new technologies, capacity development and coordination among institutions. However, due to very low implementation of this Action plan, in Mar 2018, its implementation period was extended till the end of 2021.
38. The proposed project is aligned with the Climate Change Strategy and TNC. Project activities are also in line with B&H Action Plan for Flood Protection and River Management adopted in 2014 for the period 2014-2017, extended till 2021. B&H government has been in the process of updating its NDC since 2019 in order to include priorities on adaptation to climate change.

Baseline Investments

Government investment in flood risk management³⁰ is scarce and is mainly focused on ongoing maintenance of very old flood defense structures. The condition of flood control facilities is very poor as a result of wartime damage, many years without maintenance, and minefields around some facilities (TNC). The total annual budget for climate adaptation and resilience activities in B&H related to flood risk management averages US\$40 Million comprising mainly donor grants (40%) and loans (60%). After the 2014 floods, flood risk management has been undertaken through many donor-funded projects implemented by, or in partnership with, relevant ministries. The baseline projects are listed below and elaborated in the feasibility study.

³⁰ See Section 3.6 of FS

Baseline Projects

Chapter 7 of the FS provides a description of the baseline projects and detailed discussions of the baseline project being scaled up. It is summarized in the next paragraph.

39. **Technology Transfer for Climate Resilient Flood Risk Management** (Vrbas project), USD 5 million SCCF grant implemented by UNDP in 2014-2020: This project represents the key prototype/baseline project for the proposed GCF intervention. The SCCF funds were used to enable the communities of the Vrbas basin (12% of B&H) to adapt to flood risk through the transfer of adaptation technologies for climate resilient flood management including legislative and policy amendments, upgrade and rehabilitation of the hydrometric monitoring network and the harmonisation and centralisation of the hydrometric database, development of state-of-the-art hydrological and hydrodynamic models and GIS tools for the Vrbas River Basin incorporating climate change predictions and producing flood hazard maps as the basis for spatial planning and long-term strategic FRM, development of a flood forecasting and enhanced the existing early warning system within the VRB, development of emergency response plans and provision of training in flood-specific civil protection, implementation of non-structural measures, and capacity development of over 100 practitioners and decision makers. Further, an institutional capacity development plan for the long-term development of capability and capacity in FRM was developed. The project worked closely with affected communities to introduce climate resilient community-based non-structural measures and provided training to local communities in climate resilient FRM. The Vrbas project carried out a review of land use and **spatial planning regulations** and recommended revisions to strengthen the legislation that govern activities influencing flood hazards, to provide comprehensive floodplain management and spatial planning. A floodplain zoning policy was developed which focuses on the promotion of floodplain zoning as a means of efficient flood risk management, in order to retain the natural functions of the floodplain, minimize loss of life and property damage due to flooding. It uses the hazard maps to define flood zones and vulnerability categories. Under the Vrbas project a range of non-structural measures utilizing **eco-system-based** approaches, or interventions that encourage natural floodplain functions were successfully designed and implemented for the first time in B&H. The proposed project is scaling up the successfully implemented Vrbas project and extending its many achievements to the rest of B&H. Chapter 7 (Sections 7.2 to 7.11) of the feasibility study (FS) provides detailed discussion of the Vrbas project main interventions, all aspects of which are being scaled up to other basins in B&H.

40. Other relevant projects are presented in the table below with a discussion on the relevant lessons, synergies with the proposed project, best practices that can be adopted/scaled up and the gaps that can be filled by the proposed GCF interventions:

	Project name	Key outputs / Lessons learned	Synergies with GCF proposal	Best practices, upscale potential and gap filling
1	Emergency Flood Relief and Prevention Project, EUR 55 mil EIB loan (2012-2020)	The purpose of the project is to safeguard the agriculture, industrial and housing areas prone to flood impacts and to enable a stable basis for future development. The project focused on construction and rehabilitation of hard engineering structures, mainly along the Sava River. The project has also made an inventory of damages to flood protection infrastructure within the Sava river basin.	-flood risk management measures within Sava river basin	Upscale - project will use the experience of structural flood protection and building-back better to scale up approach to implementation of flood risk management measures. Gap filling –Climate hazard and risk modelling data was not available for the design of these protection measures. Through Project activities 2.2. (FS pg. 241-247) the Project will Implement and mainstream new ecosystem-based flood risk reduction and climate change adaptation methods, while through Activity 3.3 (FS pg. 260-263) the project will Implement climate-proof structural flood risk reduction and anti-erosion interventions in Vrbas, Una-Sana, Bosna and Drina river basins. Furthermore, the Project will develop the hazard and risk modelling which will provide the climate risk data for designing climate-proofed flood protection measures, and enable damage and loss calculations by sectors to identify strategic safeguarding measures of key sectors such as agriculture (Activity 1.2, FS pg. 199-202).

2	B&H Floods Emergency Recovery Project, USD 100 mil WB loan	The goal of this project is to restore functionality of infrastructure essential for public services and economic recovery after 2014 floods. The project was focused on: - emergency disaster recovery goods - rehabilitation of key public infrastructure - capacity building.	-structural and non- structural measures within Sava river basin -capacity building for flood risk management	Upscale - The Project will build on lessons learnt in flood recovery using non-structural measures approach and use the experience and lessons learnt on capacity building. Gap filling – Climate hazard and risk modelling data was not available for the design of restoration measures for this critical infrastructure. The Project will develop the hazard and risk modelling which will provide the climate risk data for designing climate-proofed critical infrastructure (Activity 1.2). The Project will develop risk mapping, forecasting and climate information required to climate-proof future structural investments and risk reduction. The Project is developing floodplain zoning policy and legislation which will ensure that critical infrastructure with vulnerable use will be zoned away from high risk areas (Sub-Activity 2.1.2, FS pg. 235-239), and where critical infrastructure has to be placed in high risk areas, the project is embedding the use of climate risk information in the design of new and the rehabilitation of old critical infrastructure (Sub-Activity 2.1.3, FS pg. 239-240) to ensure flood resilience. Through sub activity 1.3.4 (FS pg. 220-221); project will develop and implement a capacity development plan for embedding flood hazard and risk modelling approaches and FFEWS into appropriate institutions in B&H. Through Activity 1.4 (FS pg. 221-228) project will develop and implement protocols and SoPs on data generation, data management and communication for effective impact based FFEWS and flood risk management. Through Project activities 2.2 (FS. Pg. 241-247) project will implement and mainstream new ecosystem-based flood risk reduction and climate change adaptation methods (which improve eco-system function of catchment, thus reducing flood risk), while through Activity 3.3 (FS pg. 260-263) project will implement climate-proof structural flood risk reduction and anti-erosion interventions in Vrbas, Una-Sana, Bosna and Drina river basins.
3	Advance the National Adaptation Plan (NAP) process for medium-term investment planning in climate sensitive sectors in Bosnia-Herzegovina (2018-2021)	This GCF funded project is advancing adaptation planning in B&H with a focus on sectoral approaches, upgrading the knowledge base for adaptation, prioritizing adaptation interventions for the medium term, building institutional capacities for integrating climate change adaptation and demonstrating innovative ways of financing adaptation at the sub-national/local government level. Key outputs will include:	- enhancement of climate data exchange - analyses of CC impact on HPP and agriculture	The proposed GCF investment will work in coordination with NAP project and facilitate the integration of CC adaptation and flood risk management at national and sub-national levels. The NAP project will help develop data sharing protocols for data/ information exchange between water and other sectors relevant for integrated water management and climate change (agriculture, forestry, environment, spatial planning, and hydro-energy sectors). The NAP project will also study economic impacts of climate change on the hydropower sector in Republika Srpska, and identify additional monitoring needs to support improved HPP system operations and will provide essential input observation data for the more detailed model of the basin. Knowledge gained on CC impact on HPP electricity production and agriculture will be of help for modelling and early warning system activities. Monitoring and evaluation system will set a base for data exchange protocols.

		- monitoring and evaluation system for climate change adaptation - climate change data management system accessible to all stakeholders - vulnerability assessments for key priority sectors - municipal level investment financing instruments with public/private sector engagement in selected municipalities - building national, sub-national and sectoral capacity to integrate and mainstream risk informed planning and budgeting.		Upscale – Through Activity 1.2 (FS pg. 199-202) the Project will develop hazard and risk models and maps. In particular, the methods for incorporating improved HPP operations, as well as economic studies on CC impacts will be scaled up. Through Activity 1.4 (FS pg. 221-228) project will develop and implement protocols and SoPs on data generation, data management and communication for effective impact based FFEWS and flood risk management. Gap filling - Through Activity 3.1 (FS. pg. 253-258) project will develop investment framework and budgeting mechanisms for climate induced flood risk reduction and management Through sub-activity 3.2.1 (FS pg. 259) project will develop municipal investment plan for climate resilient FRM planning for 10-12 highest risk communities and 1 canton.
4	UN Floods Recovery Programme "Danas Za Nas", USD 22.6 mln financed by 28 bilateral donors	This project was to reestablish normal living conditions in more than 60 communities most affected by 2014 floods via infrastructure rehabilitation, short-term employment opportunities, restoring water, sanitation and sewage facilities, education of citizens etc.	- structural and non -structural measures - education and public awareness	Upscale - The Project will use the experience and lessons learned on reduction of flood risk using non-structural approach and will build on the public awareness activities. Through Activity 1.4 (FS pg. 221-228) project will develop and implement protocols and SoPs on data generation, data management and communication for effective impact based FFEWS and flood risk management. Gap filling - Under Activity 2.1.2 (FS pg. 235-238) floodplain zoning will be developed to zone future development away from high risk areas while under 2.1.3 (FS pg. 239-240) climate proofing methods, standards and guidelines will be developed for critical infrastructure. Through Project activities 2.2. (FS. pg. 241-247) project will Implement and mainstream new ecosystem-based flood risk reduction and climate change adaptation methods, while through Activity 3.3 (FS pg. 260-263) project will Implement climate-proof structural flood risk reduction and anti-erosion interventions in Vrbas, Una-Sana, Bosna and Drina river basins.

5	EU flood recovery program, EUR 43.52 mln, implemented in 2014-2016 by UNDP, UNICEF and IOM	This Programme contributed to the recovery after the 2014 flood and included rehabilitation of 4,000 dwellings for approximately 14,000 people, 100 local roads and bridges, 90 educational institutions (including pre-school facilities), 10 water and sanitation facilities, three municipality buildings, four Centres for social welfare, and four healthcare facilities. The Program recognized the importance of investing in future risk informed decision making and it initiated the development of a Flood and Landslide Risk Assessment for the Housing Sector in B&H. The Assessment focuses on the flood and landslide risk for the housing sector in Bosnia and Herzegovina, prioritizes locations based on risk ranking and makes recommendations for risk reduction	- structural and non-structural measures -capacity building	Upscale - The GCF investment will use the experience and lessons learned on reduction of flood risk using structural and non-structural approach and will build upon the approach of risk informed decision making. Gap filling - Through activity 1.2 (FS pg. 199-202) project will enhance climate-induced flood hazard, risk and vulnerability information for strategic management and sound decision making for climate induced flood risk management. Under Activity 2.1.2 (FS pg. 235-238) floodplain zoning will be developed to zone future development away from high-risk areas while under 2.1.3 (FS pg. 239-240) climate proofing methods, standards and guidelines will be developed for critical infrastructure. Furthermore, through activities 2.2. (FS pg. 241-247)) project will Implement and mainstream new ecosystem-based flood risk reduction and climate change adaptation methods, while through activity 3.3 project will Implement climate-proof structural flood risk reduction and anti-erosion interventions in Vrbas, Una-Sana, Bosna and Drina River basins.
6	West Balkans Drina River Basin Management Project, USD 8.74 mln funded by GEF/SCCF and implemented by the World Bank (2017-2020)	This regional project improved planning and implementation for integrated, cooperative management of the trans-boundary Drina River basin via capacity development, public awareness campaign, small grant schemes. It strengthened regional network of hydro-meteorological stations and implemented pilot investments for increased climate change resilience to floods and droughts.	-integrated river basin management approach -flood risk management - capacity building - public awareness raising	Upscale - The proposed GCF intervention will use the experience on integrated water management approach with multiple stakeholders. Build up on strengthening of Hydro-meteorological institutes and awareness raising on flood risk management. Through activity 1.1 (FS pg. 192-198) project will upgrade and expand the coverage of the hydrometric network for enhanced monitoring of climate variables in Category I catchments and torrential streams.
7	Flood Hazard and Flood Risk Maps of B&H, WBIF	The purpose of the project was to develop flood hazard and flood risk maps for all river basins in B&H except Vrbas. Started in 2017, after almost two years delay due to delay in	-flood hazard and risk mapping	Upscale - Use the experience and lessons learned on reduction of flood risk by enhancing existing flood risk and flood hazard maps. Learn from the project's constraints and enhance stakeholders' engagement during the project implementation.

		finalization of Terms of Reference for map development, when they agreed to use Vrbas flood mapping methodology. Even though methodology for maps were good, developed maps have been rejected by stakeholders several times requesting further improvements, due to insufficient level of details, as well as incorrect overlapping of flood lines with settlements boundaries, therefore incorrect risk. The above issues are result of insufficient stakeholders' involvement and overreliance on external experts.		Gap filling - Through activity 1.2 (FS pg. 199-202) project will enhance climate-induced flood hazard, risk and vulnerability information for strategic management and sound decision making for climate induced flood risk management. The GCF will enhance the developed WBIF hazard models and maps by: 1) Extending the WBIF models to ensure whole basins approach and not just AFAS modelling 2) Modelling Category II tributaries and not only main river. 3) Modelling torrents flooding mechanisms 4) Enhanced hydrological modelling 5) Inclusion of reservoir operations and flooding from potential dam failure
8	Capacity Development for the Integration of Global Environmental Commitments into National Policies and Development Decision Making, USD 1.5 mln, GEF	The project facilitated cross-sectoral and participatory approaches to natural resource management planning and implementation, including developing individual and institutional capacities to better adapt and apply global environmental management indicators as a monitoring tool to assess the intervention performance and institutional sustainability. The project focused on environmental data exchange and capacity building for environmental reporting. It developed Environmental Management Information System, which has never been used in practice by local institutions. The reason for that is that the Project never identified exact data and channels for their exchange.	- improvement of climate change data exchange	Upscale - Use the experience and lessons learnt to develop clear protocols on data exchange. Through Activity 1.4 (FS pg. 221-228) project will develop and implement protocols and SoPs on data generation, data management and communication for effective impact based FFEWS and flood risk management.

41. As shown above, flood risk management in B&H has been undertaken mainly through donor-funded projects implemented by or in partnership with relevant ministries. Most baseline projects have been focused on post-event recovery and do not take an integrated ex-ante approach nor do they consider climate change impacts. Furthermore, many of the baseline projects were limited in scope and/or geographical extent and budget, which in turn, limited their transformational potential. The proposed Project is therefore scaling up the capacities established by the Vrbas project (described in para 38 above and in detail in Sections 7.2 to 7.11 of the FS), and will learn

lessons and scale up interventions from other baseline projects to implement a nation-wide integrated proactive risk informed and climate-responsive approach to flood risk management. It will develop and introduce new technologies and innovations such as Impact-based forecasting (IBF), scaled up forecast-based financing (FbF), blended risk financing and risk transfer mechanisms (such as IFI loans, state funds, commercial loans, private finance leveraged through the GCF intervention), establish key public-private sector partnerships to leverage and scale up investments, and will embed methods for, and implement climate resilient structural measures as well as eco-system based non-structural measures. This Project will represent the first time that investment will be made available at the scale that ensures nation-wide transformation in Flood Risk Management.

Barriers to climate resilient flood risk management in B&H

1. Limited institutional capacity (technical, legislative, and financial) for nation-wide Flood Forecasting and Early Warning System (FFEWS):

42. *Gaps in the hydrometric observation network and capacities:* Financial and human resource constraints coupled with a severely reduced monitoring network since the 1992-1995 war limit B&H's ability to monitor important hydrometeorological variables at the appropriate spatial and temporal scales, to provide adequate input to strategic flood risk management, and to implement an effective national FFEWS.
43. *Gaps in flood hazard and risk modelling and mapping capacities:* There is, as yet no comprehensive flood hazard, risk or vulnerability mapping for B&H. Ongoing projects undertaking EU Flood Directive (EUFD) mapping are severely delayed and will only be done for selected high-risk areas within river basins. Technical capacities related to hazard and risk identification, assessment, modelling and mapping, risk prevention/reduction and risk transfer are weak across institutions and governance levels. Hence climate risk information for effective risk management is lacking.
44. *Gaps in FFEWS capacities:* FFEWSs for most of the main watercourses are under development, but these systems do not cover tributaries or torrents. Existing FFEWSs are not integrated into a country-wide system, and there is limited relevant institutional capacities and community-based approaches. Interagency communications during and after natural disasters are cumbersome and inefficient. The regulatory framework for FFEWS is incomplete as it lacks protocols/SOPs on climate data generation, data management and communication. There is no national protocol on early warning system. The lack of horizontal (between entities and among sectors within the same entity) and vertical (among state, entity, cantonal and municipal levels) communication within the water management sector further limits effective cross-sectoral climate risk management.

2. Lack of a strategic integrated FRM approach, insufficient technical knowledge, and capacities on ecosystem-based and non-structural approaches to climate resilient flood risk reduction:

45. There is a lack of strategic integrated flood risk management approaches in B&H that identify the best combination of structural and non-structural measures to address climate induced risks. Flood protection currently relies on limited structural measures that are becoming increasingly ineffective in terms of the Standards of Protection they provide in the face of climate change, have high cost, can sometimes result in the problem being moved elsewhere, lack basin-scale appraisal-led approach to the identification of flood defense needs, and lack detailed assessment of risk and consequence of failure. Due to the predominant use of traditional engineered structural measures, inherited from a pre-war communist system, there is insufficient knowledge about non-structural measures and limited capacity to design and implement them. Hence there is currently no strategic integrated flood risk management approach in B&H that identifies the best combination of structural and non-structural measures to address climate induced flood risk. There is a lack of knowledge and application of non-structural measures and ecosystem-based approaches (EbA) to flood risk management, such as the use of agro-forestry, community-based early warning systems, floodplain zoning, reforestation, and financial instruments such as flood disaster insurance and other risk transfer mechanisms as a means of minimizing the impact of flooding on communities. There is no sector specific flood risk management which embeds gender responsive climate resilience measures into sectoral development plans. The existing standards and guidance for the design and implementation of all categories of critical infrastructure, do not currently include climate change considerations nor are they based on the most-up-to-date climate risk information or methods for identification, prioritization, design and implementation of climate proofed infrastructure. Activities within key sectors such as water management, energy, transport, agriculture, forestry, spatial planning, are not climate risk informed.

3. a) Outdated non-climate-risk informed approach to flood defenses design and b) lack of capacity, knowledge and information for enhanced climate-proof public and private investments into flood risk reduction and community response:

46. At the municipal and community levels, there is a lack of knowledge, tools, and investment to systematically respond and build resilience to flood risk. The population, in particular in rural areas, is lacking general understanding of climate induced risks and community response capacities. There is limited knowledge and capacities among local communities to apply climate resilient livelihoods and adaptive strategies for coping with climate-induced hazards. Limited awareness, and response and resilience capacities of communities is due to a lack of climate risk information for building such awareness and capacities.
47. There is a lack of financial resilience of communities due to a lack of climate risk-informed risk transfer instruments such as flood and disaster insurance, or similar financial instruments. There is a lack of engagement with private and productive sectors in climate responsive FRM. In order to ensure sustainability of the FRM measures, particularly related to the costs of operation and maintenance of equipment, information provision services, regular maintenance of flood defenses, and all other flood risk management activities, there needs to be an engagement and willingness to pay for such, from a wide range of beneficiaries including private sector, public sector, and the public at large. Information on the interest in, and willingness to pay is currently lacking, as are the mechanisms, instruments, and tools to engage the private sector in climate risk financing.
48. Insufficient, outdated and poorly planned and managed flood defenses increase exposure of communities, to increasing flood risk. Investment in flood defenses for the protection of communities is done in a reactive manner and as budgets allow. The limited annual budgets are used to address urgent repairs to existing defenses, many of which have long exceeded their design life due to climate change and lack of maintenance. Flood defense design and implementation does not currently take a strategic approach (e.g., river basin approach) nor does it systematically take account of climate change. There is also a lack of technical and financial capacity to design, operate and maintain climate resilient flood protection infrastructure. The Post-Disaster Rapid Needs Assessment (PDRNA) following the 2014 floods revealed that US\$49.2 Million in damages to flood protection and control infrastructure was incurred. In addition, damages and losses forecasting and accounting, and cost-benefit analysis is not systematically used to inform and develop municipal investment plans, nor to undertake appraisal-led FRM analysis, due to the lack of methods tools, and guidelines. Hence local level FRM investment planning is currently inefficient and ineffective.

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

49. The Project has been developed to address the barriers to integrated climate-informed flood risk management and is aimed at supporting the commitment of the B&H government to avoid loss of lives and to reduce economic and infrastructure losses caused by climate-induced flood disasters.
50. In the **baseline scenario**, without the Project the lack of technical, institutional, and financial capacities to implement and maintain a fully integrated flood risk management framework will continue to result in limited (in terms of accuracy, skill and spatial extent) flood warnings not reaching the communities who are most affected by flood hazards, in a timely manner. The absence of comprehensive and definitive flood risk information and legislative and policy framework will continue to exacerbate weak land use, spatial planning and climate risk management, leading to increased exposure of communities to damages, losses and loss of lives. In addition, a lack of institutional and financial capacities and modern technologies will prevent the design of climate risk informed mitigation measures. The large proportion (26%) of the population at risk from flood hazards (902,906) will continue without coping capacities and adaptation strategies to adapt to climate change and the capacity to manage and minimize their exposure and resilience to intensifying flood hazards. Hence, in the baseline scenario without GCF investment, the B&H population and economy will face increasing pressures from more frequent and severe climate change induced flooding, including losses of lives and economic losses due to climate change-induced flood disasters and associated impacts on GDP. In addition, a continued lack of financial resilience will further limit recovery from more frequent flooding, thus keeping the affected population in a cycle of loss and diminishing recovery capacity, eroding and limiting further sustainable development gains. Scaling-up of tested integrated FRM approaches will not be possible due to the existing financial gap between DRR needs and required investments, as well as an underdeveloped national capacity for flood hazard and risk knowledge.
51. In the **adaptation alternative**, this Project will support the commitment of the B&H government to avoid losses of lives and to reduce economic and infrastructure losses caused by climate-induced flood disasters by implementing a Theory of Change described below. The project will scale-up technological, institutional and financial innovation

in flood risk management through nation-wide implementation of fully integrated FFEWS based on state-of-the-art modeling technologies, mainstreaming of flood risk into key sectoral policies, mainstreaming nature-based solutions to flood risk management into FRM policy and practice, engaging private sector, promoting new risk financing tools, embedding risk-informed municipal investment planning approaches and implementation of priority flood risk reduction measures in the highest risk communities.

Theory of Change

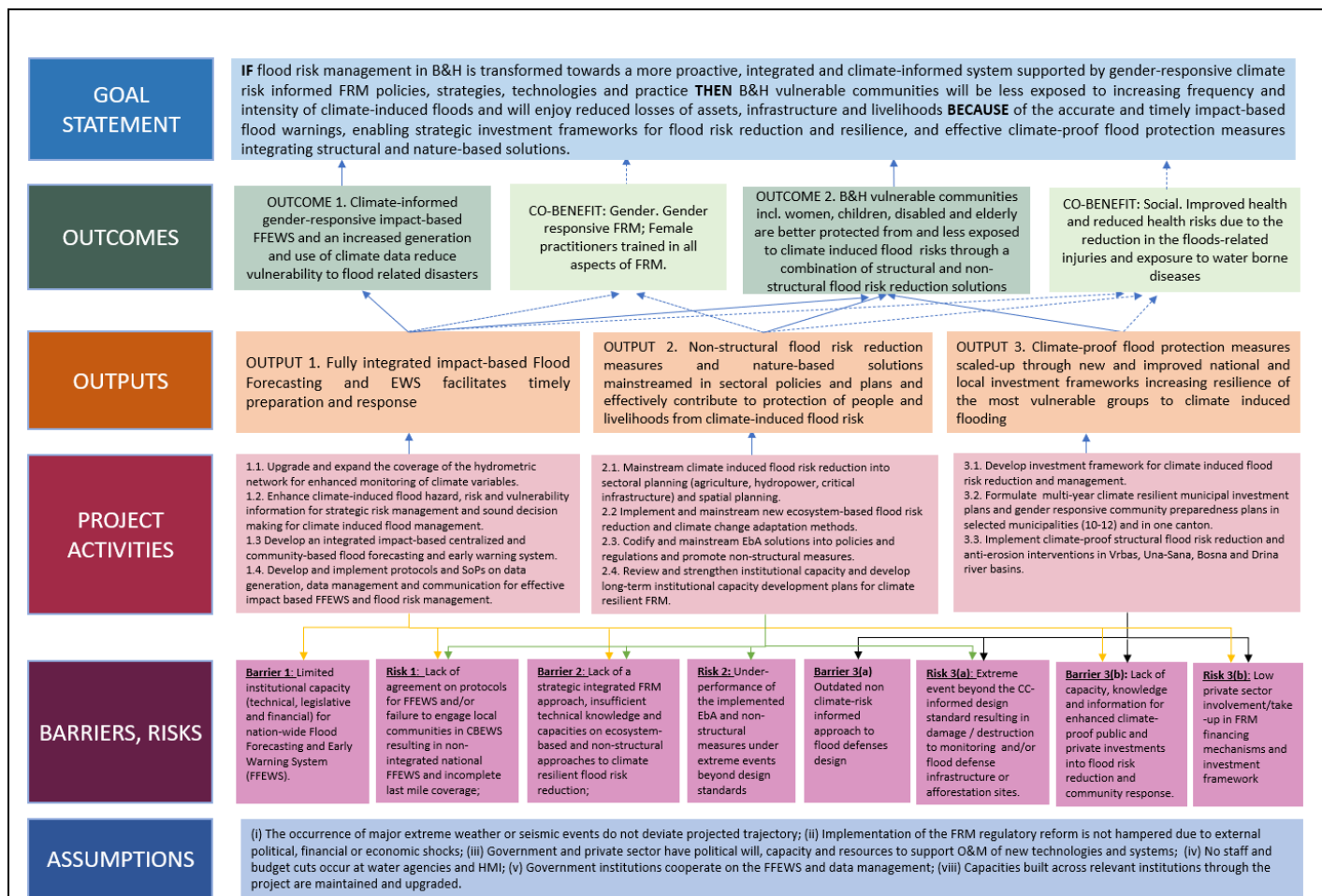
52. Project goal statement: **IF** flood risk management in B&H is transformed towards a more proactive, integrated and climate-informed system supported by gender responsive climate risk informed FRM policies, strategies, technologies and practice **THEN** vulnerable communities in B&H will be less exposed to increasing frequency and intensity of climate-induced floods and will face reduced loss of assets, infrastructure and livelihoods **BECAUSE** of access to accurate and timely impact-based flood warnings, and effective climate-proof flood protection measures integrating structural and nature-based solutions supported with long-term investment frameworks.
53. As outlined in the goal statement, the project will catalyze a **paradigm shift** in flood risk management in B&H by transforming the existing FRM practices and shifting from an ex-post facto i.e. reactionary post-disaster recovery approach to ex-ante i.e. risk informed preparedness approach based on adequate climate information and risk knowledge, as well as on climate-informed FRM policies, strategies, technologies and investment. These elements of the paradigm shifts are further described in Section D.2 below.
54. To achieve the GCF paradigm shift and deliver the project goal, the following two high-level Outcomes are planned: **Outcome 1:** Climate informed gender-responsive impact-based FFEWS and increased generation and use of climate data reduce vulnerability to flood related disasters; and **Outcome 2:** Vulnerable communities in B&H, including women, children, disabled, and elderly, are better protected from and less exposed to climate induced flood risks through a combination of structural and non-structural flood risk reduction solutions. Through these outcomes, the project's paradigm shift will be pursued by embedding innovation and technology transfer through scaling up of state-of-the-art technologies in floods monitoring, modelling, impact-based forecasting and early warning, including "last mile" communication (Outcome 1), as well as by identification, selection and implementation of cost-effective EbA measures to complement traditional structural solutions, mainstreaming FRM into sectoral policies and introducing innovative risk financing and transfer mechanism as well as investment planning (Outcome 2). Through its Outcome 1, the project will also contribute to the implementation of the new UN initiative EWS4All announced in 2022 with the aim of promoting universal coverage of early warning.
55. While pursuing these Outcomes, the project will also leverage several important **social co-benefits**. Co-benefits for health and well-being will include improved overall health and reduced health risks within communities due to the reduction in the number of flood-related injuries and a reduction in the exposure to water borne diseases. The health co-benefits will occur primarily for the vulnerable B&H communities currently lacking access to sanitation and drinking water: 59% of the population in B&H are currently not connected to sewerage systems and 25% are without access to drinking water, and therefore at a higher risk of suffering from illness due to water borne diseases that arise due to flooding. In addition, the project will deliver substantive gender co-benefits as it will train and empower female practitioners in all aspects of FRM and establish gender-responsive risk assessments and FRM framework. The gender-responsive system of data collection and modelling of vulnerability to floods will underpin the socio-economic risk model which will guide the identification and prioritization of risk management, planning, preparedness and response measures in order to ensure that the most vulnerable groups are prioritized.
56. The outcome-level change will be pursued through three interconnected Outputs to be delivered by the Project.
Output 1: Fully integrated impact-based Flood Forecasting and EWS facilitates timely preparation and response;
Output 2: Non-structural flood risk reduction measures and nature-based solutions mainstreamed in sectoral policies and plans and effectively contribute to protection of people and livelihoods from climate-induced flood risk;
Output 3: Climate-proof flood protection measures scaled-up through new and improved national and local investment frameworks increasing resilience of the most vulnerable groups to climate induced flooding.
57. The project's **Output 1** will contribute to the delivery of Outcome 1 by enhancing the capacity for flood forecasting and early warning through the development and implementation of a nationwide sustainable impact based FFEWS which will provide improved warning times and reduced losses. This path has been chosen by the project among the other outputs based on the analysis of successful comparable efforts (EWS, climate information, and community-based DRM) which have shown effective impact related to saving of lives, assets, and livelihoods. This output will be achieved through a set of four interconnected and sequenced **activities** required to deliver a

functional integrated impact-based FFEWS. First, the Project will significantly **improve the hydrometeorological network** which was damaged during the 1992-1995 war and currently cannot not provide adequate spatial and temporal coverage for an effective impact based FFEWS. Across all river basins the project will expand the hydrometeorological monitoring network to cover a greater range of hydrometric variables that climate change is bringing to enable effective monitoring, management, forecasting and early warning of all flooding process including pluvial and fluvial flooding, groundwater flooding, floods from dam operations, and torrential floods (**Activity 1.1**). Further, the project will address the lack of **comprehensive hazard and risk mapping of flood risk and vulnerability** by enhancing technical capacities related to hazard and risk identification, assessment, modelling and mapping. The project will develop hydrological modelling and extend hydraulic modelling for all main basins, undertake hazard modelling for torrents, include HPPs and groundwater modelling in flood hazard, risk and vulnerability modelling and mapping for all relevant basins, and develop a GIS-based socio-economic risk modelling (**Activity 1.2**). To address the lack of a country-wide flood forecasting and early warning system and lack of comprehensive, timely, specific and actionable flood warnings to communities and high-risk sectors, the project will implement a **countrywide FFEWS based on an impact-based forecasting** approach and following the WMO standards, the FFEWS will include provisions for effective “last-mile” communication and the community-based EWSs for the highest risk communities (**Activity 1.3**). Finally, the project will strengthen climate risk information sharing and institutional coordination and communication between different agencies, through **SOPs, Communication Protocols and Codes of Conduct** to be developed for each of the institutions responsible for impact based FFEWS (**Activity 1.4**). Implementation of the Output 1 will address barriers preventing the effective integrated FRM in B&H related to the limited institutional capacity (technical, legislative, and financial) for nation-wide FFEWS. The results of Output 1 form the basis of activities under Outcomes 2 and 3.

58. The project's **Outcome 2** will be achieved through the Outputs 2 and 3 which together will enhance the capacity to identify, plan and implement long-term flood risk management strategies at the basin and sub-basin scale by introducing combined structural and non-structural methods including EbA measures.
59. More specifically, the **Output 2** will mainstream climate-informed integrated flood risk management into key sectoral policies and plans, enhance institutional technical capacities in all aspects of FRM, and facilitate scaled-up implementation of nature-based and non-structural flood protection solutions, using the hazard and risk modelling, maps and assessments resulting from Output 1 as the basis. Through the application of nature-based flood risk management solutions, the output will above all result in improvements in watershed ecosystems and restoration of ecological flood protection functions. This output will be delivered through four complementary activities. First, in order to address the institutional, legal and policy framework barriers to the effective integrated FRM, the project will **enhance the existing legislative and regulatory FRM framework** by mainstreaming climate-induced flood risk management into sectoral policy and planning for agriculture, forestry, environment, hydropower, critical infrastructure and spatial planning (**Activity 2.1**). Then, the project will promote and scale up nature-based approaches to flood management which can provide added benefits such as enhanced ecosystem services, to be implemented in the highest risk basins in B&H. In particular, the project will **implement 21 non-structural and EbA measures that will enhance the ecosystem services** through afforestation, regressing and soil protection/sediment management measures (**Activity 2.2**). These solutions will complement structural measures to ensure that the combination of both structural and ecosystem-based measures provides the best solution to climate-resilient flood management and protection. In addition, mainstreaming climate-induced flood risk considerations into forestry policy is expected to establish policies which will protect the forest from the expected impacts of climate change and promote reforestation, afforestation, and sustainable forest use. The long-term impact will be enhanced forest eco-systems which will provide improved services to flood risk management and other biophysical functions. The project will therefore result in improvements in watershed ecosystems and restoration of ecological function through the use of innovative EbA strategies which will reverse the deleterious effects of catchment degradation and enhance livelihoods of rural communities. To fully embed these new EbA methods into FRM in BiH, the project will establish a knowledge framework for the application of climate resilient EbA methods, including **guidelines, technical specifications, standards and protocols** (**Activity 2.3**). Finally, to **enhance institutional capacity and embed continued capacity development** in all aspects of flood risk management the project will implement country-wide training programmes in the technical and practical areas of flood risk management for practitioners, decision-makers, communities, emergency responders based on the assessment of technical and functional capacities and the development of the long-term capacity development plan (**Activity 2.4**). Implementation of the Output 2 will address the gaps in FRM sectoral policy in B&H, the lack of a strategic, policy-led integrated FRM approach, and insufficient technical knowledge and capacities on ecosystem-based and non-structural solutions for climate resilient flood risk reduction.
60. Finally, **Output 3** will support the enhanced long-term flood risk investment planning at the national and local levels, engage private sector in risk management, and implement priority climate-proof structural flood protection

measures in high-risk areas, thus contributing to the enhanced flood protection and reduced losses from flooding. This output will be achieved through the implementation of three activities. First, the project will **enhance the climate risk financing and risk transfer framework** in BiH shifting from largely public sector and donor financing towards blended finance and public-private partnerships. The project will support development of FRM investment frameworks, risk transfer schemes and enabling regulatory framework (**Activity 3.1**). The project will develop a strategy for private sector engagement in long-term climate change-induced flood risk management. It will develop the most appropriate mechanisms for risk transfer, risk financing and supporting cost recovery mechanisms from a variety of sources. This will include the development of policy and legislation for private sector engagement in climate risk financing and for risk transfer mechanisms such as flood insurance. The project will develop a tool for appraisal-led design for structural and non-structural FRM measures, FRM investment planning, climate risk financing mechanisms and for appraisal-led FRM options design and decision-making, based on cost benefit analysis (CBA) approaches, building upon flood hazard and risk modelling tools to be developed under Activity 1.2. Further, the project will develop and implement the national methodologies, guidelines and tools for municipal climate-resilient investment planning based socio-economic risk and vulnerability modelling. **Investment planning will include Early Action plans identifying humanitarian funding (Forecast based Financing FbF)** for early action based on impact-based FFEWS (**Activity 3.2**). The project will build upon municipal level climate risk-informed budgeting mechanisms developed under NAP project and examine opportunities to introduce climate risk-informed budgeting into B&H's existing public financial management (PFM) framework leveraging on existing strengths and addressing any limitations of the existing budgeting processes that currently hinder climate risk-informed budgeting for flood risk management. This will include using the damage and loss and risk modelling tools (developed under Activity 1.2) to provide the evidence basis of existing cost of disasters to enable allocation of funding to different activities within the FRM framework and the investment required FRM/disaster budget ring-fencing mechanisms. Based on the risk financing and risk transfer strategies and tools developed in Activity 3.1, the project will develop municipal investment plans for climate-resilient investment planning for flood risk management in 10-12 municipalities and 1 canton in the Vrbas, Una-Sana and Bosna basins. Finally, the project will embed climate-change responsive design of flood defenses which will be supported by new standardized methodologies, tools, guidelines and capacity for **climate resilient flood defense design and implementation (Activity 3.3)**. Hence the project Output 3 will address barriers to the effective integrated FRM in B&H related to the outdated not climate-risk informed approach to flood defenses design, and the lack of capacity, knowledge and information for enhanced climate-proof public and private investments into flood risk reduction and community response. Under this Activity, utilizing appraisal-led CBA and tools to be developed under Activity 3.1 the project will embed CBA and appraisal-led prioritization methods in developing and implementing structural measures that maximize benefit, make allowance for climate change and which consider environmental impacts. Under this activity, 10 structural measures have been identified by Water Agencies, (See Section 2.6 and Annex A of FS) will be implemented in high-risk river basins - Vrbas, Una-Sana and Bosna, Neretva and Trebisnjica and Drina River basins.

61. This flow of outcomes, outputs and activities has been designed with the view of the following **assumptions** about the external environmental, political, economic, and institutional circumstances. Success of the project in reducing the losses of lives, assets, livelihoods, and infrastructure relies on the assumption that the occurrence of major extreme weather or seismic events do not deviate from projected trajectory. Implementation of the FRM regulatory reform and investment frameworks promoted by the project is based on the assumption that there will be no external political, financial or economic shocks that may hamper implementation capacities and priorities of the government and private sector partners. Consequently, sustainability of the technological solutions and infrastructure, including FFEWS and flood protection structures, depends on the capacities and resources of key stakeholders to support O&M of new technologies and systems, and to maintain adequate level of staff and budgets at water agencies and HMI. Finally, the effectiveness of the FFEWS in improving the timeliness and accuracy of flood warning is based on the assumption that government institutions cooperate on the FFEWS and data management.
62. Please refer to the Theory of Change (TOC) diagram below, which illustrates how the proposed sets of funded activities will deliver the planned Outputs, which address the barriers and risks to the desired adaptation alternative and contribute to the achievement of Outcomes and co-benefits.



B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Fill in the GCF results area table below to map each project/programme outcome identified in section B.2(a) to the contributing GCF results area(s) by referring to the description of eight results areas provided in the guidance note.

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities, industries, appliances	MRA 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and water security	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services
Outcome 1	☐	☐	☐	☐	☒	☒	☐	☐
Outcome 2	☐	☐	☐	☐	☒	☒	☒	☒
Outcome...	☐	☐	☐	☐	☐	☐	☐	☐
Outcome ...	☐	☐	☐	☐	☐	☐	☐	☐
Outcome ...	☐	☐	☐	☐	☐	☐	☐	☐
Outcome ...	☐	☐	☐	☐	☐	☐	☐	☐

If any co-benefits have been identified in section B.2(a), fill in the Co-benefit table below to map each co-benefit to the corresponding category as defined in the FP guidance note.

Co-benefit number	Co-benefit					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
Co-benefit 1: Improved health and reduced health risks due to the reduction in the number of floods-related injuries and exposure to water borne diseases	☐	☒	☐	☐	☐	☐
Co-benefit 2: Female practioners trained in all aspects of FRM and gender-responsive FRM	☐	☐	☐	☒	☐	☐

B.3. Project/programme description (max. 2500 words, approximately 5 pages)

63. The **project objective** is to address increasing vulnerability of B&H communities and livelihoods to intensified climate-induced flood-related disasters. The project will reduce vulnerability of the population to floods across B&H (pluvial, fluvial, torrential and groundwater flooding) through improved climate information and establishment of the impact-based flood forecasting and early warning system (FFEWS). Improved generation and use of climate information will enable sound decision making and investment into climate resilient flood risk reduction measures, including ecosystem-based, non-structural and structural measures. The proposed project aims to address the above barriers and shift the baseline scenario towards climate resilience through three outputs (See Chapter 10 of the FS for a more detailed description of the project Outputs and Activities).

Output 1: Fully integrated impact-based Flood Forecasting and EWS facilitates timely preparation and response

Activity 1.1: Upgrade and expand the coverage of the hydrometric network for enhanced monitoring of climate variables in Category I catchments and torrential streams. (GCF Financing = \$1,535,980; Co-financing = \$ 2,869,966)

64. Under this activity, the project will expand the hydrometeorological monitoring network to cover a greater range of hydrometric variables that climate change is bringing. Previous upgrades to the hydrometeorological observation network have not been realized at the appropriate spatial and temporal scale to enable effective flood monitoring, management, forecasting and early warning. In addition to pluvial and fluvial flooding in the biggest catchments (category I catchments), climate change is increasing the risks from torrential streams, which are currently not monitored or studied but cause great damage. The project will provide an increased density of the hydro-meteorological network for the areas at highest risk i.e., category I rivers and torrents³¹. Technical assistance to review and integrate the HPPs hydrometric network into the national network will also be provided. The project will also assist in setting-up sustainable operations and maintenance (O&M) systems for the network, by identifying long-term financing mechanisms for the maintenance of the hydrometric network that will include private and public sector financing. The expansion of the observation network will greatly enhance the ability to provide station data to GBON and will enable the use of observation data from stations to be integrated with regional datasets for more effective monitoring and forecasting.

³¹ Chapter 5 of the FS provides a detailed assessment of the existing hydrometric network and page 193-195 provides WMO key technical considerations to be taken into account when establishing the optimized observation network.

65. The **optimization of the hydrometric network** will include upgrading of the ICT infrastructure and capacity to support the telemetered and automated stations for effective data transmission, storage, sharing and management. It will also include strengthening of the existing systems and protocols for the collection, transmission, sharing, storage, management and use of the observed data and the digitization, systematization and storage of all paper format data within the relevant hydrometric databases.

66. The following sub-activities will be implemented:

Sub-Activity 1.1.1 – Develop the optimised network specification, develop ICT strategy and plan for hydrometric network.

Sub-Activity 1.1.2 – Procure and install equipment to increase density of the hydro-meteorological observation network and expand monitoring to include greater range of hydrometric variables that climate change is bringing.

Sub-Activity 1.1.3 - Set-up sustainable operation and maintenance O&M for the network.

Sub-Activity 1.1.4 - Develop long-term financing mechanisms for the maintenance of the hydrometric network.

Activity 1.1 Deliverables:

- Hydro-meteorological observation network expanded with 9-Groundwater monitoring stations; 24-rehabilitated piezometers; 18-new piezometer 100 and 200 m deep;
- 10- boreholes with average depth up to 50 m; 30- boreholes with average depth up to 30 m; 26-Meteorological stations; 23-Hydrological stations;
- 2 strategic plans, methodologies, guidelines and manuals for the operation and maintenance of the optimized hydrometric network;
- 1 prototype mechanism for private sector cost recovery;
- 2 Public Sector O&M financing strategies action plans, budgetary allocation mechanisms and tools.

67. The following is the proposed indicative **list of stations and equipment to be acquired by the project**³².

Table B-1: Indicative Hydrometric monitoring stations and equipment to be acquired by the project

	Quantity	Total Capital Cost (USD)
Groundwater network requirements		
Groundwater monitoring stations for measuring profiles of the most significant karst springs	9.00	163,800.00
Installation of the appropriate measuring equipment into existing piezometers which are out of function or never were installed with measuring equipment	24.00	172,800.00
Performing piezometer 100 and 200 m deep, with installation of equipment on following locations, average depth of 180 m	18.00	327,600.00
boreholes with average depth up to 50 m, with casing and installation of equipment	10.00	34,000.00
boreholes with average depth up to 30 m, with casing and installation of equipment	30.00	63,000.00
Hydrometric network requirements		
Meteorological Stations: <u>GMS program</u> Measurement: Temperature (t, tmin, tmax, tmin5), Humidity, Wind, pressure, precipitation, insulation, snow depth, soil temperature per depth:(2 cm, 5 cm, 10 cm, 20 cm, 30 cm, 50 cm, 100 cm) <u>AMS sensors:</u> precipitation, temperature, wind, humidity, pressure, global radiation, snow depth, percentage of water in snow. <u>AAS sensors:</u> Soil temperature up to 1m of (5 cm, 10 cm, 50 cm i 100 cm) Humidity of soil up to 1 m (5 cm, 10 cm, 50 cm i 100 cm) humidity of leaves	26.00	403,000.00

³² Section 5.4. of FS

Automatic Hydrological stations: The level and depth of the water, pressure, temperature.	23.00	121,900.00
Total cost		1,286,100.00

68. The expanded hydrometric network will be supported by **enhanced institutional arrangements**, which will be assessed in terms of existing roles and responsibilities, the capacity of staff to operate and maintain the optimized hydrometric network. Section 5.2 of the FS provides a discussion of the existing institutional capacity for hydrometric network management. The project will identify the manpower, resourcing and financial requirements as well as training needs, for the efficient O&M of the optimized network, develop the strategic plan, capacity development and training plan, methodologies, technical guidelines and manuals for the long-term O&M and will provide training for hydrometric staff in the O&M of up-graded hydrometric stations.
69. Total operations and maintenance costs of the expanded hydrometric network during the project 6-year implementation period have been assessed as \$1,633,500 and \$5,445,000 for 20 years. Relevant entity governments and water agencies will provide O&M during the project implementation period and for up to 20 years after project start (See Annex 13 for co-financing letters and Annex 21 for the O&M Plan). The project will identify and develop a long-term **financing mechanisms** for the maintenance of the hydrometric network, based on a review of existing public and private sector financing mechanisms for climate and flood risk management, and identification of key private sector players at risk from flooding with an interest in flood risk reduction. These will include critical Infrastructure providers/managers, such as utility companies' communications, telecommunications, transportation sector (airlines/air traffic, road and rail operators), energy (hydropower) and private enterprises such as hotel groups, chamber of commerce or business community, insurance and micro-finance companies. The project will undertake awareness raising on flood risk, flood risk management approaches to fully engage private and public sector.
70. **Gender Mainstreaming Activities:** The upgrade and expansion of the coverage of the hydrometric network will specifically focus on ways in which climate variables affect and impact women, assess options and propose ways in which women can benefit from hydrometric network expansion in terms of employment and income making opportunities. Participation of both men and women in consultation process during the installation of the new hydrometric equipment at catchments and torrents will be ensured through community consultation. The Project will aim to have 40% participation of women in consultation and capacity building under this activity.

Activity 1.2: Enhance climate-induced flood hazard, risk and vulnerability information for strategic risk management and sound decision making for climate induced flood management. (GCF Financing = \$1,067,833; Co-financing = \$ 4,528,756)

71. WBIF³³ project developed EUFD flood hazard and risk maps for all basins in B&H³⁴. These maps were completed in 2021. . These flood hazard and risk maps only cover high risk (AFAs³⁵) areas of category I rivers, will not have holistic river basin approach and will not include detailed hydrological modelling, which will be required for climate-informed flood forecasting and development of climate-resilient river basin flood risk management plans. Category II rivers pose a significant risk to populations, particularly where there is torrential flash flooding, but there is currently no hazard or risk mapping of torrents, except for Vrbas basin, and WBIF project will not develop the torrents flood maps. In addition, the IPA2016 has allocated funds for development of flood risk management plans for all basins in B&H, and this work started last year. While the WBIF hazard mapping will meet the minimum requires for the EUFD with regard to flood hazard and risk mapping for B&H, it will not fully enable the strategic assessment of risk to population, to economic activity and to future development under conditions of climate change which is a government priority to support and guide local municipalities to wisely and rationally manage risk exposure to acceptable levels.
72. The project will enhance the WBIF EUFD hazard and risk modelling and maps by a) extending the WBIF models to ensure whole basins approach and not just modelling of high risks areas; b) modelling Category II tributaries and not only main river; c) modelling torrents flooding mechanisms³⁶; d) enhancing hydrological modelling; and e)

³³ West Balkans Investment Framework

³⁴ Except Vrbas which is already done

³⁵ Areas for Further Assessment

³⁶ Torrents or Torrential flooding is the term widely used in the Balkans and Southern Europe to describe a flashflood with significant erosion power and which carries heavy debris/sediment load. Such catchments are normally located in the headwaters of major rivers

inclusion of reservoir operations and flooding from potential dam failure. The modelling will include future climate change projections. An essential task of flood management in HPP modified catchments is determination of an effective reservoir operation strategy that minimizes downstream flood risk and damage, while maintaining dam safety within reasonable limits and optimizing water availability for power generation. To minimize disastrous impact to downstream property, human life or even to the dam itself, there needs to be an integrated approach to basin flood risk management which includes adequate representation of HPP operations in the basin flood hazard and risk modelling and mapping, as well as in flood forecasting models, to enable real-time operational decisions during a developing flood. To this end, the project will incorporate HPP operations into flood hazard and risk modelling (to strengthen the role of HPPs in flood alleviation and maximize their capacity to utilize flood forecasts in their operations) in all relevant basins. The project will additionally develop sector-specific climate risk information products for hydropower sectors with a particular focus on the Neretva and Trebisnjica sub-basins of the Adriatic Sea basin³⁷ where there are large numbers of HPPs.

73. The project will also develop the institutional capacity of HPPs operators to use climate risk information and forecasts in their operations by providing the necessary supporting protocols, guidance, tools and training. In addition, the ability to assess flood risks posed by dams and reservoirs in the case of overtopping or breaching will be strengthened by reviewing and updating dam breach modelling and mapping approaches in B&H and development of updated unified technical guidance and specification for dam breach modelling.

74. This activity will also build on the bespoke **GIS-based socio-economic flood risk and vulnerability model** (which is based on EU Flood Directive methodology for flood socio-economics assessment) developed for Vrbas basin. The developed socio-economic flood risk model will be enhanced with improved receptor and socio-economic datasets to be acquired/established by the project, integrate various spatial socio-economic data with the hazard maps, to produce risk maps, which will include economic losses and damages and loss of life estimates. In addition, the project will develop the technical methodology and tools for the systematic collection of socio-economic datasets that are important to flood risk and vulnerability modelling and assessment.

75. **Gender Mainstreaming Activities:** Datasets for socio-economic risk and vulnerability assessment that support gender disaggregated data as well as data on other vulnerabilities will be embedded into the GIS-based socio-economic risk model to be developed for flood risk analyses and decision-making in the future. Secondary datasets from census and other existing GIS databases will be complemented by primary datasets which will be collected using gender-responsive socio-economic survey methods that will be designed to ensure participation of women (at least 40%) and all other vulnerable and socially excluded groups. A gender Advisor will be engaged to support gender mainstreaming activities.

76. The following sub-activities will be implemented:

Sub-Activity 1.2.1: Develop hydrological modelling for all main basins (category I rivers) to complement WBIF project's EUFD hazard and risk models.

Sub-Activity 1.2.2: Develop full basin hydraulic models.

Sub-Activity 1.2.3: Produce country-wide maps of torrents hazard, risk and vulnerability.

Sub-Activity 1.2.4: HPP modelling for development of enhanced operating rules for HPPs. Incorporate HPP operations into flood hazard and risk modelling.

Activity 1.2 Deliverables:

- Enhanced WBIF EUFD hazard and risk modelling by a) developing hydrological modelling for all main basins (category I rivers), b) undertaking torrents hazard modelling, c) including HPPs and groundwater modelling in flood hazard,
- GIS-based risk and vulnerability modelling and mapping based on enhanced hazard maps for all relevant basins.

and contribute significantly to the flood risk and erosion risk. See for example "Assessment of torrential flood susceptibility using GIS matrix method: case study - Vrbas River basin (B&H) in Carpathian Journal of Earth and Environmental Sciences, August 2018, Vol. 13, No. 2, p. 369 - 382

³⁷ This will build upon the current NAP project "Development of study on economic impact of climate change on the hydropower sector in Republika Srpska – multipurpose hydrosystem Trebišnjica",

Activity 1.3: Develop an integrated impact-based centralized and community-based flood forecasting and early warning system (FFEWS). (GCF Financing = \$ 2,156,040; Co-Financing = \$ 1,955,438)

77. B&H is a member of WMO and implements all of its activities in line with WMO standards and guidelines. Forecasting and early warning within B&H currently includes the key principles of GFCS. Hydro Meteorological Institutes in both entities collaborate on the development and delivery of forecasts to ensure a country-wide perspective, tailoring and delivering science-based climate predictions and services for key sectors: agriculture and food security, disaster risk reduction, energy, health and water management. Both entities have access to the same Water Information System (server and cloud computing) and Forecasting platform server, which, in addition to collaboration, provides physical redundancy of the systems. This Water Information System and Forecasting platform have been developed under UNDP's Vrbas project. The project will build on the FFEWS which was developed for the Vrbas and Una-Sana River Basins and will enhance the ICT systems to support an integrated impact-based FFEWS platform, for near real time forecasting and warning dissemination. Under the project the nation-wide impact-based FFEWS will cover the two hydro-energy intensive basins of Neretva and Trebisnjica, and the Una-Sana basin forecasting model will be updated. EU funded technical assistance for development of the hydrological flood forecasting system for Sava River Basin (Phase 1. Bosna River) project is implementing the Vrbas prototype for Bosna River basin, while Drina forecasting model is being developed under the World bank Drina project. The forecasting system to be implemented will update and integrate all existing and new FFEWS models covering the whole territory of B&H. Under the Vrbas project, meteorological forecasting capabilities were enhanced and implemented within the upgraded forecasting platform based on two different Numerical Weather Predictions (NWP) models, namely WRF and COSMO. The project will further enhance these capabilities (through the use of additional global or regional meteorological forecasting products within the forecasting platform for boundary conditions for the local meteorological models. The Global Forecasting Models (from NOAA) are used to force respectively the WRF and the COSMO local implementations). The project is supporting and facilitating the application of B&H to European Centre for Medium-Range Weather Forecasts (ECMWF) membership and will pay the application and annual fee, cost of data and access to cloud computing for the project duration (Ministry of Water Management will pay thereafter, see co-financing letter – Annex XIII) (Sub-activity 1.3.1 pgs. 203-213 of FS). The project is also enabling the inclusion of satellite precipitation data into meteorological forecasts (MPE, TRMM/GPM). This increased utilization of data from national, global and regional observation networks, models and RS data, will be enabled and underpinned by capacity development (Sub-activity 1.3.4 pg. 220-221 of FS), which will be done through partnerships with WMO regional training centers, UNESCO/ITU ICT4D and other relevant DRR institutions and partners. Importantly, the project will develop a link between the flood model and the socio-economic risk model to be developed in Activity 1.2, to provide Impact-based forecast. FFEWS will therefore be based on an impact-based forecasting (IBF) approach (Pgs. 199 to 202 of FS) to enable the provision of forecast information about the actual impact that each event is going to cause in particular locations, and for particular sectors, thus enabling impact specific warnings. This will include likely loss of life, likely damages to individuals and communities, likely damage and loss of critical infrastructure and likely failure of protective hazard defenses. The Community Based Early Warning System (CBEWS) will be developed on the basis of the prototype implemented in Vrbas basin and will support the centralised forecasting and early warning system to effectively minimize or prevent the damages from any disaster by implementing action at the local level the strengthening the last mile communication. Incorporating the active involvement of the people in the community with an early warning system aims to increase the effectiveness and sustainability of such systems, by providing better understanding of the value of these systems for safeguarding their lives and livelihoods. Selection of communities where CBEWS will be implemented will be based on the following considerations: relative risk and vulnerability, lead time of the extreme events (and the need for localised monitoring and warning), willingness of communities to participate, potential technical constraints for the central system to effectively service the community (e.g. due to remote location or connection problems) and cost-benefit ratio of the implementation. CBEWS' will be managed and coordinated at the level of municipalities. The components of the FFEWS is described in Section 10.1 (Pgs. 202 to 220 of FS) of the FS. The project will also develop sector specific FFEWS products for the agricultural and hydro power sectors.
78. For the **agricultural sector**, this activity will examine opportunities to enhance and support the capacities of the Agricultural agencies and extension services through the generation and provision of tailored climate and weather information and advice to farmers. Through this activity the project will implement ICT-based innovations in the communication of forecasts and advisories; improve the use of historical data and derivations; improve medium

and shortrange weather forecasts and longer-term predictions for agriculture (seasonal, decadal etc.) and develop partnerships with the private sector including internet providers and mobile companies, which were consulted during the project preparation and are willing to design tailored information delivery services in collaboration with agribusinesses, farmers' groups and other clients to deliver timely forecasts and advisories to farmers who will benefit from this service.

79. The project will engage the **hydropower sector** in the flood risk management of the basins in which they operate through agreement on operations of their systems during flood events. This agreement will be between hydro power sector, water agencies who are responsible for water management and hydro-meteorological institutes who provide forecast and warning information that will feed into HPP operations. In parallel, the project will provide support to the HPP sector to enable more climate resilient operations. Extended consultations with the hydropower sector stakeholders have been carried out during the project design, and co-financing leveraged from the hydropower companies to the project (see Annex 13). Energy planning and operations in the hydropower system are markedly affected by meteorological events and hence climate change, on both the demand and supply side. Thus, the project will assess and identify feasible tailored climate services that can support increased efficiency in HPP operations and will develop enhanced tools and systems that provide decision-makers with the ability to analyze and manage risks in the face of climatic variability and change. The development and application of targeted climate products and services will help improve efficiency and reduce risk associated with hydro-meteorological hazards affecting energy systems. -
80. The project is developing a **last mile communication system** (Sub-activity 1.3.2, Pgs. 213-218 of FS), which will include design of the "last-mile" communication models, work with the government to update the communication protocols and SOPs (Sub-activity 1.4.1 below), work with the mobile operators on the technical implementation of the system, work with the NGOs and media and will provide training and capacity building in dissemination and warning. According to statistics of B&H Regulatory Agency for Communications (RAK), 97% of citizens in B&H are mobile network subscribers, which is excellent coverage of population under risk, for the purpose of potential warnings by SMS in the case of a flood disaster. The main mobile network operators in B&H provide standard telecommunication services to their users, with possibility to introduce common/simple SMS flood alerts. The Vrbas project worked with mobile providers to develop an SMS warning system, which the project will scale up. The project will also leverage and enhance existing partnerships with mobile operators, to utilize additional technologies such as IoT (Internet of Things) to send data in real time, connected to the smart phone, sensor and web service. This ability can be used for smart application to control and manage an early warning system. The project will also explore opportunities to use IoT technology to monitor and control the early detection system of flood disaster. By using IoT technology, sensor data can be sent in real time from sensor to smartphone through the internet network
81. UNDP B&H established a cooperation with the International Federation of Red Cross and Red Crescent Societies (IFRC) in order to introduce **Forecast based Financing (FbF)**. Based on jointly developed selection criteria, FbF has already been piloted in B&H, in two municipalities - Dobož and Odžak - where the UNDP supported the enhancement of DRR activities within the municipal plans. For the first time in B&H the FbF approach was presented at the community level. Both municipalities have monitoring stations referenced to their area and can obtain data from hydrological stations to define thresholds for action and danger levels necessary for triggering FbF. Also, their budgets consist of accounts that can be easily used in case when natural disaster (flood) is forecasted and FbF mechanism is triggered. The conclusion of municipal representatives was that implementation of FbF concept is feasible. These pilots showed that in order to implement the FbF in the municipalities additional efforts are needed in order to ensure reliable and timely forecast information, define triggers for action and perform risk analysis, dedicate FbF specific fund within the budget, develop standard operating procedures (SOP) to ensure that FbF funding will not be used ad hoc for measures that might not be worth taking. The project is developing the methodologies, guidelines, and tools for municipal and cantonal climate-resilient investment planning (Under Activity 3.2 see below and also pgs. 259-260 of FS), utilizing the impact-based forecasting system being developed under this output, to identify the thresholds, and impacts of flood events. These thresholds will include thresholds for forecast based financing (FbF) of early action to reduce flood impacts and respond to flood disasters. The forecast thresholds that will trigger the release of resources for early action will be developed as part of the development of the thresholds for warning and will be embedded in the early warning protocols and SoPs also being developed (under Activity 1.4.1 below). The project will leverage existing relationships with Red Cross Society of Bosnia and Herzegovina and the IFRC to understand the short-term needs for humanitarian funding for early action based on in-depth forecast information and risk analysis. This will enable authorities to anticipate disasters, prevent their impact, and if possible, reduce loss of human life and loss of assets. The forecast thresholds that trigger the release of resources will be developed as part of the development of the thresholds for warning under Activity 1.3 and will be embedded in the early warning protocols and SoPs also being developed (Activity 1.4).

82. To consolidate and sustain the significant technical capacity development that will be achieved under this activity, the project will develop and implement a capacity development plan for embedding flood hazard and risk modelling approaches and FFEWS into appropriate institutions in B&H. Technical capacities related to risk identification and assessment, prevention, risk reduction, risk transfer, preparedness, climate risk management and climate change adaptation require further strengthening across institutions and governance levels. The institutional capacity gaps and training needs will be assessed, and a revised institutional capacity building plan will be developed under the project to address gaps in human, technical and financial FRM capacities. The project will also enhance the institutional arrangements for FFEWS by implementing the recommendations from the Vrbas project which identified the optimal institutional arrangement for an integrated, people-centred impact-based FFEWS for all of B&H (See Section 6.5 of FS).

83. **Gender Mainstreaming Activities:** The Project will secure equal participation and active role of women and men, in developing of FFEWS. Based on the detailed surveys and assessment of vulnerabilities and intersectional vulnerabilities of gender and socially excluded groups, the Project will secure at least 40% participation to identify appropriate solutions in terms of types and methods of warnings, to ensure that no one is left behind in terms of access to the early warning system. The design of the early warning system (centralised and community-based), will take the following key considerations for gender and social inclusion into account:

- *Participation and Representation:* Involve women, men, and diverse social groups in the design and decision-making processes of early warning systems. Ensure their meaningful participation, representation, and inclusion in planning, implementation, and evaluation stages. This can be done through engaging local community leaders, women's groups, civil society organizations, and marginalized communities.
- *Gender-Responsive Communication:* Design systems that use gender-responsive and culturally appropriate communication strategies to disseminate early warning information. Consider different communication channels, languages, and formats to reach diverse groups. Tailor the messages to specific needs and preferences, considering literacy levels, disabilities, and cultural norms.
- *Addressing Barriers:* Identify and address barriers that prevent certain groups from accessing and understanding early warning information. This includes addressing language barriers, low literacy levels, lack of access to technology, and physical or mobility challenges. Ensure that information is provided in formats that are understandable and accessible to all, including audio formats, visual aids, and local languages.
- *Consideration of Intersectionality:* Recognize that individuals may face multiple intersecting forms of discrimination and vulnerability based on factors such as gender, age, disability, ethnicity, and socio-economic status. Consider the specific needs and challenges faced by individuals who belong to multiple marginalized groups and develop inclusive strategies to ensure their participation and protection in early warning systems.
- *Safe Spaces and Privacy:* Ensure that early warning systems take into account the privacy and safety concerns of marginalized and vulnerable individuals. Create safe spaces for women and other marginalized groups to access information, ask questions, and voice their concerns without fear of discrimination or reprisal.
- *Training and Capacity Building:* Provide training and capacity-building initiatives to enhance the knowledge and skills of community members, including women and marginalized groups, on early warning systems. This will empower individuals to understand and respond effectively to warnings, take appropriate actions, and support others within their communities.
- *Monitoring and Evaluation:* Integrate gender and social inclusion indicators into the monitoring and evaluation frameworks of early warning systems. Regularly assess the system's effectiveness in reaching and benefiting different groups. Engage with communities to gather feedback and make necessary adjustments to ensure inclusivity and effectiveness.

84. The following sub-activities will be implemented:

Sub-Activity 1.3.1: Develop and implement impact-based FFEWS system (centralised and Community-based).

Sub-Activity 1.3.2: Implement CBEWS for high priority communities.

Sub-Activity 1.3.3: Review, identify and develop sector specific FFEWS products.

Sub-Activity 1.3.4: Develop and implement a capacity development plan for embedding flood hazard and risk modelling approaches and FFEWS into appropriate institutions in B&H.

Activity 1.3 Deliverables:

- 1 FFEWS model and platform integrating all basin FFEWS models;

- CBEWS for 60 high priority communities;
- develop sector specific FFEWS products for HPP and agriculture;
- capacity development plan for embedding flood hazard and risk modelling approaches and FFEWS into appropriate institutions in B&H

Activity 1.4: Develop and implement protocols and SoPs on data generation, data management and communication for effective impact based FFEWS and flood risk management. (GCF Financing = \$ 336,000; Co-Financing = \$ 200,000)

85. The project will revise and implement data access protocols which will be supported by extensive training and capacity building to ensure sustainability. Based on the work undertaken by the Vrbas project, institutional strengthening, coordination, communication will be enhanced by establishing clear communication lines between different agencies, which will avoid any duplication and inefficiencies. Based on the Vrbas recommendations SOPs, Communication Protocols and Codes of Conduct will be developed for each of the institutions responsible for impact based FFEWS. In addition, roles of regional and local authorities will be clarified and detailed. "Last-mile" communication protocols will be developed and implemented as part of the FFEWS. This will include protocols and SoPS defining the forecast thresholds for FbF which will trigger the release of resources for early action. Operational maintenance procedures for hydrometric network will be established. The project will assess and further develop specific data sharing/communication protocols for impact based FFEWS and climate resilient FRM in coordination with the NAP project and will embed these protocols across relevant institutions.
86. To enable access and sharing of climate risk information, a centralized information system and knowledge sharing platform will be developed based on the Vrbas GeoPortal GIS-based tool which integrates various spatial socio-economic data with the flood hazard and flood risk maps, vulnerability maps including loss/damage models, real time hydro meteorological data, torrents sensitivity model, cadaster of torrents etc. The flood hazard information system to be developed under the project will consist of an integrated e-Library, databases, information systems and knowledge portal (web knowledge portal to increase awareness, provide interactive hazard maps, with integration with social media and possible mobile application to increase community engagement and allow two-way flow of information. The Vrbas Geo-portal is mainly designed to be used by the municipalities for local level flood risk management to interact with communities through the PGIS tool within the platform. The expanded Geo-portal will continue to function at the local level and will be placed in the ownership of the entity and municipal civil protection units who will have responsibility for maintenance and update of the system in the future. The project will build capacity with civil protection at all levels in the use, and O&M of the portal. UNDP has also developed a Disaster Risk Analysis System (DRAS) which is an online data sharing platform for disaster data. The platform has already been populated with flood hazard maps for Vrbas basin and these will be updated to include all flood hazard maps in the future. The project will contribute to the development of the DRAS platform and the embedding of flood hazard and risk information as well as expansion of the platform functionality required for using the flood hazard and risk information at the entity and national levels.
87. As required by the Sendai Framework "to establish and strengthen government coordination forums composed of relevant stakeholders at the national and local levels, such as national and local platforms for DRR", the flood risk management platform will be a country-wide mechanism for cross-entity, multi-sectoral and inter-disciplinary coordination and policy guidance on flood risk management and risk reduction with public, private and civil society participation involving all concerned entities within B&H. The FRM platform will contribute to such coordination forums. The data sharing platform will coordinate all stakeholder engagement at the entity and local level and will pursue an all-of-society engagement in FRM. It would also have mechanisms for effective dialogue with Local Platforms in place in order to influence, encourage and coordinate local action. In parallel, the country-wide Platform will seek to understand local priorities and issues.
88. **Gender Mainstreaming Activities:** The Project will review national protocols and SOPs from gender point of view; Ensure that flood risk information and warnings are adapted to the needs and capabilities of women and other vulnerable groups (Roma women, children, elderly people, disabled, returnees, economically disadvantaged); Develop communication products considering specific needs of man, women and other vulnerable groups, aiming to share information and raise knowledge on FFEWS; Use NGOs networks to reach women and the most vulnerable groups.
89. The following sub-activities will be implemented:

Sub-Activity 1.4.1: Implement SoPs for “last-mile” warning, dissemination, and communication system. Specify and implement the “last-mile” system including dissemination and warning technologies.

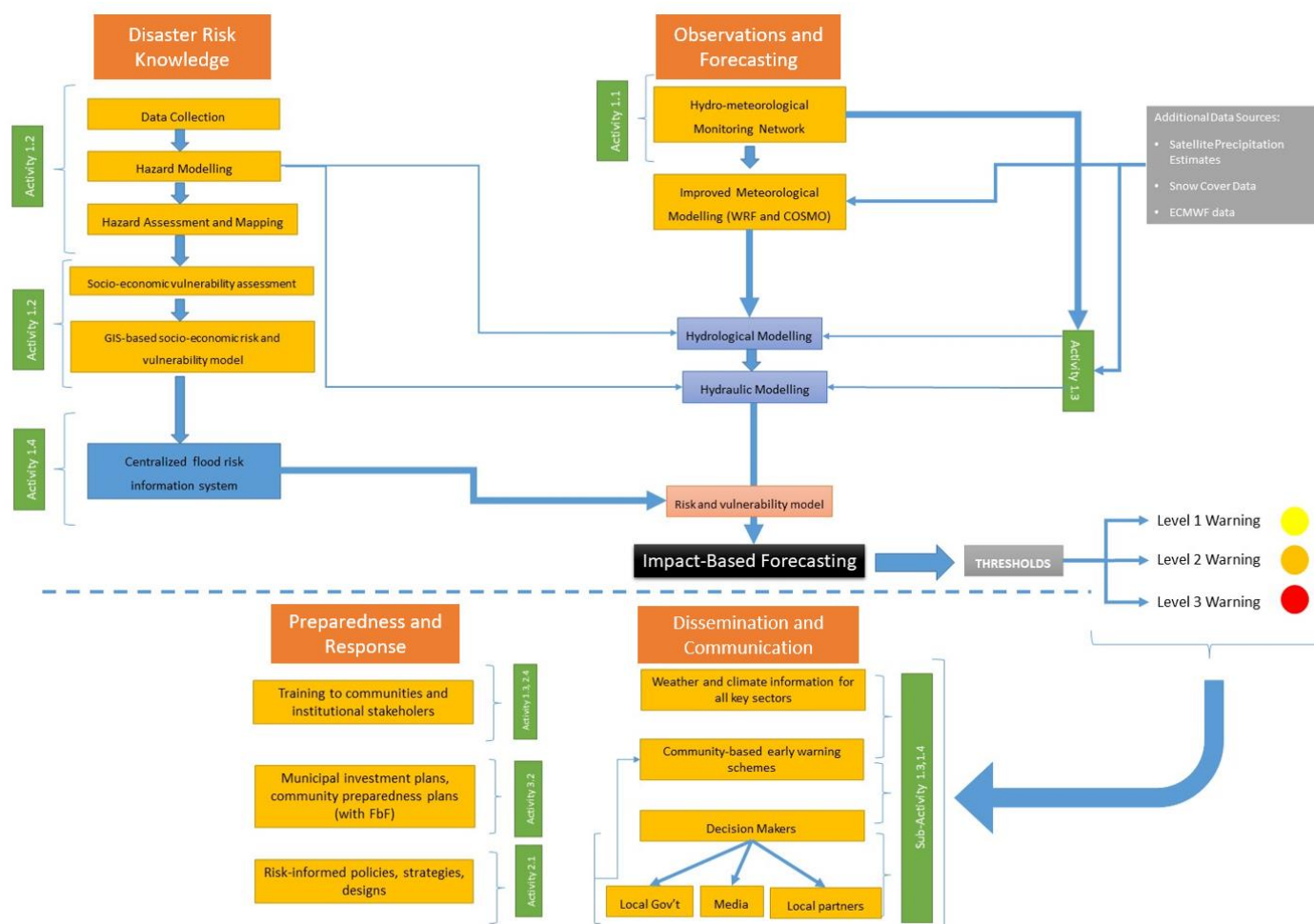
Sub-Activity 1.4.2: Develop protocols and SOPs on data generation and data management for FRM.

Sub-Activity 1.4.3: Develop data sharing protocols and platform for climate data, across all government institutions in both entities. Develop/enhance spatial data infrastructure to ensure climate data flow/exchange.

Activity 1.4 Deliverables:

- SoPs for “last-mile” warning and dissemination and communication system, Protocols and SOPs on data;
- At least 12 data sharing protocols between water management and other relevant sectors (agriculture, forestry, environment, spatial planning, and hydro-energy);
- Cross-sectoral flood risk management platform for climate data, across all government institutions

The following diagram presents conceptual designs of the impact based FFEWS to be implemented through the proposed project. The diagram presents a holistic and fully integrated scope of project activities leading to the establishment of the impact based FFEWS as a key element of the overall flood risk management framework which includes Disaster Risk Knowledge, Observations and Forecasting, Dissemination and Communication, and Preparedness and Response.



Result 1: Fully integrated flood forecasting EWS facilitates timely preparation and response

- Climate data (hydro-meteorological, spatial and damage/loss data) gathered and analysed in line with international standards
- Flood forecasting and EWS in place and operational for the whole country

- Data management and communication protocols and SOPs for effective FFEWS and risk management established at all levels
- Vulnerable communities have access to “last-mile” EWS

Output 2: Non-structural flood risk reduction measures and nature-based solutions mainstreamed in sectoral policies and plans and effectively contribute to protection of people and livelihoods from climate-induced flood risk

Activity 2.1: Mainstream climate induced flood risk reduction into sectoral planning (agriculture, hydropower, forestry, environment), critical infrastructure and spatial planning. (GCF Financing = \$366,000; Co-Financing = \$90,000)

90. The project will enhance the existing legislative and regulatory FRM framework by mainstreaming climate-responsive flood risk management into sectoral planning for agriculture, forestry, environment, hydropower, critical infrastructure and spatial planning. At the state level, a cross-entity approach will be taken to integrate flood risk reduction into sectoral planning to ensure that a holistic approach is taken, particularly where basins are shared between two entities. This requires the involvement of key stakeholders, the improvement of coordination, and the implementation of related multilateral and regional environmental agreements. Relevant regulations and standards will be reviewed and adjusted to account for the impacts of climate change induced flood risk. The project will help the entity level governments to develop climate risk informed key strategies, policies and plans by using evidence-based climate induced flood risk impact, vulnerability and adaptation assessments, analyses of the costs and benefits of adaptation options.
91. In the **Hydropower sector** the project will support the development of climate risk informed sector strategies, policies and plans for the flood safety of dams and will examine the potential role of the existing and planned dams in flood alleviation. The policy will strengthen dam safety guidelines for B&H in line with international best practice. This will include development or enhancement of guidelines for the risk-based categorization of dams, the development of improved climate risk informed methodologies for assessing spillway discharge capacities, the establishment of dam safety inspection intervals, guidelines on the assessment and quantification of risks associated with dams, including risk of overtopping, exposure to landslides and increased sedimentation, and the development of appropriate risk management plans for individual dams. The policy will also embed the requirement to develop dam operating rules that ensure that sufficient flood storage is provided at the start of large events, to provide some attenuation of the flood wave where possible, without compromising dam safety. This will require the involvement of dam owners and operators in the development and eventual implementation of the overall flood management plans for the basins and the optimization of the dam operations for the dual uses of power generation and flood alleviation. The policies and strategies will ensure that dams are maintained and operated in a manner which avoids exacerbation of the flood risk, plays a role in alleviation of flooding where possible, and which takes account of the increasing risks they pose due to climate change.
92. With respect to the **agricultural sector** the project will ensure that climate change is taken into account in the reduction of flood risk to agriculture. Flood risk informed agricultural master planning will be introduced and will include the consideration of risks and opportunities for flood management in the design and development of agricultural infrastructure. The project will develop and mainstream a strategy for climate-smart agriculture which integrates climate risk and adaptation priorities into the agriculture sector plans, investments and budget frameworks, including the investment appraisal skills, economic valuation of climate change impacts, based on sector model, trade off analysis and cost-benefit assessments for a range of plausible adaptation options in agriculture. The project will also develop a range of incentives, such as financial, policy or environmental gains, that will catalyze adoption of proposed FRM agricultural measures and will consider a range of different instruments to achieve uptake. The project will also develop guidance documents, methodologies and technical regulations for the agricultural sector on flood risk assessment and management and the use of climate change induced flood risk information and agricultural risk management.
93. In relation to the **forestry** sector, the project will develop catchment management strategies for climate resilient catchment management to reduce the exposure of communities and their physical assets, to climate-induced flood hazards. The project will provide technical assistance to entity forestry divisions (with agriculture and water management) to develop and mainstream basin afforestation/reforestation and agro-forestry strategies and other forestry strategies to enhance flood risk management role of forests, as well as flood risk adaptation priorities into

the forestry sector plans, investments and budget frameworks. The project will mainstream climate and flood risk consideration into existing Forestry legislative and policy frameworks to ensure that Forestry sector practices do not increase flood risk in the future and that afforestation, reforestation and rehabilitation are prioritized to strengthen and ensure the eco-system services that forests provide to flood mitigation and alleviation. The project will also develop guidance documents, methodologies and technical regulations for the forestry sector on flood risk assessment and management and the use of climate change induced flood risk information in forestry management.

94. In the area of **land use and spatial planning**, the project will scale up and embed nation-wide floodplain risk zoning policy based on risk and hazard maps (to be produced under activity 1.2) and develop the relevant regulations and guidance documents based on the reviews and recommendations developed by the Vrbas project.
95. With regard to the **critical infrastructure**, the project will review climate proofing design standards, methodologies, procedures and practice and will introduce climate risk screening methods and embed climate risk reduction criteria across critical infrastructure planning, prioritization, design, construction and maintenance (for -engineering facilities, protective dikes, dam spillways, highways, and railway bridges). The intervention will embed the systematic use of climate hazard and risk information in the identification and prioritization of critical infrastructure projects. At the municipal level, the project will also introduce climate risk criteria into the prioritization process and include other methods of measuring benefits of projects based on the introduction of appraisal-led project prioritization using socio-economic cost-benefit analysis methods and tools to be developed under Activity 1.
96. **Gender mainstreaming Activities:** The Project will ensure participation of both men and women, but also representation of vulnerable groups into consultative processes organized to mainstream climate induced flood risk reduction into sectoral planning. Conduct gender mainstreaming training and build capacity of national stakeholders targeted by project and responsible for various disaster risk reduction functions.
97. The following sub-activities will be implemented:
 - Sub-Activity 2.1.1: Mainstream climate induced flood risk reduction into sectoral strategies, plans and technical guidelines for agriculture, hydropower and critical infrastructure, forestry and environment.
 - Sub-Activity 2.1.2: Develop country-wide floodplain zoning policy and legislation, based on hazard, risk and vulnerability mapping, flood resilient building codes and embed climate change considerations in the design and construction standards of critical infrastructure.
 - Sub-Activity 2.1.3: Update climate risk informed methodologies, standards for infrastructure design, construction material, use and maintenance of critical infrastructure in B&H.
 - Sub-Activity 2.1.4: Deliver training and technical advice on climate resilient infrastructure design, construction and O&M approaches.

Activity 2.1 Deliverables:

- 5 Sectoral strategies, plans and technical guidelines (for agriculture, hydropower, critical infrastructure, forestry and environment);
- National floodplain zoning policy and legislation implemented in all basins at entity, cantonal and municipal levels; Existing local/cantonal Spatial Plans updated by integration of Flood Zoning Policy and flood hazard and risk information into the local/cantonal Spatial Plans; Existing, local/cantonal Spatial Plans updated by integration of Flood Zoning Policy and flood hazard and risk information into the local/cantonal Spatial Plans for 10-12 municipalities and cantons; CD in new floodplain zoning policy;
- Updated climate risk-informed methodologies, standards for infrastructure design, construction material, use and maintenance of critical infrastructure in B&H;
- Capacity development plan for climate resilient infrastructure design, construction and O&M approaches

Activity 2.2: Implement and mainstream new ecosystem-based flood risk reduction and climate change adaptation methods. (GCF Financing = \$ 5,868,747; Co-Financing = \$20,247,428)

98. Nature-based approaches which can provide added benefits such as enhanced ecosystem services will be implemented in the highest risk basins in B&H. These solutions will complement structural measures to ensure that

the combination of both structural and ecosystem-based measures provides the best solution to climate-resilient flood management and protection. Non-structural measures (See Annex A of FS) will include: floodplain reconnection, selective bed raising/riffle creation, wash lands, wetland creation, two-stage channels, re-meandering straightened rivers, land and soil management activities to retain / delay surface flows, woody debris dams on streams and tributaries, flood plain woodland, re-forestation, agro-forestry, creation or re-instatement of a ditch network to promote infiltration (swales, interception ditches, etc), in-channel vegetation management growth to maximise channel roughness.

99. Under this activity the project will implement 21 non-structural measures which were identified out of 69 structural and non-structural measures proposed by WA's in the course of feasibility analysis (See **FS Annex A for list of measures and Annex E for description of appraisal and selection process**). Of those, 8 non-structural measures will be funded by GCF resources, and 13 measures will be co-funded by the Government of B&H. Cost benefit analysis was carried out on all 69 measures based on the methodology described in the Economic Analysis annex of the proposal and projects were ranked on the basis of the benefit to cost ratio. The two highest priority measures from each target basin were selected (except Vrbas where several non-structural measures have already been implemented). A summary table of non-structural measures to be implemented by the project is provided in Annex A of the FS.
100. The project will include community based EbA as part of the solution, which will be particularly important to reduce the impact of torrential floods. The project will scale up successful EbA approaches piloted by the UNDP/SCCF Vrbas River project. The highest priority eco-system-based measures will be implemented in the Vrbas basin which is the only basin in B&H with already developed torrential flood hazard maps and prioritized and costed EbA measures to reduce torrential flood risk. In the Vrbas River Basin (and all other basins), one of the main sources of sediment production is a network of forest roads and paths for hauling logs. During the war, roads were cut in the forest for illegal logging and as a result, most of the deforestation happened then. The likelihood of expanding the roads for timber extract now is low, as there are entity legislations which prohibit cutting of such path (Federal law on forest, article 13 and 14 and Law on Forest Republika Srpska, article 82). During the construction of the road, most of the excavation material is simply pushed from the route down the slope most of which reaches the river network and is transported all the way to a recipient, usually the Vrbas main river. Forest paths for extraction of timber are usually designed with high angle of slope (over 50%), which causes significant erosion. Density of these forest paths is approx. 150-200m/ha, which sums up length of approx. 21,000 km in basin area. This area is exposed to extensive erosion and creates torrential floods which will produce significant amounts of sediment. Sediment in watercourses increases the risk of floods, and rapid drainage of water from slopes reduces concentration time. It is necessary to take measures to prevent erosion on forest roads and paths and to rehabilitate areas already affected. In the Josavka River sub-basin of the Vrbas Basin the following EbA flood risk management measures will be implemented: 9 partitions, 16 wicker fields, afforestation of the basin in the amount of 200,00 ha and grassing of the areas of approximately 80,00 ha³⁸. Detailed design documentation will be developed for all elements but outline design drawings are provided in Annex A of the FS.
101. **Gender mainstreaming Activities:** Review of the community based EbA and EbA approaches piloted by the UNDP/SCCF Vrbas River project by gender expert to identify gender gaps, locate mainstreaming opportunities and uncover specific ways in which flood risk and climate change impact women, taking into account gender specific vulnerabilities and intersectional issues (single parenthood, age, income levels, disability, etc.); Mainstreaming of gender throughout all the new ecosystem-based flood risk reduction and climate change adaptation methods being implemented (e.g., afforestation, agroforestry), the project will conduct an assessment to ensure adequate opportunities for women to equally benefit from the work, employment and income making opportunities associated with implementing these measures. Gender mainstreaming into EbA design methods will make following considerations:
- *Participation and Representation:* Engage and involve women, men, and diverse social groups in the design and decision-making processes of EbA initiatives. Ensure their meaningful participation, representation, and inclusion in planning, implementation, and evaluation stages. Consider local knowledge, priorities, and perspectives, and involve community leaders, women's groups, and marginalized populations.
 - *Gender-Sensitive Vulnerability and Capacity Assessments:* Conduct gender-sensitive vulnerability and capacity assessments to understand the specific needs, roles, and experiences of women, men, and different social groups in relation to flood risk and ecosystem services. This will help to identify their specific

³⁸ Afforestation will be conducted using only native species. Ownership of land is within government, therefore it will be closely monitored by its representatives. For the grassing, it will be restoration of grasslands that were previously degraded.

vulnerabilities, capacities, and priorities, and inform the detailed design of EbA interventions that address their unique circumstances.

- *Access and Benefit-Sharing:* Ensure equitable access and benefit-sharing of ecosystem services and resources. Recognize and address potential gender and social disparities in access to and control over natural resources, such as land, water, and forests, that are essential for implementing EbA measures. Promote inclusive mechanisms for sharing benefits derived from EbA initiatives, considering the needs and rights of different groups.
- *Knowledge and Capacity Building:* Provide training, knowledge-sharing, and capacity-building initiatives to enhance the understanding and skills of community members, particularly women and marginalized groups, in relation to EbA and flood risk management. This empowers individuals to actively participate in decision-making, contribute their knowledge, and take leadership roles in the implementation and maintenance of EbA measures.
- *Gender-Responsive Communication:* Ensure gender-responsive communication and information sharing throughout the design and implementation of EbA initiatives. Use inclusive communication strategies and diverse channels to reach different groups, considering language preferences, literacy levels, and accessibility requirements. Tailor messages to address specific gender and social norms, roles, and interests.
- *Safety and Accessibility:* Address safety and accessibility concerns in the design and implementation of EbA interventions. Consider the specific needs of women, children, elderly individuals, and people with disabilities, ensuring that the infrastructure and access routes associated with EbA measures are safe, inclusive, and accessible to all
- *Monitoring and Evaluation:* Incorporate gender and social inclusion indicators into the monitoring and evaluation frameworks of EbA initiatives. Regularly assess the social, economic, and environmental impacts of the interventions on different groups, including their implications for gender equality and social equity. Engage with communities to gather feedback and ensure that the benefits and risks associated with EbA are distributed fairly.

102. These considerations will be integrated into the design of ecosystem-based approaches for flood risk management. For the sites to be implemented with EBA solutions the above considerations will be ensured through the ESIA and ESMP and the ESMP monitoring plan. This will ensure that EbA interventions address the diverse needs and priorities of communities, promote social equity, and contribute to long-term resilience and sustainable development.

103. The following sub-activities will be implemented:

Sub-Activity 2.2.1: Implement catchment management measures for reduced erosion

Sub-Activity 2.2.2: Implement 21 selected non-structural flood risk reduction measures

Activity 2.2 Deliverables:

- Catchment management for reduced erosion;
- 5 Government-financed EbA measures.
- 13 Government-financed non-structural measures.
- 8 non-structural measures co-funded by GCF and government

Please see Chapter 10, Annex A and Annex (Section 18.9) of the FS for detailed description of the measures (Chapter 10), Design details and drawings for each measure (Annex A) and description of method of socio-economic assessment and options appraisal of EbA and non-structural measures. Note: there is a 79% Government to 21% GCF co-funding ratio of EBA and non-structural measures.

Activity 2.3: Codify and mainstream “ecosystem-based adaptation” (EbA) solutions into policies and regulations and promote non-structural measures among decision makers and communities. (GCF Financing = \$208,680 ; Co-Financing = \$90,000)

104. The project will set up a framework for the application of climate resilient EbA approaches to flood risk management in B&H and will facilitate a transfer of well-established flood management practices from countries³⁹

³⁹ For example: Space for the Rivers Policy – Netherlands; Making space for Water Policy in England; LIFE projects (Donau-Auen and DonauUfer) along the Danube downstream from Vienna,

that have been using these methods for decades⁴⁰. The project will collect lessons learned from all activities and facilitate codification and dissemination of successful approaches in B&H. To this end, the project will develop country-wide best-practice guidelines, technical specifications, standards and protocols for the application of ecosystem based non-structural climate-resilient FRM measures which will be mainstreamed into policy, and will undertake capacity development in the identification, prioritization, assessment appraisal, design and implementation of such solutions.

105. **Gender mainstreaming Activities:** Review of the existing policies and regulations by a gender advisor to identify gender gaps and mainstreaming opportunities to embed the above gender and social inclusion elements into policy and regulations; Include gender expert in the teams working on the preparation of new policies, plans and regulations and in promotion of new FRM concepts.

106. The following sub-activities will be implemented:

Sub-Activity 2.3.1: Develop best-practice guidelines for non-structural measures

Sub-Activity 2.3.2: Develop technical specification, standards and protocols for design and implementation of non-structural measures

Sub-Activity 2.3.3: Review and implement training required for new non-structural measures

Activity 2.3 Deliverables:

- Best-practice guidelines, technical specification, standards and protocols for the application of ecosystem based non-structural measures including methodology for system evaluation; methodology and guidance for undertaking risk assessment and appraisal of EbA measures;
- Knowledge portal and common resource center to collect, collate and share knowledge and fill these gaps to advance uptake of nature-based solutions and for collaboration between ecologists, or specialists to allow to co-working and knowledge sharing.
- Capacity development in the application of methods for the identification, prioritization design and implementation of basin appropriate EbA solutions;
- Training for 200 FRM practitioners at all levels and all relevant types of stakeholders (entity, municipal, community)

Activity 2.4: Review and strengthen institutional capacity and develop long-term institutional capacity development plans for climate resilient FRM. (GCF Financing = \$300,000; Co-Financing = \$500,000)

107. The activity will implement country-wide training programmes in the technical and practical areas of flood risk management for practitioners, decision-makers, communities, emergency responders. The project will develop the long-term capacity development plan and will embed training in FRM and the use of climate risk information in the relevant entity training institutions to improve the technical capacity and knowledge base for hazard and flood risk management. The project will support Training of Trainers (ToT) through the development and establishment of training curricula and rosters of trainers and the development of mechanisms for delivering long-term capacity development such as development of key partnerships with universities and other academic research institutions that can be forged to ensure long-term FRM training, following initial cooperation established during Vrbas project. The project will develop training plans, curricular, training material, guidance documents for each technical area of expertise related to climate-induced hazard risk assessment and management and consolidate into an overall capacity development plan. Under this activity the immediate priorities of the capacity development and training plans will be implemented. The list of proposed training and capacity building courses is presented in the Feasibility Study (Annex 2).

108. The project will also assist the government of B&H in shifting from ad-hoc project-based awareness and education efforts to a planned, consistent and sustainable awareness raising and education for enhanced flood

⁴⁰ According to the European Environment Agency: Future extreme floods are likely to be the cost drivers for the adaptation of infrastructure. Ecosystem-based adaptation, nature-based solutions and green infrastructure are, in many cases, key in ensuring a cost-effective approach to an uncertain scenario by delaying or avoiding lock-ins to classic infrastructure-building water management. The report also notes that although strategies for flood risk management require measures that are specific to the local circumstances, using a river basin approach avoids passing on negative consequences further downstream. <https://www.eea.europa.eu/highlights/floodplain-management-reducing-flood-risks>

and disaster risk management. To this end, it will develop and implement a comprehensive long-term awareness raising campaign.

109. **Gender mainstreaming Activities:** Gender sensitive review of institutional capacity. Participation of Gender expert in development of national training programmes in the technical and practical areas of flood risk management for practitioners, decision-makers, communities, emergency responders;

110. The following sub-activities will be implemented:

Sub-Activity 2.4.1: Embed FRM training in existing government training programmes for long-term FRM training in B&H

Sub-Activity 2.4.2: Develop a costed and prioritised institutional capacity development plan to address technical, functional, and human capacity gaps identified through the completed institutional capacity assessment

Sub-Activity 2.4.3: Implement community-based training and awareness raising initiatives at community level

Activity 2.4 Deliverables:

- Capacity Development plan for Flood Risk Management
- FRM CDP and training plan and programme embedded in existing government training programmes for long-term FRM training in B&H in the following topics: Flood hazard and risk assessment, hydrometry, forecasting and modelling, EWSS; Vulnerability and Risk Assessment; River basin flood risk management planning and the use of climate information

Output 3: Climate-proof flood protection measures scaled-up through new and improved national and local investment frameworks increasing resilience of the most vulnerable groups to climate induced flooding

Activity 3.1: Develop a country-wide investment framework for climate induced flood risk reduction and management including provisions for private sector engagement in climate risk financing. (GCF Financing = \$480,440; Co-Financing = \$603,174)

111. Climate **risk financing** and **risk transfer** in B&H is currently largely undertaken by the public sector with support from the international donor community and lacks the financing mechanisms and tools that would enable long-term comprehensive and consistent financing and implementation of risk management and risk reduction and financial resilience building measures. Willingness to pay (WTP) surveys undertaken during feasibility study found that, for a number of public and private sectors and different types of beneficiaries, there was willingness to contribute to risk financing and risk transfer, to ensure the effective long-term management of flood risk and risk reduction and to build financial resilience to flooding. The survey identified key private sector players with an interest in sector-specific climate risk information products, and their willingness to pay for tailored products that will be used in their operations, and their willingness to support or partially support the O&M of hydrometric monitoring and early warning systems, equipment and information products has been assessed and determined through consultation during proposal development. The WTP studies also identified key sector players with a desire for natural disasters' insurance as an adequate alternative to compensation for the damage in terms of their own resources, credits and donations (Annex H of FS). Significant aspect defining the demand of businesses for natural disasters' insurance have been identified.

112. The project will develop the Investment framework for climate induced floods risk reduction and management including provisions for public and private/productive sector engagement in climate risk financing and risk transfer mechanisms. No GCF proceeds will be used to provide insurance.

113. Under this activity the project will develop **risk financing and transfer mechanisms** based on detailed socio-economic risk, damages, and losses assessment. The work will be largely co-financed by the partners from national insurance sector. This activity will include:

- The identification of public-sector risk financing mechanisms for flood risk management;
- Identification and development of risk financing and transfer mechanisms products and tools, based on detailed socio-economic risk, damages and losses assessment;
- Development of natural disasters' insurance products for public institutions, businesses and agriculture
- Technical advice/training to financial institutions and financiers, including micro-financing institutions, for the development of lending products to mainstream the use of climate-risk information into their lending

processes. that maximises performance and minimises risks and to provide loans to climate responsive/resilient enterprises in B&H (i.e., loans that offer better conditions than the current market practices to borrowers - the conditions are interest rate, loan period and easier credit criteria). These loans will service Small and Medium Enterprises and will provide loans for start-up or expanding climate resilient enterprises, or climate proofing the enterprise.

114. With regard to public sector **risk financing**, the project will build upon municipal level climate risk-informed budgeting mechanisms developed under NAP project and examine opportunities to introduce climate risk-informed budgeting into B&H's existing public financial management (PFM) framework leveraging on existing strengths and addressing any limitations of the existing budgeting processes that currently hinder climate risk-informed budgeting for flood risk management. This will include using the damage and loss and risk modelling tools (Sub-activity 1.2) to provide the evidence basis of existing cost of disasters to enable allocation of funding to different activities within the FRM framework and the investment required, FRM/disaster budget ring-fencing mechanisms (such as introduction of specific budget codes), introduction of value for money (VfM) analysis and reporting based on cost-benefit analysis and data to demonstrate the advantages (and higher VfM) of greater spending on FRM, preparedness and mitigation. Similarly, mechanisms to monitor whether funds indeed achieved their intended purpose through monitoring to enable performance-based-budgeting, integrating FRM performance objectives and establishing a reporting system that enables tracking disaster related expenditure on an annual basis (e.g. as part of the Spending Review process). Such evaluations of key programmes will generate key evidence which should feed back into the budget allocation process.
115. The project will develop and implement a programme of engagement of relevant state institutions, insurers and reinsurers, financial institutions, capital market institutions, forming key **public-private sector partnerships** to leverage and scale up investments. The project is also making interventions at the legislative and governance levels and strengthening the financial institutions to provide climate responsive financing options and products within the existing markets, which will ensure long-term sustainability of the mobilized investments, with the opportunity for future scaling up. The project will provide technical advice and training to financial institutions and financiers, including micro-financing institutions, for the development of lending products and to enable them to mainstream the use of climate-risk information into their lending processes that maximizes performance and minimizes risks and to provide loans to climate responsive/resilient enterprises in B&H (i.e., loans that offer better conditions than the current market practices to borrowers - the conditions are interest rate, loan period and easier credit criteria). These loans will service Small and Medium Enterprises and will provide loans for start-up or expanding climate resilient enterprises, or climate proofing the enterprise.
116. A key **risk transfer** mechanism which will be supported by the project is natural disaster risk insurance. The project will further develop an enabling regulatory environment and products for natural disasters insurance for individual households that were initiated in Vrbas project as well as explore opportunities and identify other products and tools. The Vrbas project identified and developed the type of scheme that would be most appropriate for B&H and considered *ex-ante*, *ex-post*, index-based, and indemnity-based schemes, with various configurations of the roles of both private insurance sector and government. Please refer to the Feasibility Study (Chapter 10 and Chapter 21, Annex H) for the detailed recommendations. In addition, the draft Terms and conditions and tariff for mandatory property insurance from damages caused by natural disasters, was developed. Initial consultation with key stakeholders (community, municipality, private sector companies, insurance companies) on the proposed flood insurance scheme and the insurance zone designations revealed general acceptance of the proposed scheme (see Annex 7 for stakeholder consultations and Annex 2, Feasibility Study for the results of the WTP survey). The Ministry of Finance has been actively engaged in the consultations. Also, the Ministry of Agriculture, Forestry and Water Management is committed to develop a flood insurance framework within EU Flood directive, which was included into the Law on Waters in the RS. Moreover, private sector insurance companies as well as reinsurance companies are providing 75% co financing to support this activity. The project will provide technical advice to identify potential sector-specific natural disaster (including floods) insurance product packages for each of the defined sectors, and provide technical advice and training of insurance sector practitioners in the development of insurance products packages from natural disasters adjusted to the needs of the assessed sectors and insurers. The project will support the introduction of micro-insurance for agricultural producers, entrepreneurs and small and medium-sized enterprises in different industries based on feasibility of connecting micro-insurance and microfinance. The micro-insurance concept is largely unknown in B&H, while the microfinance sector, is quite well-developed and regulated, which is often prerequisite for the successful launch of microinsurance in a new market.

Hence the project will leverage and enhance existing micro-finance capacity in B&H to extend to include micro-insurance. In order for the micro-insurance program to be implemented in practice, the project will facilitate changes to the current regulations in the field of insurance. In addition, the project will develop three feasibility studies which include, a) micro-insurance for agricultural producers, entrepreneurs, and small and medium-sized enterprises in different industries; b) regional disaster risk insurance program to increase the degree of diversification of risk; c) catastrophe bonds to transfer disaster risk to the capital market.

117. By the end of the project the following will be achieved: public sector risk financing and budgeting mechanisms developed, public-private sector partnerships established, and public-private sector climate risk financing mechanisms and tools developed, natural disasters insurance model for public institutions; natural disasters insurance model for agriculture, natural disasters insurance model for businesses.

118. As part of the investment framework, under this activity the project will also develop a methodology and a tool for appraisal-led design for structural and non-structural climate-informed FRM measures, FRM investment planning and decision-making, based on cost benefit analysis (CBA). The project will scale up the bespoke GIS-based socio-economic risk model that was developed for the Vrbas project to provide a country-wide tool for flood socio-economic risk modelling, CBA and the identification and appraisal of climate resilient intervention measures. The tool will support strategic planning, development of risk financing, transfer and management strategies and investment planning. The GIS-based tool will be extended to include an additional module to enable identification of flood risk financing requirements for anticipatory, short, medium, and long-term measures, and strategies from which annual investment plans can be developed. A key element of FbF is the allocation of financial resources in advance for the anticipatory response to disasters, based on specific forecast threshold that triggers the release of those resources for the implementation of early actions based on an Early Action Plan (EAP). Under this output the project will develop mechanisms and tools for ensuring financing for early action interventions to be included in the EAP and the municipal investment plans (See Activity 3.2 below). FRM practitioners will be trained in the use of the appraisal and financing methods and models and importantly, capacity will be built to enable the use of updating and maintenance of the models. Municipal planners and managers will also be trained in the use of the models for appraisal led FRM planning.

119. Hence under this activity the project is developing innovative financing options across risk financing and risk transfer, enabling blended finance (state funds, loans, micro-finance, insurance, micro-insurance, regional insurance pool, catastrophe bonds), forming key public-private sector partnerships to leverage and scale up investments, and is making interventions at the legislative and governance levels and strengthening the financial institutions to provide climate responsive financing options. This proposal therefore represents a good example of finance blending and innovative adaptation finance.

120. **Gender mainstreaming Activities:** Representatives of different genders and socially excluded groups will be included in the teams for development of a national investment framework for climate induced floods risk reduction and management; Development of a gender responsive strategy and gender sensitive tools for private sector engagement in long-term climate-proof risk management. Development of risk financing and transfer mechanisms based on detailed gender sensitive assessment on socio-economic risk, damages, and losses.

121. The following sub-activities will be supported:

Sub-Activity 3.1.1: Develop investment framework for climate induced floods risk reduction and management including provisions for public and private/productive sector engagement in climate risk financing

Sub-Activity 3.1.2: Develop risk financing and transfer mechanisms based on detailed socio-economic risk, damages, and losses assessment

Sub-Activity 3.1.3: Design natural disasters insurance scheme and explore and identify other risk financing products and tools

Sub-Activity 3.1.4: Develop tool for appraisal-led design for structural and non-structural FRM measures, FRM investment planning, climate risk financing mechanisms and for appraisal-led FRM options design and decision-making, based on cost benefit analysis (CBA) approaches

Activity 3.1 Deliverables:

- 5 strategies and plans for long-term ongoing investment frameworks that would be required to address flood risk to each sector;
- Sector-specific information dissemination portals as part of the information platform to be developed under 1.4.3;
- 2 risk financing and risk transfer strategy for private sector engagement in risk financing and risk transfer mechanisms for entity-level flood risk financing and resilience strategy;
- 3 natural disasters products developed;
- 3 Feasibility studies on micro-insurance, regional disaster risk insurance program and catastrophe bonds;
- 4 sector specific awareness campaigns;
- 2 tools for appraisal-led design for structural and non-structural FRM measures based on cost benefit analysis (CBA) approaches.

Activity 3.2: Formulate multi-year climate resilient municipal investment plans and gender responsive community preparedness plans in selected municipalities and in one canton in Vrbas, Una-Sana, Bosna, Drina, Neretva and Trebišnjica basins. (GCF Financing = \$ 525,000; Co-Financing = \$ 3,097,884)

122. The project will develop municipal multi-year investment plans for climate resilient FRM planning for 10-12 highest risk communities and for one canton. It will develop new approaches to investment planning to include: annual and periodic maintenance of FRM interventions; identification of financing models for investment maintenance costs (e.g. of community-based scheme that involve the use of tariffs or in-kind contributions to establish municipal maintenance programmes or engagement of private sector in infrastructure maintenance financing); event-based forecasting of financing that will be required to respond to disaster events (FbF) and development of municipal FRM investment plans based on risk-informed project designs, including maintenance, and CBA.. For FbF, the investment plans will define the financial resources that will need to be made available in advance of flood events, based on the specific forecast threshold (See Activity 1.3) that would trigger the release of those resources for the implementation of early actions. The Early Action Plan (EAP) will be part of the preparedness response plan, and will outline the roles and responsibilities of those involved in implementing the early actions. The project will work with the IFRC to develop the FbF components of the municipal investment plans and the EAP.
123. The project will support embedding the use of municipal investment plans in the technical justification for budget allocation for FRM activities; identifying and prioritizing flood risk financing, based on the principles of portfolio risk assessment (PRA) and associated cost-benefit analysis. Furthermore, the CBA tools to be developed by the project will be embedded in municipality as a standardized requirement for developing annual infrastructure investment plans as well as Forecast based Financing. As torrents management comes directly under municipal jurisdiction, these infrastructure investment plans will include detail design for torrents management.
124. Using the example of Community Emergency Flood Preparedness plans (CEFP) the project will also support the development of community preparedness plans for the most vulnerable 10-12 communities in target basins, (excluding Vrbas) in line with the procedure, methodology and content of CEFP in B&H as defined by Law of Protection and Rescue in RS and B&H including cantons. According to the Law, the CEFP will consist of the following documents: Vulnerability and risk assessment, Prevention plan, Preparedness plan, Mobilization plan, Emergency plan, Evacuation plan.
125. **Gender mainstreaming Activities:** Both, men and women are adequately represented in municipal investment planning process; Men and women actively participate in formulation of community preparedness plans in selected municipalities; Needs of women and other vulnerable (elderly, disabled, children, Roma) groups are incorporated in developed plans.

126. The following sub-activities will be implemented:

Sub-Activity 3.2.1: Develop municipal investment plans for climate resilient FRM planning for 10-12 highest risk communities and 1 canton.

Sub-Activity 3.2.2: Develop preparedness plans for 10-12 highest risk communities and 1 canton

Activity 3.2 Deliverables:

- 2 financing models for investment maintenance costs;
- Standardized CBA tools embedded in municipality for developing annual infrastructure investment plans;
- 10-12 municipal FRM investment plans based on risk-informed project designs, including maintenance, and costs-benefit analysis based on CBA methods and models;
- Multi-year climate resilient municipal investment plan and gender responsive community preparedness plan implemented in the highest risk municipalities (10-12) in Vrbas, Una-Sana and Bosna basins;
- Community preparedness plans for the 10-12 communities in line with the procedure, methodology and content of CEF in B&H as defined by Law of Protection and Rescue in RS and FB&H

Activity 3.3: Implement climate-proof structural flood risk reduction and anti-erosion interventions in Vrbas, Una-Sana, Bosna and Drina River basins. (GCF Financing = \$875,280; Co-Financing = \$ 20,939,153)

127. This activity will focus on implementation of priority structural flood risk reduction interventions for target basins at highest risk. The structural measures will be co-financed through EIB loan resources and Water Agencies' budgets. GCF funds will provide technical assistance to develop climate proofed designs based on new climate risk information and design methods. The project will embed climate-change responsive design of flood defenses and will develop standard methodologies, tools, guidelines and capacity for climate resilient flood defense design and implementation. It will utilize flood hazard and risk modelling and will embed CBA and appraisal-led prioritization methods in developing and implementing structural measures that maximize benefit, make allowance for climate change and which consider environmental impacts. **10 structural measures** have been identified by Water Agencies, for implementation (See Annex A of FS). The risk reduction interventions will be implemented in high-risk river basin, which have been prioritized during the project design phase as detailed in Section 2.6 of the feasibility study. Based on flood socio-economic risk assessment and identification of the highest risk river basins, the target sites are located in Vrbas, Una-Sana and Bosna, Neretva and Trebisnjica and Drina River basins.

128. **Gender mainstreaming Activities:** Both, women and men, and other vulnerable groups (elderly, disabled, children, Roma) will be included in processes of stakeholders' consultation related to implementation of climate-proof flood risk measures through the site-specific ESIA and ESMPs to be conducted for each structural measure. The Project will ensure that specific ways in which flood risk and erosion impact women and men are analyzed and considered, and that women and vulnerable group members can equally benefit from technical assistance ensured with GCF funds, to develop climate proofed designs based on new climate risk information and design methods. The Project will assess and make provisions for women to equally benefit from the work, employment and income making opportunities associated with implementing these measures.

129. The following sub-activities will be implemented:

Sub-Activity 3.3.1: Finalize detailed design of climate resilient flood protection structural measures;

Sub-Activity 3.3.2: Implement new flood defences and the rehabilitation and upgrade of existing flood defences with climate proofing.

Activity 3.3 Deliverables:

- Detailed design of 10 climate-proofed structural measures;
- 10 EIB-funded climate-proofed structural measures implemented.

COVID-19 Project co-benefits and opportunities

130. The current COVID-19 global pandemic has highlighted the need for a more holistic approach to hazard identification and risk management when faced with systemic risks which are complex, interlinked and non-linear. According to UNDRR, with reference to COVID-19, there is an urgent need to change our perception of risks and their governance mechanisms, and apply a multisectoral and interdisciplinary approach as these mechanisms

become more urgent when confronted with climate-related hazards during a pandemic crisis⁴¹. This in line with the approach that is being proposed for this project to address FRM in B&H. Hence, to ensure that the project meets the additional challenges and opportunities presented by COVID-19, the following table is list of opportunities and co-benefits of the project in addressing some aspects of the impacts of COVID-19 on communities of B&H. A risk management matrix will be developed for the project and will include close monitoring of the risks and opportunities that COVID-19 will present to the project.

Output/Activity	COVID-19 co-benefits/opportunities
Output 1: Climate-informed impact-based FFEWS and an increased generation and use of climate data to reduce vulnerability to flood related disasters	
Activity 1.1: Upgrade and expand the coverage of the hydrometric network for enhanced monitoring of climate variables in Category I catchments and torrential streams	Expansion of the hydrometric observation network will underly the impact based FFEWS which will provide timely information about occurrence of flood which will enable people to take early measures to minimise flood impact as well as measures to secure drinkable water, food and medicine, thus enabling health and sanitation conditions and reducing potential exposure to virus in emergency centers.
Activity 1.2: Enhance climate-induced flood hazard, risk and vulnerability information for strategic risk management and sound decision making for climate induced flood management.	Flood protection of critical infrastructure such as municipal health centers and hospitals, to ensure full operational capacity of those institutions in the time of pandemic. The project, through hazard and risk modelling and mapping is identifying critical infrastructure by type including health facilities that might be at risk from flooding. Information collected about the facilities that relates to the capacity to treat COVID-19 will also be included in the hazard and risk modelling and mapping. The global COVID pandemic has increased the vulnerability of communities in B&H. Under this activity, the project is undertaking socio-economic risk and vulnerability modelling and mapping based on socio-economic factors derived from socio-economic secondary data and primary data from survey methods to be developed by the project. Hence the relevant COVID-19-related vulnerability factors can be built into the surveys and reflected in risk modelling and risk assessment. Opportunities will be sought to add a COVID-19 factor/factors to the risk and vulnerability modelling methodology to ensure that the risk assessment methodology reflects COVID impacts. Based on vulnerability surveys, modelling and mapping, the project will help identify what additional resources Water agencies and Civil protection units may need to allocate to priority areas to protect at risk critical infrastructure (including health facilities, utilities like water supply, power stations etc), and to prevent cascade impacts of floods and spread of COVID-19. New approaches will require more resources of financing, human, material than ones before the COVID-19 era.
Activity 1.3: Develop an integrated impact-based centralized and community-based flood forecasting and early warning system (FFEWS).	Proper response to flooding would reduce the spread COVID-19 and reduce disaster damage at the same time, thus minimising further human losses and economic damage. Response activities that include social distancing will be developed as part of the FFEWS and response activities, at community level. This will be particularly important when developing the Community-based EWS. Community engagement will include hygiene promotion with a focus on handwashing and social distancing requires full engagement of communities. Opportunities will be sought for tailoring warning to include any relevant additional COVID-19 relevant warnings.

⁴¹ Raúl Salazar, Chief of the United Nations Office for Disaster Risk Reduction (UNDRR) – Regional Office for the Americas and the Caribbean.
<https://www.undrr.org/news/understanding-complexity-risk-post-covid-19-challenge>

	This activity will also improve notification of flood risk/impacted areas so that emergency services, including frontline workers dealing with COVID, are less likely to be adversely impacted (i.e. less likely to get caught by floods or take routes that are likely/already to be flooded). Overall this will help in planning responses to emergencies, which include COVID outbreaks.
Activity 1.4: Develop and implement protocols and SoPs on data generation, data management and communication for effective impact based FFEWS and flood risk management.	Under this activity, the project is developing the data sharing platforms (scaling up the Vrbas Geo-Portal), one of which will be used for community climate risk information sharing and dissemination. This presents an opportunity to include relevant information. Appropriate risk information is critical to avoid rumors and misinformation. The project will tailor information to include most accurate advice on the likely combined impacts of floods and the likely additional COVID-19 risks.
Output 2: Non-structural flood risk reduction measures and nature-based solutions mainstreamed in sectoral policies and plans and effectively contribute to protection of people and livelihoods from climate-induced flood risk;	
Activity 2.1: Mainstream climate induced flood risk reduction into sectoral planning (agriculture, hydropower, forestry, environment), critical infrastructure and spatial planning	A key aim of this activity, is to strengthen cross-sectoral cooperation and use of climate risk information in sectoral planning, With B&H's complex governance structure there are opportunities to enhance the level of cooperation between agencies and thereby reduce 'gaps' in services/administration. The project includes capacity building/engagement of agencies which will help increase cooperation and develop greater transparency between agencies.
Activity 2.2: Implement and mainstream new ecosystem-based flood risk reduction and climate change adaptation methods	Protection from flood risk will enhance protection of drinking water infrastructure, ensuring communities' access to clean drinking water and adequate hygienic conditions, that will help reduce the chance of potential spreading of COVID-19. Importantly, flood protective structures that built resilience will ensure that economic stress of the impact of COVID-19 are not further exacerbated during flood events.
Activity 2.3: Codify and mainstream EbA solutions into policies and regulations and promote non-structural measures among decision makers and communities	The types of works proposed can be delivered in a COVID safe manner, such that workers do not need to be in close proximity to each other to undertake tasks – workforce will be mostly outdoors and can be deployed such that contact is limited. Any safety briefings, toolbox meetings, project briefings etc will be undertaken in a COVID safe manner i.e. either adopting social distancing or utilising electronic communications.
Activity 2.4: Review and strengthen institutional capacity and develop long-term institutional capacity development plans for climate resilient FRM.	Trainings and knowledge transfers to be conducted by the project will include training/awareness in epidemic recommendations, while ensuring all participants will have proper social distancing, wearing masks and gloves in well ventilated spaces. Training and awareness raising will include, promotion of good hygiene practices, for example by leading awareness campaigns about COVID-19 transmission and care or providing material support for handwashing in high-risk locations.
Output 3: Climate-proof flood protection measures scaled-up through new and improved national and local investment frameworks increasing resilience of the most vulnerable groups to climate induced flooding	
Activity 3.1: Develop a country-wide investment framework for climate induced flood risk reduction and management including provisions for private sector engagement in climate risk financing	The investment framework which will be based on risk modelling, will be COVID-19 responsive once the socio-economic surveys on which risk modelling will be based, identify the new emerging vulnerabilities due to COVID-19. Opportunities would be sought to widen the type and number of Private and public sector players who could participate in FRM investment. The project will seek opportunities to build broader cross-sectoral partnerships when developing the investment framework that will underpin FRM in the future to include partners with capacity to

	provide co-benefits to the impacts of COVID-19 pandemic as well as FRM.
Activity 3.2: Formulate multi-year climate resilient municipal investment plans and gender responsive community preparedness plans in selected municipalities and in one canton in Vrbas, Una-Sana, Bosna, Drina, Neretva and Trebišnjica basins	Ministries, Water agencies and Civil protection units should allocate resources to priority areas. New approaches require more resources of financing, human, material than ones before the COVID-19 era. Importantly municipality investment plans and community preparedness plans will be fully COVID-19 responsive and will take account of the need for municipalities to allocate resources differently.
Activity 3.3: Implement climate-proof structural flood risk reduction and anti-erosion interventions in Vrbas, Una-Sana, Bosna and Drina river basins.	Investment in new infrastructure will create jobs and deliver tangible assets that will fuel long-term economic growth, reducing current impact of COVID-19 on local economies. By reducing impacts of floods on communities and infrastructure, the burden on hospitals and emergency services is reduced thereby maintaining capacity to deal with COVID pandemic – it is well recognised that overloading medical facilities is a key risk associated with the pandemic, therefore any activities that help reduce other disasters that could overwhelm hospitals are of benefit in managing the ongoing COVID threat. Protection from flood risk will enhance protection of drinking water infrastructure, ensuring communities' access to clean drinking water and adequate hygienic conditions, that will help reduce the chance of potential spreading of COVID-19. Importantly, flood protective structures that build resilience will ensure that economic stress of the impact of COVID-19 are not further exacerbated during flood events When assessing the vulnerability of communities that will benefit from the flood structural measures the project will ensure the vulnerable get priority attention during the planning and implementation of measures. The elderly and infirm are most at risk to succumbing to the virus so consideration will be made to ensure adequate hygienic conditions. Low-income communities and informal settlements may be particularly vulnerable to spreading because of crowding and inadequate services. The project will take full account of such vulnerabilities in the site-specific ESPs and implement the necessary mitigation measures.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

131. The project will be implemented by UNDP following UNDP's Direct Implementation Modality (DIM), according to the Standard Basic Assistance Agreement between UNDP and the Government of B&H (SBAA, 1995) and the policies and procedures outlined in the UNDP Programme and Operations Policies and Procedures (POPP) (see <https://info.undp.org/global/popp/ppm/Pages/Defining-a-Project.aspx>). UNDP will assume full responsibility and accountability for the overall project management, including monitoring and evaluation of project interventions, achievement of project outputs and specified results, the efficient and effective use of resources, and reporting to GCF.
132. The Government of B&H through the GCF NDA requested UNDP to deliver this project through the DIM modality. DIM modality is justified given the complexity of B&H's administrative set-up, that includes two self-governing entities and applies multi-layered administrative procedures. The United Nations Development Assistance Framework (UNDAF) for 2015–2019 and UNDP's Country Programme Document (CPD) for 2015 – 2019 for Bosnia and Herzegovina, as well as the Draft CPD for Bosnia and Herzegovina (2021-2025) stipulate that all projects be implemented under the direct implementation modality (DIM). The DIM will be applied in a way to maximize cost-effectiveness and tailored capacity development of counterpart government institutions.
133. UNDP provides a three – tier oversight and quality assurance role involving UNDP staff in Country Offices and at regional and headquarters levels. This includes management of funds, programme quality assurance, fiduciary risk management, timely delivery of financial and programme reports to GCF and other requirements as per the AMA. The quality assurance role supports the Project Board by carrying out objective and independent project oversight and monitoring functions. This role ensures appropriate project management milestones are managed

and completed. Project Assurance must be independent of the Project Management function; the Project Board cannot delegate any of its quality assurance responsibilities to the Project Manager. The project assurance role is covered by the accredited entity fee provided by the GCF. As an Accredited Entity to the GCF, UNDP is required to deliver GCF-specific oversight and quality assurance services including: (i) Day-to-day oversight supervision, (ii) Oversight of project completion, (iii) Oversight of project reporting. UNDP's responsibilities are outlined in the AMA that has been entered into between GCF and UNDP and will also be outlined in the FAA for this project. The FAA and AMA will govern UNDP's responsibilities for GCF. The 'Development Partner' role of UNDP is to represent the interests of the parties, which provide funding and/or technical expertise to the project (designing, developing, facilitating, procuring, implementing) and is covered by the accredited entity fee provided by the GCF. The Development Partner's primary function within the Board is to provide guidance regarding the technical feasibility of the project.

134. The implementation of the project will be carried out in collaboration and consultation with B&H GCF Focal Point the Ministry of Foreign Trade and Economic Relations (MoFTER) and UNFCCC Focal Point, the Ministry of Spatial Planning, Construction and Ecology of the Republic of Srpska (MSPCEE).
135. The **Project Board (PB)** is comprised of the following organizations: UNDP, Ministry of Foreign Trade and Economic Relations of Bosnia and Herzegovina (MOFTER, GCF Focal Point), The Ministry for Spatial Planning, Civil Engineering and Ecology (MSPCEE, UNFCCC focal point), , Ministry of Agriculture, Water Management, and Forestry of Federation of B&H, Ministry of Agriculture, Forestry, and Water Resources of Republika Srpska, State, Entity and Canton Red Cross⁴², Association of Cities and Municipalities⁴³, Entity, Canton and municipal Civil Protection. The Project Board is responsible for making, by consensus, management decisions when guidance is required by the Project Manager. Project Board decisions will be made in accordance with standards that shall ensure management for development results, best value for money, fairness, integrity, transparency and effective international competition. In case a consensus cannot be reached within the Board, the final decision shall rest with the UNDP. The Project Board will meet as necessary and agreed, and at least once a year.
136. The Project Manager (PM), under supervision of UNDP will run the project on a day-to-day basis within the constraints laid down by the Project Board. The Project Manager function will end when the final project terminal evaluation report and other documentation required by the GCF and UNDP has been completed and submitted to UNDP. The Project Manager is responsible for day-to-day management and decision-making for the project within the Annual Work Plan endorsed by the PB and approved by UNDP. The Project Manager's prime responsibility is to ensure that the project produces the results specified in the project document, to the required standard of quality and within the specified constraints of time and cost. The annual work plan is prepared by the Project Manager and reviewed and approved by PB. The Project Manager is also responsible for managing and monitoring the project risks initially identified and submit new risks to the project board for consideration and decision on possible actions if required and update the status of these risks by maintaining the project risks log according to the DIM

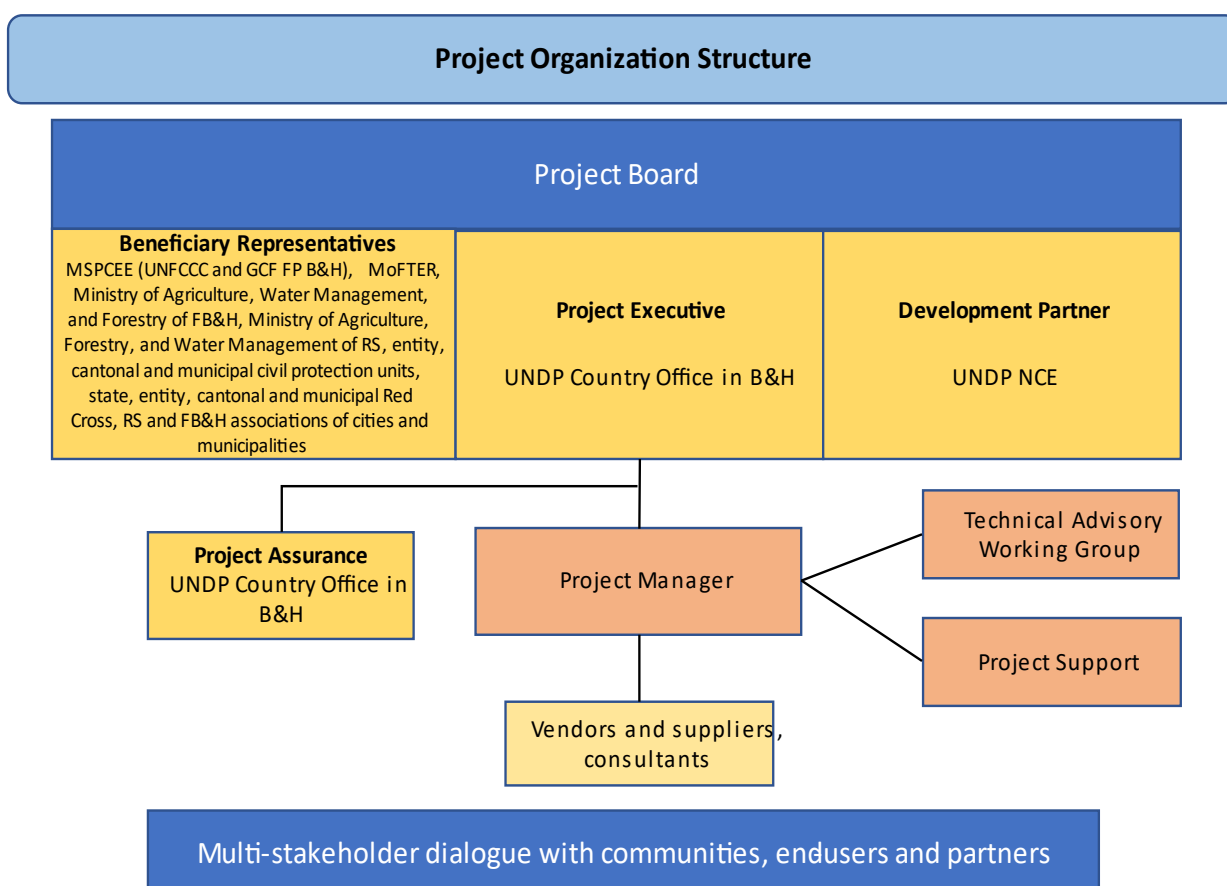
⁴² At the state level the Society of Red Cross (RC) BiH is registered as nonprofit organization and is financed by state institutions on a regular basis and consists of members of Red Cross of RS, FBiH and Brcko district. It has only overall coordination role, with competences derived from international agreements and in case of emergency, for coordination of international cooperation and support. At the sub national-entity level, the Red Cross of Republika Srpska (RS): is registered as nongovernmental organization, even though it is financed by Ministry of Health of Republika Srpska on an annual basis. It is a voluntary, humanitarian organization that operates on the territory of Republika Srpska. It is defined as a subsidiary body of public authorities in humanitarian areas. It has a coordination role over 64 Red Cross organizations registered as legal entities in local communities in RS, and 9 coordination boards (with no legal status) especially during the emergency/crisis situations. Red Cross of Federation of BiH (FBiH) is registered as association of citizens, a voluntary, humanitarian organization that operates on the territory of Federation BiH. It is financed by Federal Ministry of Labor and Social Policy, by grant on an annual basis. Additional funding is for trainings on first aid (for drivers) and small amount from lottery. However, main source of financing is through different project activities. It has coordination role over 79 Red Cross organizations registered as legal entities in local communities in FBiH and 10 cantonal organizations.

⁴³ There are two Associations of cities and municipalities in BiH: The Association of Municipalities and Cities of Republika Srpska (SOGRS), and Association of Municipalities and Cities of the Federation of Bosnia and Herzegovina (SOGFBiH). All cities and municipalities in BiH (143) including Brcko District, which is member of UDOFBiH, are members of the Associations. Both these Associations are independent organizations of voluntarily associated municipalities and cities in the Federation of Bosnia and Herzegovina and Republika Srpska, respectively, established to develop local self-government and promote and protect their common interests. The mandate of SOGRS and SOGFBiH is to represent the interests of their members before higher authorities, other bodies and organizations, in BiH and abroad, represent and advocate the common interests of membership; provide services and support to members for the development of their capacities; Information sharing, communication and mutual cooperation of local authorities and cooperation with other actors; support to local authorities in international cooperation and in the process of BiH's accession to the European Union. The legal basis for the establishment of SOG is the Law on Local Self-Government, which regulates that local self-government units may join the Union of Municipalities and Cities, to promote and protect their common interests, and that SOG may join international associations of local authorities and cooperate with appropriate associations of local authorities in BiH and abroad, in accordance with the law. SOGs are members of recognized international organizations, such as the Council of European Municipalities and Regions (CEMR), the United Cities and Local Authorities (UCLG), and the Network of Associations of Local Authorities of Southeast Europe (NALAS).

Guidelines. UNDP contracts the Project Manager, who should be different from UNDP's representative in the Project Board.

137. The Project Management Unit (PMU) will be hosted by the UNDP Country Office in BiH. The PMU will also include a safeguards officer who will be responsible for overseeing and coordinating the implementation of safeguards across the whole of the project, ensuring consistency of approach and reporting from the separate Water Agencies/contractors. The Safeguards Officer will support the application of the ESMP and other project safeguard documents so as to meet UNDP SES Standard requirements. The Safeguard officer will also be responsible for ensuring that SEP/GRM is functional.

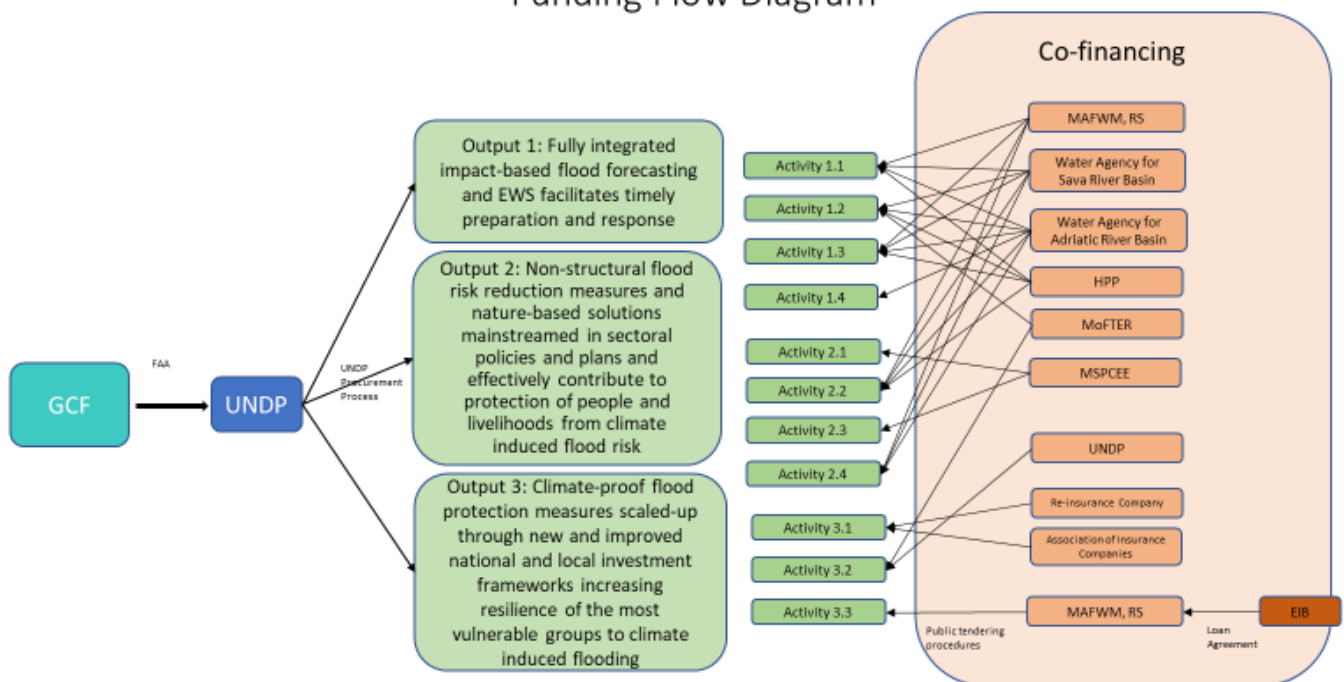
138. The project organisation structure is as follows:



139. The International Chief Technical Advisor (CTA) will provide regular technical guidance to the project management and technical teams in managerial and technical issues. He/she will be hired for a long-term during the entire project implementation period by UNDP based on UNDP recruitment procedures.
140. The International Safeguard Specialist will ensure implementation of ESMP, develop, and implement site specific environmental and social safeguards plan, as well address grievances.
141. Technical Advisory Working Groups (TAWG) will support the CTA and PM and will provide inputs to and endorsement of the design and quality of the project outputs. TAWGs members will be drawn from government, private sector, academia and civil society to provide guidance and technical advice on the project (hydro-meteorological institutes, statistics institutes, agricultural institutes, local governments together with associations of cities and municipalities of Republika Srpska and Federation of B&H Environmental Protection Funds, the civil society and other local and international NGOs). A balanced representation of women and men in the TAWGs will be ensured.

142. A Gender Advisor will be engaged by the project and will be a member of all TAWGs to ensure that gender is adequately mainstreamed in all technical discussions.
143. Local stakeholders and community members have a key role in the implementation and monitoring of the project. Local community consultations have been conducted during the project preparation phase (see Stakeholder Assessment and Engagement Plan). During the inception phase of the project, the project will additionally consult with all stakeholders, including vulnerable community members and local government, etc. and facilitate an understanding of the roles, functions, and responsibilities within the Project's decision-making structures, including reporting and communication lines, and conflict resolution mechanisms. Local community consultation forums will be established at community levels to maintain dialogue with the local beneficiaries and stakeholders throughout the project implementation. The stakeholders will also be engaged during the mid-term and final evaluations to assess the progress of the project and enable adaptive project management in response to the needs and priorities of the communities. Community forums will be coordinated and managed through the partnerships with the Society of Red Cross (RC) B&H and its members - Red Cross of RS, FB&H and Brcko district (see footnote 39 above), Association of Cities and Municipalities (for all cities and all 143 municipalities in B&H), Civil Protection (CP) and a network of volunteer groups⁴⁴ and existing civil society partners who provide flood risk management, preparedness and response services to municipalities. These partnerships will support the implementation of the community activities and interventions of the project. As a minimum, bi-annual community forums will be organised in each entity, with more frequent meetings/forums as needed.

Funding Flow Diagram



B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

144. The total annual budget for climate adaptation and resilience activities in B&H related to flood risk management averages US\$40 Million, comprising mainly donor grants (40%) and loans (60%) while the damages due to flooding is expected to reach US\$ 749.3 Million Expected Annual Damages (EAD) for households, businesses, agriculture, and public buildings by mid-century due to climate change. The shortfall in expenditure on climate change adaptation is due to a current lack of financial capacity of the B&H Government to fund such activities

⁴⁴ Both, CP and RC in critical situations engage volunteers from local communities. They have updated lists with contacts of volunteers, regularly updated and organize trainings and education for them. In addition they contract specialized volunteer groups (such as mountaineering, rafting clubs), specifically for protection and rescue.

without assistance⁴⁵. Bosnia can access loans to undertake structural measures and will take a loan with EIB to finance the highest priority structural measures, but it needs technical assistance through timely complementary intervention to shift the current paradigm from piecemeal, *ex-post facto*, project-based efforts to the adaptation alternative which will bring long-term transformative change. In so doing the large expected financial burden of damages and losses due to the climate-change induced impact of extreme hazardous flood events can be avoided in the long-term.

145. The proposed project represents a blended finance package that leverages international climate finance from GCF (19.8% of the total project value), EIB loan to the Government of B&H for flood protection measures (29% of the total project value), B&H public funds provided by the Entities and municipalities for structural and non-structural measures and O&M (42.3% of the total project budget), and the national private sector funds (HPP, insurance sector, 7.7% of the total project budget). The GCF grant element has been critical to leverage and consolidate the overall project financing package and to secure scaling-up of the investment after the project completion. The Government of B&H and HPP partner is committed to cover O&M costs of equipment and flood defenses installed within the project over the 20 years investment lifetime (Annex 13). As part of the GCF-supported intervention, the project will develop risk finance instruments and leverage partnerships with the productive sectors to secure inflow of private investment into flood risk management, including O&M of FFEWS and flood defenses beyond the GCF-funded measures. During the project development, UNDP has conducted a series of willingness to pay studies which confirmed the willingness of productive sectors (HPPs, agriculture, tourism) to invest in the climate resilient FRM provided an adequate enabling environment is established. Thus, the proposed project will catalyze a shift in FRM financing in B&H from a purely public sector and donor-dependent funding towards a more elaborate financing strategies engaging private sector. GCF technical assistance is required to catalyze such transition.
146. GCF grant resources will provide for the technical assistance and capacity building to ensure that flood risk management is undertaken in an integrated manner that secures the lives, livelihoods and assets of the most vulnerable people in B&H and reduces their susceptibility to climate -induced flood risk that they have limited coping mechanisms to withstand. GCF investment will not result in revenue generation or financial returns to private sector, rather GCF will finance public goods and will assist the government of B&H to meet its obligations providing flood protection and early warning services to vulnerable communities in the face of climate change. Thus, GCF funding will cover additional costs to the Government associated with the increasing climate change impacts and risks, including additional monitoring of catchments susceptible to torrential flash flooding, where flood risk is increasing, and which is currently missing from the monitoring network. It will also cover the additional costs of enhanced climate data and information for effective impact based FFEWS, enhanced capacities of public sector to deliver climate information services, enhanced capacities to integrate climate risks into the design and implementation of structural and non-structural flood protection measures, and enhanced awareness of climate risks and preparedness among vulnerable communities. These project activities are essential for improving climate risk information availability to the general public, improving community resilience, and strengthening institutions with mandates to strategically manage climate risks.
147. There is currently limited participation of private sector in climate risk resilience building, and risk financing, despite the high exposure of the private sector to flood risk (US\$ 3,222 billion assets at risk). The project is aiming to develop the mechanisms for risk sharing between public and private sector, including development of public-private collaboration on building climate resilience in critical flood impacted sectors and technical advice to financial institutions to mainstream climate change risks into their business models to manage the effects.
148. Through the FRM project on Vrbas basin and other past and on-going initiatives, B&H has made an important step toward piloting successful solutions for increased resilience to climate induced floods. However, single-project interventions represent only partial solutions that cannot offer the much-needed systematic approach without nation-wide scaling up. Currently available funds, public/private or donor driven, are not sufficient to address flood risk at the appropriate scale of whole territory of B&H. GCF funding offers the possibility for such systematic intervention by enabling different types of measures that in synergy will effectively address flood risk in the main river basins in B&H.

Financial Analysis recommendations and conclusions

⁴⁵ Without the project and in the current situation government spend on FRM is less than the expected EAD and currently there is low investment in infrastructure maintenance. Without action and the gap between EAD and FRM financing will widen. Government has committed to O&M for 20 years, for all project implemented activities and infrastructure and, in addition, the FRM financing mechanisms we will be identified and established by the project, to ensure longer-term risk financing needs for O&M will be met.

149. Government investment in flood risk management is mainly focused on ongoing maintenance of very old flood defence structures, which are in poor condition as a result of wartime damage, lack of maintenance, and the presence of minefields around some facilities (TNC). Funding sources for water management (including FRM) activities are defined by the legal regulations in effect in each entity, which set out the water use, the amount of charges, beneficiaries and special water charges (See Section 3.6 of FS). Revenues generated for water charges are allocated to various priority flood risk management activities. The total annual budget for climate adaptation and resilience activities in B&H related to flood risk management averages US\$40 Million which currently comprises mainly donor grants (40%) and loans (60%). This is well below the damages due to flooding which is expected to reach US\$ 749.3 Million Expected Annual Damages (EAD) for households, businesses, agriculture, and public buildings by mid-century due to climate change. It should be noted the 2014 flood resulted in \$2.7 Million USD in damages or 15% GDP.
150. Financial analysis of this project's outputs and activities has been carried out in accordance with the Guidelines for the Financial Analysis of Projects of United Nations Development Program. These guidelines clearly mandate that a financial analysis of project cash flows be computed, and Financial Internal Rate of Return (FIRR) calculated only for those proposed project activities or outputs that can clearly result in direct and quantifiable financial revenue generation (incremental earnings from baseline) or a direct and quantifiable financial savings potential to the project owners or to the project beneficiaries. These guidelines stand to ensure that GCF's minimum concession policy is always protected in the proposal.
151. This project's proposed structure includes 4 outputs and 11 activities. However, none of the activities or outputs clearly result in direct and quantifiable earnings or direct and quantifiable savings to the project owners or the project beneficiaries. All the activities and outputs of this proposed project result only in non-attributable savings that are of public good in nature to the larger community of the at-risk population in B&H. Hence, a financial modelling-based analysis has not been conducted and an FIRR has not been calculated for any of these activities.
152. However, considering the GCF's minimum concession policy, this project's proposed activities have been analysed from a macro-economic and government perspective to assess the need for GCF grant as the only feasible financial instrument to fund the project activities. Hence, taking into consideration factors such as the public good nature of benefits arising out of the project, climate change being the key driver, economic, geographic and demographic distribution of the target population, impact of COVID-19, B&H's current economic situation, persistent fiscal risks, the World Bank's observations and the catalytic nature of GCF grants, we recommend the following:
- GCF funds will leverage co-financing in a form of grants and loan from government and private sectors and, at the same time, will ensure additionality in the area of climate changes while simultaneously avoiding diversion from development needs. There is significant co-financing from Government of B&H and there is no incremental ability to stretch their contributions owing to budgetary and fiscal consolidation reasons.
 - The nature of the benefits does not accommodate repayment of capital in whatever form or serviceability of a loan instrument.
 - The nature of the beneficiaries and the level of essentiality of the service do not accommodate repayment of capital in whatever form or serviceability of a loan instrument.
 - Increase in government spending to address COVID-19 pandemic has strained public finances.
 - Private sector willingness to pay will only cover a small percentage of costs for those services directly related to private sector interests.
 - Hence, it is recommended that in order to reduce / close the existing financing and knowledge gaps and barriers to improve resilience of B&H's population to climate change-induced hazards, catalytic capital in the form of GCF grants are essential.
153. GCF grants are critical for this proposed project given the impact of COVID-19, B&H's current economic situation and the persistent fiscal risks. B&H is a transitional economy with 51.8% of the population being rural. Due to the war time devastation and a slow transition of the economy, a large part of B&H's population still lives in poverty. The relatively slow rate of the post-war economic recovery has been compounded by the negative impacts of climate change on key sectors such as agriculture, energy (hydropower), the environment, in particular, due to the increased frequency and magnitude of flood disasters, which have tripled in the last decade.
154. B&H's economy and its state's budgets continue to be under stress due to continued tepid growth rates and insurgence of COVID-19 pandemic. Due to COVID-19, GDP growth is expected to contract between -3.2% and -

5% from a previously projected GDP growth of 3.4% for 2020 . This is expected due to revenue shortfalls viz-a-viz higher health, social, and economic-support spending. As per IMF's Country Note on B&H in response to "Request for Purchase Under the Rapid Financing Instrument", the impact of COVID-19 necessitates a large fiscal expansion to increase spending on containment and mitigation of the disease. Consequently, from a previously projected surplus of 2% of GDP, the fiscal position in 2020 is now estimated to be a deficit of 4.5% of GDP . The sovereign credit rating of B&H is "B with stable outlook" as per S&P and "B3 with stable outlook" as per Moody's which makes B&H fall in speculative credit rating category with high credit risk . In the situation that has brewed since COVID-19's spread, there has been a depreciation of 3% in local currency of B&H. This can be indicative of withdrawal of investments from domestic markets and outflow of capital.

155. Before the COVID pandemic hit the country in 2020, B&H economy growth rate slowed from 3% per annum in 2016-2018 to 2.7% in 2019. The Industrial recession in the last years is largely related to spillovers from German industrial recession in 2018-19, induced decline of B&H industrial production by 5.5% in 2019. As a consequence, B&H reached a deficit of foreign trade in 2019 of 8.0 billion KM, which is a record value since 2008. The EU is the target of 73% of B&H exports and B&H is vulnerable to output contraction in EU and CEFTA markets. In addition, the fundamental weakness of inward orientation / lack of export orientation with total exports to GDP ratio of 40%, B&H lags behind the average for small and open economies in Europe. Relatively low level of development is reflected in GDP per capita at PPP (purchasing power parity) stands at 32% of EU-27 average. Public sector output represents particularly large share of B&H economy which in structural terms represents a weakness. Private remittances associated with large diaspora and a large number of workers abroad represented around 8.5% of GDP in recent years. Current account deficit represents 31.69% of GDP in 2019 and is expected to worsen due to substantial reduction in remittances in 2020, which the IMF expects at two thirds of 2019 level.
156. According to the ILO, a large part of the workforce is trapped in low productivity jobs in agriculture, construction, tourism and retail, which on average command wages that are 10 % lower than the national average. This is a signal that the structural transformation of the country has been slow. Private enterprise sector is small (15% of working age population works in the formal private sector), it is weakly capitalized and exposed to strong economic shocks. Large public sector buffer effectively serves as a shield for about a third of total employees.
157. The COVID-related shock disrupted B&H's already weak economic growth. The IMF expects real GDP contraction of 5% in 2020, which implies GDP loss of 8% compared to the pre-COVID-19 growth trend. After expected recovery at a rate of 3.5% in 2021, the GDP is expected to remain around 7% below the pre-crisis trend. B&H authorities communicated expected GDP contractions between 3% and 5% in 2020, depending on the scenario and entity. However, these forecasts may be overly optimistic as the negative demand shock during summer months has been stronger than expected."
158. Due to these economic risks that persist and the World Bank's observations on the region, B&H is currently not in a state to raise external financing in the form of loans to replace GCF's grant for this proposed project. In addition, due to the public good nature of the benefits derived from the project, the serviceability of any such loans is uncertain. The shortfall in expenditure in B&H on climate change adaptation is due to the lack of financial capacity of the B&H Government to fund such activities without assistance. The country will take a loan from EIB to finance the highest priority structural measures, but it needs technical assistance through timely complementary intervention to shift the current paradigm from piecemeal, ex-post facto, project-based efforts to the adaptation alternative which will bring long-term transformative change.

B.6. Exit strategy (max. 500 words, approximately 1 page)

159. The project's sustainability and exit strategy is rooted in the key elements of design and implementation. The following assures long-term sustainability beyond the project implementation period:
160. **Investment in human resources and institutions:** the project is focused on strengthening institutions that will have skilled human resources, information, tools and technologies to effectively pursue their mandate in climate risk management (**Sub-activity 1.3.4 and Activity 2.4**). The project investments will improve availability of climate risk information and create effective response mechanisms. This project will incorporate lessons learned and benefit from the existing collaborative relationship between UNDP and all the key flood risk management partners and will foster strong partnerships with and among relevant institutions to ensure shared goals and shared vision beyond the project. The project will help ensure sustainability by supporting all relevant government ministries in line with government policy, programme and strategic priorities. The establishment of methods and tools for developing and implementing flood risk management strategies, the introduction of risk assessment methods,

standards, tools, guidance and regulatory documents make this project highly replicable within B&H. The embedding of risk assessment, risk management and risk reduction methods for flood hazards will be particularly important. Furthermore, this project has been developed with full awareness of other climate risk and flood risk related projects in the region, which makes it highly complementary, and potentially replicable to other projects regionally. As detailed above, this project plans to attract private sector investment – particularly in cost sharing regimes for long-term O&M for hydrometric equipment. Hence, by the end of the project, the activities and monitoring network will be self-sustaining into the future. In addition, all activities will be conducted in ways that empower local stakeholders. Capacity building programs set up will ensure the ongoing transfer of skills to create knowledge sharing and better climate services for the future. Hence, Sub-Activity 1.3.4: Develop and implement a capacity development plan for embedding flood hazard and risk modelling approaches and FFEWS into appropriate institutions in B&H, which in particular will conduct FFEWS training on the new system for all institutions involved in FFEWS as identified by institutional mapping and institutional arrangement plan for FFEWS, Activity 2.4 which will conduct training on proposed FRM policy and sectoral FRM, Sub-Activity 1.4.2 which will develop protocols and SOPs on data generation and data management for FRM Work Instructions / Directives / Standard Operating Procedures and Sub-Activity 2.1. which will develop country-wide floodplain zoning policy and legislation, based on hazard, risk and vulnerability mapping, flood resilient building codes and embed climate change considerations in the design and construction standards of critical infrastructure, will all ensure that strengthen institutional capacities and institutional arrangements, tools, technologies, enabling environments and policies, are in place within organizations when the project exits, thus ensuring sustainability, scalability and replicability of project interventions.

161. **Investment in development of long-term climate risk financing:** The project is developing fully articulated long-term municipal risk management plans for 10-12 highest risk municipalities and 1 canton, and sector resilience plans for highest risk sectors, which will set the direction for systematic risk informed sectoral flood risk management in the future. In addition, the project will develop long-term risk financing mechanisms and investment plans. Cost-benefit assessment methodologies and a socio-economic risk model will be embedded into the identification and design of climate proofed structural and non-structural measures and will form the basis of municipal investment planning. Risk financing mechanisms to be developed by the project, will ensure that long-term financial and economic resources are identified and available after GCF assistance ends and will ensure cost-effective financing of flood risk reduction measures, and investments over time. This will result in sustainable financing of FRM in B&H.
162. **Investment in social and economic assets of vulnerable communities:** The damaging effects of extreme events and changes in weather patterns exacerbated by climate change will limit the potential for socioeconomic development in vulnerable and isolated communities. However, investing in climate resilient structural and non-structural measures (**Activities 2.2 and 3.3**) can help reduce the vulnerability of these areas. Flood defenses that are “climate proofed” will have structural stability and functional longevity under climate change. Furthermore, the development and implementation of climate-proofing standards for the design and construction of critical infrastructure, will strengthen the reliability of infrastructure and, in turn, reduce the impact on communities during flood events. Investment in EbA measures provides cost efficiency and long-term sustainability as EbA investments are self-sustaining in the long run, are adaptable, and require reduced OM costs. Hence, EbA solutions are an investment in social and economic assets that will provide sustained ecosystem services. This project will embed climate risk information and enhance the capacity to prioritize, budget, locate and deploy critical infrastructure that is functionally stable under climate hazards. This will help ensure the financial sustainability of critical infrastructure climate proofing efforts for the long-term.
163. **Investment in natural capital:** Securing long-term resilience and safeguard investments and communities against climate induced flood risk, by ensuring functional and protected watersheds, stable, climate resilient and well-managed natural capital (Activities 2.2 and 2.3), the project will be investing in long-term sustainability of communities and their social and economic assets through the investment in EbA solutions, and the derived ecosystem service attendant benefits of improved functions of catchments through effective flood risk management.
164. The project will develop an integrated river-basin development approach and will implement some of the structural and non-structural intervention measures in selected high priority areas. These will provide strong replicability impact as they will establish the methods, standards and approaches that will work across B&H and define these in guidance, legal and policy documents. The potential for scaling up these approaches is therefore significant. There is at least a threefold replication potential for the EbA measures, community based EWS, municipal risk management plans, natural disasters’ insurance schemes etc. Having in mind institutional and geographical similarities of all Western Balkans countries, there is a regional replication potential for all aspects of the project.

165. **Post-project O&M:** Co-financing from government and private sector will include budgeted O&M costs for a period well beyond the project duration. In terms of maintenance of the expanded hydrometric network, during and after the end of the project (please refer to co-funding letters, Annex 13).
166. In addition, the project will assist relevant government institutions in developing long-term O&M financial planning to ensure hydrometric network equipment, FFEWS and all systems implemented under the project will be maintained in the long-term. The project interventions have also been designed to strengthen financing and implementation of O&M for implemented structural and non-structural measures in the long-term. To this end, the project will address current and future requirements for maintenance by developing and embedding CBA, asset management and portfolio risk management methods that ensure the systematic identification and prioritization of the maintenance costs over the lifespan of the infrastructure.
167. Specifically, the project will provide the following safeguards to **financial sustainability**:
- Long-term investment planning for FRM including identification of long-term O&M financing needs
 - Review budgetary requirements for long-term maintenance of hydrometric network and structural and non-structural interventions based on climate risk information to be developed by the project, and development of a financing model for the long-term maintenance (using cost-benefit modelling to support the financing model)
 - Strengthening the approach to O&M for a harmonized O&M framework by implementing capacity development in O&M of the optimized hydrometric network
 - Improve private sector involvement and partnership in the area of flood risk management.

C. FINANCING INFORMATION						
C.1. Total financing						
(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount			Currency		
	14.4 million			million USD (\$)		
GCF financial instrument	Amount	Tenor	Grace period	Pricing		
(i) Senior loans	<u>Enter amount</u>	<u>Enter years</u>	<u>Enter years</u>	<u>Enter</u> %		
(ii) Subordinated loans	<u>Enter amount</u>	<u>Enter years</u>	<u>Enter years</u>	<u>Enter</u> %		
(iii) Equity	<u>Enter amount</u>	<u>Enter years</u>		<u>Enter</u> % equity return		
(iv) Guarantees	<u>Enter amount</u>					
(v) Reimbursable grants	<u>Enter amount</u>					
(vi) Grants	14.4					
(vii) Results-based payments	<u>Enter amount</u>					
(b) Co-financing information	Total amount			Currency		
	58.288			million USD (\$)		
Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
Ministry of Agriculture, Forestry and Water Management of Republika Srpska (MAFWM RS)- EiB Loan	Senior Loans	21.11 ⁴⁶	million USD (\$)	<u>Enter years</u> <u>Enter years</u>	<u>Enter</u> %	Options
Ministry of Agriculture, Forestry and Water Management of Republika Srpska (MAFWM RS)	<u>Grant</u>	3.89 ⁴⁷	million USD (\$)	<u>Enter years</u> <u>Enter years</u>	<u>Enter</u> %	Options
Water Agency for Sava River Basin, Sarajevo (WA Sava)	<u>Grant</u>	16	million USD (\$)	<u>Enter years</u> <u>Enter years</u>	<u>Enter</u> %	Options
Water Agency for Adriatic River Basin, Mostar (WA AB)	<u>Grant</u>	5	million USD (\$)	<u>Enter years</u> <u>Enter years</u>	<u>Enter</u> %	Options

⁴⁶ EIB loan is EUR 19 mil; exchange rate 1USD=0.9 EUR

⁴⁷ Co-financing provided EUR 3.5 mil; exchange rate 1USD=0.9 EUR

Ministry of Foreign Trade and Economic Relations (GCF Focal point) (MOFTER)	<u>Grant</u>	5.56 ⁴⁸	million USD			
Power Company of Republic of Srpska, Hydro Power Plant (HPP)	<u>Grant</u>	5.0	million USD			
Re-insurance company Bosna RE (Bosna RE)	<u>Grant</u>	0.33 ⁴⁹	million USD			
Association of Insurance Companies FB&H (AOIC)	<u>Grant</u>	0.3	million USD			
Ministry of Spatial Planning, Civil Engineering and Ecology of Republika Srpska (UNFCCC Focal point) (MSPCEE)	<u>Grant</u>	0.30	million USD			
UNDP	<u>Grant</u>	0.8	million USD			
(c) Total financing (c) = (a)+(b)	Amount			Currency		
	72.69			million USD (\$)		
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	UNDP projects in B&H are exempted from VAT payment in line with conditions stipulated in the Standard Basic Assistance Agreement (SBAA). For activities related to procurement of goods and services through UNDP, according to the SBAA taxes are not applicable. Section 7 of the Convention on the Privileges and Immunities of the United Nations provides, inter alia, that the United Nations, including its subsidiary organs, is exempt from all direct taxes, except charges for utilities services, and is exempt from customs duties and charges of a similar nature in respect of articles imported or exported for its official use.					
C.2. Financing by component						
168.The Government of the B&H is requesting GCF funds to cover the additional costs (required due to the impacts of climate change) of technical assistance and capacity building for enhanced flood risk knowledge, monitoring, forecasting and warnings, for scaling up ecosystem-based and other non-structural risk management approaches, and for facilitating climate risk-informed flood protection investments (Activities 1.1, 1.2, 1.3, 1.4, 2.1, 2.2, 2.3, 2.4, 3.1, 3.2). The Governments of FB&H and RS will co-finance from the Entity budget funds the upgrading and maintenance of the hydrometeorological observation network, implementation of FFEWS, regulatory improvements, resilience and investment planning and implementation of non-structural flood risk reduction measures (Activities 1.1, 1.2, 1.3, 1.4, 2.1, 2.2, 2.3, 2.4, 3.1, 3.2). Implementation of climate-proof structural risk reduction measures (Activity 3.3.) will be co-financed through loan resources (EIB). Private sector (HPPs) will co-finance enhanced						

⁴⁸ Co-financing provided EUR 5 mil; exchange rate 1USD=0.9 EUR

⁴⁹ Co-financing provided EUR 0.3 mil; exchange rate 1USD=0.9 EUR

flood risk monitoring, forecasting and warning and structural risk reduction investments (Activities 1.1, 1.2.1.3, 1.4, 2.2).

Component	Output/Activity	Indicative cost Options	GCF financing		Co-financing		
			Amount Options	Financial Instrument	Amount Options	Financial Instrument	Name of Institutions
Component 1	1.1 Upgrade and expand the coverage of the hydrometric network for enhanced monitoring of climate variables in Category I catchments and torrential streams	4,405,946	1,535,980	Grants	2,869,966	Grants	MAFWM RS WA Sava WA AB HPP
	1.2 Enhance climate-induced flood hazard, risk and vulnerability information for strategic risk management and sound decision making for climate induced flood management	5,596,589	1,067,833	Grants	4,528,756	Grants	WA Sava WA AB MOFTE R HPP
	1.3 Develop an integrated impact-based centralized and community-based flood forecasting and early warning system (FFEWS)	4,111,478	2,156,040	Grants	1,955,438	Grants	MAFWM RS WA Sava WA AB HPP
	1.4 Develop and implement protocols and SoPs on data generation, data management and communication for effective impact based FFEWS and flood risk management	536,000	336,000	Grants	200,000	Grants	WA AB
	2.1 Mainstream climate induced flood risk reduction into sectoral planning (agriculture, hydropower, critical infrastructure) and spatial planning	456,000	366,000	Grants	90,000	Grants	MSPCE E RS
	2.2 Implement and mainstream new ecosystem-based flood risk reduction and climate change adaptation methods	26,116,175	5,868,747	Grants	20,247,428	Grants	MAFWM RS WA Sava WA AB HPP
	2.3 Codify and mainstream EbA solutions into policies and regulations and promote non-structural measures among decision makers and communities	298,680	208,680	Grants	90,000	Grants	MSPCE E RS
	2.4 Review and strengthen institutional capacity and develop long-term institutional capacity development plans for climate resilient FRM	800,000	300,000	Grants	500,000	Grants	MAFWM RS WA Sava WA AB

3.1 Develop investment framework for climate induced flood risk reduction and management	1,083,614	480,440	Grants	603,174	Grants	BOSNA RE AOIC
3.2 Formulate multi-year climate resilient municipal investment plans and gender responsive community preparedness plans in selected municipalities (10-12) and in one canton in Vrbas, Una-Sana and Bosna, Drina, Neretva and Trebišnjica basins	3,622,884	525,000	Grants	3,097,884	Grants	UNDP MOFTE R
3.3 Implement climate-proof structural flood risk reduction and anti-erosion interventions in Vrbas, Una-Sana, Bosna and Drina river basins	21,814,433	875,280	Grants	20,939,153	Senior Loans Grants	MAFWM RS-EIB Loan MAFWM RS
4.1 Project Management	3,847,090	680,000	Grants	3,167,090	Grants	UNDP (TRAC) MOFTE R MAFWM RS WA Sava WA AB BOSNA RE AOIC HPP MSPCE E
Indicative total cost (USD)	<u>72,688,889</u>	<u>14,400,000</u>		<u>58,288,889</u>		

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?	Yes ☒ No ☐
C.3.2. Does GCF funding finance technology development/transfer?	Yes ☒ No ☐

Capacity Building

169. Under **Activity 1.1** the project will provide training to 50 hydrometric practitioners of HMIs and WAs in O&M of up-graded hydrometric stations.

170. Under **Activity 1.3** the project will undertake an institutional capacity assessment of FRM institutions and develop a capacity development plan for embedding flood hazard and risk modelling approaches and FFEWS into relevant institutions in B&H. Institutional capacity gaps and training needs will be assessed, and a revised institutional capacity building plan will be developed to address gaps (human, technical and financial). The recruitment and training need to fill capacity gaps will be addressed. The project will develop training plans for each technical area of expertise related to climate-induced flood risk assessment and management, and FFEWS, and consolidate into an overall capacity development plan. The long-term capacity plan for B&H will be developed and will consider options such as the development of apprenticeships, internships and voluntary schemes for University students. To address issues of skills shortage, skills retention and succession planning, the project will develop approaches which will include examining the role of the private sector (consultants, contractors, research institutes) in filling these gaps, the use of Continuous Professional Development (CPD) methods involving cross-fertilizing of staff with skills across all organizations (e.g. through training in all technical areas before specializing), development of standardized and country-wide accepted guidance documents, codes and standards which will enable consistency and uniformity of technical approaches.

171. Under **Activity 1.3** the project will also undertake capacity building and training for HPP practitioners, including specific training on the use of climate information products and services and the use of the new HPP models and climate information products in HPP operations. For the agricultural sector, the project will support improvement of existing agrometeorological advisory services to farmers with the focus on flood risk management by training and embedding training (ToT) with agricultural specialists who will work with local farmers to teach them how best to use the weather/flood forecasts within their farming methods, and help increase their understanding of how to apply it to their agricultural practices.
172. Under **Activity 2.3** the project will undertake training in the methodology and guidance for undertaking risk assessment and appraisal of EbA measures; eco-system analysis and evaluation; and eco-system valuation methods, and eco-system benefit assessment as part of CBA.
173. Under **Activity 2.4**, the project will develop and fully embed the long-term capacity development strategy for all aspects of FRM in B&H. It will establish training curricula, and rosters of trainers, mechanisms and financing strategies for delivering long-term capacity development such as development of key partnerships with universities that can be forged to ensure long-term FRM training. The project will deliver some of the prioritized training in the areas of: flood hazard and risk assessment, hydrometry, forecasting and modelling, EWSs; vulnerability and risk assessments; river basin flood risk management planning and the use of climate information; Under this activity the project will also deliver an awareness raising campaign for capacity building at municipal and community levels; Networking and advocacy; Country-wide media campaign on FRM and FFEWS. Under Sub-activity 2.4.3 (See pages 251-252 of FS) the project will assist the government of B&H in shifting from ad-hoc project-based awareness and education efforts to a planned, consistent and sustainable information and communication system for enhanced flood and disaster risk management at the community level. To this end, it will develop an awareness raising campaign which will include the following: Capacity building at municipal and community levels: Emergency response trainings for first responders in cooperation with Civil Protection; Strategic FRM planning and FRM intervention identification approaches for municipality staff; Development/adaptation of existing guidelines on community DRR/DRM; Development of FRM, DRR and DRM guidelines for specific target groups including teachers, municipal authorities, media, women's groups; Capacity building and awareness raising of municipal authorities, local NGOs, CBOs or non-CBO community members in Community-based Risk Assessment approaches, Community-based Early Warning Systems and gender-responsive Community-based Flood Risk Management; Development of 'training of trainers' programme and a syllabus for topics such as first aid, search and rescue and warning dissemination, and example standard specifications for external training providers; Working with the appropriate agencies to develop generic educational material and a 5year training programme on flood and disaster risk and emergency response for use in school outreach programmes, university courses, communities etc. (training material, course notes, plays etc.); Training on how to use the hazard and risk maps provided (paper and electronic) to raise awareness but also on how to plan development and other activities locally. In addition the project will implement networking and advocacy which will include: Organizing annual community forums on FRM and CBEWSs with participation of target communities and representatives of vulnerable groups to exchange information, lessons learned, successes and impediments; Organizing community-government and public-private dialogues around local risks and risk reduction strategies and their financing. In addition a country-wide media campaign on FRM and FFEWS will be developed to include: Develop and deliver awareness raising messages through a number of methods including using social media, art and creativity to communicate FRM and FFEWS concepts to broader society and to disseminate project lessons and successes; Organizing TV and radio talk shows around FRM and FFEWS; Media coverage of project activities; Production of footages, Public Social Announcements (PSAs) of short documentaries around FRM and project achievement; Training for media on FRM and FFEWS and annual essay/media article awards competitions. The project will utilize its partnerships with Society of Red Cross (RC) B&H and IFRC to develop, deliver and embed the community-based training and awareness raising.
174. Under **Activity 3.1** the project will provide technical advice and training of insurance sector practitioners in the development of insurance products packages from natural disasters adjusted to the needs of the assessed sectors and insurers.
175. Capacity development for each of the interventions/components will leverage existing and establish new partnerships with organizations such as the WMO regional training centers, UNESCO/ITU ICT4D and

other DRR institutions, as well as national and regional research institutions, universities and training centres. Total GCF requested funds for capacity development is approximately **\$300,000 USD**.

Technology Development/Transfer

The main technology transfer and development activities under the project are as follows:

176. Under **Activity 1.1** the project is enhancing the technologies for hydrometric monitoring. This will include the upgrade and modernization (automation) of stations, the development of the ICT systems that underpin the hydrometric network databases and enables the efficient use of data from all sources in the modelling, mapping and forecasting of floods.
177. Under **Activity 1.2**, the project will enhance the WBIF EUFD hazard and risk modelling by a) developing hydrological modelling for all main basins (category I rivers), b) undertaking torrents hazard modelling, c) including HPPs and groundwater modelling in flood hazard, risk and vulnerability modelling and mapping for all relevant basins. These activities will rely on the transfer and scaling up of technologies and methodologies developed and applied under the Vrbas basin project. Furthermore, the project will introduce new technologies for modelling and development of climate risk-informed reservoir operation rules that minimizes downstream flood risk and damage, while maintaining dam safety within reasonable limits and optimizing water availability for power generation. This will be based on an integrated approach to basin flood risk management which includes adequate representation of HPP operations in the basin flood hazard and risk modelling and mapping, as well as in flood forecasting models, to enable real-time operational decisions during a developing flood. To this end, the project will incorporate HPP operations into flood hazard and risk modelling (to strengthen the role of HPPs in flood alleviation and maximize their capacity to utilize flood forecasts in their operations) in all relevant basins. The project will additionally develop sector-specific climate risk information tools and products for hydropower sectors with a particular focus on the Neretva and Trebisnjica sub-basins of the Adriatic Sea basin where there are large numbers of HPPs.
178. This activity will also build on the bespoke GIS-based socio-economic flood risk and vulnerability model (which is based on EU flood Directive methodology for flood socio-economics assessment) developed for Vrbas basin which will be further enhanced with improved receptor and socio-economic datasets to be acquired/established by the project, integrate various spatial socio-economic data with the hazard maps, to produce risk maps, which will include economic losses and damages and loss of life estimates. In addition, the project will develop the technical methodology and tools for the systematic collection of socio-economic datasets that are important to flood risk and vulnerability modelling and assessment
179. Under **Activity 1.3**, the project will establish a fully-integrated impact-based flood forecasting and early warning system across the country, using state-of-the-art hydrometeorological modelling (hydrological and hydraulic models, DEM, etc.), forecasting (meteorological forecasting, use of ground weather radar, satellite imagery and aerial photography), EWS platform, telecommunications (GMS, internet, SMS, CCTV, TV and radio, walkie talkies, etc.) and information (GIS, web-based information portal and meta-databases, etc.) technologies. It will also create new climate and weather products (e.g. climate advisories) for various users (HPP and agriculture. Participatory “Last Mile” communication solutions tailored to the needs of local communities, including CBEWSs and demonstration of cost-effective risk reduction and community resilience measures through the combination of structural and non-structural protection measures, which are relatively new for B&H. Under this activity, the project will also enhance the ICT systems that will support the national integrated FFEWS platform, for near real time forecasting and warning dissemination.
180. Under **Activity 2.2** the project will scale up successful EbA approaches piloted by the UNDP/SCCF Vrbas River project and embed the tools and technologies that were developed to support long-term design and implementation of EbA measures.
181. Under **Activity 3.1**, the project will expand and further develop flood insurance scheme initiated under Vrbas project and explore and identify other risk financing products and tools. This represents technology

transfer which will embed risk-based natural disasters insurance and the relevant insurance models, tools and products in B&H.

182. The project will develop a unified methodology and relevant technical assessment tools for appraisal-led design of structural and non-structural measures using climate risk information and cost-benefit appraisal methods and application of methods to the detailed design of prioritized structural and non-structural measures.

183. Total GCF requested funds for technology transfer/development is approximately \$3.4 Million USD.

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's [Initial Investment Framework](#).

D.1. Impact potential (max. 500 words, approximately 1 page)

184. This project has the potential to prevent loss of life, property and critical infrastructure assets for 799,445 people currently at risk of floods (25% of the total population of the country) and to 902,906 people projected to be at risk by 2050 representing approximately 26% of the entire population of B&H and 44% of the population of the 69 municipalities covered by the targeted sub-basins. The project is targeting those most vulnerable to flood risk and the impact potential of the enhanced flood risk management and intervention measures is high and will safeguard vulnerable communities and their economic assets from these climate induced disasters. In addition, community resilience and livelihoods will be increased through the EbA strategies which will additionally enhance the environmental and ecological protective functions of the river basins.

185. The project will increase resilience and enhance livelihoods of the most vulnerable segments of population, particularly women, children, and the elderly. Women represent 51.7% of the total affected population. The population under 15 years of age and of above 65 years of age represent 15.2% and 13.7% respectively of the affected population. Through the project interventions economic damages from climate-induced floods will be reduced. As shown in the detailed economic analysis (Annex 3A and Annex 3B), expected *annual* benefits as a result of the project activities are expected to reach \$21 million once the project is fully implemented. The largest benefit (measured as avoided flood loss and damages) arises from the protection of household assets. In terms of geographic distribution, the benefits arise mostly in the Bosna and Neretva sub-basins. The project will also result in more resilient physical assets in 3 B&H river basins.

186. The project's adaptation benefits will also include a scaled transfer of climate resilient FRM technologies and mainstreaming of climate adaptation and the use of climate information in the institutional and regulatory systems in B&H that will result in improved incentives for climate resilience across sectors. The project will implement floodplain zoning policy which will be based on the matching of permitted land use to category of flood risk with vulnerability of occupants/users of the land as a key consideration of permissible development (based on the principle of avoidance of high-risk areas). This will have a direct impact in terms of reduced exposure to flood risk among the most vulnerable communities.

187. The project design has been informed through consultation with various stakeholders and has sought to reflect the gender differentiated aspects of flood risks. Information on the needs of vulnerable groups (women, ethnic minorities, disabled, elderly) has been collected and will continue to be collected through ongoing and inclusive stakeholder engagement and the embedding of tools and methodologies for the collection of gender responsive socio-economic data for development of the flood socio-economic risk and vulnerability model to be developed under output 1.2. Through the implementation of a Gender Action Plan (GAP) the project will assist in fulfilling B&H's commitment to gender equality and international obligations on gender responsive climate change and disaster risk measures. The GAP will draw on lessons learnt from the UNDP Vrbas project including the need for gender responsive planning and implementation. It will include gender analysis as part of risk and vulnerability and support capacity building of staff and implementing partners to collect sex and age disaggregated baseline data;

develop specific performance indicators to monitor, report or track progress, inform decision making; and strengthen accountability on commitments for gender equality.

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

188. The project demonstrates **ambition** in its goals and objectives in addressing climate change challenges facing B&H, by going beyond incremental changes and striving instead for significant and transformative impacts in terms of adaptation benefits, and sustainable development outcomes. The project will change the paradigm of existing flood risk management in B&H in the face of climate change through enhanced institutional capacity at all levels, improved climate information and understanding of risk, mainstreaming non-structural ecosystem-based risk management approaches, engaging private sector and climate-proofing public and private investments in flood risk management. Without project interventions there will be a progressively increasing risk of catastrophic loss of life, property and economic assets and livelihoods throughout B&H. The alternative pathway reduces the risk to 799,455 people through introduction of an integrated approach, which will embed new skills, methods, and technologies to assess the flood hazard, risk and vulnerability, to better understand the risk, and to develop a range of risk reduction and management strategies.
189. The project is enhancing the availability of climate risk information on which all FRM will be based by scaling up hazard and risk modelling and assessment to include flood risk from torrential streams, and HPPs, thus extending the technical scale/scope of hazard, risk and vulnerability information. The project will enhance the capacity for flood early warning and implement a national sustainable impact based FFEWS resulting in improved warning times and reduced losses. Technological innovations brought by the project include implementation of fully integrated impact based FFEWS based on state of the art observation, modeling, forecasting and communication platforms and technologies that will empower local communities to undertake local disaster risk reduction (DRR) and response measures, that complement the centralized EWS system, and can be replicated to other communities. The project will develop impact based FFEWS for main rivers which will be scalable to cover sub-basins as necessary. Importantly, the flood forecasting will include all relevant flooding mechanisms and will also be scaled to provide specific sector flood forecasting products for hydropower and agriculture sectors. Through the development and implementation of protocols and SoPs on data generation, data management and communication for effective impact based FFEWS, the project will ensure the uptake and replication of technical and institutional FFEWS solutions across B&H. The project is therefore contributing to the geographic and technological scaling up of the FFEWS in B&H and will provide the technical and institutional basis for FFEWS and services to all at risk communities. It also provides the basis for any future development of FFEWS to areas that may experience flooding in the future and for development of FFEWS products for other sectors in the future. In doing so, the project will contribute to the implementation of the new UN initiative “Early Warnings for All (EW4All)” announced by the UN Secretary-General in March 2022 aiming at facilitating partnerships for and promoting universal coverage of early warnings.
190. The project will enhance the capacity to identify, plan and implement long-term flood risk management strategies at the basin and sub-basin scale by introducing combined structural and non-structural methods including EbA measures. Long-term flood risk investment planning will also be enhanced, and policy and legislation strengthened to enable long-term sustainability of interventions, by engaging the private sector to shoulder some of the increasing costs of required observation networks and risk information and early warning services. Also, climate risk information will enable a better understanding of the risks to, and impact of private sector activities on increased flood risk, and their potential role in flood alleviation, enabling private sector to incorporate FRM activities into their regular operations. The project is developing the capacity and systems whereby the current gap between the demand and supply of vital climate risk information is closed. Capacity building programs that will be established and implemented by the project will ensure the ongoing transfer of skills to create knowledge sharing and better climate services for the future. The project will help ensure sustainability by supporting all relevant government ministries in line with government policy, programme and strategic priorities. The establishment of methods and tools for developing and implementing flood risk management strategies, the introduction of risk assessment methods, standards, tools, guidance and regulatory documents make this project highly replicable within B&H. This investment in human capital will strengthen institutions that will have skilled human resources, information, tools and technologies to effectively pursue their mandate in climate risk management, thus ensuring sustainability and replicability of the intervention.

191. Through the project effective Nature-based FRM solutions will be embedded in B&H to address the complex interactions amongst multiple ecosystem degradation processes that are contributing to flood risk and affecting people's vulnerabilities to climatic and other factors. In B&H, terrestrial ecosystems are under increasing pressure from changes in the use of land and natural resources, driven by socio-economic pressures. While climate change is impacting the intensity and frequency of flood risk as well as shifting natural eco-systems, such as distribution of forest cover, anthropogenic activities that increase deforestation, soil erosion and land degradation are combining to increase risk of floods due to the reduced ability of eco-systems to provide services important to protecting people, their property, and livelihoods. This, in turn, puts people under increased pressure to resort to unsustainable ecosystem use and management, further degrading ecosystems and their capacity to deliver services. While climate change affects the various terrestrial ecosystems in different ways, and each ecosystem is accompanied by its unique set of socio-economic and political issues, this vicious cycle of degradation is common to most. Adaptation to climate-induced flood risk in B&H, if tackled by hard structural measures alone, is likely to require increasingly larger investments in flood protection infrastructure, such as flood embankments, and dams and reservoirs to buffer against increased variability in rainfall and run-off. However, such traditional engineered solutions may not always be the best option. Nature-based FRM solutions, due to their potential value for risk reduction and adaptation, have been recognised as viable solutions (European Commission 2015)⁵⁰. Natural flood management (NFM) and natural water retention measures (NWRM) that alter, restore or use landscape features to manage flood risk including (1) interception (retaining water in and on plants), (2) increased plant transpiration, (3) improved soil infiltration, (4) ponds and wetlands for increased floodplain storage, and (5) reconnecting the floodplain, have the potential to reduce extremes in the flow discharge and thus help to reduce flood risk. Nature-based solution (NBS) encompassing EbA and non-structural FRM measures have been proven to be a viable substitute for, and complement to traditional hard infrastructure, and has provided additional eco-system services by enhancing the natural biophysical services of the rivers and river basins. In B&H, torrential flooding which normally originates in steep upland catchments, cannot be addressed by traditional hard engineering measures, but rather needs to be managed by restoration of the vegetation cover and re-establishment of the natural eco-system functions of the catchment. Nature-based solutions are often less capital intensive and more easily reversible than engineered alternatives, thereby providing an additional "option" value. In addition to protection from climate change impacts, they also provide many other benefits to communities, for example through the maintenance and enhancement of ecosystem services crucial for livelihoods and human well-being, such as clean water, water regulation and habitat, recreational opportunities, and often provides employment opportunities. Appropriately designed NBS measures can also contribute to climate change mitigation, for example, by reducing emissions from ecosystem loss and degradation. To reduce flood risks, it is vital to build the resilience of the natural resource base, and to promote sound environmental and natural resource management practices and the sustainable use of ecosystems. Healthy and diverse ecosystems are more resilient to hazards. In particular, healthy catchment eco-systems, reduce soil erosion and runoff that causes flooding. For example, in Switzerland, it is estimated that forests save between USD 2–3.5 billion per year equivalent in disaster damage (including flooding) through restoration of key forest ecosystems⁵¹. Forests can be used as shelterbelts and windbreaks, and also play an important role in protecting against natural disasters including floods. Trees stabilize riverbanks and mitigate soil erosion. Wetlands serve to store water, provide storm protection, flood mitigation, land stabilization and erosion control. As part of the Vrbas project, several EBA and non-structural measures have been implemented throughout the basin. Over 15 EbA measures were implemented in the mountainous part of basin (See Section 7.11 of FS), as well as in middle and lower plains and they consist of the use of natural materials in the protection of riverbanks from erosion and degradation, the reconstruction of old meanders and abandoned river beds, restoration of the riverbed to its natural state and agroforestry measures. In the most upstream part of the basin, in the municipalities of Bugojno and Gornji Vakuf, a measure of rehabilitation of the old riverbed was implemented, which enabled water retention in the upstream part, thus ensuring reduced and controlled runoff in the downstream part of the basin. The added value of this measure is that the river island created by the reconstruction of the old riverbed is now significantly used for agricultural purposes as a highly fertile land. During the heavy rains in September 2019, several news portals in Banja Luka pointed out that the projects and interventions made through the Vrbas project gave results and reduced the flood risk compared to the floods from 2014, in the largest city in the basin⁵². Based on the results of the Vrbas project the B&H government has committed to embedding nature-based approaches and are seeking to rebalance the priorities and relative value of "hard" infrastructure versus green solutions to manage flood risk. NBS solutions are embedded in the Flood Risk Management Plan for the Vrbas basin, for the first time in B&H. This move to adopt NBS is in line with many similar

⁵⁰ European Commission (2015) Towards an EU Research and Innovation policy agenda for Nature-Based Solutions & Re-Naturing Cities. Final report of the Horizon 2020 Expert Group on 'Nature-Based Solutions and Re-Naturing Cities'. European Commission, Brussels

⁵¹ <https://nature4climate.org/ecosystem-services-and-green-infrastructure/>

⁵² <https://www.atvbl.com/vijesti/drustvo/minimalna-materijalna-steta-projekti-dali-rezultata-27-5-2019>

tried and proven initiatives in Europe (e.g., “Making Space for Water” in England⁵³ and similar⁵⁴). During feasibility, 21 EbA and non-structural measures were prioritized for implementation based on CBA analysis of an original 69 such measures identified by WA’s (See paras 85 & 86 above). Government then selected the next 5 for financing. Therefore, of the 26 measures, 5 EBA and 13 non-structural measures will be fully financed by government, while 8 non-structural measures will be co-funded by GCF and government. Hence there is a 79% Government to 21% GCF co-funding ratio of EBA and non-structural measures. This shows a very serious commitment of the Government to invest in nature-based solutions. In addition, to address the impact of climate change on forest ecosystems (See para 25 above and Section 10.2 of FS) the project is mainstreaming climate-induced flood risk considerations into forestry policy (See pgs. 233-235 of FS), which is expected to establish policies which will protect the forest from the expected impacts of climate change and promote reforestation, afforestation, and sustainable forest use. The long-term impact will be enhanced forest eco-systems which will provide improved services to flood risk management and other biophysical functions. The project will therefore result in improvements in watershed ecosystems and restoration of ecological function through the use of innovative EbA strategies which will reverse the deleterious effects of catchment degradation and enhance livelihoods of rural communities. This investment in EbA is an investment in social and economic assets of vulnerable communities. The GCF project will be investing in long-term **sustainability** of communities and their social and economic assets through the investment in EbA solutions, and the derived eco-system service attendant benefits of improved functions of catchments through effective flood risk management.

192. The GCF project **will leverage additional investments** and create a **multiplier effect** through the mobilization of private sector investment, and co-financing from EIB, government, and other sources. The intervention will attract and unlock additional financial flows for climate action in the future through the establishment of risk financing and risk transfer mechanisms which will be developed. This will have a catalytic effect on the broader climate finance and investment landscape in BiH.
193. The project is investing in the development of long-term climate risk financing. The project is developing fully articulated long-term municipal risk management plans for 10-12 highest risk municipalities and 1 canton, and sector resilience plans for highest risk sectors, which will set the direction for systematic risk informed sectoral flood risk management in the future. In addition, the project will develop long-term risk financing mechanisms and investment plans. Cost-benefit assessment methodologies and a socio-economic risk model will be embedded into the identification and design of climate proofed structural and non-structural measures and will form the basis of municipal investment planning. The project is also developing risk transfer mechanisms including disaster insurance. Risk financing and risk transfer mechanisms to be developed by the project, will ensure that long-term financial and economic resources are identified and available after GCF assistance ends and will ensure cost-effective financing of flood risk reduction measures, and investments over time. This will result in sustainable financing of FRM in B&H.
194. Forecast Based Financing mechanisms will be developed and implemented and will identify financial and other resources that will be needed during a disaster. Forecast-based finance mechanisms have emerged as **innovative approaches** in flood risk management. It will enable early action by utilizing reliable and timely weather forecasts and early warning systems. By using advanced forecasting technologies and models being developed by the project, these mechanisms will provide lead time to prepare and implement preventive measures before a flood event occurs and will allow for timely mobilization of resources, reducing the response time and increasing the effectiveness of interventions. By acting in advance based on accurate forecasts, forecast-based finance mechanisms allow for targeted risk reduction measures. This includes activities such as pre-positioning emergency supplies, reinforcing infrastructure, conducting evacuation plans, and implementing community-based early warning systems. By focusing on risk reduction, these mechanisms help minimize the potential damage and losses caused by floods, ultimately enhancing resilience. Forecast-based finance mechanisms can lead to cost savings by avoiding or reducing the need for expensive emergency response and recovery efforts. By investing in preventive measures, such as early evacuation or protective measures for critical infrastructure, the costs associated with emergency relief, damage repair, and post-flood recovery can be significantly reduce. These mechanisms contribute to building and enhancing community resilience by promoting a proactive approach to flood risk management and by empowering communities with information, resources, and early action, forecast-based finance mechanisms strengthen their capacity to respond effectively to floods and reduce their vulnerability. This, in turn, helps communities recover faster and build resilience against future flood events. Forecast-based finance

⁵³ Johnson CL, Priest SJ (2008) Flood risk management in England. A changing landscape of risk responsibility. Int J Water Resour Dev 24(4):513–525

⁵⁴ <http://nwrm.eu/guide/files/assets/basic-html/index.html#1>

mechanisms will also support evidence-based decision-making by providing accurate and reliable weather forecasts and early warnings. This enables authorities, policymakers, and communities to make informed decisions on resource allocation, preparedness measures, and response strategies. Having access to timely and reliable information enhances decision-making processes, enabling more efficient and effective flood risk management. Furthermore, it will encourage collaboration and partnerships among various stakeholders involved in flood risk management, including government agencies, meteorological services, humanitarian organizations, local communities, and financial institutions. The collaborative approach fosters knowledge sharing, coordination, and joint efforts, leading to better outcomes in terms of flood preparedness, response, and recovery.

195. Forecast-based finance mechanisms offer a proactive and risk-informed approach to flood risk management. By leveraging weather forecasts and early warning systems, these mechanisms provide opportunities to take early action, reduce risks, enhance resilience, and optimize resource allocation, ultimately leading to more effective and efficient flood risk management strategies. This contributes directly to financial sustainability by reducing costs of responding to flooding, is highly scalable and replicable to all flood prone areas identified by hazard and risk maps.
196. In summary, the project is implementing **innovative approaches, technologies, and practices** that have the potential to **catalyze systemic change** in FRM in B&H. It is introducing novel solutions combined with existing approaches in innovative ways to address climate-induced flood risk effectively. This includes implementing technological innovations such as impact-based flood forecasting using technologically advanced data products and models, development of innovative climate risk financing mechanisms that promote public and private sector participation, new approaches to policy and governance through the mainstreaming of climate-induced FRM into sectoral plans, strategies and tools and the combination of Nature Based Solutions with existing structural and non-structural measures with enhanced environmental co-benefits. The project interventions will provide strong **scalability** and **replicability** impact as they will establish the methods, standards and approaches that will work across B&H and define these in guidance, legal and policy documents, and ensuring that the long-term institutional capacity for FRM is strengthened. There is at least a threefold replication potential for all project interventions and bearing in mind institutional and geographical similarities of all Western Balkans countries, there is also a regional replication potential.
197. The project will shift the current paradigm by transforming the existing flood risk management practices in B&H from an ex-post facto i.e., reactionary post-disaster recovery approach to ex-ante i.e., risk informed preparedness approach based on adequate climate information and risk knowledge.
198. The Theory of Change section of this proposal demonstrates how the current barriers can be removed through the project activities to achieve transformational change.

Contribution to the enabling environment

199. The project is creating three main categories of enabling conditions for climate resilient communities and their physical assets. First, innovative tools and methods will be introduced for hazard, risk and vulnerability assessment, modelling and mapping to enhance WBIF modelling and mapping to produce definitive hazard maps will be provided for key flood hazard and risk information that will enable climate resilient strategies, policies and practices in all sectors in B&H. Development of sector specific operational and flood forecasting modelling tools and products will enable sector development along climate resilient development pathways, for the HPP and agriculture sectors.
200. The establishment of a fully-integrated impact-based flood forecasting and early warning system across the country (Activity 1.3), is also contributing to the enabling environment, through the creation of new climate and weather products (e.g. climate advisories) for various users (HPP and agriculture and the establishment of participatory “Last Mile” communication solutions tailored to the needs of local communities, including CBEWSs and demonstration of cost-effective risk reduction and community resilience measures through the combination of structural and non-structural protection measures are relatively new for B&H.
201. Secondly, the project will combine best available science and local knowledge for vulnerability assessment, hazard and risk mapping, and will introduce appraisal-led flood risk management and investment planning processes at river basin and community levels by introducing cost-benefit analysis methods and tools to embed climate-risk informed, appraisal-led methods of strategic planning for Flood Risk Management and climate change adaptation

and investment planning thus enabling risk informed budgeting and resource allocation. Furthermore, technologies will be utilized to create efficiency in this process. The project will enable the application of EBA and non-structural solutions by embedding the new methodologies and national technical guidance documents to ensure long-term incorporation of such methods alongside more traditional structural measures.

202. Thirdly, the project will create the conditions to unlock both public and private sector finance for scalable investments into flood risk reduction. Municipal investment plans based on cost-benefit analysis and appraisal-led identification of climate-resilient flood risk management investment requirements will identify both public and private finance mechanisms. Such investment planning will enable evidence-based advocacy for public sector investment into climate-resilient FRM. The project will develop a strategy and tools for private sector engagement in long-term climate-proof risk management and provide technical assistance to financial institutions to mainstream climate change risks into their business models. The project will develop public-private partnerships for long-term implementation of risk financing mechanisms for entity-level flood risk financing and resilience strategy. The project is developing such risk financing and transfer mechanisms based on detailed socio-economic risk, damages and losses assessment.

203. Importantly, the project will improve regulatory environment and tools for the promotion of flood insurance alongside seeking solutions for other risk financing and transfer products and tools. This will include assessment of feasibility for micro-insurance for agricultural producers, entrepreneurs and small and medium-sized enterprises in different industries; feasibility for a regional disaster risk insurance program to increase the degree of diversification of risk; exploration opportunities for use of catastrophe bonds to transfer disaster risk to the capital market; sector-specific natural disaster (including floods) insurance product packages for insurance in each of the defined sectors, based on studies to estimate cost-benefit.

Contribution to Regulatory Framework and policies

204. The project will revise and implement several protocols, SOPs, and Codes of Conduct for each of the institutions responsible for FRM and FFEWS. These will include SoPs for “last-mile” warning and dissemination and communication system, Protocols and SOPs on data generation, data management for FRM, data sharing protocols and platform for climate data, across all government institutions in both entities including spatial Data Infrastructure to be developed/enhanced to ensure climate data flow/exchange.

205. The project will enhance the existing legislative and regulatory FRM framework by mainstreaming the climate-induced flood risk management into sectoral planning for agriculture, forestry, environment, hydropower, critical infrastructure and spatial planning. A draft floodplain development policy based on detailed climate-risk informed flood hazard, risk and vulnerability maps, will be scaled up to spatial planning and development control for all other basins. This policy will underpin the floodplain planning and development control to achieve basin-wide climate resilience to increasing flood risks and will result in a paradigm shift from uncontrolled floodplain development to climate-risk informed controlled floodplain usage. Hence the project will develop the country-wide floodplain zoning policy and legislation, based on hazard, risk and vulnerability mapping.

206. The project will develop strategies for the management of critical infrastructure in the floodplain in order to improve the resilience of critical infrastructure, based on a shared, consistent, proportionate and risk-based approach to reducing vulnerability over time, and embedding flood resilience measures into the planning, design, construction, and long-term management of critical infrastructure. The project will enhance/updated climate risk informed methodologies, standards for infrastructure design, construction material, use and maintenance of all categories of critical infrastructure.

Potential for Knowledge Sharing and learning

207. The project has particularly strong emphasis on knowledge generation, learning and information sharing/dissemination. First, the project will enhance and introduce new hazard and risk assessment modelling and mapping tools and generate definitive flood hazard, risk, and vulnerability maps for all basins in B&H. The project will train WA and HMI staff in the use and maintenance of these new tools and technologies. Furthermore, the project will develop damage and loss accounting methods, tools, and technologies.

208. The project is generating climate risk information which will be used to increase knowledge and provide information for all stakeholders including decision makers in progressing with climate resilient development. To enable wide reaching use and application of climate risk information the project is developing data sharing protocols to enable

data/information exchange between water and other sectors relevant for integrated water management and climate change (agriculture, forestry, environment, spatial planning and hydro-energy sectors).

209. To enable access and sharing of climate risk information, a centralized information system and knowledge sharing platform will be developed, based on the project SDI and GeoPortal established by the Vrbas project, which comprises GIS database for all spatial data related to the basin flood risk management. The GeoPortal is a GIS-based tool which integrates various spatial socio-economic data with the flood hazard and flood risk maps, vulnerability maps including loss/damage models, real time hydro meteorological data, torrents sensitivity model, cadaster of torrents etc. The database was designed to be expandable to include other basins. The flood hazard information system to be developed under the project will consist of an integrated e-Library, databases (including the GIS database previously noted), information systems and knowledge portal. The web knowledge portal will help increase awareness, provide interactive hazard maps, with integration with social media and possible mobile application to increase community engagement and allow two-way flow of information. The GeoPortal will serve the municipalities for local level flood risk management to interact with communities through the PGIS tool within the platform and will be placed in the ownership of the entity and municipal civil protection units who will have responsibility for maintenance and update of the system in the future. In addition, UNDP has developed a Disaster Risk Analysis System which is an online data sharing platform for all disaster data. The project will contribute to the development of the platform and the embedding of flood hazard and risk information as well as expansion of the platform functionality to provide the specific functionality that will be required or using the flood hazard and risk information.
210. The flood risk management platform will be established as a country-wide mechanism for cross-entity, multi-sectoral and inter-disciplinary coordination and policy guidance on flood risk management and risk reduction with public, private, and civil society participation involving all concerned entities within B&H. The data sharing platform will coordinate all stakeholder engagement at the entity and local level and will pursue an all-of-society engagement in FRM. It would also have mechanisms for effective dialogue with Local Platforms in place in order to influence, encourage and coordinate local action. In parallel, the country-wide Platform will seek to understand local priorities and issues.
211. Under Activity 2.4, the project will implement country-wide training programmes in the technical and practical areas of flood risk management for practitioners, decision-makers, communities, emergency responders based on previous institutional capacity needs and gap assessment of state, entity. The project will embed training in FRM and the use of climate risk information in the relevant entity training institutions to improve the technical capacity and knowledge base for hazard and flood risk management, based on a Training of Trainers (ToT) through the development and establishment of training curricula, and rosters of trainers and the development of mechanisms for delivering long-term capacity development such as development of key partnerships with universities and other academic research institutions that can be forged to ensure long-term FRM training.
212. The project will implement a knowledge management (KM) framework (See Chapter 11 of the FS for detailed description of the KM framework). It will have the following key aims:
 - To ensure access to data and information generated by the project as well as long-term access to data on which stakeholders' essential institutional functions rely and/or data and information that can be used for evidence for policy and practice advice (connecting people to information and knowledge)
 - Connect key stakeholder groups, practitioners, and experts to ensure that key learning and experience is shared within and across sectors (connecting people to people)
 - Ensure staff in the stakeholder institutions know about effective and relevant KM techniques so that knowledge is shared, captured, and retained by the institutions and shared within and across the sector (institutional KM improvement)
213. By developing and promoting KM as a tool for continuous and sustainable improvement and ensuring that KM tools generated by the project will be systematically used and maintained within the stakeholder institutions (Developing and embedding KM tools and practices).
214. **Knowledge Brokerage:** The Technical Advisory Working Group (to be comprised of representatives from government, private sector, academia and civil society (hydro-meteorological institutes, statistics institutes, agricultural institutes, local governments together with associations of cities and municipalities of Republika Srpska and Federation of B&H Environmental Protection Funds, the civil society and other local and international NGOs) will additionally act as a convener and facilitator linking between government, academia, and private sector to create conducive environments to enable the take up of scientific expertise and evidence to be developed by the

project. Research institutes, academic networks, learned societies, think tanks and charities will be invited, when necessary, as secondary members of, or consultees to the working group during the development of evidence-based policies. Thus, the TAWG will fulfil multiple complimentary functions, such as fostering networks and synthesizing evidence, and working at different levels of influence. It will provide the platforms and channels for promoting the project results and evidence, through targeted workshops, and information sharing (through many of the knowledge management structures discussed above). In addition, the project will seek to engage with other organizations and people who can potentially act as knowledge brokers within the TAG

D.3. Sustainable development (max. 500 words, approximately 1 page)

215. The project will deliver sustainable development co-benefits including enhanced ecosystem services, gender responsive livelihood enhancement and job creation, and improved resilience of communities' physical assets. The main **socio-economic co-benefits** from the project investment are derived from the avoided socio-economic losses from floods. Under climate change, economic losses from single extreme hazard events are expected to reach US\$ 750 Million⁵⁵ Expected Annual Damages (EAD) for households, businesses, agriculture, and public buildings, which could significantly impact and reverse socio-economic development gains of B&H. Avoided losses to sectors such as agriculture and hydropower could be significant. The project interventions that provide sector-specific risk information and risk management strategies will be important for the long-term development of the hydropower sector; such information as water flow forecasts will be critical for planning, risk information will assist with optimizing operations and avoid unplanned and ill-timed releases of water during flood events that have deleterious impacts downstream. Climate risk informed planning of the hydropower sector is important nationally as it will secure hydropower development and help continue a shift to clean energy, leading to reduction of greenhouse gases emission. Climate risk information will also safeguard assets such as transportation (roads and bridges) which are critical to the economic development and functioning of communities. By reducing and avoiding flood-related losses, the project will help the Government of B&H to reduce and optimize budget spending on recovery, improve the use of public funds for disaster preparedness and recovery, and thus releasing public funds for socially significant projects and infrastructure. Economic co-benefits will also be realized in the agriculture sector which comprises 7.8% of GDP, but on which 20% of population relies and which is at high risk from flood hazards. Prevention of agricultural losses, coupled with enhanced agricultural livelihoods through agro-forestry opportunities, will result in overall enhanced productivity of the agricultural sector.
216. This project will directly benefit the most vulnerable parts of the population and will have significant benefits to **gender equality**. The project will engage local gender experts to ensure that gender considerations are a key part of any consultation or activity planning process. Flooding and disasters in general, impact women disproportionately. In B&H there are significant gender inequalities, fueled by traditional gender roles. Men are more than twice as likely as women to be employed, self-employed or engaged in contract work. For effective flood risk management, the project will ensure that women are primary stakeholders and will therefore need to be involved in decisions on the types of solutions that are implemented. The project will therefore safeguard local communities and their assets from climate disasters with particular attention to women, and other vulnerable groups.
217. **Environmental co-benefits** mainly relate to EbA strategies and nature-based solutions such as riparian plantings and agro-forestry which will provide water retention functions; regulation of hydrological flows (buffer runoff, soil infiltration, groundwater recharge, maintenance of base flows); natural hazard mitigation (e.g. flood prevention, peak flow reduction, soil erosion and landslide control); increased streambed stabilization resulting in decreased erosion, habitat preservation, and reforestation. The project will increase the resilience of vulnerable people, properties, infrastructure, and economic sectors. Further, the project will enhance the resilience of forest ecosystems, including protected areas, as well as protect streams, reservoirs and lakes from siltation and thus, aquatic fauna from increased turbidity. The project is addressing climate-induced flood risks by introducing FRM and CCA measures. In general, the project is providing key environmental protection benefits through the introduction of disaster risk reduction, climate change adaptation and protection of people, property, and the environment from major hazards. In the long-run the project will bring about significant environmental benefits by

⁵⁵ Socio-economic impact of floods under current and climate change conditions in B&H was done as part of Feasibility Study (See Section 2.6 Annex 2)

increasing the country's resilience to climate-induced flood disasters and thus, enabling its population to better protect national assets, including environmental assets (land, forest and land resources).

D.4. Needs of recipient (max. 500 words, approximately 1 page)

218. B&H is a middle-income country with an estimated 3.5 million inhabitants. The 1992-1995 war has had a devastating impact on its human, social and economic resources, leading to enormous challenges of the post-war reconstruction and economic and social recovery. This challenge has been further compounded by the transition towards market economy requiring structural reforms and improved governance. B&H has a transitional economy with limited market reforms. The economy relies heavily on the export of metals, energy, textiles, and furniture as well as on remittances and foreign aid. Poor coordination among various government levels hampers economic policy coordination and reform, while excessive bureaucracy and a segmented market discourage foreign investment. The economy is among the least competitive in the region. The national currency (BKM) introduced in 1998 - is pegged to the euro through a currency board arrangement, which has maintained confidence in the currency and has facilitated reliable trade links with European partners. In 2016, B&H began a three-year IMF loan program, but it has struggled to meet the economic reform benchmarks required to receive all funding instalments. Interethnic warfare in Bosnia and Herzegovina caused production to plummet by 80% from 1992 to 1995 and unemployment to soar, but the economy made progress until 2008, when the global economic crisis caused a downturn. Since 2013, B&H has posted positive economic growth, though severe flooding in 2014 significantly hampered the recovery.
219. Due to the war time devastation and a slow transition of economy, a large part of B&H's population still lives in poverty (population living below the poverty line is 16.9% (2015 est.)). Between 2000 and 2019, B&H's HDI value increased from 0.672 to 0.780, an increase of 14.9 percent. However, when discounted for inequality, the B&H HDI falls to 0.667, a loss of 14.5 percent due to inequality in the distribution of the HDI dimension indices. The Human inequality coefficient for B&H is equal to 15.3 percent. B&H has a Gender Inequality Index value of 0.149, ranking it 38 out of 162 countries in 2019. The slow rate of the post-war economic recovery has been compounded by the negative impacts of climate change on key sectors such as agriculture, energy (hydropower), the environment, in particular, due to the increased frequency and magnitude of flood disasters, which have tripled in the last decade. In 2010 the second largest flood on record caused damages of US\$ 200 million or approximately 1% of GDP. In May 2014, B&H experienced its worst flooding in 150 years which resulted in 26 deaths and US\$2.7 Billion worth of damages (15% of GDP).
220. The most damaging floods have had devastating impacts on the most vulnerable groups including the rural poor, war returnees and displaced persons. Based on indicative information from local authorities, a significant proportion of the flood victims belong to one of these vulnerable groups. For example, in several municipalities in the Vrbas basin up to 100% of affected households have been identified as war returnees or displaced persons who are least equipped to cope with and recover from floods. This has led to a deepening of poverty in flood affected areas. Climate change and climate-induced floods have a significant impact on the most important economic sectors such as agriculture, water management, hydropower, and limits country's already low adaptive capacity. Please refer to Annex 7 (stakeholder consultations) for the evidence of consultations with beneficiary communities on their vulnerability and needs.
221. Annex G of the FS provides a detailed assessment of the institutional capacity in all key aspects of flood risk management. B&H institutions and professionals do not have adequate experience in identifying hazard and assessing vulnerability and risk from natural disasters in line with international standards. Complex and fragmented institutional structure, and a lack of coordination and cooperation hinders successful implementation of those actions. The lack of horizontal (between entities and among sectors within the same entity) and vertical (among state, entity, cantonal and municipal levels) communication within the water management sector further limits effective cross-sectoral climate risk management. Institutional capacity gaps in B&H include: The need to strengthen the hydrometric monitoring on which the strategic assessment and forecasting of flood risk relies, develop, and disseminate definitive flood risk maps, and climate risk data products and develop the institutional technical and financial and human resources to develop and implement a range of modern flood risk management strategies including EbA, non-structural and structural measures. The project is addressing these institutional gaps by i) implementing FFEWS according to revised and agreed institutional arrangements, roles, and responsibilities for FFEWS (See Section 6.5 of FS); ii) developing the protocols for effective FFEWS and FRM to strengthen coordination and eliminate institutional fragmentation; iii) Implementing a number of state-of the art technologies, equipment and tools, developing a capacity building plan and framework and undertaking training.

D.5. Country ownership (max. 500 words, approximately 1 page)

222. The proposal development is initiated at the request of the Government of B&H and builds upon a very strong national ownership. B&H became a party to the UNFCCC on 6 December 2000, and since then has worked to establish an appropriate political, institutional, and legal framework to meet its obligations under the Convention. B&H adopted the 'Intended Nationally Determined Contributions (INDCs)' under the Paris Agreement in October 2015 for the period until 2030. The Nationally Determined Contribution (NDC) of B&H was prepared in November 2020. B&H NDC includes a section on climate change adaptation and the updated section on contribution to climate change mitigation. The NDC identifies the sectors that are most vulnerable to climate change including: agriculture, water resources, forestry, energy, tourism, biodiversity and sensitive ecosystems, and human health with the two most vulnerable sectors being agriculture and water resources management. The NDC highlights the importance of detailed planning of short, medium and long-term adaptation measures, development of climate risk financing mechanisms, and strengthening the responsible institutions in monitoring, analysis and modelling of climate risk. B&H's first Climate Change Adaptation and Low Emission Development Strategy (CCALEDs), adopted by the Council of Ministers of B&H on 8th October 2013 is a key policy document informing the national adaptation priorities and details measures for adaptation to climate change and capacity building needs, with the aim of providing a strategic and programme basis for effective international support. The Strategy includes four outcomes covering climate change risks, vulnerabilities and opportunities for evidence-based policy development, effective institutional and regulatory framework, mainstreaming CCA approaches into decision making, and effective resourcing with timely and effective implementation. Increasing flood risk is identified in the Strategy among the key climate change impacts, the Strategy highlights an increasing vulnerability of population and ecosystems as well as agricultural sector and HPPs to climate-induced floods. The CCALEDs was updated in 2020 and includes adaptation priorities for 2020 to 2030⁵⁶. The Strategy has the following high level outcomes: **Outcome 1:** Climate change risks, vulnerabilities and opportunities are reliably identified, quantified and effectively communicated, supporting evidence-based policy development; **Outcome 2:** An effective institutional and regulatory framework supporting adaptation, coupled with delivery capacity, is addressing climate change risks and opportunities across sectors; **Outcome 3:** Adaptation approaches are mainstreamed into decision-making, implemented and their objectives and planned delivery are understood by an informed society; **Outcome 4:** A well-resourced Adaptation Strategy being implemented on time, effectively, and delivering the expected outcomes. The proposed project directly contributes to the stated outcomes of the CCALEDs 2020-2030 strategy. In particular, Output 1 of the project is developing the risk knowledge base through monitoring modelling, mapping and forecasting flood risk. It is also developing the data sharing platforms and coordination mechanisms for dissemination and use of climate risk information while under Output 2 the project is mainstreaming climate risk information into sectoral policies, thus contributing to evidence-based policy development. GCF Output 2 is strengthening the institutional and regulatory framework and building capacities for climate risk informed FRM, adaptation and resilience building. In addition, the project is supporting the development of the risk financing and risk transfer mechanisms which will ensure a well-resources investment framework for adaptation to flood risk. The proposed project activities aligns with CCA-LEDs 2020-2030 adaptation priority outputs 1.1, 1.3, 2.1, 2.2, 2.3, 2.4, 3.2, 3.12, 3.13, 3.16.

223. The European Floods Directive on the assessment and management of flood risks (2007/60/EC) (EUFD) was adopted to address flood risk which was not included in the European Water Framework Directive (2000/60 / EC) (WFD), and an important issue within integrated water resources management. The directive entered into force with the main aim of reducing the risk of floods and taking into account the impact of future climate change on flood risk, in order to reduce the harmful effects on human health, economic activities, as well as the environment and cultural heritage. The EUFD provides a framework for flood risk assessment and management and reaffirms the public's right to access information and to participate in the planning process. The EUFD framework is comprised of 3 stages – Stage 1 - Carry out a preliminary flood risk assessment (PFRA); Stage 2 - Prepare flood hazard maps and flood risk maps and Stage 3 - Establish flood risk management plans (FRMP). The proposed project activities under "Output 1: Climate-informed impact-based FFEWS and an increased generation and use of climate data to reduce vulnerability to flood related disasters" are fully aligned with Stages 1& 2 of EUFD. Namely, project activities under Activity 1.2 aim to review and enhance development of hydrologic and hydraulic modeling for the development of continuous full scale country flood hazard maps which will be used to develop flood risk maps, therefore increasing its alignment with EUFD. Under Activity 1.3 the project aims to upgrade existing FFEWS to include all remaining major basins, therefore improving flood risk management for whole country. All activities under "Output 2: Non-structural flood risk reduction measures and nature-based solutions mainstreamed in sectoral policies and plans and effectively contribute to protection of people and livelihoods from climate-induced flood risk"

⁵⁶ The 2020-2030 Climate Change Adaptation and Low Emission Development Strategy for Bosnia and Herzegovina, December 2020.

have complementary elements to EUFD which includes implementation of non-structural measures recognized in PFRM as well as develop long-term institutional capacity development plans for climate resilient FRM.

224. B&H is working on the project 'Advance the National Adaptation Plan (NAP) process for medium-term investment planning in climate sensitive sectors in Bosnia-Herzegovina'. The NAP builds on the country's CCALEDS of 2013 and advances adaptation planning in B&H with a focus on sectoral approaches, upgrading the knowledge base for adaptation, prioritizing adaptation interventions for the medium term, building institutional capacities for integrating climate change adaptation, and demonstrating innovative ways of financing adaptation at entity, cantonal and local government levels.
225. As a response to devastating 2014 floods, B&H Council of Ministers in November 2014 adopted an Action Plan for Flood Protection and River Management (2014-2021). The project is a direct response to the priorities that have emerged from the B&H National Communications, CCALEDS and Action Plan for Flood Protection and River Management, which indicate that problems of floods has been exacerbating in the last years and is expected to become progressively more serious due to climate change.
226. Strong country ownership is also demonstrated by significant government co-financing commitments. Co-financing support for the project will be provided by the Republika Srpska government through an EIB loan of US\$21.5 Million for structural measures and additionally through US\$36.0 Million of the public funds to be provided by both Governments of FB&H and RS for co-financing of EbA measures, the O&M of structural measures, hydrometric equipment, and modelling and FFEWS systems.
227. To initiate the development of this project, a UNDP-GEF technical mission was undertaken in March 2018. Since then regular consultations with the key national and regional authorities and with B&H GCF NDA have been conducted. The NDA, as well as other relevant institutions in charge of environment and water/flood management expressed an interest in developing a new project proposal with UNDP on flood risk management based on the outcomes of the existing Vrbas River project. B&H is about to initiate development of its first GCF country programme. However, the B&H GCF NDA has already confirmed that the proposed project is a priority for the GCF programming in B&H through its letter to GCF submitted in May 2019. This project has been given priority in near-term programming.
228. Further consultations were conducted with all key stakeholders in July and September 2018 when the Idea Note was presented and finalized at joint workshops. The Project idea was also presented in detail to the Project Boards of Vrbas and NAP projects consisting of GCF NDA and representatives of the ministries in charge of water management and environment. These discussions informed the development of the Concept Note, which involved all relevant institutions in B&H: the GCF NDA, ministries in charge of water management, water agencies and hydro-meteorological institutes. Validation meeting for the Concept Note engaging all relevant stakeholders was held on 5 February 2019, all stakeholders expressed their support to the proposal. In addition, separate meetings were held with HPP representatives who showed a great interest and willingness to be part of the project. Validation meeting for the Full Proposal engaging all relevant stakeholders was held on 16 October 2019, all stakeholders expressed their support to the proposal. In addition, separate meetings were held with HPP representatives, entity ministries of agriculture and forestry and municipalities, who showed a great interest and willingness to be part of the project (See Annex 7).

D.6. Efficiency and effectiveness (max` . 500 words, approximately 1 page)

229. The project addresses the fundamental barriers to achieving resilience to climate-induced floods through the provision of technical capacities, policies and legislative enhancements, and by investing in structural and non-structural risk reduction measures, including EbA approaches. The combination of interventions related to ecosystem-based approaches and structural measures, as proposed by this project, has been shown to lead to significantly larger improvements in resilience to communities, compared to an approach focusing on one of those strategies. By addressing the capacity to undertake risk assessment and monitoring and to generate missing climate risk information, the project will lay the foundation for efficient and risk-informed flood risk management in the future. In addition, by addressing the legislative and policy gaps, the project is ensuring long-term sustainability of project interventions. Furthermore, the adoption of this holistic and integrated approach that addresses all the root causes, will have long-term efficiency and effectiveness benefits, compared to the *ad hoc*, *ex-post*, *non-climate-responsive* approach to flood risk management that is currently undertaken. Climate induced extreme events are estimated to result in \$21.1 Million USD Expected Annual Damages (EAD) for households, businesses,

agriculture and public buildings annually or \$ 422.7 Million USD Expected Annual Damages (EAD) for 20 years of project, while the cost of the project is \$72.7 Million USD.

230. Of the project's total proposed budget of \$72.69 million, GCF funding represents 19.8% of the estimated project cost and co-financing 80.2% (please see section B.5 for details on financing structure), 42% or \$30.74 Million is proposed to be provided by the Government of B&H and 29% of the necessary budget resources is to be provided by the EIB in the form of senior loans. Government of B&H, in spite of its economic and budget constraints mentioned above, bringing in substantial resources to the project demonstrates its commitment to the project and more strongly its moral responsibility to provide flood forecasting and early warning system (FFEWS) as an essential service to the population.
231. The most damaging floods in B&H (2010 – 1% of GDP and 2015 – 15% have had devastating impacts on the most vulnerable groups including the rural poor, war returnees and displaced persons. The proposed project has the potential to prevent loss of life, property and critical infrastructure assets for 902,906 people representing approximately 26% of the entire population of B&H and 44% of the population of the 69 municipalities covered by the targeted sub-basins. This project will increase resilience and enhance livelihoods of the most vulnerable segments of population, particularly women (51.7%), children (15.2%) and the elderly (13.7% of the affected population). At a total project cost of \$72.69 million, this project aims to provide climate resilient flood risk protection and early warning system support at a cost of \$80.5 per direct beneficiary, which is comparable to other projects of similar nature.
232. Based on detailed economic analysis (Annex 3), the project has an economic internal rate of return (**EIRR**) of **19.3%**⁵⁷. The economic analysis of the project was carried out in accordance with the *Guidelines for the Economic Analysis of Projects of United Nations Development Program*.⁵⁸ The economic efficiency of the investment was determined by computing the economic net present value (NPV) with an assumed 10% discount rate, and the economic internal rate of return (EIRR). The economic analysis was conducted at both the national level, and then for each of the sub-basin included in the proposed investment project. The analysis shows that the proposed investment project is economically efficient as measured by NPV and IRR. The analysis also shows the proposed interventions are economically efficient in each of the sub-basins targeted by the project, with the exception of the Vrbas sub-basin which shows an EIRR slightly lower than 10%. Three different types of sensitivity analysis were conducted and demonstrated that the economic efficiency of the project as a whole is robust. There is thus a great level of confidence that the proposed investment project is economically efficient in each sub-basin of the target area, and as a whole.
233. GCF investment will catalyze US\$ 57.5 million of co-financing from public and private sector. The project partners and beneficiaries will also secure O&M costs against all project supported infrastructure and hydrometric equipment. The public-good nature of this project outputs doesn't entail significant revenue generation or cost recovery from the project although the project will support engagement of the public sector for limited cost-recovery to cover the operation and maintenance of monitoring and early warning systems. Where income generation opportunities might occur indirectly because of project outcomes, these would apply directly to the beneficiaries (for instance, improved agricultural productivity), primarily as household income.
234. Overall, the proposed solutions are based on successful best practices, community ownership, and synergies across the three outputs and inter-related activities and build upon ongoing efforts to ensure their efficiency and cost-effectiveness. The proposed activities and interventions will reach 902,906 people or 26% of the population at risk from flooding and will indirectly benefit 100% of the population through better economic resilience and stable GDP in the face of flood events.

⁵⁷ The full Economic Analysis is provided in Annex 3a and b.

⁵⁸ UNDP. 2015. Guidance on the conduct and reporting of the Economic and Financial Analysis of Climate Change Adaptation and Mitigation Projects and Programmes. UNDP.

E. LOGICAL FRAMEWORK

E.1. Project/Programme Focus

- ☐ Reduced emissions (mitigation)
☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	FRM is currently done in a largely non-integrated manner by the relevant institutions in B&H, FRM covers mainly main river fluvial flooding and does not take account of torrential floods which is prevalent on small flashy upland catchments, nor groundwater flooding, nor HPP operation flood risk. FRM is currently limited in approach and is mainly focused on response, and limited implementation of costly, non-climate risk-informed structural measures. Spatial coverage of the hydrometeorological monitoring network is incomplete and needs technological upgrades. Availability and access to flood risk information is limited. The flood forecasting and early warning system is incomplete, fragmented and non-	Low	A nation-wide integrated risk-informed and climate-responsive approach to flood risk management will be established. The FRM capacities and coverage of the FFEWS established by the Vrbas project and other baseline projects will be scaled-up to national scale. The FFEWS will be extended to cover torrential floods and HPPs flood risk and will be impact-based. FRM will be undertaken in an integrated manner, including the national integrated flood forecasting and EWS with effective last-mile communication in place. All FRM stakeholders	The following project activities will contribute to the paradigm shift through scaling-up FRM solutions: Activity 1.1. Upgrade and expand the coverage of the hydrometric network for enhanced monitoring of climate variables in Category I catchments and torrential streams contributes to the geographic and technological scaling up of the FFEWS in B&H Activity 1.2 . Enhance climate-induced flood hazard, risk and vulnerability information for strategic risk management and sound decision making for climate induced flood management - Under this activity, the EUFD maps will be enhanced to include flood risk from torrential streams, and HPPs, thus extending the technical scale/scope of hazard, risk and vulnerability information. Activity 1.3 Develop an integrated impact-based centralized and community-based flood forecasting and early warning system.- The project will develop impact based FFEWS for main rivers which

	impact-based, covering only Vrbas and Una-Sana basins. Baseline projects (see para 39 above) were limited in scope and/or geographical extent and budget, which in-turn, limited their transformational potential.		will have access to flood risk information for decision making and sectoral planning. Flood risk will be mainstreamed into key sectoral plans. Combination of structural, non-structural and EbA approaches to FRM with sufficient investment for implementing necessary measures.	will be scalable to cover sub-basins as necessary. Importantly, the flood forecasting will include all relevant flooding mechanisms and will also be scaled to provide specific sector flood forecasting products for Hydropower and agriculture sectors. Activity 2.1 Mainstream climate induced flood risk reduction into sectoral planning (agriculture, hydropower, critical infrastructure) and spatial planning - The project is mainstreaming climate risk reduction into key sectors, which can be scaled up to cover more sectors in the future. Activity 3.1 Develop investment framework for climate induced flood risk reduction and management and will lay the ground and identify financing framework for further replication and scaling up of climate-informed FRM at the national level.
Replicability	Currently there is limited capacity to replicate successful FRM project-based initiatives (such as Vrbas) due to lack of sustainable financing for FRM, despite the high replicability value of Vrbas project, which established a Vrbas basin-level integrated FRM framework.	Low	Strengthened institutional and technical capacities, financing frameworks, and enabling environment for climate informed FRM will enable replication of results throughout B&H. The project will replicate successful technologies and practices piloted by the earlier project and will further extend these by introducing new technologies and innovations such as Impact-based forecasting (IBF), forecast-based financing (FbF), blended risk financing and risk transfer mechanisms. The project will establish new partnerships with the private sector to leverage and scale up climate informed FRM investments, and will embed methods for, and implement	Activity 1.3 Develop an integrated impact-based centralized and community-based flood forecasting and early warning system. - Under this activity CBEWS will be developed for 60 high-risk communities which can be replicated to other communities. Activity 1.4: Develop and implement protocols and SoPs on data generation, data management and communication for effective impact based FFEWS and flood risk management contributes to the uptake and replication of technical and institutional FFEWS solutions across B&H. Activity 2.2 Implement and mainstream new ecosystem-based flood risk reduction and climate change adaptation methods. – The project is implementing 21 prioritised non-structural and EBA measures with significant potential for replicating in other basins using the standardised designs being developed under Activity 2.3. Codify and mainstream EbA solutions into policies and regulations and promote non-structural measures among decision-makers and communities. Activity 3.3 Implement climate-proof structural flood risk reduction and anti-erosion interventions in Vrbas, Una-Sana,

			climate resilient structural measures as well as eco-system based non-structural measures. The project will develop/update, codify and embed new SOPs, institutional guidelines and methodologies for the adoption and replication of new methods and technologies. These initiatives will ensure continued practice of embedded integrated FRM and enable the project to be highly replicable within B&H and across the region.	Bosna and Drina River basins – GCF funds are being used to climate proof the designs of structural measures which will be financed by the Government through the EiB loan. The methods for using climate risk information in the design of flood structural measures will be developed under Activity 2.1 and will provide the basis for replication in the future. Replication will be possible as a result of Activities 3.1 and 3.2 , which build financing and investment framework/plans for the FRM at the national, cantonal and municipal level.
Sustainability	Flood risk management in B&H has been undertaken mainly through donor-funded projects implemented by or in partnership with relevant ministries. In some cases, the sustainability of these initiatives has been limited due to the limited technical and geographic scope of such projects and lack of sustainable government financing to continue the initiatives beyond the life of the projects. Long-term O&M financing for equipment, technologies and FRM structural and non-structural measures is lacking. Long-term financing for the regular update, generation, sharing of climate risk information and for development of regular update of technical standards and guidelines is lacking. Long-term continuous capacity development and continued professional development in the field of FRM is missing.	<u>Low</u>	Long-term technical, financial and institutional capacities for FRM are strengthened and mechanisms in place for long-term O&M of equipment and technologies, continued capacity development. O&M plans and Capacity development plans established and updated regularly. Method, tools and technology embedded into the relevant institutions, for the long-term generation, updating and availability of climate risk information. Platforms for sharing climate risk information and collaboration on FRM established and functional and include all-of-government and all-sectors stakeholders.	Activity 1.1 Upgrade and expand the coverage of the hydrometric network for enhanced monitoring of climate variables - Under Sub-activity 1.1.3 the project will set-up sustainable operation and maintenance O&M for the network and under Sub-Activity 1.1.4 will develop long-term financing mechanisms for the maintenance of the hydrometric network. This will ensure the sustainability of the rehabilitated hydrometric monitoring network on which all FRM activities will rely in the future. Activity 1.2. Enhance climate-induced flood hazard, risk and vulnerability information for strategic risk management and sound decision making for climate induced flood management - This activity will ensure the definitive climate risk information which will ensure that climate risk activities and decision making will result in sustainable solutions. It will also embed the method, tools and technology into the relevant institutions, to ensure long-term generation, updating and availability of climate risk information. Activity 1.4 Develop and implement protocols and SoPs on data generation, data management and communication for effective impact based FFEWS and flood risk management, will provide the basis for institutional collaboration, and data sharing on FRM in the future.

				Sub-Activity 1.3.4: Develop and implement a capacity development plan for embedding flood hazard and risk modelling approaches and FFEWS into appropriate institutions in B&H, which in particular will Conduct FFEWS training on the new system for all institutions involved in FFEWS as identified by institutional mapping and institutional arrangement plan for FFEWS and Activity 2.4 Review and strengthen institutional capacity and develop long-term institutional capacity development plans for climate resilient FRM will develop the institutional capacity development plan for long term FRM in B&H and will conduct training on proposed FRM policy and sectoral FRM. The long-term CDP will ensure sustainability and continuous upgrading of capacities for FRM. Activity 2.1 Mainstream climate induced flood risk reduction into sectoral planning (agriculture, hydropower, critical infrastructure) and spatial planning - mainstreaming climate risk reduction into key sectors will strengthen the policy framework for FRM and result in sustainable FRM in the future. Activity 2.2 Implement and mainstream new ecosystem-based flood risk reduction and climate change adaptation methods and 2.3. Codify and mainstream EbA solutions into policies and regulations and promote non-structural measures will build the natural capital of B&H and provide improved eco-system services which will be more sustainable than traditional hard structural measures which inevitably have a limited lifespan.		
E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)						
GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ⁵⁹	

⁵⁹ The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

<u>Choose an item.</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Periodic condition inspection and performance evaluation reports of Water Agencies and Hydrometeorological Institutes of structural, non-structural and EBA measures and of the FFEWS. Independent evaluation of the operations and performance of the impact based FFEWS; independent evaluation reports; project reports	0	Direct: 399,723 (203,858 women, 195,864 men) Indirect: 998,622 people (0.505 million women; 0.485 million men)	Direct: 799,445 people (407, 717 women, 391,728 men) Indirect: 1,977,244 people (1.01 million women; 0.969 million men)	Methodology for estimating adaptation beneficiaries is summarised in Annex 22. Based on PFRA hazard maps combined with physical and socio-economic data for all receptors, socio-economic risk modelling has been undertaken to estimate the beneficiaries of the project under baseline and climate change (2050) scenarios for each of the five most vulnerable sub-basins in B&H (Una-Sana, Vrbas, Bosna, in the Sava basin, and Neretva and Trebišnjica in the Adriatic Sea basin)⁶⁰. The modeling shows that 799,445 people and 272,811 households are currently at risk from floods. Under climate change, by 2050 the number of affected people and households will increase to 902,906 people and 306,907 households. A conservative estimate of 799,445 beneficiaries (i.e. present day population at risk) is used as this is the number that will benefit by the end of the 6- year GCF project. The maximum indirect beneficiaries is estimated as the rest of the population, given that people in low risk areas will also have access to the early warnings via the same media and can act accordingly when in at risk areas. The rest of the population also will benefit from reduced flood economic damages and losses, as this has an indirect impact on GDP when there is a need to rebuild/rehabilitate infrastructure after every flood etc. Therefore,
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⁶⁰ See Section 2.3 and Annex E of the FS.

						population at large will benefit from the reduced disruption of businesses, logistics and infrastructure, reduced budget spendings on recovery and compensations. However, a conservative estimate of 80% of the rest of the population is used as the number of indirect beneficiaries. It is assumed that the occurrence of major extreme weather or seismic events do not deviate project trajectory and that Relevant government agencies cooperate on the implementation of all project interventions.
<u>ARA1 Most vulnerable people and communities</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Independent evaluation of the operations and performance of the structural, non-structural and EBA measures in protection the most vulnerable communities. Periodic condition inspection and performance evaluation reports of Water Agencies and Hydrometeorological Institutes of structural, non-structural and EBA measures Independent Impact Evaluation surveys and reports which will assess change in vulnerability of communities at project mid-term and end-term compared to baseline.	0	Direct: 11,258⁶¹ people (5,513 men; 5,745 women) protected from extreme floods through structural and EbA measures in 5 basins (Vrbas, Una-Sana, Bosna, Neretva and Trebisnjica river basins) Indirect: 370,670 (189,041 women, 181,628 men) people living in the municipalities	Direct: 22,516 people (11,026 men; 11,490 women) protected from extreme floods through structural and EbA measures in 5 basins (Vrbas, Una-Sana, Bosna, Neretva and Trebisnjica river basins) Indirect: 741,339 (378,083 women, 363,256 men) people living in the municipalities	Methodology for estimating adaptation direct beneficiaries is summarised in Annex 22. Modelling and consultation with Water Agencies was undertaken to identify the most vulnerable communities where structural, non-structural measures and EBA measures could provide protection and save lives, in addition to the early warning system. Number of beneficiaries is defined through the modeling and preliminary design of the proposed flood protection measures. Direct beneficiaries are those who will be directly protected by flood protection interventions. Indirect beneficiaries are remaining present day population at risk in the municipalities where the interventions will be implemented and who will benefit through reduced loss and disruption to

⁶¹ Please see sections 18.9 and 18.10 of the FS which provide detailed descriptions of the methodologies used to calculate the benefits of the 21 non-structural and 10 structural measures that will be implemented under the project.

				where structural and EBA protection measures will be implemented	where structural and EBA protection measures will be implemented	critical infrastructure assets and reduced disruption of logistics/businesses by the end of the 6-year project. Assumptions: The occurrence of major extreme weather or seismic events do not deviate projected trajectory Relevant government agencies cooperate on the implementation of the FFEWS, data management and EbA measures
<u>ARA2 Health, well-being, food and water security</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Project reports Independent evaluation of the operations and performance of the impact based FFEWS	0	Direct: 388,465 (198,117 women, 190,348 men) with access to early warnings; Indirect: 997,628 people (0.51 Million women; 0.49 Million men)	Direct: 776,929 (396,234 women, 380,695 men) with access to early warnings Indirect: 1,995,257 people (1.02 Million women; 0.978 Million men)	Methodology for estimating adaptation beneficiaries is summarised in Annex 22. The total direct beneficiaries are people at risk of flooding whose health, wellbeing, food and water security will be improved as a result of enhanced access to timely early warning and climate information. It is calculated as the present-day population at risk minus beneficiaries of the structural, non-structural and EbA measures to avoid double counting of beneficiaries. Indirect beneficiaries are estimated as the rest of the population of BiH, given that people in low risk areas will also have access to the early warnings and climate information via the same media and can act accordingly when in at

						risk areas. They will therefore also have indirect health, wellbeing, livelihoods, food and water security benefits. In order to avoid double counting of direct and indirect beneficiaries, people who will benefit from both structural/EbA flood protection measures (included under ARA 1) and also from the FFEWS were excluded from the estimation of direct beneficiaries under ARA 2. In addition, a conservative estimate of 80% of the rest of the population is used as the number of indirect beneficiaries.
<u>ARA2 Health, well-being, food and water security</u>	<u>Supplementary 2.4: Beneficiaries (female/male) covered by new or improved early warning systems</u>	Project reports Independent evaluation of the operations and performance of the impact based FFEWS	Currently 156,973 at risk people (80,056 women and 76,917 males) in Una-Sana basin and 213,740 (109,007 women and 104, 733 males) in Vrbas basin (41% of total at risk population) covered by FFEWS.	Direct: 388,465 (198,117 women, 190,348 men) with access to early warnings;	Direct: 776,929 (396,234 women, 380,695 men) with access to early warnings	Methodology for estimating adaptation beneficiaries is summarised in Annex 22. The number of beneficiaries by the end of the 6-year project is calculated as the present-day population at risk minus beneficiaries of the structural, non-structural and EbA measures to avoid double counting of beneficiaries. The occurrence of major extreme weather or seismic events do not deviate projected trajectory Relevant government agencies cooperate on the implementation of the FFEWS, data management and EbA measures

<u>ARA3 Infrastructure and built environment</u>	<u>Core 3: Value of physical assets made more resilient to the effects of climate change and/or more able to reduce GHG emissions</u>	Independent evaluation of the standard of protection provided by the project-implemented measures.	0	\$250.9 Million USD value of household assets and \$54.5 Million USD value of business assets made more resilient to the effects of climate change in 6 target basins by structural, non-structural and EbA measures; Una/Sana, Vrbas, Bosna, Sava/Ukrina, Drina, Neretva/treb	\$501.8 Million USD value of household assets and \$109 Million USD value of business assets made more resilient to the effects of climate change in 6 target basins by structural, non-structural and EbA measures; Una/Sana, Vrbas, Bosna, Sava/Ukrina, Drina, Neretva/treb	Structural and non-structural measures meet their design standards in reducing the risks to population and reduction in agricultural and other economic losses. The occurrence of major extreme weather or seismic events do not deviate projected trajectory
<u>ARA3 Infrastructure and built environment</u>	<u>Supplementary 3.1: Change in expected losses of economic assets due to the impact of extreme climate-related disasters in the geographic area of the GCF intervention</u>	Project reports Municipality post-flood assessment reports data	0 (Economic loss exposure equivalent to 15% of GDP).	\$41.8 Million USD reduction in expected economic losses to households and 9.5 million in business assets in 6 target basins by structural, non-structural and EbA measures; Una/Sana, Vrbas, Bosna, Sava/Ukrina, Drina, Neretva/treb	\$83.5 Million USD reduction in expected economic losses to households and 19 million in business assets in 6 target basins by structural, non-structural and EbA measures: Una/Sana, Vrbas, Bosna, Sava/Ukrina, Drina, Neretva/treb	Structural and non-structural measures meet their design standards in reducing the risks to population and reduction in agricultural and other economic losses. The occurrence of major extreme weather or seismic events do not deviate projected trajectory

<u>ARA3 Intrastructure and built environment</u>	<u>Supplementary 2.6: Beneficiaries (female/male) living in buildings that have increased resilience against climate hazards</u>	Independent evaluation of the standard of protection provided by the project-implemented measures.	0	Direct: 11,258⁶² people (5,513 men; 5,745 women), whose houses are protected from extreme floods through structural and EbA measures in 5 basins (Vrbas, Una-Sana, Bosna, Neretva and Trebisnjica river basins)	Direct: 22,516 people (11,026 men; 11,490 women) whose houses are protected from extreme floods through structural and EbA measures in 5 basins (Vrbas, Una-Sana, Bosna, Neretva and Trebisnjica river basins)	Structural and non-structural measures meet their design standards in reducing the risks to population and reduction in agricultural and other economic losses. The occurrence of major extreme weather or seismic events do not deviate projected trajectory
<u>ARA4 Ecosystems and ecosystem services</u>	<u>Core 4: Hectares of natural resources brought under improved low-emission and/or climate-resilient management practice</u>	WA's and HMI's periodic and post-event survey reports and performance evaluation reports on hectares of land protected by structural, EbA and non-structural measures.	0 Institutional and technical capacities for the design and implementation of non-structural and EbA FRM measures are absent or very limited. No EbA FRM measures in place Non-structural flood protection measures	2,540 ha of land protected from extreme floods through non-structural and EbA measures in 4 basins (Neretva, Una-Sana, Bosna, Vrbas) with GCF funds;	5,081 Ha of land Protected from extreme floods through non-structural and EbA measures in 4 basins (Neretva, Una-Sana, Bosna, Vrbas)	WA's, Municipalities, and community members are willing and able to implement climate-resilient non-structural and EbA design and development practices.

⁶² Please see sections 18.9 and 18.10 of the FS which provide detailed descriptions of the methodologies used to calculate the benefits of the 21 non-structural and 10 structural measures that will be implemented under the project.

			implemented in Vrbas basin only			
<u>ARA4 Ecosystems and ecosystem services</u>	<u>Supplementary 4.1: Hectares of terrestrial forest, terrestrial non- forest, freshwater and coastal marine areas brought under resoration and/or improved ecosystems</u>	WA's and HMI's periodic and post-event survey reports and performance evaluation reports on hectares of land protected by structural, EBA and non-structural measures	0 Institutional and technical capacities for the design and implementation of non-structural and EbA FRM measures are absent or very limited. No EBA FRM measures in place Non-structural flood protection measures implemented in Vrbas basin only	100 ha afforested to meet anti-flood and anti-erosion requirements. 40 ha of re- grassing as part of EBA measures	200 ha afforested to meet anti-flood and anti-erosion requirements. 80ha of re- grassing as part of EBA measures	WA's, Municipalities, and community members are willing and able to implement climate- resilient non-structural and EbA design and development practices.

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)

Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
<u>Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and</u>	Lack of protocols, updated methodologies, standards, and tools. 15% of protocols in place;	<u>low</u>	Rating: High 80% of protocols, methodologies, standards for FRM in place;	Implemented protocols – Activity 1.4 Updated methodologies, standards, and tools for FRM – Activities, 1.2, 1.3	<u>National level (one country)</u>

<u>regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner</u>	Lack of climate risks mainstreamed into sectoral policy. CC-induced FRM not mainstreamed into policies. Technical regulations and guidance on ecosystem based FRM are largely lacking.		Policies, regulations, technical guidance exist and enforced. 6 policies and technical guidelines developed and implemented and enforced in 6 target basins, flood risk reduction mainstreamed into Hydropower, Agriculture, and Forestry sectoral policies; floodplain zoning policy and development control. Critical infrastructure technical guidelines. EbA technical guidelines	Climate-induced flood risk mainstreamed into sectoral policies (Hydropower, Agriculture, and Forestry sectoral policies; floodplain zoning policy and development control) – Activity 2.1 Technical guidelines for climate proofed critical infrastructure design – Sub-Activity 2.1.3 Technical guidelines EbA technical guidelines developed – Activity 2.3	
<u>Core Indicator 6: Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer and innovation</u>	Incomplete hydrometric network with only 20% of the network using modern automatic observation equipment. Limited climate-induced flood hazard and risk modelling and mapping using modern methodologies, models and technologies (EUFD modelling, and mapping completed for Vrbas basin only). Modern FFEWS systems using updated data products and technologies to provide climate risk information in operational decision making on flood risk response available for Vrbas and Una_Sana basins (2 of 6 basins) only.	<u>low</u>	Rating: High Rehabilitated and complete hydrometric observation network using updated automatic observation stations and latest technologies in place in all basins. Enhanced EUFD hazard and risk models and maps available for 6 basins and used in decision making in those 6 basins. Impact-based FFEWS operational for all 6 basins and provides climate risk forecasts and warnings to population. At least 2 sectors specific FFEWS products are used in decision-making (Sector-specific climate information	Rehabilitated and complete hydrometric observation network - Activity 1.1 Enhanced EUFD hazard and risk models and maps - Activity 1.2 Impact-based FFEWS operational for all 6 basins - Activity 1.3	<u>National level (one country)</u>

	No sector-specific climate risk information or products available An FRM data sharing portal is developed for Vrbas basin (Vrbas Geo-Portal)		product/services for HPP and agric); Operational cross-sectoral platform for sharing and use of CC FRM data with relevant data sharing protocols	At least 2 sectors specific FFEWS products (HPP and agric) - Sub-Activity 1.3.3 Cross-sectoral platform established and operational – Sub-activity 1.4.3	
<u>Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards</u>	Limited knowledge and capacities on hazard, vulnerability and risk assessment and modelling. 15% of practitioner with relevant capacities Limited knowledge and capacities in forecasting and EW. Communities lack timely flood forecast and early warning. Lack of experience, knowledge, practice methods and good practice guidelines on EbA . Limited knowledge and capacities on flood risk management	<u>low</u>	Rating: Medium 75% of hazard, vulnerability and risk modelling practitioners with relevant capacities; Capacity development plan for embedding flood hazard and risk modelling approaches and FFEWS into appropriate institutions in B&H Best-practice guidelines, technical specification, standards and protocols, and knowledge portals established for the application of EbA and non-structural measures. At least 50% increase over baseline in institutional capacity within WA'. HMIs, municipalities (in 6 basins); FRM training embedded in existing government training	Strengthened hazard and risk monitoring, modelling,– Activities 1.1, 1.2, Capacity development plan for embedding FFEWS into relevant institutions in B&H - Activity 1.3 Best-practice guidelines, technical specification, standards and protocols for the application of EbA and non-structural measures; knowledge portal and common resource center to collect, collate and share knowledge and fill these gaps to advance uptake of nature-based solutions; Knowledge portal for collaboration between specialists to allow to co-working and knowledge sharing - Activity 2.3 FRM training embedded in existing government training	<u>National level (one country)</u>

			programmes for long-term FRM training in B&H	programmes for long-term FRM training in B&H in the following topics: Flood hazard and risk assessment, hydrometry, forecasting and modelling, EWSS; Vulnerability and Risk Assessment; River basin flood risk management planning and the use of climate information - Activity 2.4	
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E.5. Project/programme specific indicators (project outcomes and outputs)

Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Output 1. Fully integrated impact-based flood forecasting and EWS facilitates timely preparation and response	1.1 Number of new hydro-meteorological monitoring equipment purchased, installed and operational	Project reports, evaluation reports, site observations	Hydrometric monitoring network inadequate. Existing stations by basin: Una/Sana - 17 Meteo, 28 hydro. Vrbas- 25 (meteo), 18 (hydro); Bosna – 35 (meteo), 65 (hydro);	Hydro-meteorological observation network expanded with 9- Groundwater monitoring stations; 24- rehabilitated piezometers; 18- new piezometer 100 and 200 m deep. 10- boreholes with average depth up to 50 m; 30- boreholes with average depth up	Hydro-meteorological observation network expanded with 9- Groundwater monitoring stations; 24- rehabilitated piezometers; 18- new piezometer 100 and 200 m deep. 10- boreholes with average depth up to 50 m; 30- boreholes with average depth up	<u>Assumptions:</u> Government commitments to secure adequate O&M of monitoring equipment, relevant software and databases are fulfilled on a continuous basis both during the project implementation and afterwards Capacities built across relevant institutions through the project are maintained and periodically updated

			Sava/Ukrina- 1 (meteo), 1 (hydro); Drina – 15 (meteo), 28 (hydro); Neretva/treb – 3 (meteo), 38 (hydro);	to 30 m; 26- Meteorological stations; 23- Hydrological stations GCF-funded Stations by basin: Una/Sana - 6 Meteo; 12 hydro; Vrbas- 5 (meteo), 12 (hydro); Bosna – 7 (meteo), 11 (hydro); Sava/Ukrina- 1 (meteo), 21 (hydro); Drina – 1 (meteo), 11 (hydro); Neretva/treb – 6 (meteo), 47 (hydro);	to 30 m; 26- Meteorological stations; 23- Hydrological stations GCF-funded Stations by basin: Una/Sana - 6 Meteo; 12 hydro; Vrbas- 5 (meteo), 12 (hydro); Bosna – 7 (meteo), 11 (hydro); Sava/Ukrina- 1 (meteo), 21 (hydro); Drina – 1 (meteo), 11 (hydro); Neretva/treb – 6 (meteo), 47 (hydro);	Relevant government institutions cooperate on the implementation of the FFEWS and data management. <i>Risks/Externalities: Failure of institutions to agree protocols for FFEWS and/or failure to engage local communities in CBEWS resulting in non-integrated national FFEWS and incomplete last mile coverage</i>
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	1.2. Number of river basins for which enhanced flood hazard and risk maps and FFEWS are available and in use by institutional beneficiaries (Water Agencies, Hydrometeorological Institutes, and sectors: agriculture, hydropower, critical infrastructure and spatial planning)	Project reports, evaluation reports, monitoring and risk assessment products Knowledge system: e-library, created databases, information systems, knowledge portal Project surveys Performance assessment reports for FFEWS.	0 Lack of key data sets for development of hazard and risk models; flood risk maps and vulnerability assessment available only for Vrbas Basin; FFEWS platforms available for only two river basins (Vrbas and Una-Sana)	All 6 river basins (3,859,800 ha) with enhanced flood hazard and risk maps All 6 river basins with impact based FFEWS 50% of surveyed institutional users water agencies, HMI, and sectoral institutions use FRM hazard and risk information in sectoral planning, practice and decision-support functions; All relevant institutions have access to timely impact based FFEWS warnings. Institutions responsible for warning and response have 50% improvement in warning lead times and accuracy.	All 6 river basins (3,859,800 ha) with enhanced flood hazard and risk maps All 6 river basins with impact based FFEWS 90% of surveyed institutional users water agencies, HMI, and sectoral institutions use FRM hazard and risk information in of sectoral planning, practice and decision-support functions; All relevant institutions have access to timely impact based FFEWS warnings. Institutions responsible for warning and response have 50% improvement in warning lead times and accuracy.	<u>Baseline data sources:</u> WA and HMI Strategic plans Sectoral institution covering the following sectors at all levels of governance: Agriculture, hydropower, forestry, critical infrastructure and spatial planning. Ministry of Foreign Trade and Economic Relations of Bosnia and Herzegovina (MoFTER, GCF focal point), The Ministry for Spatial Planning, Civil Engineering and Ecology (MSPCEE, UNFCCC), Ministry of Agriculture, Water Management, and Forestry of Federation of B&H, Ministry of Agriculture, Forestry, and Water Resources of Republika Srpska, State, Entity and Canton Red Cross, Association of Cities and Municipalities, Entity, Canton and municipal Civil Protection, Hydropower Companies.
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	1.3 Status of implementation of protocols and SoPs on data generation, data management and communication for effective FFEWS and flood risk management introduced by the project	Project reports, evaluation reports, monitoring and risk assessment products	Data sharing platform protocol developed for Vrbas and <u>Una-Sana</u> .	WA's, HMIs and relevant institutions in 50% of basins with access to and using data sharing protocols, including: a) SoPs for "last-mile" warning and dissemination and communication system, b) Protocols and SOPs on data generation, data management for FRM, c) data sharing protocols and platform for climate data, across all government institutions in both entities including spatial Data Infrastructure to be developed/enhanced to ensure climate data flow/exchange	WA's, HMIs and relevant institutions in 100% of basins with access to and using data sharing protocols , including: a) SoPs for "last-mile" warning and dissemination and communication system, b) Protocols and SOPs on data generation, data management for FRM, c) data sharing protocols and platform for climate data, across all government institutions in both entities including spatial Data Infrastructure to be developed/enhanced to ensure climate data flow/exchange	
Output 2. Non-structural flood risk reduction measures and nature-based solutions mainstreamed	2.1. Number and level* of enforcement of sectoral policies, regulations and technical guidelines with climate induced flood risk reduction, including EbA,	Project reports – policy reviews, evaluation reports, independent evaluation report.	Level 1 CC-induced FRM not mainstreamed into policies. Technical regulations and	Level 2 4 policies, and technical guidelines developed and	Level 3 6 policies and technical guidelines developed and	Government has political will to implement relevant regulatory reform to embed CC-induced FRM in relevant institutions and are willing to conduct regulatory and institutional reform

in sectoral policies and plans and effectively contribute to protection of people and livelihoods from climate-induced flood risks.	mainstreamed as a result of the project	B&H CCA M&E system under the NAP. Sector-specific compliance reports for Energy, Forestry and Agriculture.	guidance on ecosystem based FRM are largely lacking. Floodplain zoning policy developed for Vrbas river basin.	implemented in 3 target basins; flood risk reduction mainstreamed into Hydropower, Agriculture and Forestry sectoral policies; floodplain zoning policy and development control;	implemented and partially enforced in 6 target basins flood risk reduction mainstreamed into Hydropower, Agriculture and Forestry sectoral policies; floodplain zoning policy and development control. Critical infrastructure technical guidelines EbA technical guidelines	Capacities created as a result of project are maintained and periodically upgraded; Women involvement is encouraged Various Sectoral Ministries; willingness to engage in the programme and transfer received know-how to stakeholders; Government engages in each step of the development of the sectoral policies, endorses it officially and takes a commitment to implement it through allocation of relevant funds for implementation of priority measures and/or resource mobilization
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	2.2. % increase in the level of institutional capacity for implementation of FRM and FFEWS among key government agencies over baseline	Institutional capacity assessment scorecard for all institutions (entity ministries, two Water Agencies, and two HMIs) Project reviews, independent evaluation report.	Lack of coherent institutional capacity development framework that embeds FRM and FFEWS capacities into relevant institutions Baseline level of institutional capacities to be established during Year 1 of the project	At least 25% increase over baseline in institutional capacity (technical capacity: knowledge, skills, systems, technologies), within WA'. HMIs, municipalities (in 3 basins);	At least 50% increase over baseline in institutional capacity (technical capacity: knowledge, skills, systems, technologies) within WA'. HMIs, municipalities (in 6 basins); FRM training embedded in existing government training programmes for long-term FRM training in B&H	*Level 0 - Policies do not exist; Level 1 – Some fragmented policies designed but the policy framework is incomplete; Level 2 - policy and regulatory framework designed but not enforced Level 3 - policies, regulations, technical guidance exist and partly enforced; Level 4 – policies, regulations, technical guidance enforced Detailed institutional capacity assessment will be undertaken based on scorecards which will provide a baseline at the start of the project and short-medium and long-term targets. The Detailed Capacity Development plan will be developed and priority short- and medium-term capacity development will be undertaken by the project. Capacity development delivered by the project will be measured against the baseline capacity assessed at the start. Activity 2.4 is embedding Long-term FRM training and CD into relevant government training institutions and programmes. (FS pgs. 249-252. Please also see Chapters 3-7 of FS for a detailed description of all aspects of existing FRM
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						institutional capacity and practice in BiH).
	2.3 Number of people protected through ecosystem-based flood risk reduction measures	Site visit report; Project reports, evaluation reports, independent evaluation report, Mid-term and terminal evaluation reports; IE surveys and reports; site-specific ESPs	0 Institutional and technical capacities for the design and implementation of non-structural and EbA FRM measures are absent or very limited; No EBA FRM measures in place; Non-structural flood protection measures implemented in Vrbas basin only.	6,799 ⁶³ people (3,293 men; 3,506 women)	13,598 people (6,586 men; 7,012 women)	<u>Risks/Externalities:</u> Under-performance of the implemented EbA and non-structural measures under extreme events beyond design standards Method for calculating beneficiaries of the structural and EBA measures to be implemented in the 5 basins is provided in Section 18.9 and 18.10 of the FS. Please see all of Annex E for full socio-economic CBA methodology.

⁶³ Please see sections 18.9 and 18.10 of the FS which provide detailed descriptions of the methodologies used to calculate the benefits of the 21 non-structural and 10 structural measures that will be implemented under the project.

			Structural measures design are not climate risk informed.			
Output 3. Climate-proof flood protection measures scaled-up through new and improved national and local investment frameworks to increase resilience of the most vulnerable groups to climate induced flooding.	3.1. Number of country-wide investment frameworks, new private sector partnerships and/or risk transfer mechanisms for climate induced flood risk reduction and management;	Project reports, evaluation reports, independent evaluation report	Insurance product developed for Vrbas basin but not implemented No other private sector investment frameworks in place	4 strategies/mechanisms developed (including 2 Private sector engagement strategies and 1 flood risk financing mechanism developed and FbF mechanism developed);	7 (including 2 Investment frameworks for climate induced floods risk reduction and management; 3 flood risk financing mechanism in place; 2 MoUs with private/productive sector actors)	WA's, Municipalities, and community members are willing and able to implement, municipal investment plans, response plans, climate-resilient structural design and development practices. WA's have the financial, technical and resourcing capacity to maintain structures
	3.2. Number of Multi-year climate resilient municipal investment plans and gender responsive community preparedness plans developed with project support in Municipalities and cantons following investment and preparedness plans completed by project.	Project reports, evaluation reports, independent evaluation report	0 No multi-year climate resilience municipal investment plans and gender responsive community preparedness plan in place	At least 6 Multi-year climate resilient municipal investment plans and gender responsive community preparedness plans (include EAPs for FbF) developed in 5-6 municipalities and 1 canton in Vrbas, Una Sana and Bosna, Drina, Neretva and Trebisnjica river basins	At least 12 Multi-year climate resilient municipal investment plans and gender responsive community preparedness plans ((include EAPs for FbF)) developed in 10-12 municipalities and 1 canton in Vrbas, Una Sana and Bosna, Drina, Neretva and Trebisnjica river basins	

	3.3. Number and effectiveness (nu. of people/ha of land protected) of climate-proof structural flood risk reduction and anti-erosion interventions implemented in Vrbas, Una-Sana and Bosna, Drina, Neretva and Trebisnjica river basins with the project support	Project reports, evaluation reports, independent evaluation report	0	5 climate-proof structural flood risk reduction and anti-erosion interventions implemented in Vrbas, Una-Sana and Bosna, Neretva and Trebisnjica river basins (approx.. 16,000 meters total length) 4,459 people (2,220 men; 2,239 women); and 1,154 ha of agricultural land protected from flooding	10 climate-proof structural flood risk reduction and anti-erosion interventions implemented in Vrbas, Una-Sana and Bosna, Neretva and Trebisnjica river basins (approx.. 32,131 meters total length) 8,918 people (4,440 men; 4,478 women); and 2,138.7 ha of agricultural land protected from flooding	See Annex A (table 14-3) of the FS for design drawings and details of the 10 structural measures, and Annex E (section 18.10) for the socio-economic assessment and options appraisal of the structural measures.
Project/programme co-benefit indicators						
Co-benefit 1: (Social/Health) Improved health and reduced health risks due to the reduction in the number of floods-related injuries and exposure to water borne diseases.	Number of people with reduced risk of water-borne diseases as a result of the project flood risk reduction intervention	Project reports, evaluation reports, independent project impact evaluation report	0 59% of the population of B&H currently not connected to sewerage systems and 25% are without access to drinking water, these people are at higher risk of suffering from illness due to water borne disease that arise due to flooding.	235,836 people (120,277 women; 115,560 men)	471,673 people (240,553 women; 231,120 men)	Estimation method: Present day population at risk = 799,445; % of population not connected to sewerage = 59% Therefore, number of beneficiaries = 471,673 people The occurrence of major extreme weather or seismic events do not deviate historic trajectory Relevant government agencies cooperate on the implementation of the FFEWS, data management and EbA measures

						Structural and non-structural measures meet their design standards in reducing the risks to population.
Co-benefit 3: (Gender) Female practitioners trained in all aspects of FRM; gender-responsive risk assessments and FRM framework established	<i>% of trained practitioners are female; Gender-responsiveness of CDP; gender-responsiveness of data collection and modelling.</i>	Project reports, evaluation reports, independent project impact evaluation report	Limited training in FRM through all institutions. No long-term capacity development plans in place; gender-responsive systematized system of collection of vulnerability and modelling vulnerability to floods is fully developed for Vrbas basin only	Minimum of 40% trained practitioners are female; Capacity development plan is gender-responsive; gender-responsive systematized system of collection of vulnerability and modelling vulnerability to floods is fully developed for all basins.	Minimum of 40% trained practitioners are female; Capacity development plan is gender-responsive; gender-responsive systematized system of collection of vulnerability and modelling vulnerability to floods is fully developed for all basins.	GAAP effectively implemented; Government counterparts are committed to gender responsive FRM

E.6. Project/programme activities and deliverables

Activities	Description	Sub-activities	Deliverables
Activity 1.1: Upgrade and expand the coverage of the hydrometric network for enhanced monitoring of climate variables in Category I catchments and torrential streams	The Project will expand the hydrometeorological monitoring network to cover a greater range of hydrometric variables that climate change is bringing. Previous upgrades to the hydrometeorological observation network have not been realized at the appropriate spatial and temporal scale to enable effective flood monitoring, management, forecasting and early warning. In addition to pluvial and fluvial flooding in the biggest catchments (category I catchment), climate change is increasing the risks from torrential streams, which are currently	1.1.1. Develop optimized hydrometric network specification, develop ICT strategy and plan for hydrometric 1.1.2. Procure and install equipment to increase density of the hydro-meteorological observation network and expand monitoring to include greater range of hydrometric variables that climate change is bringing 1.1.3. Set-up sustainable O&M for the network 1.1.4. Develop long-term financing mechanisms for the maintenance of the hydrometric network	Hydro-meteorological observation network expanded with 9-Groundwater monitoring stations; 24- rehabilitated piezometers; 18-new piezometer 100 and 200 m deep. 10- boreholes with average depth up to 50 m; 30- boreholes with average depth up to 30 m; 26-Meteorological stations; 23-Hydrological stations

	not monitored or studied but cause great damage		2 strategic plans, methodologies, guidelines and manuals for the operation and maintenance of the optimized hydrometric network. 1 prototype mechanism for private sector cost recovery 2 Public Sector O&M financing strategies action plans, budgetary allocation mechanisms and tools
Activity 1.2: Enhance climate-induced flood hazard, risk and vulnerability information for strategic risk management and sound decision making for climate induced flood management	The Project will enhance the WBIF EUFD hazard and risk modelling by a) developing hydrological modelling for all main basins (category I rivers), b) undertaking torrents hazard modelling, c) including HPPs and groundwater modelling in flood hazard, risk and vulnerability modelling and mapping for all relevant basins.	1.2.1: Develop hydrological modelling for all main basins (category I rivers) to complement WBIF project's EUFD hazard and risk models. 1.2.2: Develop full basin hydraulic models 1.2.3: Produce country-wide maps of torrents hazard, risk and vulnerability 1.2.4: HPP modelling for development of enhanced operating rules for HPPs. Incorporate HPP operations into flood hazard and risk modelling. Upgrade models in Adriatic Sea basin to include groundwater modelling	: • Topographic surveys in target basins • Relevant topographic data sets procured • Technical methodology and survey completed for the systematic collection of gender-sensitive socio-economic data to support natural disaster risk and vulnerability assessment. • Enhanced WBIF EUFD hydrological models for 6 river basins. • Enhanced WBIF EUFD hydraulic flood hazard models for 6 river basins(Category I) • HPPs and groundwater flood models incorporated into modelling in flood hazard models, • Reservoir operation models for all target basins • Updated Neretva and Trebišnjica basin model and optimised operating rules for HPP • Torrents hazard models for 6 basins • GIS-based socio-economic risk and vulnerability model • Country-wide flood hazard, risk and vulnerability maps for 6 basins. • Guidance, methodologies and specifications for dam breach modelling and mapping;

			• Strategy and programme of updating dam breach modelling and mapping for all medium and high hazard dams in B&H; • Guidance, methodologies and specifications for the inclusion of HPP operations in flood hazard modelling and mapping; • Practitioners/experts/modelers trained in Flood hazard and risk modelling •
Activity 1.3: Develop an integrated impact-based centralized and community-based flood forecasting and early warning system (FFEWS)	Under the Project the Vrbas prototype will be scaled up to cover the two hydro-energy intensive basins of Neretva and Trebisnjica, and the Una-Sana basin forecasting model will be updated. The project will Implement CBEWS for 60 high priority communities and develop sector specific FFEWS products and develop and implement a capacity development plan for embedding flood hazard and risk modelling approaches and FFEWS into appropriate institutions in B&H	1.3.1: Develop and implement National impact-based FFEWS system (integrated and Community-based). 1.3.2: Implement CBEWS for high priority communities 1.3.3: Review, identify and develop sector specific FFEWS products for agriculture and hydropower/energy sectors 1.3.4: Identify and implement a capacity development plan for embedding flood hazard and risk modelling approaches and FFEWS into appropriate institutions in B&H	• Assessment Report on legal, institutional, organisational, financial and technical environment, issues and gaps at HMSs. • A logical and physical database model for new and historic meteorological and hydrological data collections. • The tools and services for online and offline data and metadata acquisition, quality and quantity control, database feed, post-processing and reporting are fully functional. • 6 FFEWS models and 1 FEWS platform integrating all basin FFEWS models; • Equipment procured for CBEWS (60 water level sensors; 120 staff gauges; 2 GSM/GPRS modem device; 40 generators; 40 sirens; 400 portable radio stations; 100 mobile radio stations; 120 boards. • Community Based Early Warning Systems for 60 high priority communities; • Sector-specific FFEWS products for HPP and agriculture; • Capacity development plan for flood hazard and risk modelling and FFEWS developed and institutionalized in B&H • Training in partnership with WMO regional training centers, delivered to all HMI, WA an other staff and practitioners on FFEWS

Activity 1.4: Develop and implement protocols and SoPs on data generation, data management and communication for effective impact based FFEWS and flood risk management	The project will revise and implement a number of protocols, and SOPs, and Codes of Conduct for each of the institutions responsible for FRM and FFEWS. These will include SoPs for “last-mile” warning and dissemination and communication system, Protocols and SOPs on data generation, data management for FRM, data sharing protocols and platform for climate data, across all government institutions in both entities including spatial Data Infrastructure to be developed/enhanced to ensure climate data flow/exchange.	1.4.1: Update and implement SoPs for last-mile warning and dissemination and communication system. 1.4.2: Develop national protocols and SOPs on data generation, data management for FRM. 1.4.3: Develop data sharing protocols and platform for climate data, across all government institutions in both entities	• SoPs for “last-mile” warning and dissemination and communication system; • Protocols and SOPs on data; • FRM, at least 12 data sharing protocols between water management and other relevant sectors (agriculture, forestry, environment, spatial planning and hydro-energy); • Cross-sectoral platform for climate data, across all government institutions
Activity 2.1: Mainstream climate induced flood risk reduction into sectoral planning (agriculture, hydropower, critical infrastructure) and spatial planning	The project will enhance the existing legislative and regulatory FRM framework by mainstreaming the climate-induced flood risk management into sectoral planning for agriculture, forestry, environment, hydropower, critical infrastructure, and spatial planning	2.1.1: Mainstream climate induced flood risk reduction into sectoral strategies, plans and technical guidelines for agriculture, hydropower, critical infrastructure, forestry and environment 2.1.2: Develop national floodplain zoning policy and legislation, based on national hazard, risk and vulnerability mapping, flood resilient building codes and embed climate change considerations in the design and construction standards of critical infrastructure 2.1.3: Update climate risk-informed methodologies and standards for infrastructure design, construction material, use and maintenance of critical infrastructure in B&H 2.1.4: deliver training and technical advice on climate resilient infrastructure design, construction and O&M approaches	5 Sectoral strategies, plans and technical guidelines (for agriculture, hydropower, critical infrastructure, forestry and environment); National floodplain zoning policy and legislation implemented in all basins; Updated climate risk-informed methodologies, standards for infrastructure design, construction material, use and maintenance of critical infrastructure in B&H Capacity development plan for climate resilient infrastructure design, construction and O&M approaches
Activity 2.2: Implement and mainstream new ecosystem-based flood risk reduction and climate change adaptation methods	The project will implement catchment management measures including 8 non-structural measures	2.2.1: Implement catchment management measures for reduced erosion 2.2.2: Implement 21 selected non-structural flood risk reduction measures	Catchment management for reduced erosion; 8 Non-structural measures implemented;
Activity 2.3: Codify and mainstream EbA solutions into policies and regulations and promote non-structural measures among decision makers and communities	The project will document best practice non-structural approaches that can be adapted to the B&H context and collect lessons learned and facilitate dissemination of successful approaches in the country. The successful practices will be codified in a form of guidelines	2.3.1: Develop best-practice guidelines for EbA and non-structural measures	• EbA and non-structural measures incorporated into relevant entity FRM policies and regulations. • Best-practice guidelines, manuals technical specification, standards and protocols that provide clear guidance

	and tools. This activity will document and mainstream non-structural measures into FRM policies and practices.	2.3.2: Develop technical specification, manuals, standards and protocols for design and implementation of non-structural measures 2.3.3: Review and implement training required for new non-structural measures	on the design, implementation, and maintenance of EbA solutions and non-structural measures for flood risk management. • Methodology for system evaluation; • Methodology and guidance for undertaking risk assessment and appraisal of EbA measures; • Knowledge portal and common resource center to collect, collate and share knowledge and for interdepartmental collaboration, coordination and joint decision-making to ensure the integration of EbA solutions and non-structural measures across different policy domains, such as water management, land-use planning, and environmental conservation.; • Capacity development: Training in the application of methods for the identification, prioritization, design and implementation of basin appropriate EbA solutions; • Training for 200 FRM practitioners at all levels and all relevant types of stakeholders (entity, municipal, community) • Awareness raising: Workshops, and knowledge-sharing platforms to build awareness, technical skills, and understanding of the benefits and implementation considerations of nature-based approaches and non-structural measures
Activity 2.4: Review and strengthen institutional capacity and develop long-term institutional capacity development plans for climate resilient FRM	The activity will implement country-wide training programmes in the technical and practical areas of flood risk management for practitioners, decision-makers, communities, emergency responders.	2.4.1: Embed FRM training in existing government training programmes for long-term FRM training in B&H. 2.4.2: Develop a costed and prioritized Institutional capacity development plan to address technical, functional and human capacity gaps identified through the completed institutional capacity assessment	FRM training embedded in existing government training programmes for long-term FRM CD in B&H in the following topics: • Flood hazard and risk assessment, hydrometry, forecasting and modelling, EWSs; • Vulnerability and Risk Assessment;

		2.4.3: Implement community-based training and awareness raising initiatives at community level	- River basin flood risk management planning and the use of climate information Key deliverables: - Training curricula, and rosters of trainers developed and mechanisms for delivering long-term capacity development (e.g. development of key partnerships with universities and institutions) for FRM training. - Costed and prioritised institutional capacity development plan to address technical, functional and human capacity gaps in FRM. - Community-based training and awareness raising initiatives implemented at community level - Emergency response training of first responders - Guidelines on community DRR/DRM - Local NGOs, CBOs or non-CBO community members trained in Community-based Risk Assessment approaches, Community-based Early Warning Systems and gender-responsive Community-based Flood Risk Management; - training of trainers' programme and a syllabus for topics such as first response, search and rescue and warning dissemination, and example standard specifications for external training providers - Educational material and a 5year training programme on flood and disaster risk and emergency response for use in school outreach programmes, university courses, communities etc. (training material, course notes, plays etc.) - Annual community forums on FRM and CBEWSs with participation of target communities and representatives of vulnerable groups
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			to exchange information, lessons learned, successes and impediments; • Community-government and public-private dialogues around local risks and risk reduction strategies and their financing. • Develop and deliver awareness raising messages using social media, art and creativity to communicate FRM and FFEWS concepts to broader society and to disseminate project lessons and successes; • Media partnerships: TV and radio talk shows around FRM and FFEWS; media coverage of project activities; • Production of footages, Public Social Announcements (PSAs) of short documentaries around FRM and project achievement • Training for media on FRM and FFEWS and annual essay/media article awards competitions.
Activity 3.1: Develop investment framework for climate induced flood risk reduction and management	The project will develop a strategy and tools for private sector engagement in long-term climate change-induced flood risk management. The project will develop the most appropriate mechanisms for risk transfer, risk financing and supporting cost recovery mechanisms from a variety of sources. The Activity will also include the development of policy and legislation for private sector engagement in climate risk financing and for risk transfer mechanisms such as flood insurance.	3.1.1: Develop investment framework for climate induced floods risk reduction and management including provisions for public and private/productive sector engagement in climate risk financing ; 3.1.2: Develop risk financing and transfer mechanisms based on detailed socio-economic risk, damages and losses assessment; 3.1.3: Design natural disasters' insurance scheme and explore, and identify, other risk financing and transfer mechanisms products and tools. 3.1.4: Develop tool for appraisal-led design for structural and non-structural FRM measures, FRM investment planning, climate risk financing mechanisms and for appraisal-led FRM options design and decision-making, based on CBA approaches	5 strategies and plans for long-term ongoing investment frameworks that would be required to address flood risk to each sector; Sector-specific information dissemination portals as part of the information platform to be developed under 1.4.3; 2 risk financing and risk transfer strategy for private sector engagement in risk financing and risk transfer mechanisms for entity-level flood risk financing and resilience strategy; 3 natural disasters products developed; 3 Feasibility studies on micro-insurance, regional disaster risk insurance program and catastrophe bonds; 4 sector specific awareness campaigns;

			2 tools for appraisal-led design for structural and non-structural FRM measures based on cost benefit analysis (CBA) approaches.
Activity 3.2: Formulate multi-year climate resilient municipal investment plans and gender responsive community preparedness plans in selected municipalities (10-12) and in one canton in Vrbas, Una-Sana and Bosna, Drina, Neretva and Trebišnjica basins	The project will develop municipal investment plans for climate-resilient investment planning for flood risk management in 10-12 municipalities and 1 canton in the Vrbas, Una-Sana and Bosna basins.	3.2.1: Develop municipal investment plans for climate resilient FRM planning for 10-12 highest risk communities and 1 canton 3.2.2: Develop preparedness plans for 10-12 highest risk communities and 1 canton based on Vrbas methodology	2 financing models for investment maintenance costs; Standardized CBA tools embedded in municipality for developing annual infrastructure investment plans. 10-12 municipal FRM investment plans based on risk-informed project designs, including maintenance, and costs-benefit analysis based on CBA methods and models. Multi-year climate resilient municipal investment plan and gender responsive community preparedness plan implemented in the highest risk municipalities (10-12) in Vrbas, Una-Sana and Bosna basins; Community preparedness plans for the 10-12 communities in line with the procedure, methodology and content of CEFP in B&H as defined by Law of Protection and Rescue in RS and FB&H
Activity 3.3: Implement climate-proof structural flood risk reduction and anti-erosion interventions in Vrbas, Una-Sana, Bosna and Drina river basins (co-financed)	This activity will focus on implementation of priority structural flood risk reduction interventions for areas at highest risk. The structural measures will be financed through loan resources and Water Agencies.	3.3.1: Finalize detailed design of climate resilient flood protection structural measures identified using CBA methods and appraisal-led optioneering to identify and prioritize critical flood protection structures. 3.3.2: Implement new flood defenses and the rehabilitation and upgrade of existing flood defenses with climate proofing	Detailed design of 10 structural measures; 10 structural measures implemented.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

235. Risk factors associated with the project implementation include institutional, policy, financial, technical and operational aspects of establishing an integrated climate resilient flood risk management and introducing an impact-based MHEWS both national-wide and at community level; and social and environmental due to the implementation of climate-resilient nature-based solutions and construction of protective infrastructures against flood risks. The latter risks may affect the sedimentation movement during construction of flood defence infrastructures. Other risks may affect the lack of commitment from communities where restoration activities, alternative livelihoods and EWS are planned. The proposed project includes several mitigation measures to address these risks.

Selected Risk Factor 1

Category	Probability	Impact
<u>Governance</u>	<u>Low</u>	<u>Medium</u>

Description

Political situation becomes unstable due to local upheavals or regional conflicts

Mitigation Measure(s)

The project will develop and implement emergency management/contingency plan in line with UNDP CO's crisis management requirements. This may reduce the level of impact of the risk to medium to low level

Selected Risk Factor 2

Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Medium</u>

Description

Extreme hazardous event resulting in damage or destruction to hydrometeorological and/or flood defense infrastructure, damage to beneficiary communities, and disruption to project implementation, requiring reprogramming of government FRM budgets or implementation delays.

Mitigation Measure(s)

The project will develop and implement emergency management/contingency plan in line with UNDP CO's crisis management requirements. During the planning, design and construction of relevant infrastructure, disaster risks will be taken into consideration for all stages, and relevant risk mitigation/reduction measures will be put in place. Importantly, the project is designing and implementing climate proofing of infrastructure which should ensure that designs are robust against flood damage and destruction up to the design Standard of Protection. These risk reduction measures will reduce the level of impact and probability that the infrastructure will be destroyed, to minimum level.

Selected Risk Factor 3

Category	Probability	Impact
<u>Governance</u>	<u>Low</u>	<u>Medium</u>

Description

Climate adaptation, FRM and DRR do not remain a government priority and therefore political, financial and technical support to the project is reduced

Mitigation Measure(s)

The project will leverage (and strengthen if necessary) already established strong partnerships with high-level government representatives and will carry out lobbying and advocacy campaigns in support of CC adaptation, FRM and DRR. This will reduce the impact of the risk to the minimum level.

Selected Risk Factor 4

Category	Probability	Impact
<u>Technical and operational</u>	<u>Low</u>	<u>Medium</u>
Description		
Absorption and operational capacities of project beneficiaries (WAs and HMIs, municipalities etc.) are inadequate to properly run and maintain FRM technologies and systems implemented by the project, after completion		
Mitigation Measure(s)		
The project will pay close attention to the capacity building of all relevant agencies through carrying out training of trainers, on-the-job and field trainings of the staff of relevant agencies, introducing/strengthening internship mechanisms, developing technical guidelines, methodologies and sustainable operations and maintenance plans and embedding ongoing training capacity within relevant institutions. These actions, altogether will reduce the probability and impact of the risk to minimum level.		
Selected Risk Factor 5		
Category	Category	Category
<u>Governance</u>	<u>Low</u>	<u>Medium</u>
Description		
Government and private sector unable to meet co-financing and/or O&M commitments to the project (e.g. due to unforeseen budget cuts).		
Mitigation Measure(s)		
The project will assist the government authorities to develop and implement sustainable long-term investment planning and will develop risk financing mechanisms which will secure financing for activities in the long-term.		
Selected Risk Factor 6		
Category	Probability	Impact
<u>Technical and operational</u>	<u>Low</u>	<u>Medium</u>
Description		
Local communities are not interested to be engaged in CBEWS and community based EbA processes		
Mitigation Measure(s)		
The project will conduct extensive awareness campaign at municipal and community level on the climate-induced flood hazards, vulnerabilities and risks and benefits for reducing these risks. It will also make significant efforts to mobilize and empower local communities in CBEWS and CB-EbA. This will reduce the impact and probability of the risk to the minimum.		
Selected Risk Factor 7		
Category	Probability	Impact
<u>Other</u>	<u>Medium</u>	Select <u>High</u>
Description		
COVID 19		
Mitigation Measure(s)		
The world is currently suffering from the COVID19 pandemic, and B&H is no exception. As a highly transmissible infectious virus, COVID 19 represents a risk whenever mass gatherings, close working conditions or physical contact occurs involving people infected with the virus. 'COVID safe' practices are to be adopted by the project. Compliance with directives of WHO and Ministry of Health must occur and all efforts to reduce the risk of spread will be taken. The project itself does not represent a significant		

risk, but given the high transmissibility of COVID, any activity that brings people together needs to be treated as risky and appropriate precautions taken.

Selected Risk Factor 8

Category	Probability	Impact
Governance	Medium	<u>Medium</u>
Description		
The complex institutional structure and division of authorities and responsibilities between the state government, the two entities in B&H, the line ministries of both entities, and the municipal authorities and the different management arrangements for FRM according to the category of river and type, might present implementation and coordination challenges to the targeted project activities.		
Mitigation Measure(s)		
Close collaboration with relevant stakeholders from the outset and by determining collaborative strategies and focal points in each of the key institutions for the Project Board. The project implementation team and UNDP will use the relevant experience from the previous projects and will rely on the Stakeholder Engagement Plan to make sure that the institutional barriers are tackled timely and efficiently. The project activities (1.3,1.4,2.4) will improve institutional capacities, information and coordination for FRM thus reducing institutional vulnerability and reducing the risk.		

Selected Risk Factor 9

Category	Probability	Impact
Sanctions	Low	<u>Medium</u>
Description		
Project engage with entities or individuals listed on UN sanctions		
Mitigation Measure(s)		
This project is implemented by UNDP under the Direct Implementation Modality (DIM) and will be guided by UNDP's policy framework. UNDP, by virtue of being an organ of the United Nations and therefore being part of the same entity, is also beholden to the principles on countering the financing of terrorism. UNDP systematically screens all entities being considered to be contracted by or partner with UNDP against the Consolidated UN Security Council Sanctions List (https://www.un.org/securitycouncil/content/un-sc-consolidated-list). UNDP Country Office in B&H will be responsible for screening all entities or individuals against the Consolidated UN Security Council Sanctions List before engaging with them for the project.		

Selected Risk Factor 10

Category	Probability	Impact
ML/FT	Low	<u>Medium</u>
Description		
Money laundering and terrorist financing risks		
Mitigation Measure(s)		
This project is implemented by UNDP under the Direct Implementation Modality (DIM) and will be guided by UNDP's policy framework. UNDP, by virtue of being an organ of the United Nations and therefore being part of the same entity, is also beholden to the principles on countering the financing of terrorism. UNDP systematically screens all entities being considered to be contracted by or partner with UNDP against the Consolidated UN Security Council Sanctions List. Please refer to the UN Security Council Consolidated Sanctions List available at https://www.un.org/securitycouncil/content/un-sc-consolidated-list for the latest information on sanctions.		

Selected Risk Factor 11

Category	Probability	Impact
Prohibited practices	Low	<u>Medium</u>
Description		
Prohibited Practices Risks		
Mitigation Measure(s)		
UNDP applies a zero tolerance policy in relation to fraud and corruption. The guiding principles of UNDP's commitment to prevent, identify and address all acts of fraud and corruption have been laid out in the UNDP Policy against Fraud and Corrupt Practices (the "FCP Policy"), which applies to all activities and operations of UNDP, including projects and programmes. The fundamental principles of UNDP's FCP Policy serve as the basis for and are integrated in UNDP's policy frameworks in relation to procurement, financial management, internal control and accountability and staff rules and regulations. Under the terms of the FCP Policy and related policies, UNDP has a commitment, when developing a new programme or project, to ensure that risks related to fraud and corrupt practices are fully identified and considered in the programme/project design and processes and that adequate and effective measures to mitigate such risks are put in place. In this light, the processes and requirements for UNDP's partnership capacity assessment (PCAT) require UNDP to carry out a thorough due diligence of the risks in relation to the programme/project it intends to engage in, as well as in relation to the potential partners who will be involved in the implementation of such programmes/projects. Such capacity assessments include, among other things, a screening of all entities being considered to be contracted by or partner with UNDP against the Consolidated UN Security Council Sanctions List (to which UNDP, by virtue of being an organ of the United Nations and therefore being part of the same entity, is also beholden), an assessment of any history of fraud, corruption or other fraudulent practices and/or potential conflicts of interest. Based on this capacity assessment, UNDP then builds an internal control component into the design of the project and ensures that the financial management of the programme/project contains adequate safeguards to prevent, monitor and address any risks and acts of fraud, corruption or AML/CFT that may be identified.		

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

236. The project was screened against UNDP's Social and Environmental Standards using the UNDP Social and Environmental Screening Procedure (SESP), to identify any potential social and environmental risks that could arise from project activities. An Environmental and Social Assessment Report (ESAR) was prepared to assess the potential risks in more detail (see Annex 6). The ESAR has been prepared using publicly available information, project reference documents, consultation with stakeholders such as government, NGOs and project development team members, and field-visits by a safeguard specialist to representative sites.
237. The risk profile of the project has been determined to be moderate (Category B). The risks are considered acceptable and manageable through the application of mitigation measures. The ESAR includes an Environmental and Social Management Plan (ESMP), which provides an outline of the types of mitigation measures that are likely to be required when implementing the project.
238. Through delivery of the project's inter-related outputs, communities in B&H will be better equipped to adapt to climate impacts, particularly flooding. The project will deliver sustainable development co-benefits including enhanced ecosystem services, gender responsive livelihood enhancement and job creation, and improved resilience of communities' physical assets.

Environment

239. Physical impacts will be primarily associated with construction activities for the river works, and access to land for both river works and forestry activities, particularly the implementation of the priority risk reduction interventions at the sites proposed under Activity 3.3. Construction impacts will be temporary in nature, such as localised erosion/sedimentation, noise and dust from construction equipment, and possible changes to people's ability to move within the locale during construction. Permanent infrastructure will be designed to minimise long-term impacts. The implementation of the ESMP will ensure that these impacts are satisfactorily managed.
240. The majority of the construction works involve building river infrastructure. It is anticipated that most materials, particularly rock, will come from existing quarries near the sites. Other material requirements will be minimal. The proposed river works will not impact on critical ecosystems and/or habitats. The rivers have been significantly impacted by both anthropogenic activities and climate induced events in the past. Some riparian vegetation will be lost at some of the sites.
241. The project is addressing climate-induced flood risks by introducing FRM and CCA measures. In general, the project is providing key environmental protection benefits through the introduction of disaster risk reduction, climate change adaptation and protection of people, property and the environment from major hazards. In the long-run the project will bring about significant environmental benefits by increasing the country's resilience to climate-induced flood disasters and thus, enabling its population to better protect national assets, including environmental assets (land, forest and land resources).
242. Environmental co-benefits mainly relate to EbA strategies such as riparian plantings and agro-forestry which will provide water retention and functions; regulation of hydrological flows (buffer runoff, soil infiltration, groundwater recharge, maintenance of base flows); natural hazard mitigation (e.g. flood prevention, peak flow reduction, soil erosion and landslide control); increased streambed stabilization resulting in decreased erosion, habitat preservation, and reforestation. Environmental impacts associated with the project are considered acceptable due to the nature of the interventions and the ability to manage impacts through implementation of mitigation measures, while social benefits are significant and can provide long-term improvements to the lives of communities in the target areas (see Section D).

Social

243. The project is expected to directly benefit approximately 26% of the population. Project interventions will be undertaken in areas of B&H where communities are particularly vulnerable to the impacts of climate change. The project is targeting those most vulnerable to flood risk and the impact potential of the enhanced flood risk management and intervention measures is high and will safeguard vulnerable communities and their economic assets from these climate induced disasters. Government land will be used for structural interventions, so the project does not require any land acquisition and/or resettlement.
244. The project will also improve the capacity of government and communities to plan, design and manage rural infrastructure that is climate resilient. By enhancement of the existing legislative and regulatory FRM framework in order to mainstream the climate-induced flood risk management into sectoral planning for agriculture, hydropower, critical infrastructure and spatial planning, the project will directly protect and save lives, increase food production safety, and flood risk informed local development and land use.

245. The approach applied within the project will underpin the floodplain planning and development control to achieve basin-wide climate resilience to increasing flood risks and will result in a paradigm shift from uncontrolled floodplain development to climate-risk informed and controlled floodplain usage.
246. It is anticipated that based on the communities' needs, the projects will be fully accepted. None the less, the project has developed a Grievance Redress Mechanism (GRM) to deal with any complaints and issues that may arise because of the project. The GRM has been designed to ensure that an individual and/or group are not financially impacted by the process of making a complaint and/or grievance. The GRM complies with Bosnian and UNDP Safeguard procedures. Furthermore, the GRM allows individuals and/or groups to also file a complaint with the Office of Anticorruption and Integrity within the UNDP or utilise the GCF Independent Redress Mechanism should they have any concerns as to corruption, unethical behaviour or where they believe their complaint or grievance has not been adequately addressed.

Gender and social inclusion

Gender

247. The project will directly benefit the most vulnerable parts of the population, particularly women who are disproportionately affected by flooding and disasters in general. In B&H there are significant gender inequalities, fuelled by traditional gender roles. The project has been designed with attention paid to gender and social inclusion. UNDP⁶⁴ has identified that the most vulnerable women in the country are: the women living below the poverty line, single mothers, Roma women, internally displaced women, minority returnees and LGBTI women. The gender analysis undertaken at the project design phase, acts as an entry point for gender mainstreaming throughout implementation, and builds on stakeholder consultation, existing analytical documentation and data from projects currently being implemented, and national statistics where available.
248. The project will ensure that women are primary stakeholders and will therefore need to be involved in decisions on the types of solutions that are implemented. The project will safeguard local communities and their assets from climate disasters with particular attention to women, and other vulnerable groups. The Gender Assessment and Action Plan prepared for the project outlines positive actions that will be taken by the project.
249. The project will ensure that women are empowered to benefit from the structural interventions that will mitigate flood events and will be included in consultations during design to ensure that the structures will not impact negatively on gender equality. With regard to EBA approaches, the project will implement agro-forestry, which will enable and empower women to diversify agricultural activities, thus building women's resilience to cope with climate change impacts.
250. Along with a multi-year climate resilient municipal investment plan, the project will formulate and implement gender sensitive community preparedness plans in selected municipalities (10-12) in Vrbas, Una-Sana and Bosna basins. The preparedness plans will be based on needs assessment of both genders and will comprise of gender sensitive actions including access to knowledge (trainings), information and resources that will increase capability of both genders for timely and effective actions to avoid the loss of lives and to reduce the damages of flooding during the event.
251. UNDP will request that contractors, suppliers, and partners adhere to zero tolerance for sexual exploitation and harassment (SEAH) and commit to taking adequate action if faced with SEAH allegations, in the absence of which, contractual arrangements can be terminated. UNDP has a standalone Code of Ethics which applies to all staff and contractors and defines the ethical standards applicable to all, citing the specific policy references. UNDP is committed to creating and maintaining a working environment that respects the inherent dignity of all persons, affording them the opportunity to reach their fullest potential and empowering them to deliver the best possible results for UNDP and the people we serve. To that effect, UNDP has further implemented policies for protection against Harassment, Sexual Harassment, Discrimination and abuse of Authority, as well as special measures for protection from sexual exploitation and sexual abuse. These policies are further backed by a Policy for Protection against Retaliation against individuals holding UNDP assignments/contracts (i.e., staff members, interns, UN Volunteers and contractors) who have, in good faith, properly reported allegations of misconduct, or who have cooperated with a duly authorized audit or investigation.

Other vulnerable groups

⁶⁴ <https://www.undp.org/bosnia-herzegovina/publications/intersectional-analysis-differential-impact-covid-19-crisis-and-inflation-women-bosnia-and-herzegovina>

252. In Bosnia and Herzegovina, several groups are particularly vulnerable to flood risk. These groups may face higher levels of exposure and have limited capacities to cope with and recover from the impacts of flooding. The main vulnerable groups are:

- *Socioeconomically Disadvantaged Communities:* Socioeconomically disadvantaged communities, including low-income households, marginalized groups, and those living in informal settlements or substandard housing, are often more vulnerable to flood risk. Limited resources, lack of infrastructure, and inadequate access to services make it challenging for these communities to effectively prepare for, respond to, and recover from floods
- *Rural and Agricultural Communities:* Rural communities, particularly those engaged in agriculture and subsistence farming, are highly susceptible to flood risk. Floods can lead to crop losses, damage to livestock, destruction of infrastructure (e.g., irrigation systems), and disruption of livelihoods. The dependence on agricultural activities and limited access to alternative income sources amplify their vulnerability.
- *Elderly Population:* The elderly population faces heightened vulnerability to flood risk due to physical limitations, limited mobility, and underlying health conditions. These factors can hinder their ability to evacuate, seek shelter, or access necessary assistance during and after flooding events. Additionally, the elderly population may face challenges in recovering and rebuilding their lives after a flood
- *Children and Youth:* Children and youth are a vulnerable group during floods due to their dependency on caregivers, increased health risks, and potential disruptions to education and social support networks. Displacement, loss of shelter, separation from family members, and exposure to hazardous conditions can have long-lasting impacts on their well-being, development, and future opportunities
- *Ethnic and Minority Groups:* Ethnic and minority groups, such as Roma communities, particularly those living in remote or marginalized areas, may face disproportionate vulnerability to flood risk. Limited access to resources, infrastructure, and decision-making processes can hinder their capacity to prepare for and respond to floods. Cultural and language barriers may also impede effective communication and coordination during emergencies.
- *People with Disabilities:* Individuals with disabilities may face additional challenges during floods, such as limited mobility, sensory impairments, or reliance on assistive devices. Barriers in physical infrastructure, lack of inclusive emergency plans, and limited access to information and support services can exacerbate their vulnerability and reduce their ability to safely evacuate or access necessary assistance.

Understanding the specific vulnerabilities of gender and other socially and economically excluded groups is crucial for developing inclusive and targeted flood risk management strategies. It is also important to address the intersecting vulnerabilities and promote social equity in building resilience to flood risk in B&H. During project development, a number of these vulnerabilities were included in the socio-economic assessment of flood risk modelling and assessment based on socio-economic and census data. During implementation and as part of the ESIA and ESMP, measures will be tailored to ensure their participation, access to early warning systems, evacuation plans, and post-flood recovery support.

Conclusion

253. Environmental impacts associated with the project are considered acceptable due to the nature of the interventions and the ability to manage impacts through implementation of mitigation measures, while social benefits are significant and can provide long-term improvements to the lives of communities in the target areas.

254. Budgeting for environmental interventions and the application of mitigation measures to enhance positive impacts for the project is an investment in the future as it will reduce the environmental and social liability at local, state and national levels. The end result of this expenditure will be that there will be reduced river sedimentation and erosion; stronger and improved resilience to the impacts of climate change, healthier ecosystems, more knowledgeable communities and overall improvement in the quality of life of the population as an investment in the future of the people of Bosnia and Herzegovina, which if implemented as per the project proposal, will be repaid many times over through reduced long-term operation and maintenance costs of implementing the project.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

255. The project will address specific gender-based vulnerabilities as a key requirement of enhanced climate change adaptation strategies and actions. As a result of the project implementation more lives, property and productive assets will be secured from the impacts of climate induced disasters. While all B&H population will benefit from

the project, the gender gap will decrease as women's benefits will be relatively bigger given greater dependence of their livelihoods on natural resources. During the proposal development individual gender-specific impacts and needs have been examined through the lenses of vulnerability, capacities and voices and this will continue throughout project implementation as detailed in the Gender Action Plan (Annex 8). This approach will ensure support to most vulnerable social groups among which women in B&H are specifically recognized. By enabling equal participation of men and women, the project will ensure that climate change adaptation is more effective and sustainable.

256. Assessment of the human rights within the B&H legal framework in latest decades reveals that significant progress has been made in the application of the principle of the gender equality, which is a result of activities taken by gender institutional mechanisms but also due to civil society activism. However, the Gender Assessment and Gender Action Plan (Annex 8) outline several social phenomena that are increasing gender-based vulnerability especially in conditions of climate induced disastrous floods in B&H.
257. The vulnerability of women in B&H is caused by social and economic reasons which are deeply rooted in traditional norms of the society. Representation of women in legislative and executive authorities as well as in managerial positions, at all levels, remains unsatisfactory. Hence there is reduced participation of women in decision making including economic, environmental and social issues. Due to gender stereotypes women with secondary and university education are also more likely to be unemployed than men with the same education. Out of total number of employees only 37% are women. One working hour of women in B&H is 10% less paid than men. Also, great prevalence of traditional socio-cultural gender norms supports male ownership. Men account for over 70 percent of landownership and own 74 percent of dwellings. Women have lower financial inclusion by almost 10% compared to man. Social gender stereotypes that restrict women's involvement outside of household and family are even more deeply rooted in rural areas.
258. The existing gender inequality factors (e.g. limited engagement of women in planning and decision making) and traditional distribution of gender roles in families and communities call for tailoring and targeting of the project solutions to outreach beneficiaries of both genders equally. In order to address these vulnerabilities and more effectively involve women in FRM processes the project has dedicated special attention to gender issues under activities: 1.2, 1.3, 1.4, 2.4, 3.1, 3.2. The project will assess gender-specific vulnerabilities to flood risks through gender responsive socio-economic data collection, ensure equal participation and decision making as well as information access and warning dissemination services. Further, the project will ensure women are involved in the processes of capacity building for climate induced hazard risk assessment within the FRM at all levels, but also in the development of public/private collaboration on building resilience by climate risk finance mechanisms.
259. At Project inception, additional gender analysis will be undertaken to ensure that the baseline data set is sex- and age-disaggregated and adequate for assessment of the gender impacts of the Project. Progress reports will provide periodic updates on the effect of the Project on women, and regular UNDP review missions and the midterm review mission will review and monitor the gender impacts. The Gender Action Plan includes a number of recommendations aimed at ensuring that the project:
- narrows gender inequality.
 - addresses the needs and constraints of women, girls, men, and boys.
 - avoid any risks of adverse gender impacts;
 - ensure women's participation, promotes their leadership qualities; and
 - ensure women are included as planners, co-implementers and agents of change.

As part of the implementation of the Gender Action Plan the project will:

- (i) Carry out gender responsive vulnerability assessment informed by gender analysis: When carrying out group discussions and in-depth interviews with key stakeholders, the project will ensure an equal representation of women, elderly people, disabled or family members of disabled, IDPs, community leaders and government officials.
- (ii) Ensure collection of sex- and age-disaggregated data for project indicators: Gender responsive and sex-disaggregated reporting will be further introduced beyond the project level at the national level through the multi-hazard disaster risk information and knowledge system to be designed by the project.
- (iii) Enhance gender considerations in disaster preparedness activities based on gender analysis: Men and women of different age groups, as well as those from vulnerable communities will be involved in planning and implementation. Community consultancy groups with at least 30 percent representation of less represented sex will be established. The project will ensure that information on hazards, their character, probability of occurrence, threats to life, possible impact on livelihood, houses, crops and livestock and on protecting

measures is reaching both women and men through appropriately tailored channels. Gender Mechanisms representative will be engaged in disaster preparedness activities.

- (iv) Warrant universal access to disaster warnings: The project will tailor warning to the gender-differentiated needs and capabilities of specific population groups, such as children, senior citizens, IDPs, the sick and persons with disabilities.
- (v) Mainstream gender considerations and engagement of women in resilience building activities targeting 30 percent representation of less represented sex in capacity building and training activities supported by the project. The project will ensure that women have equal access to information and livelihood generating activities facilitated by the project. The project will strengthen social capital through creating community groups for planning for, safeguarding from and mitigating adverse effects of natural disasters with at least 30 percent less represented sex participation.
- (vi) Conduct gender mainstreaming trainings and build capacity of national stakeholders targeted by project and responsible for various disaster risk reduction functions
- (vii) Engage women in decision-making: the project will ensure at least 30 percent representation and active participation of less represented sex in project stakeholder consultations, local and national decision-making bodies established and/or facilitated by the project. The project will secure participation of the Gender Institutional Mechanism representatives in the consultations and decision-making.
- (viii) Project staff: The project will ensure that men and women are adequately represented in the project implementation team, staff members will be trained on gender mainstreaming and on social vulnerability approach. The Project team leaders will ensure that all specialists are briefed on their responsibilities in relation to the Gender Action Plan upon mobilization.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

260. The financial management and procurement of this project will follow UNDP financial rules and regulations available here: https://info.undp.org/global/documents/frm/Financial-Rules-and-Regulations_E.pdf. Further guidance is outlined in the financial resources management section of the UNDP Programme and Operations Policies and Procedures available at <https://info.undp.org/global/popp/frm/Pages/introduction.aspx>.

261. UNDP has comprehensive procurement policies in place as outlined in the 'Contracts and Procurement' section of UNDP's Programme and Operations Policies and Procedures (POPP). The policies outline formal procurement standards and guidelines across each phase of the procurement process, and they apply to all procurements in UNDP. See here: <https://info.undp.org/global/popp/cap/Pages/Introduction.aspx>. Procurement of services and goods will be done in a cost effective and reliable way and by applying following principles: **Best Value for Money**, which consists of the selection of the offer that best meets the end-users' needs and that presents the best return on investment; **Fairness, Integrity and Transparency**, which ensures that competitive processes are fair, open, and rules-based. All potential vendors will be treated equally, and the process will feature clear evaluation criteria, unambiguous solicitation instructions, realistic requirements, and rules and procedures that are easy to understand; **Effective International Competition**, understood as giving all potential vendors timely and adequate information on UNDP requirements, as well as equal opportunity to participate in procurement actions; and **In the best interest of UNDP**, which means that any business transactions must conform to the mandates and principles of UNDP and the United Nations. As outlined in the Procurement Plan, procurement of goods and services will be carried out by UNDP and the UNDP procurement procedures will be applied (POPP). Additional details on the UNDP procurement thresholds and methods are provided in the Annex 10. Procurement Plan

262. The project will be implemented following the Direct Implementation Modality (DIM) following DIM guidelines.

263. The project will be audited following the UNDP financial rules and regulations noted above and applicable audit guidelines and policies. According to the current audit policies, UNDP will be appointing the auditors. In UNDP scheduled audits are performed annually, during the programme cycle as per UNDP assurance/audit plans, on the basis of the implementing partner's risk rating and UNDP's guidelines. The project financial statements will comply with international accounting standards. A scheduled audit is used to determine whether the funds transferred to the implementing partner were used for the appropriate purpose and in accordance with the work plan. A scheduled audit can consist of a financial audit or an internal control audit.

264. **Financial Sustainability.** The government's co-finance covers O&M costs during the 6 years of the project's implementation duration, and the government and HPP partner are committed to cover O&M costs of equipment

and flood defences installed during the project over the 20 years of investment lifetime, thus ensuring continued operation and financial sustainability of the project's proposed investments. Details of the financial analysis are presented in Annex XII (b). In addition, the project is developing risk-based investment planning frameworks at the municipal and contonal levels, climate risk financing and risk transfer mechanisms. These interventions will help to identify and plan the financing that is needed to address climate risks to communities, develop and implement long-term public and private sector climate financing, as well as instruments for transferring risk and building financial resilience of households and key sector players. All of these interventions will contribute to the financial sustainability of the project.

G.4. Disclosure of funding proposal

Indicate below whether or not the funding proposal includes confidential information.

- ☒ **No confidential information:** The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.
- ☐ **With confidential information:** The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:
- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
 - redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

- ☒ Annex 1 NDA no-objection letter(s) [\(template provided\)](#)
- ☒ Annex 2 Feasibility study - and a market study, if applicable
- ☒ Annex 3 Economic and/or financial analyses in spreadsheet format
- ☒ Annex 4 Detailed budget plan [\(template provided\)](#)
- ☒ Annex 5 Implementation timetable including key project/programme milestones [\(template provided\)](#)
- ☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
[\(ESS disclosure form provided\)](#)
 - ☐ Environmental and Social Impact Assessment (ESIA)
 - ☒ Environmental and Social Management Plan (ESMP/Annex 6b) or
 - ☐ Environmental and Social Management System (ESMS)
 - ☒ Others: SESP(Annex 6a)
- ☒ Annex 7 Summary of consultations and stakeholder engagement plan
- ☒ Annex 8 Gender assessment and project/programme-level action plan [\(template provided\)](#)
- ☒ Annex 9 Legal due diligence (regulation, taxation and insurance)
- ☒ Annex 10 Procurement plan [\(template provided\)](#)
- ☒ Annex 11 Monitoring and evaluation plan [\(template provided\)](#)
- ☒ Annex 12 AE fee request [\(template provided\)](#)
- ☒ Annex 13 Co-financing commitment letter, if applicable [\(template provided\)](#)
- ☒ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

- ☒ Annex 15 Evidence of internal approval [\(template provided\)](#)
- ☒ Annex 16 Map(s) indicating the location of proposed interventions
- ☐ Annex 17 Multi-country project/programme information [\(template provided\)](#)
- ☒ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
- ☐ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☐ Annex 20 First level AML/CFT (KYC) assessment
- ☒ Annex 21 Operations manual (Operations and maintenance)
- ☒ Annex 22 Beneficiary Estimation Methodology
- ☒ UNDP Responses to GCF comments on FP

* Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.

No-objection letter issued by the national designated authority(ies) or focal point(s)

**BOSNA I HERCEGOVINA
MINISTARSTVO VANJSKE TRGOVINE
I EKONOMSKIH ODNOSA**



**БОСНА И ХЕРЦЕГОВИНА
МИНИСТАРСТВО СПОЉНЕ
ТРГОВИНЕ И ЕКОНОМСКИХ ОДНОСА**

**BOSNIA AND HERZEGOVINA
MINISTRY OF FOREIGN TRADE AND
ECONOMIC RELATIONS**

Number: **01-SV-2021/23**
Sarajevo, 14 June 2023

NO OBJECTION LETTER

To: The Green Climate Fund ("GCF")

Re: Funding proposal for the GCF by UNDP regarding "Scaling-up Climate Resilient Flood Risk Management in Bosnia and Herzegovina" project

Dear Madam, Sir,

We refer to the project titled "Scaling-up Climate Resilient Flood Risk Management in Bosnia and Herzegovina" in Bosnia and Herzegovina as included in the funding proposal submitted by UNDP to us on 26 April 2023.

The undersigned is the duly authorized representative of the Ministry of Foreign Trade and Economic Relations of Bosnia and Herzegovina, the NDA/Focal Point of Bosnia and Herzegovina.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The governments in Bosnia and Herzegovina have no-objection to the project as included in the funding proposal;
- (b) The project as included in the funding proposal is in conformity with the national priorities, strategies and plans in Bosnia and Herzegovina;
- (c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Name: **Stasa Košarac**

Title: **Minister of Foreign Trade and Economic Relations of Bosnia and Herzegovina, the GCF National Designated Authority/Focal Point of Bosnia and Herzegovina.**

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Scaling up climate resilient flood risk management in Bosnia and Herzegovina
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Public
Accredited entity	United Nations Development Programme (UNDP)
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	Bosnia and Herzegovina Flood risk reduction activities will be implemented in: Neretva river basin, Čapljina municipality; Bosna river basin, municipalities Gračanica, Maglaj, Doboj, Modriča, Lukavac; Sana river Basin, Prijedor municipality; Jošavka river basin, Čelinac municipality; Vrbaš river basin, City of Banja Luka; Gradiška municipality; and Drina river basin, City of Bijeljina.
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Friday, September 8, 2023
Language(s) of disclosure	English, Bosnian, Croatian, and Serbian
Explanation on language	The official languages of Bosnia and Herzegovina are Bosnian, Croatian, and Serbian. Bosnian and Croatian is the same language, which is called Croatian in the areas with Croat majority and Bosnian within Bosniak majority. The Bosnian and Croatian language uses Latin alphabet. Serbian language uses the Cyrillic alphabet. ESMF has been published in all official national languages as well as in the English language.
Link to disclosure	English: https://www.undp.org/sites/g/files/zskgke326/files/2023-09/scaling_up_climate_resilient_flood_risk_management_bih_esar_and_esmf.pdf Bosnian: https://www.undp.org/sites/g/files/zskgke326/files/2023-09/izvjestaj_o_procjeni_utjecaja_na_okolis_i_drustvo_bos.pdf Croatian: https://www.undp.org/sites/g/files/zskgke326/files/2023-09/izvjestaj_o_procjeni_utjecaja_na_okolis_i_drustvo_cro.pdf Serbian: https://www.undp.org/sites/g/files/zskgke326/files/2023-09/proshirene_upravlana_rizikom_od_poplava_otpornog_na_klimu_u_bosni_i_khercegovini.pdf
Other link(s)	UNFCCC site in Bosnia and Herzegovina:

	English: http://www.unfccc.ba/en/site/upload/PDF_dokumenti/Scaling_up_climate_resilient_flood_risk_management_BIH-eng.pdf Bosnian: http://www.unfccc.ba/site/upload/PDF_dokumenti/Scaling_up_climate_resilient_flood_risk_management_BIH_FP_complete.pdf Croatian: http://www.unfccc.ba/site/upload/PDF_dokumenti/Scaling_up_climate_resilient_flood_risk_management-CRO.pdf Serbian: http://www.unfccc.ba/site/upload/PDF_dokumenti/Scaling_up_climate_resilient_flood_risk_management_BIH-cir.pdf Water Agency "Public institution Vode Srpske": http://www.voders.org/dokumentacija/izvjestaj_o_procjeni_utjecaja.pdf Проширење управљања ризиком од поплава 08092023.pdf (voders.org) Republic Hydrometeo institute: https://rhmzrs.com/project/prosirenje-upravljanja-rizikom-od-poplava-otpornog-na-klimu-u-bosni-i-hercegovini/
Remarks	An ESIA consistent with the requirements for a Category B project is contained in the "Environmental and Social Assessment Report and Environmental and Social Management Framework".
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Friday, September 8, 2023
Language(s) of disclosure	English, Bosnian, Croatian, and Serbian
Explanation on language	The official languages of Bosnia and Herzegovina are Bosnian, Croatian, and Serbian. Bosnian and Croatian is the same language, which is called Croatian in the areas with Croat majority and Bosnian with Bosniak majority. The Bosnian and Croatian language uses Latin alphabet. Serbian language uses the Cyrillic alphabet. ESMF has been published in all official national languages as well as in the English language.
Link to disclosure	English: https://www.undp.org/sites/g/files/zskgke326/files/2023-09/scaling_up_climate_resilient_flood_risk_management_bih_esar_and_esmf.pdf Bosnian: https://www.undp.org/sites/g/files/zskgke326/files/2023-09/izvjestaj_o_procjeni_utjecaja_na_okolis_i_drustvo_bos.pdf Croatian: https://www.undp.org/sites/g/files/zskgke326/files/2023-09/izvjestaj_o_procjeni_utjecaja_na_okolis_i_drustvo_cro.pdf Serbian:

	https://www.undp.org/sites/g/files/zskgke326/files/2023-09/proshirene upravlana rizikom od poplava otpornog na klimu u bosni i hercegovini.pdf
Other link(s)	UNFCCC site in Bosnia and Herzegovina: English: http://www.unfccc.ba/en/site/upload/PDF_dokumenti/Scaling_up_climate_resilient_flood_risk_management BIH-eng.pdf Bosnian: http://www.unfccc.ba/site/upload/PDF_dokumenti/Scaling_up_climate_resilient_flood_risk_management BIH FP complete.pdf Croatian: http://www.unfccc.ba/site/upload/PDF_dokumenti/Scaling_up_climate_resilient_flood_risk_management-CRO.pdf Serbian: http://www.unfccc.ba/site/upload/PDF_dokumenti/Scaling_up_climate_resilient_flood_risk_management BIH-cir.pdf Water Agency "Public institution Vode Srpske": http://www.voders.org/dokumentacija/izvjestaj_o_procjeni_utjecaja.pdf Проширење управљања ризиком од поплава 08092023.pdf (voders.org) Republic Hydrometeo institute: https://rhmzrs.com/project/prosirenje-upravljanja-rizikom-od-poplava-otpornog-na-klimu-u-bosni-i-hercegovini/
Remarks	An ESMP consistent with the requirements for a Category B project is contained in the "Environmental and Social Assessment Report and Environmental and Social Management Framework".
Environmental and Social Management (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), Indigenous Peoples Planning Framework (IPPF) (if applicable)	
Description of report/disclosure on accredited entity's website	Social and Environmental Safeguards Procedure (SESP) Tuesday, September 12, 2023
Language(s) of disclosure	English, Bosnian, Croatian, Serbian
Explanation on language	The official languages of Bosnia and Herzegovina are Bosnian, Croatian, and Serbian. Bosnian and Croatian is the same language, which is called Croatian in the areas with Croat majority and Bosnian with Bosniak majority. The Bosnian and

	Croatian language uses Latin alphabet. Serbian language uses the Cyrillic alphabet. We have, however added documents indicating a language, not a letter. SESP has been published in all official national languages as well as in the English language.
Link to disclosure	English: https://www.undp.org/sites/g/files/zskgke326/files/2023-09/social_and_environmental_screening_template.pdf Bosnian: https://www.undp.org/sites/g/files/zskgke326/files/2023-09/obrazac_za_socijalni_i_ekoloski_pregled_bos.pdf Croatian: https://www.undp.org/sites/g/files/zskgke326/files/2023-09/obrazac_za_socijalni_i_ekoloski_pregled_cro.pdf Serbian: https://www.undp.org/sites/g/files/zskgke326/files/2023-09/obrazac_za_socijalni_i_ekoloski_pregled.pdf
Other link(s)	UNFCCC site in Bosnia and Herzegovina: English: http://www.unfccc.ba/en/site/upload/PDF_dokumenti/FP-UNDP-20220421-6360-Annex%20a%20SESP_20230522_clean.pdf Bosnian: http://www.unfccc.ba/site/upload/PDF_dokumenti/FP-UNDP-20220421-6360-Annex%20a%20SESP_20230522_clean_LAT.pdf Croatian: http://www.unfccc.ba/site/upload/PDF_dokumenti/FP-UNDP-20220421-6360-Annex%20a%20SESP_20230522_clean_CRO.pdf Serbian: http://www.unfccc.ba/site/upload/PDF_dokumenti/FP-UNDP-20220421-6360-Annex%20a%20SESP_20230522_clean_CIR.pdf
Remarks	The SESP is a UNDP document that is used for identifying social and environmental risks and mitigation measures. It does not replace or conflict with GCF ESS requirements and policies.
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Tuesday, September 12, 2023
Place	Physical copies of the ESMF are available to the public at regional sub-offices of the Public Institution "Vode Srpske" in Bosna, Sana, Vrbas, Drina and Trebisnjica river basins at their local offices as follows: - Bijeljina: Miloša Obilića 51, 76300 Bijeljina - Doboj: Vojvode Mišića 22, 74000 Doboj, - Prijedor: Aleja Kozarskog odreda 4, 79101 Prijedor

	- Banja Luka: Milana Radmana 10, 78000 Banja Luka - Trebinje: Srpska 2, 89101 Trebinje - Zvornik: Svetog Save bb, 75400 Zvornik In the Federation of Bosnia and Herzegovina, physical copies of the ESMF are available -in the Water Agency for Sava River Basin (Hamdije Čemerlića 39a, Sarajevo 71000) and -Water Agency for Adriatic River Basin (Dr. Ante Starčevića bb. 88000 Mostar).
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, October 23, 2023

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP216

Proposal name:	Scaling up climate resilient flood risk management in Bosnia and Herzegovina
Accredited entity:	United Nations Development Programme
Country/(ies):	Bosnia and Herzegovina
Project/programme size:	Medium

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
The direct benefits will be extended to 799,445 people, accounting for 25 per cent of the total population, with a focus on the most vulnerable population.	Political instability could pose a risk that requires careful management.
Scaling up successful innovative flood risk management (FRM) technologies and implementing a nationwide, impact-based Flood Forecasting and Early Warning System (FFEWS), which will lead to improved warning times and reduced losses.	The project's strong inclination towards scaling up and replicating proposed innovations and knowledge products beyond the project areas will require great attention to the technical soundness and flexibility of, in particular, innovative FRM technologies and FFEWS to incorporate the needs and vulnerability aspects of communities and institutions outside the scope of the project.
Engaging the private sector in climate information services and FRM activities and facilitating the increased use of climate risk information in its operations.	
GCF investment of USD 14.4 million will leverage an additional USD 58.29 million in co-financing from both public and private sectors. The government of Bosnia and Herzegovina will contribute USD 30.74 million in co-financing.	

2. The Board may wish to consider approving this funding proposal in accordance with the terms and conditions listed in the respective term sheet and addendum XXI, titled "List of proposed conditions and recommendations".

II. Summary of the Secretariat's assessment

2.1 Project background

3. Bosnia and Herzegovina, identified as a middle-income country, has an estimated population of 3.531 million and spans an area of 51,209.2 km². During the 1992–1995 conflict, the country underwent severe destruction. In subsequent years, it has been navigating a complicated landscape of post-war rebuilding, along with economic and social revitalization. The conflict significantly compromised the country's flood risk management capabilities, inflicting direct harm on flood control infrastructure and hydrological monitoring stations, while also creating hazardous minefields around these facilities. Adding to these difficulties is the country's gradual economic transition towards a market-based system, demanding enhanced governance and structural changes.

4. Around a quarter of the population of Bosnia and Herzegovina faces a high risk of flooding exacerbated by climate change, which has led to increased rainfall intensity and unpredictability, resulting in more frequent and severe flooding events that impact lives, property and essential infrastructure. Key economic sectors such as agriculture, water management and hydropower are especially vulnerable. Between 2006 and 2012, flooding accounted for 62 per cent of all natural disaster-related damages across the countries. This was followed by a significant flood event in 2010 that led to damages worth USD 200 million, or approximately 1 per cent of gross domestic product. An even more catastrophic flooding event occurred in May 2014, the worst in 150 years, affecting a quarter of the nation's land and causing a staggering USD 2.7 billion in damages, equivalent to 15 per cent of Bosnia and Herzegovina's gross domestic product. These floods have disproportionately affected vulnerable segments of the population, including the rural poor, war returnees and displaced persons, who often live in flood-prone areas.

5. The objective of the project is to seek to reduce the growing vulnerabilities of communities and livelihoods in Bosnia and Herzegovina to heightened risks of climate-triggered floods. The project is designed to establish an integrated and innovative, gender-responsive, climate-resilient approach to FRM. It aims to fortify the institutional, technical and financial frameworks for the sustained implementation of comprehensive FRM strategies, which include a blend of structural interventions, non-structural measures as well as ecosystem-based approaches. Additionally, the project will focus on improving the utilization of climate information, flood forecasting, early warning and emergency response systems to heighten the resilience and adaptive capacities of communities at risk. This initiative is aligned with the United Nations multi-partner initiative Early Warnings for All (EW4All) programme, launched in March 2022, and aims to contribute to the execution of the EWS4All Executive Action Plan, developed in March 2023. The project outlines two outcomes and three outputs:

- (a) **Outcome 1:** Climate-informed gender-responsive impact-based FFEWS and increased generation and use of climate data reduce vulnerability to flood-related disasters;
 - (i) **Output 1:** Fully integrated impact-based FFEWS facilitates timely preparation and response;
- (b) **Outcome 2:** Vulnerable communities in Bosnia and Herzegovina including women, children, persons with disabilities and older persons are better protected from and less exposed to climate-induced flood risks through a combination of structural and non-structural flood risk reduction solutions;
 - (i) **Output 2:** Non-structural flood risk reduction measures and nature-based solutions mainstreamed in sectoral policies and plans and effectively contribute to the protection of people and livelihoods from climate-induced flood risk; and
 - (ii) **Output 3:** Climate-proof flood protection measures are scaled-up through new and improved national and local investment frameworks increasing the resilience of the most vulnerable groups to climate-induced flooding.

6. Aimed at impacting a significant segment of the population, the proposed project targets 799,445 direct beneficiaries by the end of the project, representing 25 per cent of the total population. It will prevent loss of life, property, critical infrastructure and economic assets, and will build resilience to flooding in the most vulnerable communities.

7. The total project financing amounts to USD 72.69 million, with a request for GCF grants of USD 14.4 million. The remaining sum of USD 58.23 million will come from a diverse range of co-financing partners, which include the Government of Bosnia and Herzegovina and UNDP.

2.2 Component-by-component analysis

8. The project objective will be achieved through three complementary outputs that comprise its three components.

Component 1: Climate-informed impact-based FFEWS and an increased generation and use of climate data to reduce vulnerability to flood-related disasters (total cost: USD 14.65 million; GCF cost: USD 5.095 million)

9. The project aims to enhance the hydrometeorological monitoring network by covering a broader array of hydrometric variables impacted by climate change. Special focus will be on placed areas with the highest risk, such as Category I rivers and torrents. Technical support will be offered to integrate hydro power plants' hydrometric networks into the national grid. A sustainable operation and maintenance (O&M) system will also be set up, financed by both the public and private sectors.

10. This component will refine existing hazard and risk models to include a more comprehensive approach. Activities will include extending models to cover whole basins, including category II tributaries, and improving hydrological modelling. The goal is to create an effective reservoir operation strategy that balances flood risk, dam safety and power generation. The project will incorporate hydro power plant operations into flood hazard and risk modelling in key basins and will produce specialized climate risk information for the hydropower sector, especially focusing on the Neretva and Trebisnjica River Sub-basins.

11. Building on existing FFEWS, this project will advance information and communications technology systems to support real-time forecasting and warning. The new FFEWS will cover hydro-energy intensive river basins such as Neretva and Trebisnjica, and update existing models for the Una-Sana River Basin. Further enhancements will include advanced meteorological forecasting capabilities based on numerical weather prediction models.

12. To ensure the sustainability of these initiatives, the project will deliver extensive training and capacity-building programmes. It will streamline communication between different agencies to avoid duplication and inefficiencies and establish standard operating procedures (SOPs), communication protocols and codes of conduct, including "Last-mile" communication protocols, protocols and SOPs defining the forecast thresholds for forecast-based financing. The project will also develop operational maintenance of the hydrometric network. It will assess and further develop specific data-sharing/communication protocols for impact-based FFEWS and climate-resilient FRM in coordination with the national adaptation plan project and will implement these protocols across relevant institutions.

Component 2: Non-structural flood risk reduction measures and nature-based solutions mainstreamed within sectoral policies and plans and effectively contribute to protection of people and livelihoods from climate-induced flood risk (total cost: USD 27.67 million; GCF cost: USD 6.74 million)

13. The second component of the project is focused on elevating key FRM policies. This will be accomplished by enhancing existing legal and regulatory frameworks to incorporate climate-responsive FRM across various sectors, such as agriculture, forestry, hydropower and spatial

planning. At the state level, a holistic, cross-entity strategy will be adopted, particularly for shared basins. This integrated approach necessitates the participation of key stakeholders, improved coordination, and the ratification of relevant regional and multilateral environmental agreements. The project will also revise existing regulations to accommodate the challenges posed by climate-induced floods. Through these activities, local governments will be enabled to formulate evidence-based, climate-resilient policies and strategies, grounded in comprehensive impact assessments and cost-benefit analyses.

14. This project component will incorporate community-based ecosystem-based adaptation (EbA) solutions aimed at mitigating the impacts of severe flooding, particularly in the Vrbas River Basin. The project will expand upon the successful EbA methods previously piloted by the UNDP/Special Climate Change Fund Vrbas River initiative. The Vrbas River Basin, unique in Bosnia and Herzegovina for its existing torrential flood hazard maps and evaluated cost-effective EbA strategies, will be prioritized for the implementation of high-impact ecological measures. Additionally, one key issue contributing to sediment build-up is the existing network of forest roads, initially carved out during wartime for illegal logging. Though current legislation largely prevents the expansion of such roads for timber extraction, their initial construction has led to significant material run-off, affecting river networks usually the mainstream of the River Vrbas.

15. The project aims to establish a comprehensive framework in Bosnia and Herzegovina for applying climate-resilient EbA approaches to flood-risk management. It will adapt well-tested flood management strategies from other countries and compile best practices. These insights will be consolidated into countrywide guidelines, technical specifications, standards and protocols, and will be incorporated into national policies. The project will also strengthen institutional capacities in identifying and implementing these EbA approaches to flood risk management measures.

16. The project will support the Government of Bosnia and Herzegovina in transitioning from ad-hoc project-based awareness and education to a more systematic and sustainable approach to flood and disaster risk education. A long-term awareness-raising campaign will be developed and executed. This will include nationwide training programmes focused on flood risk management, targeting practitioners, decision makers, local communities and emergency responders. The project will also formulate a sustainable capacity development plan, enhancing training in FRM and climate risk information within existing educational institutions. This strategy will include a Train-the-Trainers programme, foster key partnerships with academic institutions, and establish mechanisms for long-term capacity-building, building on the groundwork laid by the initial Vrbas project.

Component 3: Climate-proof flood protection measures scaled-up through new and improved national and local investment frameworks increasing resilience of the most vulnerable groups to climate-induced flooding (total cost: USD 26.52 million; GCF cost: USD 1.88 million)

17. Component 3 aims to bolster the resilience of community livelihoods by amplifying flood protection measures that are resilient to climate change. This will be realized by formulating an Investment Framework designed specifically to reduce and manage climate-induced flood risks, which will involve both public sector and private sector participation in climate risk financing and risk transfer mechanisms. The planned activities will establish these financing and risk transfer mechanisms after conducting thorough assessments of socioeconomic risks, damages and losses. These initiatives will be largely co-financed by the national insurance sector.

18. Building on the climate risk-sensitive budgeting mechanisms developed at the municipal level under the national adaptation plan project, this component will explore avenues for integrating such budgeting practices into Bosnia and Herzegovina's existing public financial management framework. This will leverage current strengths while addressing any bottlenecks

in the budgeting process that obstruct climate risk-sensitive budgeting for flood risk management. Key tools for this effort will include damage and loss modelling to quantify the existing cost of disasters, thereby informing resource allocation within the FRM framework. The component will also introduce value-for-money analysis, utilizing cost-benefit data to underscore the advantages (and higher value-for-money) of greater spending on FRM, preparedness and mitigation.

19. This component will orchestrate a comprehensive engagement programme involving state institutions, insurance and reinsurance firms, and financial and capital market entities. The goal is to forge pivotal public-private partnerships aimed at augmenting and scaling up investments in climate-resilient initiatives. Legislative and governance interventions will be executed to reinforce financial institutions, enabling them to offer climate-responsive financing options and products within existing markets, thereby ensuring the long-term viability of the investments with the potential for future scaling. Additionally, financial institutions, including microfinance entities, will receive specialized training and technical guidance for the development of lending products. This will allow them to integrate climate risk data into their lending processes, thereby optimizing performance and reducing risks. Furthermore, loans with favourable terms will be provided to climate-responsive enterprises in Bosnia and Herzegovina, improving upon current market conditions in aspects such as interest rates, loan periods and credit criteria.

Project management (total cost: USD 3.85 million; GCF cost: USD 0.68 million)

20. The portion of the project management cost covered by GCF constitutes less than 5 per cent of the total GCF funding requested and is in compliance with GCF's policy on fees.

III. Assessment of performance against investment criteria

3.1 Impact potential

Scale: High

21. The initiative aims to mitigate flood risks for about 799,445 people, or 25 per cent of the nation's total population, who are currently vulnerable to climate change. By 2050, it is estimated that this will extend to 902,906 people or roughly 26 per cent of the entire population. This project is particularly aimed at safeguarding the most flood-vulnerable populations in 69 specified municipalities, making up 44 per cent of their inhabitants. The implementation of enhanced flood risk management is expected to have a high impact, preserving not only lives but also economic assets and infrastructure. Furthermore, EbA strategies will bolster both community resilience and ecological safeguards within targeted river basins.

22. The project is geared towards fortifying resilience and improving the livelihoods of at-risk demographics, specifically women, children, and senior citizens. Women make up 51.7 per cent of the affected population, whereas those under 15 and over 65 account for 15.2 per cent and 13.7 per cent, respectively. Economic analyses suggest that the initiative will yield an estimated annual benefit of USD 21 million upon full implementation, with the most significant gains seen in the protection of household assets. Geographically, the Bosna and Neretva River Sub-basins stand to benefit the most. The resilience of physical infrastructure across three key river basins in Bosnia and Herzegovina will also be heightened.

23. In addition to local benefits, the project also aims for broader climate resilience through the adoption of FRM technologies and the integration of climate information into institutional frameworks in Bosnia and Herzegovina. The introduction of floodplain zoning policies, attuned to the specific vulnerabilities of local communities, aims to reduce their flood risk exposure significantly.

24. Stakeholder consultations have informed the project's design, with particular attention paid to the unique vulnerabilities of women, ethnic minorities, the persons with disabilities and older persons. The collection of gender-sensitive socioeconomic data will be an ongoing process. A Gender Action Plan (GAP) will further anchor the project in fulfilling Bosnia and Herzegovina's commitments to gender equality and international protocols on gender-responsive climate adaptation and disaster risk measures. The GAP will leverage insights from prior initiatives, like the UNDP Vrbas project, and will include metrics and tools for monitoring progress in gender equality.

3.2 Paradigm shift potential

Scale: Medium to High

25. The project aims to revolutionize existing flood risk management procedures in Bosnia and Herzegovina by enhancing institutional effectiveness at various levels, disseminating reliable climate information and early warnings, deepening the understanding of disaster risk, and integrating non-structural, ecosystem-based risk management methods. The project also plans to collaborate with the private sector and bolster the resilience of both public and private investments to climate change, aiming to make flood risk management more integral to development planning. Furthermore, the project will cultivate new skill sets by employing integrated techniques and technologies to evaluate flood-related risks and formulate innovative risk management and mitigation strategies.

26. The project aspires to make robust climate information and early warning services widely accessible. It intends to broaden climate risk information for FRM to extend the reach of existing hazard and risk modelling to include flash floods and hydroelectric power plants, thereby enriching the spectrum of hazard, risk and vulnerability information. The innovative introduction of forecast-based financing mechanisms will facilitate early, proactive measures, thereby enhancing the efficiency of response efforts and potentially reducing loss and damage.

27. This will mark the first instance of a comprehensive, large-scale FFEWS being implemented in the nation. Innovative measures of observational tools, forecasting systems and digital technologies like Internet of Things and data analytics will be used to empower local communities to undertake their own disaster risk reduction measures. This complements the national EWS and offers potential for duplication in other locales. The project also seeks to partner with the private sector to create specialized climate services for sectors like hydropower and agriculture, thereby enhancing business continuity planning and private sector engagement.

28. The plan underscores the importance of knowledge dissemination and collective learning. A nationwide flood risk management platform is envisioned to facilitate interdisciplinary coordination and offer policy guidance on flood risk management, drawing participation from public, private and civil sectors. Activity 2.4 is designed to offer extensive training programmes on flood risk management for various stakeholders, including decision makers and emergency responders. It also aims to bolster research and development in flood risk management via academic collaborations.

29. The proposal is fully aligned with the United Nations Secretary-General's EW4All initiatives, encapsulating all its four pillars: disaster risk knowledge and management; detection, observation, monitoring, analysis and forecasting; warning dissemination and communication; and preparedness and response capabilities.

3.3 Sustainable development potential

Scale: High

30. The project is poised to deliver multiple co-benefits including enhanced ecosystem services, gender-responsive livelihood enhancement and job creation, and improved resilience

of communities' physical assets. One of the notable economic advantages stems from averting the socioeconomic toll of flood disasters, which under climate change scenarios could reach up to USD 750 million in expected annual damages. This would notably hamper Bosnia and Herzegovina's socioeconomic progress. Avoiding such losses in sectors like agriculture and hydropower would be particularly significant.

31. The project's interventions offer detailed, sector-specific climate risk information, which is critical for strategic planning and operational optimization in sectors like hydropower. This targeted approach not only ensures the sustainable development of the hydropower sector but also facilitates a shift toward clean energy, contributing to greenhouse gas reduction. Similarly, safeguarding crucial infrastructure like roads and bridges will bolster economic development and community functionality. This, in turn, allows the Government of Bosnia and Herzegovina government to allocate public funds more effectively, directing them toward socially significant initiatives rather than disaster recovery.

32. In terms of social impact, the project is specifically designed to benefit vulnerable populations, with a focus on gender equality. By actively involving women – who are disproportionately affected by disasters – in decision-making processes, the project aims to counter existing gender disparities in employment and social roles in Bosnia and Herzegovina.

33. Environmentally, the project deploys EbA strategies and nature-based solutions like riparian plantings and agro-forestry. These measures serve multiple purposes: they enhance water retention, regulate hydrological flows, mitigate natural hazards like flooding and soil erosion, and provide habitat preservation and reforestation benefits. In the long run, these interventions will significantly boost the country's environmental resilience to climate-induced flood disasters, thereby aiding in the protection of valuable natural assets such as land and forest.

34. The Global Framework for Climate Services (GFCS) identifies five priority sectors for national development planning: agriculture, food security, water, health, disaster risk reduction, and energy. The potential benefit-cost of return on investment could be up to 500:1 for flood-prone areas and 2500:1 for heat waves in urban centers (Teisberg and Weiher, 2008). The proposal could explore these benefits in a multi-hazard context for flood-related hazards and linkages to the other hazards of the GFCS priority sectors. The proposal would benefit from a clearer articulation and demonstration of health benefits from flood-related risk management. The proposal could benefit from more detailed elaboration of how these interventions could lead to health benefits from basic hydro-meteorological forecasting and impact-based forecasting. It is recommended that UNDP should consider laying the foundation for the establishment of the infrastructure and tools for multi-sectoral forecasts in the other four GFCS priority sectors through the GFCS user interface platform. We also proposed the multi-hazards forecasting system could follow overall value chain of risk information sectoral risk management.

3.4 Needs of the recipient

Scale: High

35. Bosnia and Herzegovina is a middle-income nation with approximately 3.5 million residents. It is still grappling with the devastating effects of the 1992–1995 war, which decimated its human, economic, and social resources. These challenges are exacerbated by an ongoing transition toward a market economy that demands structural reforms and enhanced governance. Bosnia and Herzegovina's economy is in a transitional phase, heavily reliant on exports such as metals, energy, textiles and furniture, as well as on remittances and foreign aid. Poor governmental coordination and excessive bureaucracy have stifled economic growth and deterred foreign investment, making the economy one of the least competitive in the region.

36. Socially, poverty remains high, with 16.9 per cent of the population living below the poverty line as of 2015. While the Human Development Index showed improvement between 2000 and 2019, the gains were significantly offset when adjusted for inequality. Additionally, the country's Gender Inequality Index ranks it 38th out of 162 countries. Climate change has further strained the slow post-war economic recovery, particularly impacting sectors like agriculture and hydropower due to increasing flood events.

37. Vulnerable groups, including the rural poor and displaced individuals, have been most affected by flooding disasters. A lack of institutional capacity exacerbates these issues, with fragmented governance and poor communication hampering effective FRM. Key gaps in institutional capacity include insufficient hydrometric monitoring, outdated flood risk maps, and limited technical and financial resources for modern FRM strategies.

38. The project aims to address these institutional challenges by implementing Flash Flood Early Warning Systems (FFEWS) based on clearly defined roles and responsibilities, creating protocols for effective FFEWS and FRM coordination, and employing state-of-the-art technologies and training to build capacity.

3.5 Country ownership

Scale: High

39. The project demonstrates a strong national ownership, and has established the appropriate/relevant systems and processes (political, institutional and legal frameworks) for implementing its commitments under the United Nations Framework Convention on Climate Change. This includes the adoption of its intended nationally determined contributions under the Paris Agreement, for the period until 2030, in October 2015, and the development of its nationally determined contribution in November 2020. The nationally determined contribution provides guidance on the country's climate change adaptation and contribution to climate change mitigation. It identified the following as the country's most vulnerable/priority sectors to climate change, for intervention: agriculture, water resources, forestry, energy, tourism, biodiversity and sensitive ecosystems, and human health, with the two most vulnerable sectors being agriculture and water resource management. It outlines the role of detailed planning of short, medium and long-term adaptation measures, development of climate risk financing mechanisms, and strengthening the responsible institutions in monitoring, analysis and modelling of climate risk. The proposal is also aligned with the United Nations Sustainable Development Goals (in particular target 13) and the Sendai Framework for Disaster Risk Reduction (in particular target G).

40. The proposal is developed based on these national strategies and the application of international recommendations. Strong country ownership is also demonstrated by significant government and local institution co-financing commitments. Co-financing support for the project will be provided by the Government of Republika Srpska through a European Investment Bank (EIB) loan of USD 21.5 million for structural measures and additionally through USD 36.0 million of the public funds to be provided by the governments of both entities composing the country – the Governments of the Federation of Bosnia and Herzegovina and Republika Srpska for co-financing of EbA measures, the O&M of structural measures, hydrometric equipment, and modelling and FFEWS.

3.6 Efficiency and effectiveness

Scale: Medium

41. The total budget for the proposed project is USD 72.69 million, of which GCF funding makes up 19.8 per cent, and co-financing accounts for the remaining 80.2 per cent. The Government of Bosnia and Herzegovina is expected to contribute 42 per cent or USD 30.74 million, while EIB will provide 29 per cent in the form of senior loans. Despite facing budgetary and economic challenges, the substantial investment by the Government of Bosnia and

Herzegovina underscores its commitment to the initiative and, more importantly, to its social responsibility to provide an FFEWS as an essential service to the population.

42. The GCF contribution is projected to mobilize an additional USD 58.29 million from the public and private sectors. Both project partners and beneficiaries are expected to bear the O&M costs related to all the project-supported infrastructure and hydrological equipment. While the project aims to produce significant public benefits, it is also designed to engage the public sector in limited cost-recovery measures for the ongoing operation and maintenance of the monitoring and early warning systems.

43. Based on a detailed economic analysis (annex 3), the project estimated an economic internal rate of return (EIRR) of 19.3 per cent, and economic net present value (ENPV) of USD 44.92 million. The results were calculated over a 20-year time frame at the economic discount rate of 10 per cent. The economic analysis of the project was carried out in accordance with the GCF Economic and Financial Analysis (EFA) Guidelines, and was conducted at the national level and then for each river sub-basin included in the proposed investment project. The results of the EFA suggest that the proposed investments can be justified from the economic point of view as obtained measures, ENPV and EIRR ensure the economic viability of proposed investments (e.g. the ENPV is positive, and the EIRR is higher than the chosen economic discount rate). Furthermore, the results of the EFA suggest that the proposed investment will be beneficial to the entire economy as the benefits of the proposed interventions will outweigh the investment costs.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

44. **Project background.** The project will upgrade and expand the coverage of the hydrometric network and develop an integrated impact-based centralized and community-based flood forecasting and early warning system. The project will also implement structural and non-structural flood risk reduction measures and nature-based solutions and strengthen institutional capacity for flood risk management. In addition, it proposes to scale up flood protection measures through improved national and local investment frameworks. Some of the expected environmental co-benefits are water retention functions from riparian plantings and agro-forestry and reduced erosion. Social co-benefits of the proposal are improved resilience of communities' physical assets and protection of people from major flooding hazards.

45. **Environmental and social risk category.** The confirmed and assigned environmental and social risk category of the project is equivalent to GCF's medium/category B. It is within the accreditation level of the accredited entity (AE). This is rationalized by structural and non-structural flood reduction interventions that will be promoted by the project and are likely to have moderate risks which are generally reversible and site-specific which can also be readily managed with mitigation measures. Activity 2.2 non-structural measures include gabion walls, torrent and wicker barriers, and planting forest strips among others. Activity 3.3 structural measures include rehabilitation of river embankments, and canals and establishing dykes. Additionally, the associated risks and impacts of activities such as upgrading and expansion of the coverage of the hydrometric system and capacity-building such as training and development of guidelines, plans and frameworks are expected to be low and minimal.

46. **Safeguard instruments.** The AE provided an environmental and social assessment report (ESAR) and environmental and social and management framework (ESMF) with the funding proposal as the safeguards document. The document provides an overview of the national context for environmental and social risk assessment, potential environmental and social risks likely to be associated with the project activities, including medium risk site-specific

environment and social studies that will be undertaken during the project's implementation. Environmental impact assessment will be required for completely new facilities built by the project according to national regulations, however, other proposed activities such as the reconstruction of existing dykes and inlets fall within exemptions provided to the water authorities which will be confirmed prior to commencing implementation.

4.1.1. Compliance with GCF's environmental and social safeguards (ESS) standards.

47. **ESS 1) Assessment and management of environmental and social risks and impacts.** The ESAR was prepared based on the environmental and social risks identified through the screening of activities using the AE's procedure. It includes an assessment of the significance level of each risk that was identified under each activity. The ESMF provides an outline of a range of mitigation measures that will be undertaken when implementing the project to manage risks and impacts and the site-specific environmental and social assessments and management plans that will be prepared for moderate risk structural and non-structural flood protection measures, for which outlines of the documents are annexed into the ESAR and ESMF. Baseline environmental and socioeconomic information about Bosnia and Herzegovina is also presented in the ESAR and ESMF. The main outcomes of the environmental and social assessment undertaken, including the key risks and impacts and proposed measures to be taken to avoid, minimize and mitigate them are summarized below. A gap analysis of the AE's environmental and social standards, those of EIB, a co-financier and national policies, legislation and regulations is included in the document. The AE is recommended to include GCF standards in the analysis.

48. **ESS 2) Labour and working conditions.** There is a risk that work sites may flood during the implementation of flood protection measures. This will be mitigated by timing the works appropriately and monitoring of upstream rainfall and river levels prior and during the works. Contracts and working conditions for personnel working on the project will be in line with relevant national legislation and in compliance with UNDP Social and Environmental Standard 7. Appropriate personal protective equipment will be provided to personnel who need it.

49. The AE is advised to ensure that vendors are evaluated during the procurement process and ensure inclusion of requirements and standards concerning irregular working conditions for hydrological, meteorological and precipitation stations' solar components that may be procured by the project.

50. **ESS 3) Resource efficiency and pollution prevention.** It is anticipated that most materials for construction purposes, particularly rocks, will come from quarries near the work sites. The project will ensure proper handling of materials, and where possible, recycling and reuse of any local materials may be considered. The key waste streams likely to be generated are residual sediment and construction wastes which will be minimized, have designated areas for temporary management and disposed of according to local requirements. Potential erosion and sediment movement may also occur during works that involve physical interventions. To avoid significantly increased sediment discharges and erosion, the AE will ensure project activities are not carried out during periods of rainfall. There is a small chance of water pollution from oil, grease and fuel spills, and other materials from vehicles working on site.

51. **ESS 4) Community health, safety and security.** The project is expected to involve minor construction work impacts for the upgrade and expansion of the coverage of the hydrometric network. Other construction-related risks include noise and dust from heavy construction machinery and haul trucks. Some of the proposed mitigation measures are sourcing water for dust suppression during the implementation of activities that generate dust and restricting use of machinery and vehicles to normal working hours during the day.

52. Additionally, there is the possibility of mines and unexploded ordnance (UXO) disturbance by earthworks for which the project will update the magnetometry survey to confirm the presence or absence of UXO and the contractor will be required to detail procedures to safely handle and dispose of UXO if any UXO are encountered during works.

53. Conflict sensitivity analysis is included in the ESAR and ESMF which assesses the conflict dynamics in the country and the project activities' potential to cause conflict. Most risks are classified as low and low-medium. The AE concludes that the probability of conflicts occurring or being exacerbated due to the project interventions is slight. Recommended conflict mitigation measures include stakeholder engagement and the use of grievance mechanisms available for the project.

54. **ESS 5) Land acquisition and involuntary resettlement.** Activities involving physical interventions may result in economic displacement from construction activities that alter people's ability to move or access resources. The AE will be required to ensure that when this standard is triggered, plans for the restoration of livelihoods accompany the site-specific environmental and social assessments and management plans whether economic displacement is temporary or permanent. The AE is recommended to pay particular attention to related impacts on internally displaced persons who may be more vulnerable. The project will not result in physical displacement. Sites for physical interventions have been selected only where land tenure rests with the government, that is public land.

55. **ESS 6) Biodiversity conservation and sustainable management of living natural resources.** Implementation of structural and non-structural measures proposed by the project may potentially cause habitat loss and disturbance of fauna. This will be mitigated by revegetation where vegetation is removed by the project, avoiding in-stream works during the spawning season and minimizing water turbidity during works on waterways. Disturbed areas will be revegetated using native and locally endemic species that have high habitat value. Native and locally endemic species will be used to revegetate to mitigate against invasive species. A biodiversity action plan may be prepared for the Krupa River site in case there are likely impacts to the Hutovo Blato Ramsar site.

56. **GCF Indigenous Peoples Policy and ESS 7) Indigenous Peoples.** The AE confirmed there were no Indigenous Peoples or internally displaced persons affected by the project and that the EIB, which has a related investment and uses a similar Indigenous Peoples safeguard, had come to the same conclusion. Accordingly, the AE concluded that the Indigenous Peoples Policy, ESS 7 and its own Social and Environmental Standard 6 were not triggered.

57. **ESS 8) Cultural heritage.** Screening for the project indicated that cultural physical resources were unlikely to be at risk as a result of the project's implementation. Nonetheless, there is the possibility that unexpected cultural heritage items could be discovered during works. A chance finds procedure is annexed into the ESAR and ESMF for this eventuality.

58. **Sexual exploitation, sexual abuse, and sexual harassment (SEAH) safeguarding.** The AE has identified the project's moderate SEAH risks related to the construction work for implementing new ecosystem-based flood risk reduction and climate change adaptation methods, and when conducting regular project information and training sessions. The project's ESAR and ESMF has triggered AE's relevant policies, safeguard standards and programming principles on the prevention and elimination of SEAH. The AE will implement a SEAH Action Plan to address the potential risks and impacts of SEAH and will require the project's contractors, suppliers and partners to adhere to zero tolerance for SEAH. The AE will terminate contractual arrangements with those who cannot conduct and commit to taking adequate action if faced with SEAH allegations. As part of the project's SEAH Action Plan, the project will ensure the availability of an effective GRM with multiple channels to initiate a complaint, including confidential reporting with safe and ethical documenting of SEAH cases. The project will map out SEAH prevention and response actors in the project's adjoining communities, develop

training materials and organize SEAH training for project workers and the local community, clearly define SEAH requirements and expectations in the bid documents, and undertake regular monitoring and evaluation of progress on SEAH prevention and response activities, including reassessment of risks as appropriate. In the event of gender-based violence/SEAH complaints, the project's Gender Advisor will serve as the SEAH focal point to connect the SEAH survivor with relevant service providers.

59. **Implementation arrangements.** The Project Management Unit (PMU) will ensure timely remedial actions are taken by the contractor where necessary. The PMU will have an International Safeguards Officer responsible for overseeing and coordinating the implementation of safeguards across the project and ensuring consistent application of measures by the separate water agencies/contractors involved in the project. This officer will ensure that relevant standards are met. The Safeguards Officer will be supported by field officers who will be responsible for day-to-day inspection to ensure that the required safeguards are being implemented and monitored at the site level. Delivery organizations have the responsibility for ensuring that relevant employees and contractors are aware of the environmental and social safeguards requirements. Personnel working on the project will receive training in environmental, health and safety requirements. UNDP, as the AE, will maintain oversight on all matters related to environmental and social safeguards. A budget is included in the ESAR and ESMF with a breakdown of how resources allocated for the implementation of safeguards matters will be utilized.

60. The AE is recommended to assess the capacity of the project's delivery organizations and provide interventions that strengthen the relevant capacities that are required to undertake responsibilities related to environmental and social safeguards matters.

61. **Stakeholder engagement and information disclosure.** Consultations on the project were undertaken during the design phase with a wide range of stakeholders, including relevant national authorities, industry groups, civil society organizations, and individual community members. A report on stakeholder consultations undertaken is presented in a stakeholder engagement plan. The AE and its partners will develop and release updates on the project on a regular basis to provide interested stakeholders with information on project status and circulate this through various media. Stakeholder engagement activities that will support project outcomes are listed in the plan according to project activities and stakeholders identified, their roles and means of participation.

62. **Grievance redress mechanism (GRM).** Grievances can be registered by telephone via the project-level grievance redress mechanism that will be publicized and maintained throughout the project to serve as a point of contact for enquiries and grievances concerning the project. Complaints will be recorded on a register, including details of the response and investigation undertaken and actions taken to address the matter. The PMU Safeguards Officer will be responsible for the functioning of the grievance mechanism of the project. The process for responding to grievances is outlined in the ESAR and ESMF. Furthermore, details of how the AE's institutional-level grievance mechanism and the GCF Independent Redress Mechanism can be contacted have been included in the document.

4.2 Gender policy

63. The AE has provided a gender assessment and Gender Action Plan therefore complies with the requirements of the GCF's Gender Policy and Gender Action Plan.

64. The gender assessment describes the enabling environment and gender issues at the national level and indicates the existence of a national policy environment, supportive legislation and practices to promote gender equality. It also identifies gendered challenges and barriers that leave women more vulnerable to climate-related disasters. Despite a strong policy

and legal framework supporting gender equality and women are still largely excluded from political and decision-making processes; they have disproportionately limited access to employment and income, and many are subject to violence. There is particular concern with respect to women internally displaced persons, persons with disabilities, older persons and Roma women in Bosnia and Herzegovina, who are even more severely disadvantaged and marginalized. And institutional capacity to implement a normative framework for gender equality remains a serious challenge. The greatest obstacles to broader development and empowerment of women in rural areas are related to property rights, access to public services, finance, training, advisory services and mentorship, and limited time to engage in activities outside of their homes. Limited opportunities for employment and self-employment are perpetuated by low levels of understanding of opportunities coupled with a lack of access to information and training. A lack of affordable childcare and pre-schools in rural and smaller settlements is one of the central obstacles to women's participation in trainings, entrepreneurial activities, and consequently, employment. There is also a limited number of women associations and cooperatives which exacerbates the lack of access to relevant information and services. Moreover, discrimination against Roma women is evident from the level of their formal employment. This discrimination, especially toward Roma girls, is also reflected in their right to education. Every second woman has experienced some form of gender-based violence from the age of 15, while every fifth woman has most likely experienced this over the past 12 months. There are several vulnerabilities that increase the risk to women from natural disasters, and based on experiences following the floods in 2014, the Republika Srpska Gender Centre has developed a Gender Action Plan for natural disasters and has incorporated measures into it to address these challenges.

65. The AE has provided a programme-level Gender Action Plan, which includes activities, indicators, targets, budget, timeline and gender expertise to ensure that gender issues are incorporated into the project and GAP activities. While an initial gender analysis was conducted, further analysis will be undertaken to ensure that the baseline data set is sex- and age-disaggregated and adequate for assessment of the project's gender impacts. The gender analysis will assess gender-specific vulnerabilities to flood risks through gender-responsive socioeconomic data collection, and ensure women's participation and decision-making as well as information access and warning dissemination services. Further, the project will ensure women are involved in the processes of capacity-building for climate-induced hazard risk assessment within the FRM at all levels, but also in the development of public-private collaboration on building resilience by climate risk finance mechanism. Men and women of different age groups, as well as those from vulnerable communities will be involved in planning and implementation. Community consultancy groups with at least 40 per cent representation of women will be established. The project will ensure that information on hazards, their character, etc., reaches both women and men through appropriately focused channels. It will also tailor a warning to the gender-differentiated needs and capabilities of specific population groups, such as children, older persons, internally displaced persons, the sick and persons with disabilities. The project will also ensure that women have access to information and livelihood-generating activities. The project will strengthen social capital through the creation of community groups for planning for, safeguarding from and mitigating against adverse effects of natural disasters with at least 40 per cent women's participation. Additionally, the project will conduct gender-mainstreaming trainings and build capacity of national stakeholders targeted by the project and responsible for various disaster risk reduction functions. The project will ensure at least 30 per cent representation and active participation of women in project stakeholder consultations, and local and national decision-making bodies established and/or facilitated by the project. The project will secure participation of the gender institutional mechanism representatives in the consultations and decision-making. In developing an investment framework for climate-induced flood risk reduction and management, the project will include representatives from the gender mechanisms. A gender-responsive strategy and gender-sensitive tools for private sector engagement in long-term climate-proof risk management will be developed along with the

development of risk financing and transfer mechanisms based on detailed gender-sensitive assessment on socioeconomic risk, damages and losses. Multi-year climate resilient municipal investment plans and gender-responsive community preparedness plans will be developed in selected municipalities (10–12) and in one canton. The project will ensure men and women (40 per cent) are represented in municipal investment planning process and that they actively participate in the formulation of community preparedness plans in selected municipalities and that the needs of women and other vulnerable (older persons, persons with disabilities, children, Roma) groups are incorporated into any plans developed.

66. Finally, the project will implement climate-proof structural flood risk reduction and anti-erosion interventions in the Vrbas, Una-Sana, Bosna and Drina River Basins. Both, women (40 per cent) and men, and other vulnerable groups (older persons, persons with disabilities, children, Roma) will be included in processes of stakeholders' consultation related to the implementation of climate-proof flood risk measures. The project will ensure that specific ways in which flood risk and erosion impact women are analysed and considered. Ensure that women and vulnerable group members can equally benefit from technical assistance. The AE will aim to ensure at least 50 per cent of women participants have benefited from the employment and income-making opportunities associated with implementing these measures.

67. The project will engage a gender expert and the project implementation team, staff members will be trained on gender mainstreaming and on the social vulnerability approach. Project team leaders will ensure that all specialists are briefed on their responsibilities in relation to the Gender Action Plan upon mobilization.

68. The AE is recommended to consistently mainstream gender throughout the project as a means to meet the needs of women and other vulnerable people. There is great potential to consider gender issues significantly and sufficiently in all flood risk information and warning messages, in amendments to all sectoral planning documents and feasibility studies on microinsurance and regional disaster risk insurance programmes and catastrophe bonds. Further concerns remain with the lack of clarity on indicators that will be used to measure women's access and use of products and mechanisms such as risk financing, transfer mechanism products and responsible financial products with corresponding targets. These considerations, if taken seriously, will significantly contribute to women's empowerment.

4.3 Risks

4.3.1. Overall programme assessment (medium-high risk)

69. GCF is requested to provide a grant of USD 14.4 million, which will be invested in flood risk management by strengthening the use of climate information, flood forecasting, early warning and emergency response systems to enhance adaptive capacity and resilience of at-risk communities. The Government of Bosnia and Herzegovina will provide a senior loan of approximately USD 21.1 million which is from EIB. The government will also provide a grant along with other partners which will be a total of approximately USD 37.2 million. While the co-financing accounts for 80 per cent of the total budget of this project, some of the details around the co-financing will be further negotiated at the FAA stage.

70. The AE is requesting full grant support for this project. As the government will take out a loan from EIB to finance structural measures, the grant from GCF will finance timely complementary interventions such as non-structural measures and technical assistance.

4.3.2. Accredited entity/executing entity capability to execute the current programme (Low risk)

71. UNDP is the AE and also the executing entity (EE) for the project. The project will be implemented by UNDP following UNDP's Direct Implementation Modality (DIM). The AE has been operating in the country since 1996. As the funding proposal involves different government entities with the complex structure, UNDP's coordination role will be indispensable for successful project implementation.

4.3.3. Programme-specific execution risks (medium-high risk)

72. **Complex governance structure.** The AE has identified several governance-related risks in the funding proposal. The funded activities include a number of policy and regulation updates. For example, activity 2.1 is on financing the development of countrywide flood zoning policy and regulation. The governments of the two entities need to confirm their acceptance of the policy and regulations once they are updated. For activity 3.1, the project will facilitate the changes to the current regulations in the field of insurance. The relevant legal and regulatory provisions require adoption by the governments to enable the implementation of flood-risk financing instruments. The fragmented institutional structure of the governments and a lack of political consensus may hinder the process and limit the policy effectiveness even after the updates/changes. The AE will leverage the existing relationship and relevant experience from past projects to mitigate such risks.

73. **Co-financing risk.** the ratio of GCF funding to co-financing is 1:4. The sizable co-financing is critical for the funded activities, especially for activities which are implemented through blended funds from GCF and co-financers such as activity 2.1. If the disbursement of co-financing is delayed or reduced, it will hamper project implementation. To mitigate this risk, it is recommended that the term sheet include a proportionate disbursement between GCF and co-financing with a specific ratio which would be in line with the implementation plan each year. However, the details for disbursement and how to address the shortfall in co-financing will be further negotiated at the FAA stage.

74. As part of risk financing and as a risk transfer measure, the project will provide technical assistance and training to financial institutions for the development of lending products and the use of climate risk information into their lending process to provide loans to climate-resilient enterprises in Bosnia and Herzegovina. However, technical assessment and training will not automatically lead to the provision of loans to enterprises unless there are incentives for the financial institutions. A specific target or indicator to monitor the number of loans to be provided and/or engagement of financial institutions will enhance the impact of the funding proposal.

4.3.4. Compliance risk (medium risk)

75. UNDP confirmed that it has conducted a risk assessment in relation to money-laundering and the financing of terrorism risks. The probability of occurrence of a prohibited practice event has been assessed as low, whereas the impact severity of occurrence has been assessed as medium.

76. UNDP confirmed that it has also conducted a risk assessment in relation to sanction risks. The probability of occurrence of a prohibited practice event has been assessed as low, whereas the impact severity of occurrence has been assessed as medium.

77. UNDP noted that this project will be implemented under the DIM and will be guided by UNDP's policy framework. UNDP, by virtue of being an organ of the United Nations and therefore being part of the same entity, is also beholden to the principles on anti-money-laundering/countering the financing of terrorism.

78. UNDP systematically screens all entities being considered to be contracted by or partner with UNDP against the Consolidated United Nations Security Council List. This is built into the

capacity-screening tools for partners, as well as in the SOPs for vendor management. UNDP also secures contractual representations from its national implementing partners that address risks of the funds received for implementation of projects being used to support individuals or entities associated with terrorism and recipients of any amounts provided by UNDP hereunder appearing on the Consolidated List.

79. These internal controls are further supplemented by internal controls to mitigate the risks, including due diligence requirements for partners and private entities which UNDP intends to engage with. When developing a new project, UNDP ensures that risks are fully identified and considered in the project design and processes and that adequate and effective measures to prevent, monitor and address such risks are put in place.

80. UNDP confirmed that there will be no third parties involved in the delivery of the GCF funds (such as the implementing party or responsible parties). UNDP will deliver all activities through the UNDP procurement process/UNDP-approved vendors. Therefore, the due diligence process for third parties was not triggered at the project proposal stage. However, UNDP has carried out the first level due diligence assessment of all co-financiers, including public and private co-financing entities, which will be implementing their respective co-financed activities in the framework of the project. The assessments have been carried out in line with UNDP's policies and risk assessment tools.

81. UNDP also referred to the Harmonized Approach to Cash Transfers (HACT) applied by UNDP, which represents a common operational (harmonized) framework for transferring cash to government and non-governmental partners (both implementing partners and responsible parties). This project initially does not envisage engaging responsible parties. However, in the event that responsible parties are engaged in the future for project implementation, the HACT framework will be applied by UNDP.

82. The common objective of the HACT framework is to support the closer alignment of development aid with national priorities and to strengthen national capacities for management and accountability. It serves as a set of procedures on requesting, disbursing, providing assurance, and reporting on funds to effectively manage risks, reduce transaction costs and promote sustainable development in a coordinated manner. Cash transfers over a certain amount (above the threshold of USD 150,000 in a programme cycle, or otherwise in all circumstances where UNDP considers the operating environment high risk or has no prior experience with the partner) are presumed to present a material risk to UNDP and will require a HACT micro-assessment and risk rating by qualified third party service providers. Spot checks, programmatic output verification and auditing requirements are then determined and carried out at a frequency determined in line with the risk rating.

83. UNDP confirmed that cash, vouchers, commodities or other items of value will not be distributed or disbursed to any beneficiaries involved in the project, either directly or indirectly.

84. UNDP confirmed that it has carried out first-level due diligence assessment of all co-financiers. For public-sector partners (five entities), UNDP's Partner Capacity Assessment Tool (PCAT) has been used. For the private sector co-financiers (three entities), UNDP's Private Sector Risk Assessment Tool has been used in line with the UNDP Policy on Due Diligence and Partnerships with the Private Sector.

85. None of the co-financiers are listed on the Consolidated List, the UNDP Vendor Sanctions List or the United Nations Global Marketplace (UNGM) Ineligibility List. Assessments have not found any evidence of fraud, corruption or violation of programming principles or standards by these co-financing entities. These due diligence assessments have not revealed any violations or concerns over the partners' capacities to deliver their co-financed activities in the framework of the project. In addition, co-financiers are required to reconfirm their commitments to meet the requirements of the UNDP social and environmental safeguards policy and standards prior to the launch of the project implementation.

86. UNDP confirmed that it has conducted a risk assessment in relation to prohibited practices risks. The probability of occurrence of a prohibited practice event has been assessed as low, whereas the impact severity of occurrence has been assessed as medium.

87. UNDP stated that it applies a zero-tolerance policy in relation to fraud and corruption. The guiding principles of UNDP's commitment to prevent, identify and address all acts of fraud and corruption have been laid out in the UNDP Policy Against Fraud and Corrupt Practices (FCP Policy), which applies to all activities and operations of UNDP, including projects and programmes. The fundamental principles of UNDP's FCP Policy serve as the basis for and are integrated into UNDP's policy frameworks in relation to procurement, financial management, internal control and accountability, and staff rules and regulations.

88. Under the terms of the FCP Policy and related policies, UNDP has a commitment, when developing a new programme or project, to ensure that risks related to fraud and corrupt practices are fully identified and considered in the programme/project design and processes and that adequate and effective measures to mitigate such risks are put in place. In this light, the processes and requirements for UNDP's PCAT require UNDP to carry out a thorough due diligence of the risks in relation to the programme/project it intends to engage in, as well as in relation to the potential partners who will be involved in the implementation of such programmes/projects.

89. Such capacity assessments include, among other things, a screening of all entities being considered to be contracted by or partner with UNDP against the Consolidated List (to which UNDP, by virtue of being an organ of the United Nations and therefore being part of the same entity, is also beholden), an assessment of any history of fraud, corruption or other fraudulent practices and/or potential conflicts of interest. Based on this capacity assessment, UNDP then builds an internal control component into the design of the project/programme and ensures that its financial management contains adequate safeguards to prevent, monitor and address any risks and acts of fraud, corruption or anti-money-laundering/countering the financing of terrorism that may be identified.

90. UNDP confirmed that the project has developed a grievance redress mechanism to deal with any complaints and issues that may arise because of the project. The GRM has been designed to ensure that an individual and/or group are not financially impacted by the process of making a complaint and/or grievance. The GRM complies with safeguard procedures of Bosnia and Herzegovina and UNDP. Furthermore, the GRM allows individuals and/or groups to also file a complaint with the Office of Anticorruption and Integrity within the UNDP or utilize the GCF independent Redress Mechanism should they have any concerns as to corruption, unethical behaviour or where they believe their complaint or grievance has not been adequately addressed.

91. **Compliance risk rating:** The Secretariat has conducted a review of the project in accordance with relevant Board-approved policies and does not find any material issue or deviation with respect to compliance issues. Based on available information for this funding proposal, the Secretariat has determined a compliance risk rating of medium and has no objection to this proposal.

4.3.5. GCF portfolio concentration risk (low risk)

92. In the case of approval, the impact of this proposal on the GCF portfolio concentration in terms of results area and proposal is not material.

4.3.6. Recommendation

93. It is recommended that the Board consider the above factors in its decision.

Summary risk assessment	
Overall programme	Medium-high
Accredited entity/executing entity capability	Low
Project-specific execution	Medium-high
GCF portfolio concentration	Low
Compliance	Medium

4.4 Fiduciary

94. The UNDP will be the AE and the EE of the project. It will provide oversight and quality assurance in accordance with its policies and procedures, and any specific requirements in the accreditation master agreement (AMA) and project confirmation to be agreed with GCF.

95. The project will be implemented by UNDP following UNDP's DIM, according to the Standard Basic Assistance Agreement between UNDP and the Government of Bosnia and Herzegovina (1995) and the policies and procedures outlined in the UNDP Programme and Operations Policies and Procedures.

96. The project will be audited following the UNDP financial rules and regulations noted under funding proposal section G.3 and applicable audit guidelines and policies. According to the current audit policies, UNDP will appoint the auditors. Scheduled audits will be performed annually, during the programme cycle, as per UNDP assurance/audit plans, on the basis of the implementing partner's risk rating and UNDP's guidelines. The project's financial statements will comply with international accounting standards.

97. UNDP has procurement policies in place as outlined in the "Contracts and Procurement" section of UNDP's Programme and Operations Policies and Procedures. The policies outline formal procurement standards and guidelines across each phase of the procurement process, and they apply to all UNDP procurements.

98. As of 7 pm on 2 October 2023, there remains a divergence of views between the GCF Secretariat and the AE concerning how co-financing would come through. Co-financing accounts for more than 80% of total funding, therefore clear mechanism and undertaking by the AE on how the co-financing would come through is important for the implementation per the outlined targets and implementation plan. Despite numerous rounds of reviews, the AE has not yet been amenable to including a covenant on co-financing. Given the GCF to co-financing ratio of 1:4, the Secretariat has been hesitant to advance the co-financing covenant discussions to the FAA stage. However, since no agreement on the co-financing covenants has been reached, we have to defer relevant clauses on co-financing to the FAA stage on a condition that the language of this covenant to set out a mechanism for the co-financing to be contributed by all the co-financiers listed in the FP in line with the targets, total amount, Budget and Workplan (per Annex 1) as outlined in the FP. We propose to include the proportionality of the GCF proceeds disbursed each year with the proceeds of co-financiers, inline with the Budget.

4.5 Results monitoring and reporting

99. As an adaptation initiative, the project aims to generate various adaptation results across four GCF result areas. It is expected that 799,445 direct beneficiaries and 1,977,000 indirect beneficiaries will receive adaptation benefits in Bosnia and Herzegovina, representing 25 per cent and 80 per cent of its total population, respectively. The AE has provided a methodology for estimating numbers of direct and indirect adaptation beneficiaries based on

expected adaptation benefits. The Secretariat provided further guidance on how to disaggregate the beneficiaries against GCF results areas, which can feed into the target-setting of GCF core indicator 2 and its supplementary indicators in the logical framework.

100. The theory of change diagram adequately captures various levels of expected changes and logical linkages between them from the goal statement to the proposed activities. The barriers and risks that may hinder the achievement of these changes are identified clearly and linked to pertinent activities.

101. The logical framework is structured in alignment with the GCF integrated results management framework. It shows how the project can contribute to four GCF adaptation results areas with project outcomes and outputs. Relevant GCF core and supplementary indicators have been chosen for monitoring and reporting the key performance of the project. The project-specific indicators also seem to be adequate to measure the progress of project implementation along with project deliverables.

102. There have been discussions with AE on the project monitoring and evaluation (M&E) plan (annex 11) and project implementation timetable (annex 5). The Secretariat recommended that the AE ensure that all M&E-related budgets, including project-specific indicators, are identified in the M&E plan and these should be identical to the M&E budget in the project budget plan. The AE has been responsive and annexes 11 and 5 have been revised accordingly.

4.6 Legal assessment

103. The original AMA was signed with the Accredited Entity on 5 August 2016, and it became effective on 23 November 2016. The Accredited Entity's first term of accreditation expired on 22 November 2021. The Accredited Entity was re-accredited by the Board pursuant to decision B.30/05. The first amendment and restatement agreement amending and restating the AMA was signed on 16 March 2023 and it became effective on 3 May 2023.

104. The Accredited Entity has provided a legal opinion/certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the project.

105. The proposed project will be implemented in Bosnia and Herzegovina, country in which GCF is not provided with privileges and immunities. This means that, amongst other things, GCF is not protected against litigation or expropriation in this country, which risks need to be further assessed. The GCF Secretariat provided a draft agreement on privileges and immunities and a background note to the Government of Bosnia and Herzegovina on 11 May 2016. The latest communication was sent by the GCF Secretariat to the Government of Bosnia Herzegovina on 7 February 2018.

106. The Heads of the Independent Redress Mechanism (IRM) and Independent Integrity Unit (IIU) have both expressed that it would not be legally feasible to undertake their redress activities and/or investigations, as appropriate, in countries where the GCF is not provided with relevant privileges and immunities. Therefore, it is recommended that disbursements by the GCF are made only after the GCF has obtained satisfactory protection against litigation and expropriation in the country, or has been provided with appropriate privileges and immunities.

4.7 List of proposed conditions (including legal)

107. To address the matters raised in this section, it is recommended that any approval by the Board is made subject to the following conditions:

- (a) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board; and

- (b) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP216

Proposal name:	Scaling up climate resilient flood risk management in Bosnia and Herzegovina
Accredited entity:	United Nations Development Programme
Country/(ies):	Bosnia and Herzegovina
Project/programme size:	Medium

I. Assessment of the independent Technical Advisory Panel

1.1 Overview

1. In its assessment, the independent Technical Advisory Panel (iTAP) is guided by decision B.33/12 adopted by the Board, which explains the principles for demonstrating the impact potential of GCF-supported activities as well as other appraisal and assessment guidance in considering these investment criteria and as endorsed by the Board.
2. The Intergovernmental Panel on Climate Change (IPCC) has assessed the scientific literature regarding the impacts of climate change on Bosnia and Herzegovina and has reached several important conclusions. According to its findings, the country (and the wider region) is susceptible to a variety of adverse effects as a result of global warming.¹
3. One of the key impacts of climate change identified by the IPCC in Bosnia and Herzegovina is the alteration of precipitation patterns. It is projected that the country will experience an overall decrease in annual rainfall, coupled with an increase in the intensity of extreme weather events, such as heavy rainfalls and floods. This alteration in precipitation patterns is likely to exacerbate water-related issues, including water scarcity, droughts and potential threats to agriculture.
4. Furthermore, rising temperatures are identified as a significant concern for Bosnia and Herzegovina and the region. The IPCC anticipates an increase in average temperatures, resulting in more frequent and prolonged heatwaves. This rise in temperature could have detrimental effects on ecosystems, agriculture and public health, notably increasing the risk of forest fires and heat-related illnesses.
5. As such, recent research has indicated that climate change will likely exacerbate flood risks in Bosnia and Herzegovina. Rising temperatures have led to more intense rainfall and increased snowmelt, subsequently overwhelming drainage systems and increasing river flow. As Bosnia and Herzegovina is already prone to flooding due to its mountainous terrain and complex river network, the impacts of climate change will be particularly severe. Moreover, flooding poses a significant threat to the country's infrastructure, agriculture and public health. Increased frequency and intensity of floods may damage roads, bridges and buildings, resulting

¹ See World Bank. 2021. *Climate Risk Country Profile, Bosnia and Herzegovina*. Available at https://climateknowledgeportal.worldbank.org/sites/default/files/2021-07/15914-WB_Bosnia%20Country%20Profile-WEB%20%281%29.pdf (accessed 15 September 2023). See also Government of Bosnia and Herzegovina. 2021. *National Adaptation Plan – NAP with proposed measures*, specifically chapter 7. Available at <https://unfccc.int/sites/default/files/resource/NAP-Bosnia-and-Herzegovina%20.pdf> (accessed 13 August 2023).

in economic losses. Agriculture, a vital sector for the country's economy, will also be adversely affected by floods, leading to a decline in productivity and potential food shortages. Furthermore, flooding increases the risk of waterborne diseases, posing a threat to public health and well-being.

6. Considering these developments, the IPCC emphasizes the importance of adapting to climate change in Bosnia and Herzegovina. It suggests implementing strategies to mitigate the identified risks, including the development of resilient infrastructure, the promotion of sustainable agricultural practices, and the enhancement of early warning systems for extreme weather events.

7. The project titled "Scaling up climate resilient flood risk management in Bosnia and Herzegovina" proposed by the United Nations Development Programme (UNDP) seeks to respond to this call for a strengthening of adaptive capacity in the country. The overall objective of the project is to establish a comprehensive and innovative flood risk management approach that is responsive to gender issues and climate change impacts. The project also seeks to enhance the capacity of institutions, both technically and financially, to implement long-term flood risk management strategies that incorporate a range of measures, including structural (i.e. physical construction) and non-structural (i.e. land-use planning and zoning) measures as well as ecosystem-based adaptation (EbA, working with nature and natural processes). Additionally, the project aims to improve the utilization of climate information, flood forecasting, early warning systems, and emergency response mechanisms to strengthen the adaptive capacity and resilience of vulnerable communities in Bosnia and Herzegovina. The project aligns with the Early Warnings for All Initiative which was announced by the United Nations Secretary-General in March 2022.

8. The project is designed to scale up the Technology Transfer for Climate Resilient Flood Risk Management (Vrbas project) that UNDP implemented between 2014 and 2020 with funding from the Global Environment Facility, and that focused on introducing state-of-the-art climate-resilient flood management (including, but not limited to, an upgrade and rehabilitation of the hydrometric monitoring network and development of a flood forecasting and enhanced the existing early warning system) for the Vrbas River Basin.²

1.2 Impact potential

Scale: Medium

9. The funding proposal highlights various anticipated adaptation impacts resulting from the proposed project across various GCF adaptation results areas. Specifically, the funding proposal references two main project outcomes:

- (a) **Outcome 1:** Climate informed gender-responsive impact-based Flood Forecasting and Early Warning System and increased generation and use of climate data to reduce vulnerability to flood related disasters; and
- (b) **Outcome 2:** Vulnerable communities in Bosnia and Herzegovina, including women, children, people with disabilities, and older persons, are better protected from and less exposed to climate induced flood risks through a combination of structural and non-structural flood risk reduction solutions.

² That baseline project received a "Satisfactory" rating in the independent terminal evaluation report: "The project has achieved impressive results. The 638,600 ha VRB is under improved management for climate resilience through successful transfer of climate adaptation technology and advances to policy and regulatory frameworks. The strong emphasis on institutional strengthening has built an enhanced enabling environment for facilitating follow-up actions. The incremental benefits achieved through the GEF funding is confirmed through the replication that has occurred during the project implementation phase and the high potential and expected investments for upscaling" (annex 18, p. 90).

10. These outcomes are anticipated to be accomplished through three outputs, namely: (1) the implementation of a fully integrated impact-based FFEWS allows for prompt preparation and response; (2) the integration of non-structural flood risk reduction measures and nature-based solutions in sectoral policies and plans, effectively contributing to the protection of people and livelihoods from climate-induced flood risk; and (3) the scaling-up of climate-proof flood protection measures through new and improved national and local investment frameworks increasing the resilience of the most vulnerable groups to climate-induced flooding. These outputs are organized into eleven distinct activities and 35 subactivities.

11. In conjunction, the project interventions are expected to yield the following positive adaptation impacts in Bosnia and Herzegovina, as estimated below in relation to GCF core indicators and adaptation results areas:

- (a) 799,445 direct and 1,977,244 indirect beneficiaries that benefit from reduced vulnerability to loss of life, property and critical infrastructure (core indicator 2);
- (b) 22,516 direct and 741,339 indirect beneficiaries protected from extreme floods in those municipalities where interventions are implemented (adaptation results area 1 (ARA 1));
- (c) 77,929 direct and 1,995,257 indirect beneficiaries benefiting from improved health, well-being, food and water security as a result of enhanced access to timely early warning and climate information (ARA 2);
- (d) 776,929 beneficiaries covered by new or improved early warning systems (ARA supplementary 2.4);
- (e) USD 501.8 million of household assets and USD 109 million of business assets made more resilient to the effects of climate change (ARA 3);
- (f) USD 83.5 million reduction in expected economic losses to households and USD 19 million reduction in business assets (ARA 3 supplementary 3.1);
- (g) 22,516 direct beneficiaries living in buildings that have increased climate resilience (ARA 3 supplementary 2.6); and
- (h) 5,081 ha of land protected from extreme floods (ARA 4).

12. The funding proposal contains a detailed description of the methodology adopted for the Geographic Information System-based risk and vulnerability modelling and mapping and cost-benefit options appraisal that forms the basis of the calculation of these anticipated adaptation impacts (annex E to annex 2). It also contains the results of the socioeconomic analysis across municipalities in Bosnia and Herzegovina (annex D to annex 2) under current as well as expected future climate change conditions, and an economic analysis comparing estimated costs and benefits of the project (annex 3a and 3b). However, the results of this modelling exercise need to be interpreted with caution. First, the iTAP found it somewhat challenging to reconstruct the stated adaptation impact targets of the project. The feasibility study (annex 2) refers to the calibration and validation of the underlying flood risk model but does not contain a comprehensive description of its results to gauge the degree of its precision and thus the confidence in the extent of future floods. Second, the estimated adaptation benefits are based on the detailed climate change flood hydrological and hydraulic modelling for the Vrbas River Basin that was conducted as part of the baseline project. In the absence of detailed hazard mapping for the other river basins, the overall modelling results for the entire proposed project region across five river basins assumes that the climate change scenarios and impact of proposed interventions will be similar for these other river basins (Una-Sana, Bosna, Neretva and Trebišnjica). While this may well be a reasonable assumption, the approach taken is overly simplistic and thus the results cannot be accepted with certainty without consideration of alternative scenarios or some insights related to land cover dynamics, among others, related to these other river basins.

13. Another methodological challenge is the choice of the Representative Concentration Pathway (RCP) scenario 8.5 that is being utilized as the base scenario to inform the design of the proposed project interventions (as highlighted for example by the response of the accredited entity (AE) to the question-and-answer session (Q&A) with the iTAP). In the iTAP's view, it would have been prudent to also model the design of project interventions based on alternative, and equally relevant scenarios such as RCP 4.5 or RCP 6 that include increasingly frequent and intense climate events for example (e.g. intense precipitation resulting in floods and landslides, heatwaves causing fast melting of snow which then trigger floods). These alternative scenarios are likely to have more significant negative impact on livelihoods and business operations over time and may require different interventions to support adaptation. The RCP 8.5 scenario (as a "high business-as-usual scenario") generates a higher likelihood of large catastrophic events triggered by much more significant temperature increases towards the end of the century (akin to floods such as the 2014 flood in Bosnia and Herzegovina that represented a 1 in 500-year event). The RCP 8.5 scenario may result in higher costs for the adaptation measures at the expense of other important interventions. It may also create a false sense of security among stakeholders (e.g. communities, businesses, insurance) that the project interventions will deliver safe results as presumably they are designed to reduce loss of life and property in the event of catastrophic floods. This carries the risk of maladaptation. The iTAP recognizes the fact that the various scenarios track each other in terms of climate change impact in the medium term, and that the European Union (EU) – Bosnia and Herzegovina has EU candidate status and seeks to achieve compliance with the region's *acquis communautaire* – requires application of the RCP 8.5 scenario to comply with the EU Flood Directive; but recommends that the anticipated impact proposed project interventions be considered under alternative climate change scenarios by the AE prior to project commencement (**recommendation**).

14. A major concern when evaluating the expected adaptation impacts of the proposed project relates to the lack of recent local stakeholder involvement in its design. Experience with similar projects strongly suggests that the success and sustainability of flood risk management approaches greatly depend on effective stakeholder consultation, and integration of different views from different groups into project design. While the AE has conducted some limited stakeholder consultations in 2018 and 2019 (21 bilateral meetings with individuals, including mayors, and representatives of civil protection departments and businesses), since then, no further consultations have been conducted. In the Q&A, the AE stated that between 2019 and 2023 consultations with local stakeholders have continued via two UNDP-implemented projects (the GCF-funded "Advance the National Adaptation Plan (NAP) Process for Medium-term Investment Planning in Climate Sensitive sectors in B&H" and the Joint Swiss United Nations Development Programme "Disaster Risk Reduction for Sustainable Development in B&H") but no summaries of these consultations have been made available. There is also no indication of how any feedback received may have been included in the design of the proposed project interventions. In the iTAP's view, this presents a significant risk to the project. Thus, the iTAP recommends the Board adopt a condition as part of its approval of the funding proposal that would require the AE to conduct additional consultations covering the entire project region regarding the design of the project and how the local, entity, national and regional institutions may be involved, with a view towards future sustainability, in the production of the project's results. The AE should submit a summary of the outcomes to the Secretariat (proposed **condition 1**, see below for more information).

15. Likely the most significant challenge for implementation and thus the achievement of the anticipated adaptation impacts is related to the political and institutional complexity of the governance system in Bosnia and Herzegovina, driven (among other things) by i) ethno-political divisions (the country consists of two largely autonomous entities, Republika Srpska and the Federation of Bosnia and Herzegovina, along with the Brčko District that are formed along ethnic lines, with separate government structures and decision-making processes); ii) complicated decision-making processes (the governance structure of Bosnia and Herzegovina

involves multiple layers of authorities, including a central government, entity governments, cantonal governments and municipal governments); iii) vetoes and blocking mechanisms (entities and ethnic groups have the power to veto decisions that go against their interests, resulting in a high level of political gridlock and difficulty in implementing necessary reforms); and iv) a lack of fiscal autonomy (the entities in Bosnia and Herzegovina are heavily reliant on financial transfers from the central government, which creates a financial imbalance and dependence). The funding proposal references these challenges, refers to UNDP's experience in working successfully in this complicated environment and highlights the co-financing commitments that various Bosnia and Herzegovina entities have made as evidence that the project will have significant government buy-in and support. This highlights significant interest in the project and high likelihood of success in continued and formalized stakeholder engagement.

16. In the iTAP's view, however, the most effective mechanism to ensure that the project delivers sustainable results in this complex environment would be to implement the proposed project interventions in close partnership with the relevant entities and other institutions in Bosnia and Herzegovina on a day-to-day basis. The Direct Implementation Modality proposed by the AE, however, does not facilitate such extensive ownership of project implementation by relevant partners in the country (see also section 1.6 on country ownership). Indeed, the funding proposal makes only limited reference to how the AE will involve institutions of Bosnia and Herzegovina in the actual implementation of project interventions. The Project Management Unit will be hosted by the UNDP Country Office rather than a partner institution. (It is worthwhile noting in that context that the Project Implementation Unit supporting the implementation of the structural measures funded by the European Investment Bank (EIB) will be based at the Ministry of Agriculture, Forestry and Water Management of Republika Srpska.) In the iTAP's view, the concentration of execution, implementation and project assurance responsibilities within UNDP presents a material risk for the project and may undermine its potential to achieve the envisaged adaptation impacts in a sustainable manner. To mitigate this risk, the iTAP recommends the Board adopt a second condition (**condition 2**), to be fulfilled before second disbursement, consisting of development of a sustainability plan by the AE. The sustainability plan, building on the updated stakeholder consultations, should outline the various local/entity/national/regional institutions' mandates and roles in the production and maintenance of project results (see below for more information).

17. Finally, under output 3 the funding proposal suggests implementing activity 3.1: "Develop a country-wide investment framework for climate induced flood risk reduction and management including provisions for private sector engagement in climate risk financing". The deliverables under this activity largely consist of feasibility studies, as well as the development of strategies and tools to be developed by consultants. It remains unclear which partners the AE would work with in the development of these deliverables, and how the AE will ensure that these deliverables will translate into concrete results. This is especially relevant when the project proposes to improve the enabling environment (for example, for products related to national disaster insurance) which presumably would require engagement with policymakers and relevant regulators and government agencies, but where specific information on how such engagement will be initiated is not included. The funding proposal also references various public-private partnerships being established and public-private sector climate risk financing mechanisms and tools developed, but it remains unclear what private sector entities would be involved (aside from the co-financier) and who would ultimately utilize the various products and tools developed. In its implementation plan and reporting to GCF, the AE should provide further clarification on these matters to ensure that the project does not just result in studies and tools that do not find practical application (**recommendation**).

18. Overall, the iTAP rates the impact potential for the project as **medium**, with potential for improvement if the issues highlighted above are adequately addressed.

1.3 Paradigm shift potential

Scale: Medium

19. The project aims to transform the current flood risk management in Bosnia and Herzegovina in response to climate change by bolstering institutional capabilities at all levels, enhancing comprehension of risk through improved climate information, incorporating non-structural ecosystem-focused risk management methods, involving the private sector, and ensuring that both public and private investments in flood risk management are resilient to climate impacts. The project anticipates achieving a paradigm shift in flood risk management on five levels:

- (a) Introduction of technological innovations (implementation of fully integrated impact based Flood Forecasting and Early Warning System based on state-of-the-art observation, modelling, forecasting and communication platforms and technologies);
- (b) Capacity development and enhancement (capacity to identify, plan and implement long-term flood risk management strategies at the basin and sub-basin scale);
- (c) Mainstreaming of nature-based flood risk management solutions in Bosnia and Herzegovina (encompassing EbA approaches as well as non-structural solutions);
- (d) Development of long-term climate risk financing (including via the development of municipal risk management plans); and
- (e) Introduction of innovative approaches to flood risk management (including forecast-based financing approaches).

20. The iTAP recognizes the project's potential to substantially improve current flood risk management practices in Bosnia and Herzegovina. For a comprehensive paradigm shift to occur, however, the different project interventions will have to demonstrate sustainability over time. The various commitments that relevant partners of Bosnia and Herzegovina have made with regard to, for example, financing long-term operation and maintenance (O&M) expenses for the hydrometric network equipment procured and installed via this project are a positive demonstration of commitment. Many of the co-financing commitments included in the funding proposal reviewed by the iTAP date back to 2019/2020, however, and should be renewed and resubmitted to GCF, prior to project commencement (**recommendation**).

21. In the Q&A, the AE indicates that it will take a comprehensive approach to ensuring reliability of long-term O&M finance from different (public and private) sources, an approach that the iTAP believes is crucial. The funding proposal, however, lacks clarity in terms of how the project would concretely leverage different funding sources, and on what timeline. It remains unclear, for example, how the project's proposed activity 3.1 would directly generate private-sector funding streams in support of O&M and other flood risk management/early warning system financing requirements, except with regard to exploring the option to introduce insurance products at conceptual level via the production of feasibility studies. The AE should clarify this as part of its regular reporting obligations to GCF (**recommendation**).

22. Apart from the important commitments partners of Bosnia and Herzegovina have provided regarding long-term O&M finance, the funding proposal lacks sufficient detail on how the project will ensure that the various activities it undertakes and that will require sustained attention after project completion will be taken on by the relevant institutions within the country. The AE should generate a detailed sustainability/handover plan, specifying which institutions it will engage and work with to ensure "institutionalized operationalization" (e.g. how the institutions that ultimately will be responsible for the system will be involved in specific tasks, such as developing standard operating procedures) and continuity of the various project activities after project closure (**condition 3**, see below for more information). This plan should be completed and submitted to the Secretariat prior to the second disbursement. This should include all project activities. Thus, the plan to be submitted to the Secretariat should

make specific operational, maintenance and sustainability provisions for each activity to be funded by GCF.

23. A significant portion of the funding that GCF would provide to the project would be in support of implementation of eight non-structural flood risk reduction measures (roughly USD 6,74 million of the USD 14,4 million requested in total). The funding proposal highlights the reality that owing to the predominant use of engineered structural measures in mitigation of flood risk in Bosnia and Herzegovina historically, there is a lack of knowledge about non-structural measures and limited capacity to implement them. The iTAP welcomes the fact that this project (building on the experience with the implementation baseline Vrbas project) intends to further mainstream the utilization of non-structural (i.e. zoning and land-use planning) as well as EbA approaches to flood risk management in Bosnia and Herzegovina. It remains unclear, however, from the funding proposal how that mainstreaming is supposed to be accomplished. The proposed budget for that subactivity will be utilized entirely for consulting fees, and it is unclear how the project will engage with relevant institutions within the country to ensure mainstreaming does take place and will be reflected in relevant investment plans, etc. These aspects should be addressed in the sustainability plan (condition 2).

24. Overall, the iTAP rates the potential for the project to effect paradigm shift as **medium**, with potential for improvement if the issues highlighted above are adequately addressed and the various recommendations are implemented.

1.4 Sustainable development potential

Scale: Medium-High

25. Within the limits of the modelling exercise as discussed earlier, the funding proposal presents a credible pathway for the project to generate the following sustainable development benefits:

- (a) **Environmental co-benefits:** The funding proposal states that the project will generate environmental co-benefits largely via the implementation of EbA strategies to flood risk management. Since the specific EbA measures to be implemented by the project have not yet been defined, it is difficult to conclude with certainty what (if any) specific environmental co-benefits will be generated. However, the iTAP concurs with the AE that the generic examples of EbA strategies highlighted in the funding proposal should in theory generate environmental co-benefits.
- (b) **Economic co-benefits:** The economic co-benefits expected from the project are a function of the losses avoided from floods that are predicted to occur based on the project's flood risk modelling exercise. It should be noted that the socioeconomic modelling contained in the feasibility study is largely based on out-dated data since the study was originally constructed in 2018/2019. For example, the costs for the reconstruction of buildings following floods are based on data calculated by the Centre for Development Bills of Quantities for public buildings in 2015. In the Q&A, the AE provided some indicative information on how price levels have moved since completion of the feasibility study (e.g. based on Consumer Price Index data). Especially with the impact of coronavirus disease 2019 (COVID-19) (e.g. supply chain challenges) and inflation, the modelling may significantly underestimate potential economic losses from floods. Also, and as previously discussed, considering that the flood risk modelling and the intervention design of the project are based on the extreme RCP 8.5 scenario, the modelling generates avoided losses from expected extreme weather events, and thus the results need to be interpreted with some caution. The estimates for avoided losses on business assets is based on a damage function for the non-residential sector in the United Kingdom of Great Britain and Northern Ireland. Bosnia and Herzegovina is an upper-middle income country, and should thus have been compared with upper-middle and lower-middle income countries (e.g. Albania, Botswana). The AE clarified that at the

time the feasibility study was compiled, no data from comparable countries were available. The AE also noted that for the purposes of the damage calculation, only damages to buildings were taken into account. Overall, the iTAP concurs that if implemented successfully and sustainably, the project should generate considerable economic co-benefits for Bosnia and Herzegovina.

- (c) **Social co-benefits:** In the discussion of the Theory of Change (section B.2a) for the project, the funding proposal refers to potential health benefits that the project is expected to engender. The improved health and reduced health risks of the project's beneficiaries are presented as a function of the reduction in the number of flood-related injuries and exposure to waterborne diseases. In general, the iTAP agrees that such health benefits are likely to manifest themselves if the project is successfully implemented and sustainable over time. However, no specific data on the project region is presented in the funding proposal to support this supposition so it is not possible to quantify these potential health benefits. It is also noteworthy that these potential social co-benefits are not discussed again in section D.3 ("Sustainable development") of the funding proposal.
- (d) **Gender-sensitive development co-benefits:** The funding proposal includes an evidence-based gender assessment (annex 8), the results of which have been incorporated into the project's Theory of Change and the design of the specific proposed project activities. For each activity, the funding proposal highlights gender mainstreaming activities. The iTAP welcomes this structured approach to incorporating analysis on existing inequalities into project design across all levels.
- (e) Overall, the iTAP rates the sustainable development impact potential of the project as **medium-high**.

1.5 Needs of the recipient

Scale: Medium-High

26. Following the war (1992–1995), Bosnia and Herzegovina faced a severe economic crisis. The conflict had devastating effects on the country's economy, leading to the destruction of infrastructure, the collapse of industries, and the displacement of the country's population. In this post-war situation, Bosnia and Herzegovina heavily relied on external assistance and aid to recover and rebuild. International organizations, such as UNDP and the World Bank, played a critical role in providing financial support. The EU has also been a significant donor and partner in the country's post-war recovery. Through various financial programmes and assistance packages, the EU has supported the state's economic development, governance reforms and integration processes. The stabilization and association agreement, signed with the EU in 2008, aimed to boost the country's economic and political stability. Remittances from Bosnian citizens living abroad have also played a crucial role in the country's post-war economy. Many Bosnians migrated to Western Europe and the United States of America during and after the war, and their remittances have provided a significant source of income for their families and communities back home.

27. Bosnia and Herzegovina is an upper-middle income country and ranks 79th (out of 185) in the Notre Dame Global Adaptation Initiative Country Index (with a vulnerability score of 0.348 and a readiness score of 0.365). The index assesses the country's vulnerabilities as manageable but improvements in readiness are advisable. As also emphasized in the funding proposal, and despite the external assistance and aid, the post-war economic situation in Bosnia and Herzegovina has been, and remains, challenging. The war deeply impacted the country's industrial base, leading to a decline in manufacturing and limited economic diversification. The complex political system and disputes between the two entities, Republika Srpska and the Federation of Bosnia and Herzegovina, have contributed to a slow pace of structural reforms and hindered economic progress. The impacts of climate change are likely to further stress

Bosnia and Herzegovina's economy, with increasing risk of floods and associated damage being just one manifestation of how climate change may impact the country negatively in the years and decades to come. From that standpoint, the assistance potentially made available by GCF for this project will make an important contribution to strengthening readiness and mitigate growing vulnerabilities to the impacts of climate change, specifically in the context of increasing flood risks.

28. The AE did not include, as part of the funding proposal, an analysis of alternative sources of financing that might potentially be available. The iTAP enquired whether any assistance may be available from the EU, considering that Bosnia and Herzegovina is an accession candidate and does receive support from the regional block to achieve compliance with the EU's *acquis communautaire*. Compliance with the EU Directive on the assessment and management of flood risks (which compels EU member States and candidate countries to apply the RCP 8.5 scenario in their flood risk modelling) is one example of how Bosnia and Herzegovina tries to solidify its status as a potential EU accession country. The AE responded that the EU already makes assistance available to the project via the EIB's concessional financing for structural measures, and that no further (grant) support is available under the Instrument for Pre-accession Assistance.

29. The reliance on external assistance and aid since the end of the Bosnian conflict has also created a level of dependency that the country must reduce to achieve sustainable economic growth. While the proposed project has the potential to significantly improve flood risk management practices in Bosnia and Herzegovina, the proposed Direct Implementation Modality carries the risk of further aggravating such dependency if the relevant entities and other institutions are not properly capacitated to maintain and further extend the "soft" and "hard" flood risk management infrastructure that is being funded by the project. As noted earlier, the current lack of specifics on how the AE will engage with relevant institutions within the country during project implementation and the lack of a proper handover (or exit) plan is of concern to the iTAP.

30. Overall, the iTAP rates the potential for the project to serve the needs of the recipients as **medium-high**.

1.6 Country ownership

Scale: Medium-High

31. **Coherence with existing policies:** The proposed project is well aligned with national climate change strategies and plans of Bosnia and Herzegovina. Specifically, the funding proposal responds to the country's nationally determined contribution for 2020–2030 that specifically references the need for strengthening adaptive capacity in light of growing, climate change-induced flood risk. The funding proposal also refers to the Climate Change Adaptation and Low Emission Development Strategy of Bosnia and Herzegovina as adopted by the country's Council of Ministers in 2013 (updated in 2020) that highlights increasing flood risks and lays out various adaptation priorities which align with the overall objectives of the funding proposal. Finally, the funding proposal dovetails with the NAP concluded in 2021 and that specifically references increasing flood risks (chapter 7) and the need for more proactive flood risk management in response to increasing climate change hazards.³ The iTAP would also like to highlight the significant share of co-financing commitments received from entities and institutions in Bosnia and Herzegovina which signify buy-in and ownership (even though many of these commitments, as previously pointed out, are now dated and will have to be renewed).

³ The NAP was developed with support from the GCF-funded "Advance the National Adaptation Plan (NAP) Process for Medium-term Investment Planning in Climate Sensitive sectors in B&H" (Readiness and Preparatory Support Project).

32. **Capacity of the AE:** UNDP has implemented USD 733 million worth of programmatic activities between 1996 and 2022 in Bosnia and Herzegovina, and thus has a long and extensive track record working in the country. Through the baseline project, as well as other related engagements (“Advance the National Adaptation Plan (NAP) process for medium-term investment planning in climate sensitive sectors in B&H” (GCF Readiness and Preparatory Support Project) and “Scaling-up Investment in Low-Carbon Public Buildings”, co-financed by GCF (FP051)), UNDP has garnered significant experience working on adaptation issues within the country.

33. **Stakeholder consultations and engagement:** As previously noted in section 1.2, the iTAP notes with concern the limited local stakeholder consultations that have been conducted for the preparation of this funding proposal that potentially undermines the project’s impact proposal, but also ownership (and thus long-term sustainability). The three proposed conditions provide a route to rectify this.

34. The iTAP considers the alignment of the project with host country government priorities to be high but assesses the AE’s track record as medium. Overall, the iTAP rates country ownership of the funding proposal as **medium-high**.

1.7 Efficiency and effectiveness

Scale: Medium

35. The funding proposal highlights the following three aspects to demonstrate the economic soundness of the project:

- (a) A significant share of co-financing is from other public and private sector sources (GCF funding represents only 19.8 per cent of the estimated project cost);
- (b) The results of the economic analysis show an economic internal rate of return of 19.3 per cent and thus demonstrate that the individual project interventions are economically efficient (with the exception of one); and
- (c) A cost of USD 80.50 per direct beneficiary for climate-resilient flood risk protection and early warning system support provided by the project which the AE deems as “comparable” to other projects of similar nature.

36. Overall, the iTAP agrees that the proposed project performs satisfactorily with regard to the principles of economy and efficiency. However, the iTAP would like to make the following observations:

- (a) Regarding the first point, and as previously noted, the iTAP commends the large share of co-financing committed to this project from other sources, and particularly from Bosnia and Herzegovina’s partners. This signifies buy-in and ownership. The iTAP notes that the funding package for the project also includes a concessional loan from EIB with a volume of EUR 21.2 million (approximately USD 22.65 million at the exchange rate at time of writing), while the funding portion from GCF is expected to be made available in grant form. The AE notes that none of the interventions co-financed by GCF will result in any quantifiable earnings or savings for entities or local governments in Bosnia and Herzegovina, that they are predominantly focused on providing technical assistance, and that as a result of the current economic crisis (aggravated by the COVID-19 pandemic), the government of Bosnia and Herzegovina is not in a state to raise additional external debt. However, the iTAP would like to emphasize that i) Bosnia and Herzegovina (with a 2023 gross domestic product per capita of EUR 7,366 (projection)) is considered an upper middle-income country by the World Bank; and ii) per the International Monetary Fund’s latest 2023 Article IV Consultation Staff Report,⁴ the country’s external debt is

⁴ International Monetary Fund 2023. Bosnia and Herzegovina. 2023 Article IV Consultation – Press Release; Staff Report; and Statement by the Executive Director for Bosnia and Herzegovina. See

assessed to be sustainable over the medium-term. The external debt to gross domestic product ratio has been declining consistently over the past few years (to about 52 per cent in 2022). In that context, the International Monetary Fund states “[...] that here is room to increase IFI [international financial institution] financing to boost investment and growth potential”.⁵ Investments in adaptation to enhance the country’s capacity to avoid economic losses as a result of flood events should be considered important investments into the country’s growth potential, and as such it is not immediately obvious to the iTAP why GCF co-financing should be provided in the form of a grant, especially considering the pressing needs for adaptation and mitigation financing in least developed countries that GCF needs to respond to.

- (b) Regarding section 36 (c) above, the AE does not provide any cost data from comparable projects, and thus makes validation of its claim difficult. Since adaptation investments also tend to be highly bespoke to the specific country and regional context in which they are made, the iTAP is not clear how presentation of such average cost per beneficiary data provides a useful indication of the economic soundness of the project.

1.8 Overall remarks from the independent Technical Advisory Panel

37. The iTAP endorses this funding proposal, suggesting that the Board approve it subject to the three conditions specified below.

38. Overall, the funding proposal is comprehensive and ambitious, and has potential to generate significant adaptation as well as broader sustainable development benefits for Bosnia and Herzegovina. The project, if successfully implemented by and with local, entity and national institutions, will fundamentally upgrade flood risk management in the country across various dimensions, with significant socioeconomic, environmental and social benefits for the target population in the project region, and for the population of Bosnia and Herzegovina as a whole.

39. The iTAP welcomes the significant share of co-financing raised from other international and local funding sources which underscores the relevance of the project as well as the degree of ownership by key public and private sector stakeholders in the country. UNDP has considerable experience in planning and implementing adaptation projects in Bosnia and Herzegovina, and is thus well-positioned to lead the implementation of this project.

40. In order to ensure that the project can unfold its full impact and paradigm shift potential, the iTAP recommends that the Board include the following condition to disbursement to its potential approval of this funding proposal:

1.8.1. Stakeholder Consultation:

41. The following condition to first disbursement should be fulfilled by the Accredited Entity, in form and substance satisfactory to the GCF:

- (a) “Deliver to the GCF Secretariat a summary report of stakeholder consultations conducted pursuant to the covenant below including recommendations for project design and implementation modalities.”

42. Inclusion of the following covenants in the FAA:

- (a) “Prior to the first disbursement of GCF proceeds, the Accredited Entity shall conduct local stakeholder consultations across the entire project area, including in each of the target river basins regarding project design. The local consultations shall be conducted

<https://www.imf.org/en/Publications/CR/Issues/2023/09/05/Bosnia-and-Herzegovina-2023-Article-IV-Consultation-Press-Release-Staff-Report-and-538832> (accessed 10 September 2023).

⁵ Ibid., p.61.

with a view towards: (i) how flood risk reduction measures and nature-based solutions are to be mainstreamed into sectoral policies and plans of Bosnia and Herzegovina; and (ii) how the flood forecasting and early warning systems shall be operationalised and sustained.”

- (b) “Prior to submission of a summary report of stakeholder consultations to the Fund, deliver to the GCF Secretariat a copy of such summary report for the review and approval of the Fund.”

1.8.2. Sustainability Plan:

43. The following condition to second disbursement should be fulfilled by the Accredited Entity, in form and substance satisfactory to the GCF:

- (a) “Deliver to the GCF Secretariat a sustainability plan developed in close consultation with relevant entities and local government institutions, specifying their respective roles in project implementation, including policy mainstreaming and the operationalising and sustaining of the flood forecasting and early warning systems..”

44. Inclusion of the following covenant in the FAA:

- (a) “Deliver to the GCF Secretariat a copy of the sustainability plan endorsed by the relevant institutions in Bosnia and Herzegovina for the review and approval of the GCF Secretariat.”

1.8.3. Operationalising Sustainability:

45. The following condition to third disbursement should be fulfilled by the Accredited Entity, in form and substance satisfactory to the GCF:

- (a) “Deliver to the GCF Secretariat an updated organization structure for the project for implementation of the sustainability plan.”

46. Inclusion of the following covenant in the FAA:

- (a) “Deliver to the GCF Secretariat a detailed project organization structure, with equipment property rights assigned or handed over to the various institutions in Bosnia and Herzegovina that will operate and maintain it,for the review and approval of the GCF Secretariat.”

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP216)

Proposal name:	Scaling up climate resilient flood risk management in Bosnia and Herzegovina
Accredited entity:	United Nations Development Programme
Country/(ies):	Bosnia and Herzegovina
Project/programme size:	Medium

Impact potential

Paragraphs 12-13: UNDP acknowledges the iTAP's observations regarding the methodology for the flood risk and vulnerability modelling and mapping that forms the basis of the calculation of the anticipated adaptation impacts.

We are confident in the rationale behind the choice of scenario RCP8.5 for the estimation of impact of climate change-driven flooding in BiH, which is projected to result in increased frequency and intensity of flooding with the highest flood risk, highest flood levels and highest discharge of all CC scenarios. Large flood events such as expected under RCP8.5, are likely to result in substantial damage to infrastructure, agriculture, and communities. The approach to adapting to these projections of future flooding risks is to invest in flood defenses, early warning systems, and sustainable land use planning.

The modelled 100yr+CC (using RCP8.5) flood levels for Vrbas basin were found to be similar to the flood levels observed in the 2014 flood event. Given that the 2014 flood impacted all target basins and represented the worst flooding in each of these basins, it is reasonable to extrapolate the RCP8.5 flood impact modelled in Vrbas basin, to other basins. We are therefore confident that the use of RCP8.5 and its extrapolation to other basins, provides a robust climate change scenario for the assessment of flood risk (identification of population and assets at risk), the design of flood protection measures (structural, non-structural and NbS), and the impact of these adaptation measures. As noted by ITAP, this approach also follows the EU flood directive and was requested by stakeholders given they had already in the present day experienced what should have been an extreme flood under a future extreme scenario. We also note that since RCP8.5 gives the highest flood levels (of those tested) and that we expect it to produce the largest increases in rainfall intensity between the different RCP scenarios, the protection measures are expected to provide protection under both RCP6 and RCP4.5.

Whilst flooding under moderate (RCP4.5) and intermediate (RCP6) climate change scenarios are likely to result in smaller magnitude, lower intensity events floods, they may occur more frequently and therefore (as noted by ITAP) still cause damage due to their shorter response time for authorities and communities to prepare, respond, and recover. These types of events are likely to impact people and communities on an annual and seasonal basis and could be highly disruptive and economically costly. For these types of flood events, sustainable land use (through flood zoning, flood resilient agric. and forestry etc.), early warning systems (including CBEWS), preparedness (including FbF), insurance, investment planning (which will also factor in the economic impact and management of smaller more frequent floods), awareness raising and capacity building are most appropriate and are being implemented by the project. Their

design and implementation will, therefore, need to be fully informed by all relevant climate change scenarios to cover the full range of adaptation measures that will be required for different types of events.

During year 1 and year 2 of implementation, the WBIF hazard maps for all basins will be enhanced and, in line with flood modelling and mapping best-practice, the impact on flooding of all relevant climate change scenarios will be simulated using individual calibrated hydraulic models for all basins, the results from which will be used to inform detailed design and implementation of flood risk reduction interventions. As recommended by ITAP, this will include using RCP4.5 for the hydrological simulations, but unfortunately RCP6.0 is not available for the EU-CORDEX regional climate models used to provide the climate forcing for the hydrological modelling. For the reasons noted above, RCP8.5 is likely to remain most relevant for the design and implementation of protection measures, while the lower impact RCP4.5 scenario will additionally inform other flood risk reduction measures.

Paragraph 14: UNDP fully agrees that regular and effective stakeholder consultations and engagement are critically important for the success and sustainability of the flood risk reduction measures and for achieving the anticipated adaptation impact. This has been achieved through continuous consultations with the institutional stakeholders and local communities which have been ongoing throughout project preparation. Most recently, all institutional stakeholders (entity ministries of agriculture, forestry and water management, water agencies, civil protection units) were consulted through the GCF NDA in the process of securing the updated no-objection letter in June 2023. Consultations with the local communities to be targeted by this project were carried out during the project design and continued more recently in 2021-2023 via two UNDP implemented projects on the National Adaptation Plan (NAP) and on disaster risk reduction action.

As noted above, UNDP agrees with the iTAP's suggestions to reinforce stakeholder and community consultation processes, which are planned during the start of the project. The national and local stakeholder consultations will be carried out and reported during the project inception phase as part of the planned baseline assessments and then regularly during implementation through the stakeholder engagement mechanisms outlined in the stakeholder engagement plan. Please refer to our detailed feedback on the iTAP condition 1 in the Overall remarks Section.

Paragraph 16: We are in agreement with iTAP highlighting that the sustainability of the project's results and impacts in the complex institutional context requires implementation of the project interventions in close partnership with the relevant entities and institutions in Bosnia and Herzegovina. Please note, that unlike the EIB loan project, which operates only in one entity (Republika Srpska) and focuses on the implementation of infrastructural measures, the GCF project covers the whole territory of the country and a wide arrange of policy, regulatory, institutional, capacity building, structural and non-structural investments. Hence, the GCF PMU would need to work across entities and secure coordination across various institutions and workstreams.

UNDP's Direct Implementation Modality offers ample opportunities and concrete instruments to ensure and facilitate such extensive engagement and ownership by relevant partners in the country, while also mitigating the political and institutional risks to implementation noted by the iTAP. Over the decade of operating in the country, the UNDP Country Office in B&H has established operational practices facilitating engagement of the partner institutions in the project's implementation, which will apply to the GCF project as follows:

- National institutions participate in the development and validation of the ToR's for all activities relevant to their institutional mandates (e.g. for the procurement of hydro-meteorological stations, development of FFEWS, design of the non-structural measures, etc.).
- The entities and their subordinate agencies and institutions will be members of the project working groups for the development of regulatory and legal documents, operational and financing strategies and plans (including O&M plans and investment strategy and plans), guidelines and studies. Every relevant institution will be asked to nominate a member to the working group/s and all deliverables/products will have to be approved by the working groups, prior to being sent to institutions for official adoption/approval.
- Republika Srpska Ministry for Agriculture, Forestry and Water Management will nominate a liaison officer to work closely with UNDP PMU on the implementation of infrastructural measures, including ones financed from the EIB loan.
- Water authorities in both entities will nominate Field Officers who will be responsible for the day-to-day inspection and monitoring of sites. The Field Officers are employees from the water agencies who are experienced with field operations and infrastructure delivery, inspection of sites, ensuring implementation of laws and regulations.
- Selected municipalities will have designated officers who will be engaged in the development of community investment and preparedness plans. These plans will, in turn, be adopted by municipal authorities.

Nevertheless, we appreciate and accept the condition proposed by iTAP consisting of development of a sustainability plan, building on the updated stakeholder consultations, which will outline the various local/entity/national/regional institutions' mandates and roles in the production and maintenance of project results.

Paragraph 17: The work on the risk finance and public-private partnerships represents one of the innovative areas in B&H and requires a considerable investment into enabling environment, leveraging partnerships and capacity building before these financing instruments could be picked up by the finance/private sector. As noted by the iTAP, the GCF project will invest in such enabling environment. The following specific deliverables will be produced under the Activity 3.1:

- *5 strategies and plans for long-term ongoing investment frameworks that would be required to address flood risk to each sector;*
- *Sector-specific information dissemination portals as part of the information platform to be developed under 1.4.3;*
- *2 risk financing and risk transfer strategies for private sector engagement in risk financing and risk transfer mechanisms for entity-level flood risk financing and resilience strategy;*
- *3 natural disasters products developed;*
- *3 Feasibility studies on micro-insurance, regional disaster risk insurance program and catastrophe bonds;*
- *4 sector specific awareness campaigns;*

- *2 tools for appraisal-led design for structural and non-structural FRM measures based on cost benefit analysis (CBA) approaches.*

Additionally, these deliverables are supported by activities 1.2 and 1.3 which will develop the climate risk and vulnerability maps and impact-based forecasting which is needed to underpin an assessment of risk and related financing options. These activities will also engage stakeholders to ensure that sector-specific risks (particularly the agriculture and hydro-power sectors) and community-related risks are co-developed through local, national and regional forums (e.g. the national and regional climate outlook forums), thereby embedding these stakeholder needs in forums and processes beyond the immediate project related activities. The Logical Framework for the proposal, furthermore, includes the following indicators and targets for Activity 3.1:

- *2 Investment frameworks for climate induced flood risk reduction and management;*
- *3 flood risk financing mechanism in place;*
- *2 MoUs with private/productive sector actors*

The latter targets represent specific and tangible commitments and will be reported to GCF through the regular reporting instruments as recommended by the iTAP. In view of the above, we accept the iTAP recommendation and will be tracking and reporting on the progress and results under the Activity 3.1.

Paradigm shift potential

UNDP appreciates the recognition by the iTAP of the project's potential to substantially improve current flood risk management practices in Bosnia and Herzegovina. We agree that securing long term sustainability is critical for achieving the paradigm shift.

Paragraph 20: We are pleased to confirm that the co-financing commitments have been validated by GCF NDA with all the co-financing partners during renewal of NoL in 2022-2023. UNDP has been in regular contact with the project co-financiers. All the earlier co-financing commitments are valid. However, in response to the iTAP recommendation we will be reaching out to the co-financiers and will obtain the updated co-financing letters from them as suggested by the iTAP. Therefore, the iTAP recommendation is accepted.

Paragraph 21: We also note and accept the iTAP recommendation related to defining and reporting on the project's solutions aimed at ensuring reliability of long-term O&M finance. The project proposal defines a menu of options/financing sources for securing long term sustainable funding for the O&M. These options include above all public finance (improved budget allocations, ODA and IFIs, PPPs with e.g. hydropower operators, etc) and private sector finance (corporate sponsorships, user fees, partnerships with insurance sector, joint ventures engaging e.g. data and IT companies). The project will carefully consider the context, local regulations, and stakeholder needs when selecting financing mechanisms and developing long-term O&M financing strategies. Often, a combination of several mechanisms may be necessary to ensure a stable and diverse funding base for the long-term sustainability of the national hydrometric observation network and forecasting and early warning system. We accept the iTAP recommendation and will be reporting on this element of the project as part of our regular reporting obligations to GCF.

Paragraph 22: UNDP notes iTAP comments on the need for a detailed sustainability/handover plan specifying which institutions it will engage and work with to ensure "institutionalized operationalization" and continuity of the various project activities. We agree with the

proposed condition for the sustainability plan which will be developed in close consultation with relevant entities and local government institutions.

Paragraph 23: UNDP noted with thanks that iTAP welcomed this project's plans to mainstream the utilization of non-structural and EbA approaches to flood risk management in Bosnia and Herzegovina. Such mainstreaming will involve a coordinated effort among various institutions and government partners, all of which are key project partners. The EbA mainstreaming activity (Activity 2.3) will build upon the outcomes of several other activities which are interlinked. Therefore, the stated mechanisms of delivering Activity 2.3 (i.e. the use of consultancy services) is one part of an integrated approach to mainstreaming EbA into FRM and a culmination of other project activities. Please also note a high co-financing ratio for the activities related to the non-structural and EbA measures (79% government co-finance to 21% GCF financing), which demonstrates the strong government commitment to embedding NbS into FRM. GCF funds will be used to provide relevant technical expertise and inputs to create enabling conditions for the implementation of non-structural and EbA solutions and the mainstreaming of EbA into FRM policy and practice. The following is a general roadmap for achieving mainstreaming.

- *Policy Framework.* The entity governments will take the lead in developing the policy framework that integrates EbA into flood risk management which may involve creating or amending relevant entity laws and regulations. Key ministries, all project board members, will collaborate to ensure a holistic approach to EbA in flood risk management and the mainstreaming into policy.
- *Raising Awareness and Capacity Building.* Awareness campaigns, workshops, and training sessions to educate stakeholders about the benefits of non-structural measures and EbA in enhancing climate resilience and reducing risks (Activity 2.3).
- *Case studies.* The implementation of 21 non-structural and 5 EbA measures (Activity 2.2) will provide the basis for the development of methodologies, and best practices guidelines which would form the technical basis for policies. In addition, the project will collaborate with research institutions to gather scientific data on ecosystems, flood risk, and climate change impacts and provide evidence-based research to support policy advocacy efforts, and with environmental NGOs which can provide valuable expertise in assessing the ecological aspects of EbA solutions and can advocate for stronger EbA policies and regulations and hold the government accountable for their implementation.
- *Quantify Benefits and Co-Benefits.* Using the socio-economic modelling tool to be developed, the project will quantify the multiple benefits of non-structural measures and EbA, including flood risk reduction, improved water quality, carbon sequestration, and enhanced ecosystem services. Through the project-implemented measures the project will provide evidence of the cost-effectiveness of non-structural and EbA solutions (under Activity 2.2), based on economic analyses comparing NbS to traditional structural approaches and highlighting the potential for reducing maintenance and operational costs over time. This can demonstrate to stakeholders how non-structural measures and EbA solution can help achieve broader sustainability goals.
- *Link to Local Development Goals.* Non-structural and EbA measures will be included in the investment planning of high-risk municipalities as part of their overall FRM investment planning frameworks (under Activity 3.2), and will ensure alignment with local development goals, such as improved water management, sustainable agriculture, enhanced biodiversity, and resilient infrastructure.

- *Access to finance and Public-Private Partnerships (PPPs).* The project is developing the risk financing mechanisms which will likely include PPPs. The project will examine partnerships between government entities and the private sector to jointly invest in non-structural measures and EbA projects. Feasibility studies will be used to demonstrate how sharing risks, costs, and benefits in a balanced manner could make non-structural and EbA Solutions attractive. Risk financing feasibility studies may also examine and identify incentive mechanisms for implementing non-structural and EbA solutions, including fiscal incentives, subsidies, grants, or tax breaks for businesses and individuals that invest in non-structural measures and EbA projects and reward for entities that contribute to ecosystem restoration, conservation, and sustainable land use (Activity 3.1).
- *Enabling environment and capacity building.* The project is developing the technical specification, standards and protocols for design and implementation of non-structural measures and EbA solutions as well as a knowledge portal for EbA. The project is also building capacity, which will enable and encourage the uptake of non-structural and EbA approaches (Activity 2.3) and will work with training institutions to build the capacity of government officials, community leaders, and practitioners in EbA approaches (Activity 2.4). The project is establishing a knowledge portal and common resource centre to collect, collate and share knowledge and fill these gaps to advance the uptake of Non-structural and EbA solutions.

As noted above, UNDP agrees with the condition of developing a sustainability plan addressing above all the roles and responsibilities of various national and local actors in mainstreaming and scaling up non-structural and ecosystem-based solutions in B&H. We therefore confirm that such plan will be developed (please also see our feedback on the proposed Condition 2).

Sustainable development potential

UNDP appreciates iTAP's recognition of a credible pathway for the project to generate sustainable development benefits including environmental, economic, social, and gender mainstreaming benefits. As discussed with the iTAP and noticed in the iTAP's review, the project is likely to result in greater economic [co-]benefits than presented in the proposal because the economic benefit analysis applied conservative methodologies in assessing projected flood losses. The iTAP welcomed the structured approach to incorporating analysis on existing gender inequalities into project design across all levels. The observations from iTAP are well noted.

Needs of the recipient

UNDP appreciates the assessment of this investment criteria as Medium-High.

Paragraph 30: UNDP's Direct Implementation Modality provides ample opportunities for engaging national and local partners in the project implementation and building capacities of the relevant national entities and institutions to maintain and further extend the "soft" and "hard" flood risk management infrastructure that is being funded by the project.

Please refer to our responses in the Impact Potential Section regarding the specific ways the national institutions will be engaged in the project implementation, delivery of the project activities, and validation of the project results.

Finally, as we noted above UNDP agrees with the iTAP condition to produce a delayed sustainability/handover plan in close consultation with relevant entities and local government institutions prior to the second disbursement.

Country ownership

UNDP appreciates the iTAP's assessment of the alignment of the project with host country government priorities as high. The iTAP also highlighted the significant share of co-financing commitments received from entities and institutions in Bosnia and Herzegovina which signify buy-in and ownership.

Paragraph 34: UNDP agrees with the critical importance of stakeholder consultations and engagement for the success and sustainability of the project highlighted by the iTAP. Continuous consultations with the institutional stakeholders and local communities have been ongoing throughout the project preparation and until recently. However, as noted above, UNDP agrees with the iTAP's suggestions regarding the reinforced stakeholder and community consultation process. Please refer to our detailed feedback on the iTAP condition 1. The national and local stakeholder consultations will be carried out during the project inception phase as part of the planned baseline assessments and then regularly during the implementation through the stakeholder engagement mechanisms outlined in the stakeholder engagement plan and in the full proposal.

Efficiency and effectiveness

We appreciate iTAP's observation that the proposed project performs satisfactorily with regard to the principles of economy and efficiency. The iTAP also commended the large share of co-financing committed to this project, including a concessional loan from EIB and co-financing from Bosnia and Herzegovina's partners, which signifies buy-in and ownership.

Paragraph 37: We concur with iTAP's observations regarding the existing pressures on the B&H post-war economy described in the iTAP review for the Needs of the Recipient (paragraph 28). As we also demonstrated in the financial analysis (Annex III(c) to the full proposal), the existing economic risks and fiscal pressures further aggravated by COVID and the Ukraine-Russia conflict limit the ability of B&H to borrow. The shortfall in expenditure in B&H on climate change adaptation is due to the lack of financial capacity of the B&H Government to fund such activities without assistance.

The GCF co-financing framework includes a significant co-financing from the Government including the EIB loan resources, which are utilized by the government to implement structural flood protection measures resulting in reduced damages and losses from floods. GCF co-financing will be used to create enabling environment, establish/improve risk knowledge and information baseline, mainstream innovative EbA solutions, and deliver capacity building required to achieve transformative shift in flood risk management in the country. The GCF support secures activities that enable and enhance the current and future flood risk reduction and protection interventions in the context of climate change. Therefore, we believe that the GCF grant resource are requested and justified for this intervention.

None of the activities or outputs funded by GCF clearly result in direct and quantifiable earnings or direct and quantifiable savings to the project owners or the project beneficiaries. All the activities and outputs of this proposed project result only in non-attributable savings that are of public good in nature to the larger community of at-risk population in B&H. Hence, taking into consideration the factors such as the public good nature of benefits arising out of the project, climate change being the key driver, economic, geographic and demographic distribution of the target population, impact of COVID-19 and energy crisis, B&H's current

economic situation, the persistent fiscal risks and the catalytic nature of GCF grants, the following has been concluded:

1. There is significant co-financing from Government of B&H and there is no incremental ability to stretch their contributions owing to budgetary and fiscal consolidation reasons.
2. The nature of the benefits does not accommodate repayment of capital in whatever form or serviceability of a loan instrument.
3. The nature of the beneficiaries and the level of essentiality of the service do not accommodate repayment of capital in whatever form or serviceability of a loan instrument.
4. Increase in government spending to address COVID-19 pandemic and energy crisis has strained public finances.

Overall remarks from the independent Technical Advisory Panel:

We thank the iTAP for the endorsement of the proposal noting the conditions and recommendations provided in the review. We appreciate iTAP's acknowledgement of the comprehensive design, ambition, potential to generate significant adaptation and sustainable development benefits for Bosnia and Herzegovina, relevance and ownership of the proposal signified by the strong partner co-financing.

iTAP condition 1:

The following condition to first disbursement should be fulfilled by the Accredited Entity, in form and substance satisfactory to the GCF: "Deliver to the GCF Secretariat a summary report of stakeholder consultations conducted pursuant to the covenant below including recommendations for project design and implementation modalities."

Inclusion of the following covenants in the FAA:

- a) *"Prior to the first disbursement of GCF proceeds, the Accredited Entity shall conduct local stakeholder consultations across the entire project area, including in each of the target river basins regarding project design. The local consultations shall be conducted with a view towards: (i) how flood risk reduction measures and nature-based solutions are to be mainstreamed into sectoral policies and plans of Bosnia and Herzegovina; and (ii) how the flood forecasting and early warning systems shall be operationalised and sustained."*
- b) *"Prior to submission of a summary report of stakeholder consultations to the Fund, deliver to the GCF Secretariat a copy of such summary report for the review and approval of the Fund."*

As noted in our responses above, UNDP fully agrees with the spirit of this condition. Stakeholder consultations and engagement are critically important for the success and sustainability of the flood risk reduction measures and for achieving the anticipated adaptation impact. As detailed in our responses under the Impact Potential section (paragraph 14), continuous consultations with the institutional stakeholders and local communities on the proposed project have been ongoing throughout the project preparation. Most recently, all institutional stakeholders (entity ministries of agriculture, forestry and water management, water agencies, civil protection units) were consulted through the GCF NDA in the process of securing the updated no-objection letter in June 2023.

The next round of detailed stakeholder consultations aimed at validating and vetting the project strategy, design and implementation modalities by the national and local stakeholders

is planned as part of the baseline assessments phase. The PMU will be set up by that time and will be engaged in the stakeholder consultations. Local community consultations will be carried out in all 11 municipalities located in 6 river basins targeted by the project. During the project inception phase, the national and local stakeholder consultations will be carried out and will be documented and reported to GCF Secretariat as part of the planned “baseline assessments” milestone within twelve months after FAA Effective Date. We therefore suggest adjustment to the proposed timeline for the covenant while we fundamentally agree with its content.

iTAP condition 2:

The following condition to second disbursement should be fulfilled by the Accredited Entity, in form and substance satisfactory to the GCF: “Deliver to the GCF Secretariat a sustainability plan developed in close consultation with relevant entities and local government institutions, specifying their respective roles in project implementation, including policy mainstreaming and the operationalising and sustaining of the flood forecasting and early warning systems..”

Inclusion of the following covenant in the FAA: “Deliver to the GCF Secretariat a copy of the sustainability plan endorsed by the relevant institutions in Bosnia and Herzegovina for the review and approval of the GCF Secretariat.”

We accept the condition, as per the elements outlined. We note that the sustainability plan as proposed by the iTAP will be developed in close consultations with the relevant entities and local government institutions and will encompass, above all, respective roles in project implementation, including policy mainstreaming and the operationalizing and sustaining of the flood forecasting and EWS. The plan will ensure “institutionalized operationalization” and continuity of the various project activities after the project closure. It will be important to include solutions/tools for the long-term financial sustainability of the project results and O&M in this plan. These solutions/tools will be developed throughout the project implementation. Therefore, we suggest shifting the fulfilment of this condition to the third disbursement. We also suggest integrating the detailed implementation structure for the sustainability plan discussed below under the Condition 3 in this plan (please also see below our feedback on Condition 3).

iTAP condition 3:

The following condition to third disbursement should be fulfilled by the Accredited Entity, in form and substance satisfactory to the GCF: “Deliver to the GCF Secretariat an updated organization structure for the project for implementation of the sustainability plan.”

Inclusion of the covenant in the FAA: “Deliver to the GCF Secretariat a detailed project organization structure, with equipment property rights assigned or handed over to the various institutions in Bosnia and Herzegovina that will operate and maintain it, for the review and approval of the GCF Secretariat.”

Please note that the project organization structure is defined in the full proposal reflecting the key roles in the project governance and management in line with UNDP policies. As recommended by the iTAP, we will further detail the structure for the implementation of the sustainability plan, including roles and responsibilities for operationalization of the O&M across various national organizations. We suggest that this detailed implementation structure for the sustainability plan is developed, discussed, endorsed and shared with the GCF

Secretariat together with the sustainability plan itself. We also suggest that this structure is encompassed in the sustainability plan. We therefore recommend that conditions 2 and 3 could be merged into one condition/covenant with the same implementation timeline as proposed here for Condition 3.

However, in relation to the equipment property rights please note that the transfer of assets is regulated by the SBAA process between UNDP and the Government. Clause 23.4 of the AMA between UNDP and the GCF stipulates the following in this respect: "In relation to a Funded Activity that is a grant financed in whole or in part with GCF Proceeds, if any part of such grant is used to purchase any durable assets or equipment used to implement the relevant Funded Activity (such as vehicles or office equipment), upon completion of the Funded Activity or termination of the relevant FAA in accordance with its terms, the Accredited Entity shall take such steps in relation to such assets or equipment which it reasonably deems in the best interest of the continued operation of the Funded Activity taking into consideration the objectives of the Fund and the terms of the applicable SBAA." UNDP therefore believes that the sustainability plan, while providing a general framework for roles in project implementation, including policy mainstreaming and the operationalizing and sustaining of the flood forecasting and early warning systems, should not preempt the rights and obligations of UNDP under the SBAA. Given that this is already fully covered under the proposed condition for third disbursement, UNDP cannot agree to the additional covenant proposed under this Condition.



ANNEX 8

Gender Assessment and Gender action plan

Scaling up Climate Resilient Flood Risk Management in Bosnia and Herzegovina

July 2023

I. Introduction

The GCF project will address the increasing vulnerability of BiH communities and livelihoods, to intensified climate-induced hydro-meteorological flood-related disasters (pluvial, fluvial, and torrential). One of the root causes of increasing vulnerability and damages from floods is lack of quality climate data and flood forecasting and early warning system. As a result, the BiH authorities are unable to make qualified decisions and the affected population cannot effectively cope with the risk. Lack of climate information and risk knowledge increase the vulnerability and costs from flooding.

The GCF project will build resilience of BiH communities and livelihoods to climate-induced floods by catalyzing a nation-wide shift from ad hoc, a posteriori response approaches to holistic a priori flood risk management and adaptation approaches based on enhanced climate information, risk knowledge, flood forecasting and early warning.

The project will scale-up successful prototypes tested under the UNDP supported GEF financed project “Technology transfer for climate resilient flood management in Vrbas River Basin”.

This Gender Assessment and Gender Action Plan has been prepared for the submission to the Green Climate Fund with the proposal “Scaling up climate resilient flood risk management in BiH”. The assessment aims to provide an overview of the gender situation in Bosnia and Herzegovina, identify gender issues that may be relevant to the project context, to develop recommendations for advancing gender equality and women’s empowerment through the financed activities, and to inform the development of the project’s gender action plan. Individual needs and contexts in a disaster situation are examined through the lenses of vulnerability, visibility, and voices. The analysis follows social vulnerability approach and further defines whose voice is seen as the default one, and whose is the most visible perspective. The assessment was based upon available data from the Agency for Statistics of Bosnia and Herzegovina, RS and FBiH, studies conducted by the Government of Bosnia and Herzegovina, international organizations, civil society organizations and any other sources as UNDP Technology transfer for climate resilient flood management in Vrbas River Basin project and includes:

1. Undertaking a desktop review and aligning approaches in this proposal with the national priorities of Bosnia and Herzegovina.
2. Incorporating information and lessons learned from past studies and assessments on gender in Bosnia and Herzegovina from the Government of Bosnia and Herzegovina, the United Nations, civil society organisations, and multilateral development banks.
3. Conducting stakeholder consultations, including consultations with the Agency for Gender Equality of Bosnia and Herzegovina and the two gender centers (of Republika Srpska and Federation BiH), and engaging women/men/youth affected by the project and incorporating all points raised; and
4. Integrating gender considerations in the project indicators, targets, and activities, identifying women as leaders and decision-makers

While it is noted that the Gender Analysis and Action Plan should specifically identify and analyze the most critical issues relevant to the project and whether the needs and priorities of an equal

number of women/girls/men/boys are being assessed, this Gender Analysis and Action Plan is based on available published and grey literature, consultation with civil society organizations participated in recovery project after the floods happened in 2014 in Bosnia and Herzegovina and Gender Agency of Bosna and Herzegovina. Further an institutional gender analysis has been undertaken (although there will be need for deeper analysis and close cooperation during the project implementation). During project implementation, further stakeholder consultation will be undertaken to assess the components in relation to gender, age, and other relevant intersecting social and economic strata. There may be a need for the Gender Analysis and Action Plan to be modified depending on stakeholder consultation during the implementation of the project.

UNDP prioritizes gender mainstreaming as its main strategy to achieve gender equality and women's empowerment. Gender mainstreaming is the process of assessing any planned action in all areas and levels to determine the implication for women and men. It is a strategy for making women's, as well as men's, concerns and experiences an integral dimension of the design, implementation, monitoring and evaluation of projects so that women benefit equally. Gender mainstreaming aims to transform unequal social and institutional structures in order to make them profoundly responsive to gender, and, when realized, it ensures that both women and men benefit equally from the development process. It involves much more than simply adding women's participation to existing strategies and programmes. Special attention and action are often required to compensate for the existing gaps and inequalities that women currently face.

The [UNDP Gender Equality Strategy 2022-2025](#) is aligned with the 2030 Development Agenda and UNDP's Strategic Plan. The strategy recognizes gender equality as a human right as well as instrumental to the achievement of sustainable development. It considers women and men as active agents of change and development, not simply beneficiaries and vulnerable groups and it recognizes how working with men and boys is of critical importance to change gender norms and attitudes and achieve gender equality. The gender assessment and action plan developed for the submission to the Green Climate Fund with the proposal "Scaling up climate resilient flood risk management in BiH" has been prepared to meet the requirements and standards in UNDP's Gender Equality Strategy which meets or exceeds the requirements set forth by the Green Climate Fund Gender Policy and Gender Action Plan.

Short background on Bosnia and Herzegovina and on EU Accession Process

Bosnia and Herzegovina is a potential candidate for EU membership. The Stabilization and Association Agreement between Bosnia and Herzegovina and the EU was ratified in 2011. In February 2016, the country applied for EU membership and in September 2016 the European Council invited the European Commission to submit its Opinion on the merits of Bosnia and Herzegovina's application. In December 2016, the Commissioner for European Neighborhood Policy and Enlargement Negotiations handed over a comprehensive Questionnaire covering all EU accession criteria. The country's consolidated answers were finalized in February 2018 and the answers handed over to the President of the European Commission. The Commission has prepared its "Opinion" report in 2019, based on the country's answers to the Questionnaire and follow up inquiries, dedicated peer reviews as well as Commission's consultations with international organizations and civil society. This interim report on Bosnia and Herzegovina provides an update on the situation in the country as well as on key developments on

"fundamentals' first" areas: the rule of law and fundamental rights, public administration reform and economic development. According to the Opinion of EU the legislation and institutions on fundamental rights are largely in place; they should be fully implemented and substantially improved, including by harmonizing legislation within the country and aligning it with European standards, strengthening administrative capacity and providing adequate resources for the effective enforcement of fundamental rights. There are no countrywide strategies on human rights and on antidiscrimination. The equality of all citizens is not ensured. The laws on non-discrimination and on gender equality are in place but not sufficiently enforced, including on gender-based violence. Bosnia and Herzegovina does not have any policy areas where the country has a good level of preparation or is well advanced regarding its capacity to undertake the obligations stemming from EU membership. Overall, considerable adjustments of the legal and institutional framework and of administrative and implementation capacity are needed across all policy areas to align the legislation with the EU acquis and to implement it effectively.¹ Bosnia and Herzegovina continued to implement the Stabilization and Association Agreement and the meetings of the joint bodies under the agreement took place, except for the parliamentary committee which has yet to adopt its Rules of Procedures².

Overview on Bosnia and Herzegovina Government Construction

The current Constitution of Bosnia and Herzegovina came into existence in 1995 as an Annex to the General Framework Agreement for Peace in Bosnia and Herzegovina. While the General Framework Agreement for Peace in BiH ended the war and brought peace and stability, the extent to which the constitutional arrangement defined by its provisions is adequate under present circumstances, is widely debatable. According to the provisions of the Peace Agreement the State of Bosnia and Herzegovina is organized as a country composed of two entities with a broad set of competencies: the Federation of Bosnia and Herzegovina (FBiH) and Republika Srpska (RS). Some years after, an arbitration decision on the disputed areas of the country made Brčko District the third administrative unit of BiH. RS is a centralized entity whereas the FBiH is comprised of ten Cantons.

The State level comprises a tripartite rotating Presidency, a Council of Ministers (executive branch) and a bicameral Parliamentary Assembly consisting of a House of Representatives (lower chamber) and a House of Peoples (upper chamber). The judicial branch established by Dayton consists of a State-level Constitutional Court. The High Judicial and Prosecutorial Council, as well as a State Court and Prosecutor Office, were established later.

An international presence under UN auspices— the Office of the High Representative (OHR) — has been in place in Bosnia and Herzegovina since 1995. The Peace Implementation Council Steering Board has set objectives and conditions to be met for the closure of the OHR³. Since the decoupling of the European Union Special Representative's mandate from the OHR, the EU has reinforced its role in the country in line with the relevant Council Conclusions. The enhanced

¹ <https://ec.europa.eu/neighbourhood-enlargement/sites/near/files/20190529-bosnia-and-herzegovina-opinion.pdf>

² Bosnia and Herzegovina 2018. Report, <https://ec.europa.eu/neighbourhood-enlargement/sites/near/files/20180417-bosnia-and-herzegovina-report.pdf>

³ See details on the OHR website (www.ohr.int) under '5+2 Agenda'

presence of the combined EU Special Representative and the EU Delegation in Bosnia and Herzegovina have continued to be instrumental in communicating EU priorities to citizens and in implementing the objectives of the EU agenda in key areas.

II. Gender Equality/Inequality in Bosnia and Herzegovina

Before starting to analyze gender inequality in Bosnia and Herzegovina, it is important to make clear and to agree on the meaning of gender equality.

Gender equality refers to the equal rights, responsibilities and opportunities of women and men and girls and boys. Equality does not mean that women and men will become the same but that women's and men's rights, responsibilities and opportunities will not depend on whether they are born male or female. Gender equality implies that the interests, needs and priorities of both women and men are taken into consideration, recognizing the diversity of different groups of women and men. Gender equality is not a women's issue but should concern and fully engage men as well as women. Equality between women and men is seen both as a human rights issue and as a precondition for, and indicator of, sustainable people-centered development (UN Women, [OSAGI Gender Mainstreaming - Concepts and definitions](#))

The fundamental principle of human development is for people to enlarge their choices, to realize their potential, and to enjoy the freedom to lead lives they value or have reason to value. Equal opportunities in all spheres, for all people, women, and men alike, are at the heart of the human development. However, those options remain unequally distributed within and across societies and unfortunately many are still largely unavailable to women. Disadvantage and discrimination, lack of insights into what particularly affects women's lives, including health (physical and emotional), reinforced by social norms and values, institutions, and public policies, exist in every sphere of women's lives, manifested differently in different country settings. They limit women's capabilities, opportunities, and choices, resulting in less progress than men in terms of health, education, standard of living, empowerment, personal security, and other critical dimensions of human development. Thus, the full potential of many women's lives cannot reach optimal level and humanity as a whole does not reap its potentials (Issue papers, "Gender Equality in Human Development – Measurement Revisited", Human Development Report Office, UNDP June 2015).

According to the latest UNDP Human Development Report (2020), Bosnia and Herzegovina's HDI value for 2019 is 0.780— which ranks it at 73 out of 189 countries and territories. Between 2000 and 2019, Bosnia and Herzegovina's HDI value increased from 0.679 to 0.780, an increase of 14.9 percent. Bosnia and Herzegovina's progress in each of the HDI indicators shows that between 1990 and 2019, Bosnia and Herzegovina's life expectancy at birth increased by 6.5 years, mean years of schooling increased by 2.8 years and expected years of schooling increased by 2.2 years. Bosnia and Herzegovina's GNI per capita increased by about 737.8 percent between 1990 and 2019. Bosnia and Herzegovina's 2019 HDI of 0.780 is above the average of 0.753 for countries in the high human development group and below the average of 0.791 for countries in Europe and Central Asia. The inequality HDI (IHDI) introduced in 2010, is basically the HDI discounted for inequalities. The 'loss' in human development due to inequality is given by the difference between the HDI and the IHDI and can be expressed as a percentage. When discounted for inequality, Bosnia and Herzegovina's HDI falls to 0.667, a loss of 14.5 percent due to inequality in the

distribution of the HDI dimension indices. This is above the average loss due to inequality for high HDI for Europe and Central Asia which is 11.9 percent.

With regard to Bosnia and Herzegovina, there can be no dispute that significant advances have been made in the realization of the principle of the equality of women and men, which is a result of activities taken by all parties (Gender institutional mechanisms, as well as citizens through civil society organizations) responsible for implementing the Law on Gender Equality in Bosnia and Herzegovina. Nonetheless, in spite of these positive changes, there are still a lot to be done.

Through the years, several indices have developed to quantify the concept of gender inequality. The United Nations Development Programme uses the Gender Inequality Index (GII) and Gender Development Index (GDI).⁴ GII is a composite measure that shows inequality in achievement between women and men in reproductive health, empowerment, and the labour market while with a measures achievement in human development in three areas: health, education, and command over economic resources. The GDI considers the gender gaps on human development between men and women.

The Gender Development Index (GDI) based on the sex-disaggregated Human Development Index, defined as a ratio of the female to the male HDI is 0.937 for Bosnia and Herzegovina (female HDI = 0.753 and male HDI = 0.803). This is below the average for Europe and Central Asia, which is 0.953, and placing it in Group 3 – which comprises countries with medium equality in HDI achievements between women and men. The Gender Inequality Index (GII) reflects gender-based inequalities in three dimensions – reproductive health, empowerment, and economic activity. Reproductive health is measured by maternal mortality and adolescent birth rates; empowerment is measured by the share of parliamentary seats held by women and attainment in secondary and higher education by each gender; and economic activity is measured by the labour market participation rate for women and men. The GII can be interpreted as the loss in human development due to inequality between female and male achievements in the three GII dimensions. Bosnia and Herzegovina has a GII value of 0.149, ranking it 38 out of 162 countries in the 2019 index. In Bosnia and Herzegovina, 21.1 percent of parliamentary seats are held by women, and 74.0 percent of adult women have reached at least a secondary level of education compared to 89.3 percent of their male counterparts. For every 100,000 live births, 10.0 women die from pregnancy related causes; and the adolescent birth rate is 9.6 births per 1,000 women of ages 15-19. Female participation in the labour market is 35.4 percent compared to 58.1 for men.

The Multi-dimensional Poverty Index (MPI), which identifies multiple overlapping deprivations suffered by individuals in 3 dimensions: health, education, and standard of living. In Bosnia and Herzegovina, 2.2 percent of the population (73 thousand people) are multidimensionally poor while an additional 4.1 percent are classified as vulnerable to multidimensional poverty (135 thousand people). The breadth of deprivation (intensity) in Bosnia and Herzegovina, which is the average deprivation score experienced by people in multidimensional poverty, is 37.9 percent. The MPI, which is the share of the population that is multidimensionally poor, adjusted by the intensity of the deprivations, is 0.008. Compared to Income Poverty, the multidimensional

⁴ United Nations Development Programme. Human Development Report. <http://hdr.undp.org/en/content/table-4-gender-inequality-index>.

poverty headcount is 2.1 percentage points higher which implies that individuals living above the income poverty line may still suffer deprivations in health, education and/or standard of living.

The Global Gender Gap Index (GGGI) of the World Economic Forum examines the gap between men and women in four categories: economic participation and opportunity, educational attainment, health, and survival; and political empowerment.⁵ Out of 149 countries, Bosnia and Herzegovina ranks 16th out of 26 countries in the Europe and Central Asia Region and 76 out of 155 countries globally.:

Description	Score	Rank
Economic participation and opportunity	0.608	115
Educational attainment	0.967	110
Health and survival	0.974	64
Political empowerment	0.302	45
Gender Gap Index 2018	0.713	76

Table 1. Bosnia and Herzegovina rank based on GGGI in 2021, where Inequality = 0.00; Equality = 1.00. Source: The Global Gender Gap Report 2021

The Organization for Economic Cooperation and Development (OECD) developed the Social Institutions and Gender Index (SIGI), a composite index that scores countries (i.e., 0 to 1) on 14 indicators grouped into five sub-indices: discriminatory family code, restricted physical integrity, son bias, restricted resources and assets, and restricted civil liberties to measure the discrimination against women in social institutions across 160 countries. The 2018 SIGI value for Bosnia and Herzegovina is 22%, suggesting that discrimination against women is low.⁶

Political participation and Public Life

According to the 2013 Law on Amendments to the Election Law of Bosnia and Herzegovina, equal representation of genders exists when one of the genders is represented by at least 40% of the total number of candidates on that list (previously, the quota was 33%). Nowadays, the percentage of women in legislative and executive authorities at all levels remains unsatisfactory (even the situation is better comparing 2014 and 2018 elections) and the newly introduced 40% representation quota has not yet been reached.

The last General Elections were held in 2018 in Bosnia and Herzegovina (State, entities, cantonal level, and Brčko District level). The Governments are still not fully established, but the available figures give an overview of the gender reality in B&H. There are 9 (21.4%) female representatives in the House of Representatives of the Parliamentary Assembly of Bosnia and Herzegovina out of 42 members⁷ and 3 (20%) female delegates in the House of Peoples of the Parliamentary

⁵ World Economic Forum. The Global Gender Gap Report 2021 Country Profiles.
https://www3.weforum.org/docs/WEF_GGGR_2021.pdf

⁶ OECD (2019), Gender, Institutions and Development Database, <https://oe.cd/ds/GIDDB2019>.
<https://www.genderindex.org/wp-content/uploads/files/datasheets/2019/BA.pdf>

⁷ Parliamentary Assembly of Bosnia and Herzegovina, Representatives of the House of Representatives, available at: <https://www.parlament.ba/Content/Read/27?title=Statisti%C4%8Dkipodaci&lang=en>

Assembly of Bosnia and Herzegovina, out of 15 members.⁸ At the time, this analyze was done, (after 2018 Elections) the BiH Council of Ministers was still not established. The Presidency consists of three persons, and all three are male, as was the case, with the composition of the previous Council which is still in power. The Council of Ministers consisted of nine ministries, and during the 2014 – 2018 mandate there were only two female ministers (22.20%). But just to have a wider picture, this document has analyzed male/female ratio more deeply (ministers and deputy ministers – 18 persons), and there are only two women (11.11%) out of 18 more influential persons in the B&H Government.

In total, comparing to men, some more women are employed in state institutions (53%). However, gender imbalance is noticeable when observing the gender structure of managerial positions - the representation of men in managerial positions is considerably higher.

The House of Representatives of the FBiH Parliament consists of 98 representatives, out of which 25 (25.5%) are women⁹ and the House of Peoples of the FBiH Parliament counts 22 (24.8%) female delegates out of a total 55 (usually there are 58, but only 55 were appointed until 28.03.2019.)¹⁰. The Government of FBiH (Prime-minister and 16 ministers), appointed after the last elections, in 2014, has had 4 female ministers (25%) and 12 (75%) male ministers¹¹. The Prime minister was male.

National Assembly of the Republic of Srpska counts 23 (27.7%) female representatives out of 83 in total (Elections 2018)¹². There are 6 (37.5%) female ministers, out of 16, appointed to the Government of the Republic of Srpska¹³

Local/ municipal election were held in 2020. The electoral cycle was marked in particular by the minimal participation of women as candidates for mayoral positions. In the end, four women out of 425 mayoral candidates in total (out of which 29 were women mayoral candidates) were elected as mayors, mostly in small municipalities, which is a drop of over 50% in comparison to 2016 local elections.

All the city mayors in BiH are, currently, men. In the municipal/city councils and assemblies of municipalities/cities in BiH, every fifth term belongs to women¹⁴.

When it comes to municipal/city councils/assemblies, in FBiH, women won 21.8% of mandates, i.e., 402 out of 1,845. Two municipalities (2.53%) elected less than 5% of women, namely municipalities Tešanj and Kalesija. Most of the municipalities (70.9%) elected between 11% and

⁸ Parliamentary Assembly of Bosnia and Herzegovina, Delegates of the House of Peoples, available at: <https://www.parlament.ba/delegate/list>

⁹ Federal Parliament, Representatives of the House of Representatives, available at: <https://predstavnickidom-pfbih.gov.ba/hr/page.php?id=8>

¹⁰ Federal Parliament, Delegates of the House of Peoples, available at: <https://parlamentfbih.gov.ba/v2/hr/stranica.php?idstranica=6>

¹¹ Government of the Federation of Bosnia and Herzegovina, Ministers, available at: <http://www.fbihvlada.gov.ba/hrvatski/ministarstva/>

¹² National Assembly of the Republic of Srpska: <https://www.narodnaskupstinars.net/?q=en>

¹³ Government of the Republic of Srpska: <http://www.vladars.net/eng/Pages/default.aspx>

¹⁴ Official Gazette, Women and Men in Bosnia and Herzegovina, http://www.bhas.ba/tematskibilteni/FAM_00_2017_TB_0_BS.pdf

30% of women, and only four municipalities (5%) elected more than 40% of women, namely municipalities Drvar, Dobretići, Kiseljak and Trnovo (FBiH).

In RS, women won 17% of mandates, i.e., 214 out of 1,256. Seven municipalities (11.5%) elected less than 5% of women, namely municipalities Pelagićevo, Stanari, Ugljevik, Knežev, Bratunac, Rudo, Milići. No women were elected in the Pelagićevo municipality. Most of the municipalities (63.9%) elected between 11% and 30% of women. Two municipalities (3.3%) elected more than 40% of women, namely municipalities Kupres (RS) and Istočni Stari Grad. Kupres (RS) is the winner of the 2020 Local Elections when it comes to women's representation – the municipal council now consists of 63.6% of women, i.e., women hold seven out of 11 mandates.

Inhabitants ¹⁵				Local	Elections	2020
Town	Total	Men	Women	Elected	Men	Women
Bugojno	31,470	15,635	15,835	25	21	4
Donji Vakuf	13,985	6,922	7,063	21	17	4
Gornji Vakuf-Uskoplje	29,933	10,532	10,401	23	18	5
Jajce	27,258	13,856	13,402	25	20	5
Cazin	66,149	33,072	33,077	30	23	7
Bosanska Krupa	25,545	12,548	12,997	24	20	4
Bihać	56,261	27,041	29,220	30	21	9
Sanski Most	41,475	20,826	20,649	31	25	6
Bosanski Petrovac	7,328	3,678	3,650	17	14	3
Ključ	16,744	8,365	8,379	25	20	5
Usora,	6,603	3,266	3,337	13	11	2
Tešanj,	43,0632	21,376	21,687	25	24	1
Maglaj	23,146	11,430	11,716	25	23	2
Lukavac	44,520	21,786	22,734	29	25	4
Tuzla,	110,979	52,745	58,234	30	21	9
Kalesija	33,053	16,597	16,456	26	25	1
Živinice	54,765	28,714	29,051	31	29	2
Žepče	30,2019	15,335	14,884	25	19	6
Zavidovići	35,988	26,574	26,908	29	27	2
Travnik	53,842	26,574	26,908	30	24	6
Zenica	110,663	54,660	56,003	30	27	3
Vitez	25,836	12,767	13,069	25	16	9
Visoko	39,938	19,705	20,233	25	21	4
Kakanj	37,441	18,651	18,790	29	24	5
Ilidža	66,730	32,359	34,371	31	21	10
Prozor-Rama	14,218	7,210	7,070	21	18	3
Jablanica	10,111	5,027	5,084	19	16	3
Konjic	25,148	12,360	12,788	27	22	5
Mostar	105,797	51,210	54,587			
Čitluk	18,140	8,937	9,203	25	20	5
Čapljina	26,157	12,947	13,210	25	19	6

¹⁵ Popis stanovništva 2013 - <http://www.statistika.ba/#tab4>
<http://fzs.ba/index.php/popis-stanovnistva/popis-stanovnistva-2013/konacni-rezultati-popisa-2013/>

Table 2¹⁶. Inhabitants and elected Members of Local Municipal Councils in the selected municipalities

All the analyzed figures are fare from 40% stipulated by BiH Gender Equality Law.

Education and Economic participation of women

In general, BiH has adequate laws and regulations in place to ensure gender equality in the education sector. The right to equal education is provided by the Constitution of BiH (part II.3 (L)) and the constitutions of the entities, and the Law on Gender Equality in BiH (Article 10). The Framework Law on Elementary and Secondary Education in BiH and the Framework Law on Pre-school Upbringing and Education in BiH provide a legal framework for education. These laws empower and require that education authorities in the FBiH (including those in cantons), RS and BDBiH, implement and respect the principles and norms defined by these laws (GAP 2013-2017).

Here we first look at the working age population of BiH (15-64), consisting of employed, unemployed and inactive persons, in accordance with the Labor Force Survey standards, to identify the gender gaps in education.

Working aged women have lower educational attainment levels than men in BiH. 39.5 percent of working age population in BiH has only primary education, 51 percent has secondary or specialized education, and 9.5 percent has higher education. A considerably higher proportion of women than men have only a primary education (49 percent compared to 29 percent), with a similar achievement gap in favor of men present in secondary education (60.9 percent compared to 41.7 percent). The gender gap in higher education is considerably smaller, at only one percentage point in favor of men. The FBiH has the largest gender gap (over 20 percentage points) for primary and secondary education – more women with only primary and fewer with secondary. In the RS and BDBiH, this gender gap is in the 16 percentage point range for primary and 14 percentage point range for secondary education, respectively¹⁷.

Education achievements of working age women in the entities and the Brcko District are similar to BiH levels. In the FBiH and RS, almost 50 percent of working age women have only a primary education and 41 percent and 44 percent, respectively, have a secondary education. The FBiH has the highest ratio of women with higher education (9.7 percent) and the RS, the lowest (7.7 percent). The BDBiH has the highest ratio of women with only primary education (57.3 percent) and the lowest with secondary and tertiary (34.9 percent and 7.8 percent, respectively)¹⁸.

By contrast, in the labor force (employed + unemployed), there is a significantly higher percentage of women than men with a university education (20.7 percent of women and 12.3 percent of men), while it is the opposite for secondary education (60.4 percent of women and 70.7 percent of men)¹⁹. On average, a greater percentage of employed women than men have a higher education. Women with higher education account for a quarter of all employed women,

¹⁶ Table created by the Author

¹⁷ LFS, 2015

¹⁸ LFS, 2015: Graphs 6.1-6.3

¹⁹ LFS, 2015: Graph 7

compared to one in seven men. At the same time, a greater percentage of women than men have only a primary education (three percentage point difference).

Considering educational achievement and unemployment rates, except for primary education, women with secondary and university educations are also more likely to be unemployed than men with the same education. Among those with secondary education, women tend to be more often unemployed with a gender gap of nine percentage points. For university-educated women, the situation is slightly more favorable – the gender gap in favor of men is five percentage points²⁰. For emphasis, note these figures do not account for men and women who are inactive.

Men with higher education are less likely to be employed than women and more likely to be inactive. 58 percent of men with higher education are employed, compared to 62 percent of women with higher education. Men with higher education are more likely to be inactive than women by nine percentage points. However, women with higher education are more likely to be unemployed than men by five percentage points. One in six highly educated women is unemployed compared to one in ten men.

Comparing the labor force (employed + unemployed) in the entities and BDBiH, found that the FBiH has the best educated female labor force. In the FBiH, 23.4 percent of women in the workforce have higher education, compared to 19.1 percent in the BDBiH, and 16.5 percent in the RS. The FBiH also has the lowest percent of workforce with primary education (16.5 percent). The FBiH and RS have almost equal proportions of workforce with a secondary education, 60.1 percent, and 61.2 percent, respectively. The BDBiH has the highest percent of women with primary education (26.9 percent) and the lowest with secondary education (54 percent) (LFS, 2015: Graphs 7.1 – 7.3). Looking at employed women only (LFS, 2015: Graphs 8.1-8.3), over half of them have a secondary education. The FBiH and BDBiH have the highest percent of employed women with higher education (27.9 percent and 26.6 percent, respectively). In the RS, 17.3 percent of employed women have higher education. The RS also has the highest percent of employed women with only a primary education (25 percent) and the FBiH has the lowest (16.2 percent). Women with secondary education also account for the highest percentage of unemployed women - 70.3 percent in the RS, 68.6 percent in the FBiH, and 50.6 percent in the BDBiH. The BDBiH stands out with 43.6 percent of unemployed women with primary education, compared to considerably lower rates of 15.2 percent in the RS, and 17.3 percent in the FBiH.

Women's and men's choices in education have an impact on their earnings later in life. The "World Bank study on Bosnia and Herzegovina: Gender Disparities in Endowments, Access to Economic Opportunities and Agency (2015)" is, for the time being, the most current and relevant source of information on the gender wage gap. Their estimates show that the hourly-wage gap for the population 16-64 years of age working for salaries or wages, is nine percent. This means that the difference between men's and women's average hourly wage is nine percent of the men's wage. The gap varies across education levels and age groups, but is always in favor of men, with the exception of the 15-24 and 55-64 age groups²¹. When controlled for factors such as age, industry,

²⁰ LFS, 2015, calculations

²¹ WB, 2015: p. 52, based on LFS, 2013

occupation, gender and education, the latter being the main difference in observable characteristics between men and women, the wage gap persists²².

There is clear evidence that gender norms translate into preferences and choices when it comes to the fields of study chosen by male and female students. In a number of instances, these choices have led to a 'gender divide' and occupational segregation, creating 'feminization of professions.' Overall, these feminized professions are characterized by lower salaries and, consequently, translate into lower lifetime earnings and social benefits. Lower paid 'women's' jobs exist in both public and private sectors and in the informal economy. Sectors primarily associated with lower earnings include education, health, culture, and agriculture.

There is a serious problem of gender-based inequalities and discrimination in women's access to the labour market, combined with structural stereotyping and the problem of unpaid work, which affects women disproportionately. All these combined factors are reflected in the low number of employed women in 2019: with 20.47%, 16.85% and 62.68% of females employed in agriculture, industry, and services sectors respectively²³. It is also alarming that there are twice as many women than men with university degrees who are unemployed. There is a serious problem of gender-based inequalities and discrimination in women's access to the labour market, combined with structural stereotyping and the problem of unpaid work, which affects women disproportionately. The current employment strategies do not seriously address the issue of female participation in the labour market, the improvement of which would contribute to reducing poverty.

Access to resources

Women and men have equal legal rights to property ownership, management, and use, as well as equal legal status in relation to access to financing and services²⁴. However, local traditions and customs persist in giving preference to men. Women are known to 'voluntarily' cede their right to a male relative or simply are not included in the deeds for land or housing (Interviews). Nationally, men account for over 70 percent of landownership and own 74 percent of dwellings. Women own 15 percent of all homes in 76 percent of female-headed households. Ten percent of dwellings has joint ownership²⁵. Women hold joint ownership in 11 percent of male-headed households but only five percent of men have joint ownership in female-headed households. One possible explanation is that female-headed households are more likely to include women without partners. Across political entities, women own 16 percent of dwellings in the RS and BDBiH and 14 percent in the FBiH. However, the FBiH has the highest joint ownership – 13 percent – compared to three percent in the RS and one percent in the BD²⁶.

There is considerable difference in dwelling ownership among men and women in urban and rural settings. Men own 80 percent of dwellings in rural areas and 66 percent in urban settings. For women, the trend is reversed – dwelling ownership among urban women is higher at 19

²² WB, 2015

²³ WB Development Indices, 2021 data

²⁴ WB, Women, Business, and the Law, 2016: p. 94

²⁵ WB 2015: p. 79

²⁶ 2011 BiH Household Budget Survey, in WB 2015

percent compared with 12 percent for rural areas. The same holds true in cases of joint ownership – 13 percent in urban settings and 8 percent in rural. Among landowners, 70 percent are men²⁷.

While only approximately 16 percent of women are heads of household, more non-poor than poor women are heads of households. In the top 60th percentile of the income category, women account for 17.8 percent of heads of household compared to 13.2 percent in the bottom 40th percentile category. More women are heads of household in urban settings (20 percent) compared to rural areas (13 percent). One explanation for this is that more income generation opportunities are available in urban areas than in rural and, consequently, women who are not able to support themselves and their families may not have an alternative option and choose to remain with their husbands (interviews). An additional likely factor is greater prevalence of traditional socio-cultural gender norms that support male ownership in rural areas. The share of households headed by women is similar in the three political entities: FBiH – 15 percent; RS – 17 percent; BDBiH – 16 percent²⁸.

Level of financial inclusion in BiH is similar to the Europe and Eurasia (E&E) average but lags considerably behind upper-middle income countries. Financial inclusion has been broadly recognized as a central factor for promoting broad economic development and reducing poverty. According to 2014 FINDEX data, 53 percent of the BiH population had an account at a financial institution. This is slightly higher than the E&E average of 51 percent but considerably below upper-middle income countries at 71 percent. The percentage of account holders has declined by three percentage points in comparison with 2011, with a similar trend in E&E and upper-middle income countries. In 2014, 34 percent of the population in the 15-24 age group had an account, compared to 36 percent in E&E and 58 percent in upper-middle income countries. The rural population in BiH has higher ratios of access to a financial institution than in E&E – 52 percent compared with 46 percent. However, this falls considerably below the 69 percent in upper-middle income countries.

A considerable financial gender gap still exists in BiH. Only 47 percent of women have an account at a financial institution, compared to 59 percent of men. However, in comparison with 2011, the decline in the number of accounts among women was only one percentage point, compared to eight percentage points for men. While still low, men have higher ratios of savings than women – 12 percent compared to six percent. And women have slightly higher rates of borrowing than men – 16 percent of women compared to 12 percent of men borrowed money from financial institutions. Nationally, financial institutions are sources for 14 percent of loans, and 7 percent come from family and/or friends. Looking at all sources of funding as a group, 27.5 percent of women borrowed money in the past year, compared to 22.6 percent of men (FINDEX, 2014). Other sources of finance include informal lenders, family and friends, shops, and employers.

Women and men borrow for different purposes. According to FINDEX, women tend to borrow more for education or school fees (2.1 percent compared to 1 percent of men), health (4.5 percent compared to 3.6 percent of men) and other immediate needs, while men borrow more in order to start, operate, or expand a farm or business (2.7 percent of men compared to 2 percent of

²⁷ WB, 2015

²⁸ Ibid, WB 2015

women (FINDEX, 2014). The World Bank (WB) approach to financial inclusion relies on three main dimensions: holding an account, formal savings, and formal borrowing. This data is available sex-disaggregated, as well as per rural/urban, age group, and education categories within the Databank of the WB²⁹.

Women in Rural Areas

The agricultural sector employs slightly more men than women – 18.5 percent and 16.8 percent, respectively. For women, this represents a 2.4 percentage point decline since 2013. The RS has the highest percentage of their workforce employed in agriculture and the highest percentage of women in the sector. Twenty-nine percent of the workforce are in this sector in the RS, compared to 24.6 percent in the BDBiH, and 10.6 percent in the FBiH. Also in the RS, 28 percent of women are employed in agriculture, compared to 18 percent in the BDBiH, and 9.1 percent in the FBiH.

Men and women tend to engage in different types of farm-related activities. Women more often are engaged in lower levels of agricultural activity, own smaller business, and are more likely to be engaged in more local and informal markets. By contrast, men are engaged in a wider spectrum of activities, have different levels of ownership and are more likely to be engaged in larger, formal markets. Farm activities performed almost exclusively by women include milking, picking fruit, and activities that require greater precision. Men tend to engage in contract labor, operation/use of machinery, transport of goods, marketing, and negotiating contracts³⁰.

In the RS, women in rural areas have very few (if any) opportunities for non-work or house social interaction. In preparation for developing a new strategic program for agricultural development by 2020, the RS Ministry of Agriculture conducted a detailed analysis of the situation in rural areas, administering 800 surveys to agricultural households. One issue raised by women that stood out and came as a surprise was the lack of and need for opportunities, services, and activities, for women to engage in socially. There are very few associations or activity groups that coalesce around different interests. This was surprising given the assumptions that in the age of global communication, even in rural settings, women would have more access to such activities. This finding provides additional validation to the existence of still-deep social gender stereotypes that restrict women's involvement outside of household and family. In response to this finding, the new agricultural development plan in the RS includes strong emphasis on addressing this issue. Engagement in this area is a continuity of the 2010 rural development strategy and the RS Action Plan for the Advancement of the Position of Women in Rural Areas. The biggest limitation is lack of resources to address all needs.

In many communities, NGOs were the first to assist in the protection and implementation of women's human rights and today are active in communities in empowering women at the local level. Their activities include providing free legal assistance in case of violation of women's rights, provide protection to victims of domestic violence and human trafficking, support women through programs of credit and economic literacy, employment, and self-employment, as well as activities aimed at empowering rural women, Roma women's organizations etc.

²⁹ Financial Inclusion Data/Global Index: <http://datatopics.worldbank.org/financialinclusion/country/bosnia-and-herzegovina>

³⁰ FARMA II Validation Workshop; Oxfam meeting, 2018

A significant mechanism for establishing partnerships between institutional mechanisms for gender equality and non-governmental organizations is the Financial Mechanism for the Implementation of the BiH Gender Action Plan (FIGAP), which operates since 2010. Its main objective is to strengthen capacity of gender institutional mechanisms and relevant institutions to integrate activities from the GAP BiH into regular plans and programs, as well as to support non-governmental organizations in the implementation of projects under FIGAP.

In 2011, through FIGAP funds, 36 non-governmental organizations from all over BiH were supported to implement projects in total value of BKM 710,000. Implemented projects addressed issues of prevention and supporting victims of domestic violence, strengthening women's entrepreneurship through self-employment and employment programs, and strengthening the role of women in political and public life.³¹

The actual number and types of women's association and NGO's that focus on gender equality in BiH is not known, but good reference is Women's Network of BiH which is an informal group of civil society organizations and individuals who represent and work on women's rights and encourage gender equality, promotes anti-discrimination, freedom of decision-making, equality, non-violence and anti-militarism, through acceptance of feminist values of action – peace, solidarity, trust, fellowship, equality and diversity. The network currently counts 40-member organizations, all from urban areas, of which 9 are registered in RS and 31 in FBiH. The network is supported by Royal Norwegian Embassy.³²

Women (and men) in less developed and poorer municipalities face greater challenges than women living in more developed areas. For example, in the northern part of the RS, the socioeconomic position of women is considerably better than in East Herzegovina (RS Ministry of Agriculture, interview). More women are co-owners of land and houses, while in the more culturally traditional southern part, female ownership is negligible. Similar differentiation exists in terms of access to public services, especially to healthcare, due to distance from relevant services and inadequate transportation. In the FBiH, issues of access to services and gender norms which, in practice, lead to an unfavorable position of women compared with men, are particularly important in mountain areas, and often exacerbated by natural disasters, such as the devastating landslides of 2014.

The greatest obstacles to broader gender development and empowerment in rural areas faced by women are related to property rights, access to public services, finance, training, advisory services and mentorship, and limited time to engage in outside activities³³. Only 12 percent of rural women own their dwellings compared to 80 percent of men and only eight percent of women have joint ownership, compared to 13 percent of urban women³⁴. This likely holds true for other types of property, such as land, farm equipment, inheritance, and financial means (bank accounts, etc.). With absent support from a husband or other family members, this imbalance in

³¹ Priručnik za ostvarivanje ravnopravnosti spolova na lokalnom nivou, mart 2015 (Manual for Achievement Gender Equality at Local Level, March 2015)

³² <http://zenskamreza.ba/en/about-us/>

³³ Ibid, FARMA II validation workshop; Oxfam meeting

³⁴ WB 2015, p. 79

property rights makes it practically impossible for women who would like to undertake new entrepreneurial activity to secure credit at reasonable rates. However, the situation seems to be improving among the younger women in terms of asset inheritance, co-ownership of land and dwellings, and increased decision-making (RS MoA interview).

Women in rural areas have lower levels of employment and self-employment than women in urban areas. The UNDP 2013 report called, “Rural Development in Bosnia and Herzegovina, Myth and Reality,” indicates that self-employment rates and employment rates are lower for women in rural areas³⁵. A World Bank analysis of the LFS data also confirms that the women’s inactivity rate in rural areas is much higher than in urban³⁶.

Limited opportunities for employment and self-employment are perpetuated by low levels of understanding of such possibilities, which is linked to inadequate (or non-existent) opportunities for information and training. As part of the preparations for the new RS agricultural strategy, the RS Ministry of Agriculture undertook an agricultural household survey with tailored questions for women in the households. Eighty percent of female respondents said that they never took part in educational or training activities about business (Interview and unpublished findings). It is reasonable to assume that the situation is probably not much different in the FBiH, but this kind of information is currently not available.

Rural women have overall lower levels of education than women in urban areas, as well as limited facility with and access to the Internet. This limits their knowledge about what training opportunities might exist. In those instances that the services are available, women may not be able to access them because of the distance from their homes, times they are being offered, and the limited time they have for such activities because of their household responsibilities.

Lack of affordable childcare and pre-schools in rural and smaller settlements is one of the central obstacles to women’s participation in trainings, entrepreneurial activities, and consequently, employment. The Multiple Indicator Cluster Survey (MICS) showed that 25 percent of first-graders in urban areas attended preschool, compared to 13 percent from rural areas³⁷, and 22.8 percent of urban children aged 36-59 months attended early childhood education compared to 7.9 percent of rural children³⁸. Part of the reason for such a rural-urban gap may be the difference in awareness of the importance of pre-school education, or the fact that the level of the mother’s education impacts how she views children’s needs during the child’s formative years. However, a portion of the difference can likely be attributed to inadequate availability of accessible and affordable childcare and pre-school education facilities in the rural areas.

Women in rural areas have limited access to appropriate information and services. There is a very limited number of women associations and cooperatives which exacerbates the lack of access to relevant information and services. This undermines opportunities for women to engage, learn, and share information on more equal footing. Prevailing patriarchal/conservative social norms

³⁵ UNDP, 2013: p. 80

³⁶ WB, 2015: p. 80

³⁷ UNICEF, 2013: p.xix

³⁸ UNICEF, 2013: p. 76

are not conducive to women's participation in associations with predominantly male membership. This is particularly true for women who are at the stage of considering entrepreneurial activity or who are engaged in micro-activity. Even women who are owners or co-owners of larger agricultural production firms face gender stereotypes in the market and often decide to include a husband or a male relative during business discussions³⁹.

Women's participation in agricultural activities and decision-making varies by regions. Regional differences are along the levels of economic development, with less developed areas/municipalities showing lower levels of participation in agricultural activities and decision-making. Women noted that lack of adequate financial assets as well as under-developed market channels are the greatest obstacles to increasing levels of economic activity (Interviews).

Marginalized groups and Social Inclusion

Marginalized groups considered in this section include Roma, persons with disabilities, and internally displaced persons, although the list of marginalized groups varies across agencies and institutions.

International human rights law directly applies through the BiH Constitution, but the Constitution does not protect equal rights for all groups and individuals. Human rights and fundamental freedoms for all individuals are enshrined in the International Covenants on Civil and Political Rights, and Economic, Social and Cultural Rights, as well as the European Convention on Human Rights and Fundamental Freedoms (ECtHR), directly applicable in BiH. Beyond clear-cut, direct discrimination issues confirmed in ECtHR judgments, gender-based discrimination, especially in the cases of multi-level discrimination (women with disabilities, Roma women with disabilities, etc.) is difficult to prove in court.

Persons with disabilities in BiH find themselves in an extremely difficult situation. Due to fragmented policies, persons with disabilities face discrimination on the basis of their place of residence. Authorities and the private sector ignore affirmative measures related to their employment and rehabilitation⁴⁰. The most common forms of violation of the rights of people with disabilities in Bosnia and Herzegovina 'are in the areas of social protection, health, education, rights of access to information, labor and employment, and in the field of organized action for people with disabilities. Even where laws exist to regulate this area, there is an evident record of discrimination in practice⁴¹.

Women with disabilities face daily discrimination, both as women and as persons with disabilities. They often lack adequate health care and access to services and are often socially isolated. The latest census demonstrates that at least 294,058 out of the 3,531,159 residents of BiH (8,3% of the population) are diagnosed with some form of a disability and that 30% of the population is either directly or indirectly affected by the consequences of disability. Women make for 50.9% of the population and 54.7% of all disabled people.

³⁹ FARMA II validation workshop

⁴⁰ Institution of the Human Rights Ombudsman of BiH, 2010

⁴¹ MDG Progress Report, 2013

As the State and Gender Equality Agency recognizes, women with disabilities are especially vulnerable to being victims of multiple discrimination in BiH society, particularly in the areas of labour and employment, although BiH has ratified the UN Convention and adopted the EU standards, which require the rights of the disabled to be determined on the basis of their individual condition, not the cause of disability. However, this has never been implemented.

The rights of people with disabilities are determined differently in Bosnia's two entities and, within one of them (the Federation of BiH), in each of the ten cantons. Both entities have adopted strategies to improve the rights of disabled people, as required by the Convention, but neither has implemented them. In the Federation of BiH, 248 people with disabilities have started to work in 2015. Only 26.6% (66 out of 248) of them were women. In Republika Srpska, 547 disabled people were employed in the period from 2013 to 2016. An astounding 6.2% (34 out of 547) were women. The State has no special programs for employment, retraining or rehabilitation related to women with disabilities, nor does it have relevant statistics, data, or special measures for protection of women with disabilities in the areas of labour and employment⁴².

Almost two thirds of all disabled persons in BiH live close to or below the poverty line. One of the major obstacles is a lack of statistical data and information on the extent, nature, and characteristics of people with disabilities. There is also no central registry of people with disabilities and gender disaggregated data is not available. It is especially alarming since women with disabilities are also more likely to be victims of domestic violence. Health care is a very problematic field, especially reproductive health protection and protection and care for pregnant women.

In addition to the lack of appropriate examination tables for women in wheelchairs, the majority of health care workers are not trained for examinations or delivery when it comes to women with disabilities⁴³. As it is mentioned in a report issued by NGOs on the Application of the Convention on the Rights of Persons with Disabilities in Bosnia and Herzegovina, women with disabilities have problems getting custody of children after a divorce since there is a common belief that the child should be with the "healthy" parent, ignoring the child's best interest, even in cases where the father was violent, and taking for granted that women with disabilities are not able to raise a child⁴⁴.

The most severe and most common forms of violating the rights of people with disabilities in Bosnia and Herzegovina are found in the fields of social protection, health, education, rights of access to information, labour, and employment and in the field of organized action for people with disabilities.

⁴² BiH Combined Fourth and Fifth Periodic Report on the Implementation of the CEDAW

⁴³ Annex to the 3rd Alternative Report on the implementation of CEDAW and women's human rights in BiH (2013)

⁴⁴ Prava za sve/ICVA: Izvještaj nevladinih organizacija. Primjena konvencije o pravima osoba sa invaliditetom u Bosni i Hercegovini (Report from the NGOs on the Application of the Convention on the Rights of Persons with Disabilities in Bosnia and Herzegovina). July 2013, available at <http://www.pravazasve.ba/publikacije-bs/docs-bs/izvjestaj-nevladinih-organizacija.pdf>

Levels of discrimination against Roma are not the same for men and women. **Roma women** experience multiple vulnerability factors compared to Roma men and are even less politically represented and economically secure than Roma men. The 2009 ruling of the ECtHR to amend the BiH Constitution and allow ethnic minorities to run as candidates for the presidency or the House of Peoples of BiH, has not yet been implemented. Even when the direct discriminatory provision gets removed, Roma women will still face indirect obstacles in running for office at any level. Among Roma representatives on municipal councils and assemblies, there are no Roma women, despite campaigning by the Roma Women's Network before the 2016 local elections.

Discrimination of Roma women is very evident when one considers the levels of their formal employment (Human Rights Ombudsman Institution, 2013). The Alternative Report on the Implementation of CEDAW for BiH (Prava za sve, 2010) reported that of the 595 Roma women surveyed, almost 82 percent were unemployed, only nine percent were working in the informal sector, and seven percent were begging for subsistence. Additionally, in 2008, only two to three percent of Roma, men, or women, were employed in the public sector⁴⁵. Since the population census report did not include employment data broken down by ethnicity, it is not possible to cite more recent data on the employment of Roma women.

The discrimination toward Roma, especially toward Roma girls, is also reflected in their rights to education. Percentages of children of primary school age attending primary or secondary school in 2011-2012 varied by sex for Roma children. Gender differences were larger in the FBiH than in the RS. In the FBiH 70.8 percent of Roma boys attended school compared to 66.9 percent of Roma girls. In the RS, 74.9 percent of Roma boys attended school, compared to 73.1 percent of Roma girls⁴⁶. The number of girls who dropped out of primary school was greater than the number of boys⁴⁷.

However, 'there are still no programs in place that would systemically improve the education of Roma girls and employment of Roma women and their access to health and social protection⁴⁸. Roma women also experience very high levels of violence. Over 43 percent of surveyed Roma women self-reported being subjected to some form of physical violence⁴⁹. The same survey also identified that 76 percent of them know a Roma woman who had been a victim of physical violence and 46.5 percent know a woman who has been subjected to sexual violence. In addition, there is an increasing number of Roma girls vulnerable to trafficking for forced marriages and a high percentage of arranged or paid marriages.

Twenty-four years after signing of the Dayton Peace Agreement (DPA), 98,324 **internally displaced people** (IDPs), many of whom are women victims of war, and some 47,000 minority returnees in Bosnia and Herzegovina are still in need of solutions⁵⁰. The revised strategy for implementation of annex VII of DPA was supposed to be implemented by 2014.

⁴⁵ Ministry for Human Rights and Refugees, MHRR, 2008

⁴⁶ UNICEF and BiH MHRR, 2013

⁴⁷ BiH Combined 4th and 5th CEDAW Reports, 2011

⁴⁸ UNICEF, 2013

⁴⁹ Rights for All, 2011

⁵⁰ MfHRR, 2017

The main issues are unclear status of IDPs after relocation, source of subsidized rents, lack of livelihoods solutions and inadequate public awareness. The public gaze of the media has long since moved on elsewhere and donors have shifted their resources. Displaced peoples have specific burdens over belonging, housing, occupation, welfare, security, and loss of communities. The decision whether to return to their homes is complex, with local and international political pressures adding to their uncertainties and insecurities. In addition, there is the impact of the war, the experiences of violence, the remembering and issues of reconciliation, and a variety of mostly unevaluated psychosocial programmes aimed at helping with these. All this has a profound impact on their health and well-being.

According to Norwegian Refugee Council/Internal Displacement Monitoring Centre (NRC/IDMC), Prijedor, Sanski Most, Banja Luka, Dobo, Modriča, Bijeljina, Zvornik, Bratunac, Srebrenik, Lukavac, Tuzla, Živinice, Zenica, Sarajevo, Gorazde, Mostar, Nevesinje, Čapljina and Trebinje are municipalities with the highest level of IDPs in Bosnia and Herzegovina.

BiH Ministry for Human Rights and Refugees, in close cooperation with UNHCR developed the Strategy of Supporting Solutions for Vulnerable Internally Displaced Persons and Returnees in BiH in 2017.

International arena (international partners and donors), as well as BiH, are facing a great challenge with the global migration crisis. However, given the great disproportion between the investment and the real needs for implementing the strategic goals of the BiH Strategy for Annex VII Implementation, BiH Government hope that this challenge will not cause them to hesitate in the continuation of the work on a well-traced road of support to the return process in BiH.

Gender-based Violence

BiH ratified the Council of Europe Convention against Violence against Women and Domestic Violence in 2013 and the entities have their own strategic documents. In the RS, acts of domestic violence can be prosecuted as a misdemeanor or felony, and domestic violence is recognized in the Misdemeanor Code, and in the Criminal Code of the RS. The RS established an entity council for fighting domestic violence. In the FBiH, the acts are processed in accordance with the Criminal Code of FBiH. In this criminal code, each canton has a referral mechanism for responding to domestic violence.

Every second woman has experienced some form of gender-based violence since the age of 15, while every fifth woman has probably experienced it over the past 12 months. Working with perpetrators of violence against women, especially in cases of domestic violence, is not yet a standard practice. Financing of safe houses is problematic in both the RS and FBiH, leading to a near closure of two out of nine safe houses. Victims of conflict-related sexual violence (CRSV) still lack socio-economic rights and legal protection. Victims of sexual violence are not treated equally in the two entities. While the government in BiH has made significant efforts toward eliminating trafficking, it still does not fully meet the minimum standards for doing so.

The lesbian, gay, bisexual, transgender, and intersex (LGBTI) community in BiH is for the most part absent from the public sphere due to pervading hostile attitudes among the majority of the

population, fear for physical safety, and discrimination by family, friends, and coworkers. Following the 2014 general election, sustained advocacy on the part of civil society and changes in government led to progress on several fronts in the protection of LGBTI human rights.

Bosnia and Herzegovina has yet to implement coordination and cooperation between government institutions at all levels in the context of preventing and combating violence against women, as well as to harmonize laws and entity public policies in this area, which leads directly to the inability to ensure the equal status and protection of rights of women who survived violence and their access to justice. Safe houses run by NGOs have neither the same legal status in both entities, nor a secured system support, which directly threatens the existing limited services to help women in the period of acute violence.⁵¹

III. Mechanisms to address Gender Inequality in Bosnia and Herzegovina

The introduction of the principle of the equality of women and men in all areas of public and private life does not only constitute the realization of one of the core standards required to achieve human rights and fundamental freedoms, nor can it be reduced merely to an issue of social justice, but rather and first of all it is a necessary precondition for the sustainable social and economic development of a country, with special emphasis on poverty reduction and the improvement of the quality of life for all citizens.

Over the past twenty years significant efforts have been made in Bosnia and Herzegovina to develop a national legal, institutional, and political framework for the implementation of the principle of gender equality.

Institutional gender mainstreaming mechanisms have been established within the system of legislative and executive power, at all levels, as a part of an overall effort to improve the status of gender equality (men/boys and women/girls) in Bosnia and Herzegovina.

The BiH Law on Gender Equality (LoGE) was adopted in 2003 (BiH OG 16/03) and amended in 2009 (BiH OG 102/09), including improved definitions and improved implementation provisions. An integrated text was published in 2010 (BiH OG 32/10). The law specifically addresses education, employment, labor and access to resources, social protection, healthcare, culture and sports, public life, and media. The Law provided the framework for the gender institutional mechanisms which implement it and monitor its implementation. The Law has established the Agency for Gender Equality of Bosnia and Herzegovina and the gender centres (Gender Centre of the Federation of Bosnia and Herzegovina and Gender Centre- Gender Equality Centre of the Government of Republika Srpska) which exist on the entity level. The Agency and the entity gender centres are public institutions i.e., governmental bodies.

The BiH institutional mechanism on gender equality forms a network which includes institutions on the municipal, cantonal, entity and state level, both in the executive and the legislature. Unfortunately, those institutions on municipal and cantonal level were established through different projects supported by international community (2004, 2005, 2006) and after the projects were over the gender mechanisms did not continue with their active role.

⁵¹ Alternative CEDAW Report, 2016

At the state (legislative) level there is a Commission for Gender Equality of the Parliamentary Assembly of B&H. At the entity level, there are Commissions for Gender Equality of the House of Peoples and the House of Representatives of the Parliament of the Federation of B&H, as well as the Equal Opportunities Committee of the National Assembly of Republika Srpska.

All institutional mechanisms have precisely defined mandate, which shows governments' commitment to achieve gender equality and inclusion of gender issue in all segments of social life. It is important to underline that the Agency and the entity Gender Centres represent the real drivers of the initiatives and measures, and the key implementers and monitors of the gender mainstreaming in BiH. These three institutional mechanisms for gender equality are crucial instrument in initiating and implementing activities in the area of gender equality. They have the prominent place considering their activities, projects, and measures which they have undertaken to strengthen and ensure sustainability of institutional mechanisms for gender equality.

The LoGE prescribes those victims of discrimination may seek judicial protection in accordance with existing procedures. Special proceedings for protection from discrimination are possible in compliance with the Law on Prohibition of Discrimination (Official Gazette of BiH, No. 59/09).

RS and FBiH Laws on Protection from Domestic Violence were adopted in 2012 and 2013, respectively. The laws replicate the definitions and elements of violence from an international framework. Domestic violence (DV), therefore, includes physical, psychological, or sexual pain and suffering, as well as economic damages against a family member.

The first BiH Gender Action Plan (GAP) was adopted for the period of 2006 – 2011 and financed through the Financial Instrument for the Gender Action Plan (FIGAP) which aggregated funds from various donors.

The new GAP for the period 2018 - 2022 builds on important international documents: CEDAW Convention and Recommendations of the UN CEDAW Committee, National Plan of Action based on the Beijing Declaration and Platform for Action, CoE recommendations, EU directives and regulations as well as a number of other binding and recommended UN, CoE and EU documents. GAP also refers to domestic documents such as the state, entity, and cantonal constitutions as well as the Gender Equality Law in BiH. GAP covers similar areas as the previous GAP and places significant focus on strengthening the gender machinery as one of the strategic goals, and on monitoring and reporting on the implementation of the GAP.

GAP is a policy document that guides gender equality policies at lower levels of governance and includes direct obligations not only for the BiH institutions, but also for its entities. In accordance with the LoGE, many local self-governance units were adopting local action plans (during the period 2013 – 2017) which operationalized the GAP's high-level priorities, bearing in mind local circumstances and competence of local self-governance units. However, the first local 3-5 year policies for women's empowerment developed by the municipalities were only the first steps in changing the mindset of local authorities and were often of lower quality due to low capacities for gender analysis and limited sources of sex-disaggregated, statistically representative data.

Increasingly, and with guidance from the entity gender centers, the quality of data available for gender analysis is improving, as is the quality of gender analyses and policies themselves.

Bosnia and Herzegovina developed an Action Plan for the implementation of Security Council Resolution 1325 on Women, Peace and Security, Bosnia and Herzegovina has failed to provide real mechanisms for its implementation and monitoring, without qualitative indicators of progress, and failed to ensure the involvement of institutions from lower governing levels and did not allow a broader involvement of civil society in its preparation and monitoring⁵².

Bosnia and Herzegovina has prepared and submitted the Sixth periodic report to the CEDAW Committee on 19.04.2018.

Institutional capacity to implement normative framework for gender equality remains to be one of the most serious challenges in BiH. The research about the institutional capacity to implement the Gender Equality Law and GAP in BiH, including Resolution 1325 (Ravnopravnost spolova i institucije BiH, GEA, 2014;) has shown that there is a serious limited capacity of public institutions to implement the established legal framework. The mailed questionnaire has been sent to 22 public institutions and 2,500 respondents. However, only a small percentage (325) has replied. Among those who replied, only 40% replied that they are familiar with the Law on Gender Equality, only 28% replied that they are familiar with the mandate of GEA, and only 9% that they had been involved with gender equality issues. Since it is reasonable to assume that responses were selective, in terms that those who were not familiar with the problem were also the ones who did not respond, this research is showing serious lack of present-day institutional support to gender equality in BiH. In addition, research revealed that great majority of responses has shown that there was a lack of institutional regulations related to gender discrimination and sexual harassment. This research actually has proven that there is a serious gap between normative and institutional aspects of gender equality in BiH which needs to be addressed by concrete set of measures and recommendations which will be fully contextualized in present day BiH society.

Civil Society organizations (women/gender/DRR)

Prior to the 1990s the associations, organizations, media, and trade unions that existed at the time were mainly part of the governmental machinery and thus could hardly be viewed as elements of civil society. This means that there were only a few organizations in former Yugoslavia that were able to function as a factor able to monitor and control the socialist Government; their work was also not directly related to the needs of the country's population. These organizations generally included trade unions, chambers of commerce, sports and professional associations and women's associations, which acted in accordance with the interests of the ruling socialist system.

However, certain feminist initiatives were launched during the 1970s in the capitals of the former Yugoslav republics, in universities, academic circles and among intellectuals⁵³ The first NGOs in

⁵² Alternative CEDAW Report, 2016

⁵³ Lepa Mladenović: An article titled "Development of Feminism – a Women's Movement in the Former Yugoslavia" (<http://www.womenngo.org.yu/sajt/sajt/istorija.htm#per2>).

the region, including BiH, were established at the start of and during the 1991-1995 war; they were aimed at calling for peace and organizing protests to support the cause of peace, to address the developing humanitarian crisis and to provide aid to refugees and the victims of war. Starting in 1996, after the Peace Accords were signed and the new Constitution had been adopted, a large number of NGOs were established in BiH in an attempt to rebuild confidence in the divided country, take part in the rebuilding of the country and its institutions and to provide services and support to people in need. A significant number of these newly established organizations were women's organizations. These organizations were the first to cross the entity separation lines and to start rebuilding communication and trust. They began the process of raising awareness in society in regard to the effects of inequality and they organized activities related to the participation of women in politics, combating violence against women, the provision of assistance for the posttraumatic recovery of female victims of war, education, the need to empower women with a view to establishing groups and associations across the country, etc. Some years later, women's NGOs even began to be established in the remote and rural areas of the country.

Some civil society interventions include fragmented approaches to gender equality, but there are several areas that could be improved. Project design is rarely preceded by a gender analysis. Implementation is rarely monitored using gender sensitive indicators and sex-disaggregated data. Finally, advocacy initiatives rarely take a gender transformative approach. One of the interviewees noted that civil society actors have a limited understanding of gender mainstreaming and gender equality. (Centar za ekologiju i energiju, Tuzla, Gender Centers of RS and FBiH). Another interviewee emphasized that, as in other sectors, men are primarily the leaders of the biggest CSOs which do not specifically deal with gender issues. (Gender Agency BiH, Gender Centers RS and FBiH). At the other end, there are gender equality organizations where nearly all organization directors are women. The expertise of the most advanced women's organizations is not properly deployed in the biggest civil society initiatives for capacity building and gender mainstreaming. On a positive note, organizations are increasingly working with men on gender equality in public and in private life.

IV. Gender issue in addressing the increasing vulnerability to intensified climate and flood related disasters (pluvial, fluvial, and torrential) in Bosnia and Herzegovina

The project objective is to address the increasing vulnerability of B&H communities and livelihoods to intensified climate-induced hydro-meteorological flood-related disasters (pluvial, fluvial, and torrential). This will be accomplished by adopting an integrated approach to flood risk management, resulting in strengthened institutional, technical, and financial capacity to implement long-term flood risk management strategies, including a combination of structural and non-structural measures which protect communities and employing eco-system-based approaches. In addition, strengthened early warning and emergency response systems will provide forecasts and early warning to at-risk populations. For achieving its objectives and targeting the GCF assistance the project employs social vulnerability approach. The various elements of this approach are discussed below in the context of gender equality and gender mainstreaming.

Climate change, climate variability and weather events pose a threat to the eradication of extreme poverty and should serve as a spur to hasten efforts not only to reduce greenhouse gas emissions but also to tackle other underlying risk drivers such as unplanned urban development, vulnerable livelihoods, environmental degradation, and gaps in early warnings.

Families are displaced and victims lose shelter. This is complicated further by immense shortages of food and drinking water. Severity of impact of natural hazards are often measured by the degree to which functioning of individuals, groups or organizations is disrupted. The disruption can be caused by death of family member or friends, displacement, injury, separation from family and community, damage to vegetation and infrastructure, epidemics of communicable diseases, loss of land, house, crops, livestock, productive assets, and other property, loss of employment and trade, damage to infrastructure.

Vulnerability describes the characteristics and circumstances of a community, system or asset that make it susceptible to the damaging effects of a hazard. There are many aspects of vulnerability, arising from various physical, social, economic, and environmental factors. The vulnerability is determined by the material and social assets that individuals and communities possess. The effect of possessing resources to a different degree is evident at all phases of disaster cycle, i.e., protection, response, impact, and recovery.

The vulnerability of households also depends on their composition. Gender and age of household members, number of dependents and persons with disabilities, economic status, human, social and political capital - all have an effect on protection from, response to and impact or recovery from natural disasters. Households can be nuclear, or three or four generational, male headed, or female headed, consisting of only a woman or a man. In all phases of disaster cycle it is important to consider composition of households and the ratio of dependents, these being children, seniors, and persons with disabilities (PWD). Concept of vulnerability is closely linked with resilience - "the capacity of a system to maintain its basic functions and structures in time of shocks and perturbations" (Birdman, 2006,15)

Social vulnerability approach to disasters emphasizes the roles of social, economic, and political relations, it describes social distribution of risks: as "some groups in society are more prone than others to damage, loss, and suffering in the context of differing hazards" (Blake, et al 1994,9). In dealing with disasters next to their physical impact the social conditions that underlie different outcomes should be taken into consideration. Households rather than individuals should be taken as a unit for intervention strategies.

Gender intersects with other characteristics such as poverty, disability, age, seniority, and displacement – which rarely act alone – to exacerbate inequality and vulnerability.

Gender

During the third week of May 2014, heavy rains fell in Bosnia and Herzegovina (BiH) causing sudden and extreme flooding and landslides over one third of the country. A similar flood event also occurred again in 2019. Urban, industrial, rural areas and roads were submerged; thousands were displaced and isolated without electricity or communications. Many houses and businesses were destroyed or damaged, and the effect on agriculture and productive services was severe. More than one million people have been affected.

A government-led Post-Disaster Needs Assessment was conducted in 2014 by the World Bank, E.U. and UN, to identify damages, losses and needs, and to provide an estimate of national recovery costs⁵⁴. UN Women responded to the emergency by contributing a humanitarian gender expert to the PDNA team, to support analysis of gender issues and to ensure these will be addressed in response and recovery planning. The PDNA estimates not only the replacement cost of damaged assets, but also the cost to “build back better”, to increase resilience and mitigate the effects of future catastrophic events.

Having in mind all earlier mentioned, throughout the Gender Analysis, as well as the 2014 PDNA, it is clear that Gender inequality in Bosnia-Herzegovina has significant implications for disaster recovery. There is a widely held perception that because of prevailing norms and attitudes flood-affected women will experience greater difficulty compared to men in obtaining assistance. Therefore the PDNA 2014 Report among others, recommends: provision of training in gender-sensitive disaster preparedness and response planning for municipal governments including preparation of gender-responsive disaster plans, compliance with existing requirements for disaggregated statistics at all levels of government, ensure transparency in the distribution of recovery funds and support that is equitably provided to women and men, strengthen economic empowerment programmes for women as part of DRR, ensure meaningful participation of women, including marginalised and excluded women, in decision-making processes relevant to housing, land, property and agriculture etc. Also, the PDNA recognizes that “funding for women’s economic and political empowerment, and for equal participation in all disaster-related discussions, is key to building back better to reduce future disaster risks”.

While the country is one of the most advanced in terms of legislation and policy on gender equality, women are largely excluded from political and decision-making processes; they have disproportionately limited access to employment and income, and many are subject to violence. There is particular concern with respect to women IDPs, disabled, elderly and Roma women in BiH, who are even more severely disadvantaged and marginalized.

However, there is one promising example. Gender specific issues in state of natural disasters are recognized by RS Gender Center. Based on experiences from floods 2014 the Center has developed a Gender Action Plan for conditions of natural disaster. The Plan, among others, anticipates constant communication with relevant institutions in order to introduce gender perspective in all their activities related to rehabilitation, reconstruction and disaster relief, monitoring of disaster impact through defined gender-sensitive indicators, constant communication with women’s NGOs and local communities in order to assess needs of women, men and children during disaster, raising public awareness and information exchange to support population in need etc. ⁵⁵

Resilience-building against future shocks will require attention and funding directed toward reducing the severe economic and decision-making disparities between women and men.

Gender issues related to different phases of disaster management cycle

⁵⁴ <https://www.gfdrr.org/sites/default/files/BiH-rna-report.pdf>

⁵⁵ Gender Action Plan for conditions of natural disaster, Gender Center of RS, 2014

The impact of all mentioned vulnerabilities is revealed at all phases of disaster management cycle, i.e., at prevention and protection, response, impact, and coping. The purpose of the gender mainstreaming throughout various phases of disaster management is to empower women and see them as capable agents of change, who can manage crisis, deal with its aftermath, and take on leadership roles in the family and community. Women play important economic and community roles that help in reconstruction and resilience building. Women often take on leadership roles in family and community in the face of death or disability.

Summary of gender differences in vulnerability and adapting to disasters

Disparities that increase risks for women in disasters • Higher levels of poverty • Extensive responsibilities of caring for others • Domestic violence • Traditional women's occupations	Disparities that increase risks for men in disasters • Occupational segregation • Internalized norms of masculinity • Roles in the family and in the home
Gender experiences that can increase capacities for managing disaster situations: Women • Social networking • Caring abilities • Extensive knowledge of communities • Management of natural and environmental resources • High levels of risk awareness	Gender experiences that can increase capacities for managing disaster situations by: men • Professional and work contacts • Technical abilities • Limited childcare responsibilities

Prevention and protection:

Early warning system works as a potent protection mechanism against natural hazards. But as receivers of warnings are humans, for warranting the desired response a number of human related factors should be considered. Information on hazard risk can be delivered in time, but still the problem of understanding information, believing it, and acting according to provided recommendations and hence saving life, health and property remains problematic. That is for the messages on multi-hazard risk information are to be tailored to the needs and capabilities of vulnerable groups, targeting women, children, senior citizens, and persons with disabilities (PWDs). Education level, knowledge of language in which warning is communicated, problems of hearing and seeing constitute serious barriers for adequately grasping the threat.

Men appeared to be more confident in managing an emergency situation, including the perception that they were better prepared to take action, including physical preparedness and response. Additionally, women had fewer opportunities to maintain a high level of social networking in the community, which may lead to them being less informed. Effective targeting requires utilization of both sources.⁵⁶

⁵⁶ <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6313390/>

In general, men are less vulnerable to climate induced hazards than women, hence less prone to take self-protective actions. Men often label evacuation calls as panic and do not react. Besides, acting according to stereotypical gender roles men may decide not to evacuate to safeguard property. On the other hand, women are readier to respond to risk, but lack of social power deters them to mobilize family to respond, they also may be slow to react according to instructions until securing family members.⁵⁷ Patriarchic structures and power imbalances in many countries tend to reduce women's status in society, their access to resources, opportunities and power and subsequently lead to higher female vulnerability to natural hazards.⁵⁸ Women's vulnerability to the impact of disasters is also increased by socially determined differences in roles and responsibilities of women and men and inequalities between them in access to resources and decision-making power.⁵⁹

Children and people with low education level may encounter problems of understanding messages when they are worded in impersonal, official manner. Elderly citizens with the problems of hearing or seeing and living alone can be left out of reach in case of delivery of hazard risk messages only by printed or electronic media.

Response to disasters:

Effectiveness of response greatly depends on a well-planned emergency behaviour, preparedness, and social cohesion of community. Therefore, outlined below features should be reflected in emergency planning. Timely evacuation is a challenging issue for small children, seniors, and persons with disabilities, especially with problems of moving and of persons with poor health. People dependent on health services for survival (dialyses, cancer treatment) are faced with life threatening circumstances in disaster.

Women's, children's, and elderly's' vulnerability is greater due to the mobility constraints. Both car ownership and having driver's license is less frequent among women than men.

It is important to have list of CSOs dealing with vulnerable groups in the flooding area and include them in the phases of disaster management cycle.

Impact of disasters and coping:

Psychological reaction to natural disaster can be: withdrawal, stunning, apathy, disbelief, but also increase in community bond and social cohesion. Experience of natural hazards may result in stress, anxiety, depression, and other mood disturbances. Effect usually is not long lasting, only 25 percent of victims suffer psychological effects some months after disaster, but again a bigger percentage of women than men. Coping strategies of disaster affected population could include: raising of public awareness, information exchange and communication with citizens, improvement of preparedness, development of evacuation plans and preparation of evacuation centers, development of gender sensitive flood response plans (prior to floods), or leaving area,

⁵⁷ https://socwomen.org/wp-content/uploads/2018/03/fact_10-2006-disaster.pdf

⁵⁸ Vulnerability to Natural Hazards: A Gender Perspective in Disasters, Carolin-Anna Achberger Karl-Franzens-Universität Graz

⁵⁹ https://www.who.int/gender/other_health/genderdisasters.pdf

over-exploiting resources in order to survive (e.g. cutting down trees for wood), liquidation of assets (e.g. livestock), reducing food intake (during the flood).

As women, more so than men, depend on natural resources for livelihood (23% of woman employed in agriculture comparing to 19% men)⁶⁰, disaster has a more severe effect on women. At the same time women play key roles in the sustainable use and management of natural resources. As senior women considerably outnumber men, they are more likely to experience physical limitations that matter so much in emergencies. The burden of domestic work and care-giving to children, as well as to ill and disabled family members mainly falls on women, so caring in aftermaths of disasters becomes more challenging for women. As men can decide to migrate due to property or employment loss, family burden increases for women. But also, more and more women decide to migrate for providing for the family, as the demand for domestic labour increases in EU countries. The migration of family member also effects senior citizens as they have to look after grandchildren.

Lesser political and professional representation, low involvement in consulting process of key stakeholders make women more vulnerable as their perceptions and needs are not reflected in planning of recovery process. Gender stereotypes negatively reflect on men. Men often are overwhelmed with emotions after disaster, but are constrained to express them, which has health related consequences and often stimulates substance abuse, domestic violence, gambling, and engagement in risky behaviours.

Recommendations

The analysis above shows that in order to set up effective national and community based early warning systems, climate-informed planning and improved resilience, gender consideration need to be integrated into the project implementation. The existing gender inequality factors (e.g., limited engagement of women in planning and decision making) and traditional distribution of gender roles in families and communities call for tailoring and targeting of the project solutions to outreach beneficiaries of both genders equally. Based on the analysis of the gender aspects of vulnerability to climate-induced natural disasters a number of recommendations for the proposed GCF project have been elaborated. These recommendations and the following Gender Action Plan are aimed at ensuring that the GCF project:

- narrows gender inequality;
- addresses the needs and constraints of women, girls, men, and boys;
- avoid any risks of adverse gender impacts;
- ensure women's participation, promotes their leadership qualities; and
- ensure women are included as planners, co-implementers and agents of change.

As a result of the project implementation more lives, property and productive assets will be secured from the impacts of climate induced disasters. Beneficiaries of this outcome will be all the population, but women's benefits will be bigger as for the livelihood women more than men

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http://www.odraz.hr/media/280503/izvjestaj_zene%20u%20lokalnom%20razvoju%20bosne%20i%20hercegovine_web.pdf

depend on natural resources. They also are poorer than men and hence proportionally loose more. Moreover, after flood, majority of women is left with household and some farm works since their husbands are going out for cash. Women again have to be in charge of all the remaining stuff including restarting their productive work such as farms, preparing livestock for raising, or replanting vegetables for some small cashes. Food security, psychological stresses, households' debts, and more labor-intensive works have degraded the wellbeing of women in the post-flood scenario⁶¹. The recommendations and the Gender Action Plan as summarized below have been designed to ensure that both men and women have full and equitable access to the Project's resources and benefits, with specific actions and responsibilities aimed at ensuring the full participation of women in Project activities.

At Project inception, additional gender analysis will be undertaken to ensure that the baseline data set is sex- and age-disaggregated and adequate for assessment of the gender impacts of the Project. At this time, the Gender Action Plan will be presented to the Project Implementation Team, including Government staff and consultants. In particular, the Project team leaders will ensure that all specialists are briefed on their responsibilities in relation to the Gender Action Plan upon mobilization. Progress reports will provide periodic updates on the effect of the Project on women, and regular UNDP review missions and the midterm review mission will review and monitor the gender impacts.

Carry out gender responsive vulnerability assessment

- Ensure that risk assessments are informed by gender analysis.
- Subjecting 2013 Census data to additional analyses, providing sex- and age-disaggregated data on social vulnerability across regions, assessing employment, education, health problems connecting with adequate functioning, disability, physical abilities (to swim, climb and run), ownership of house and productive assets and farming opportunities, ratio of dependent persons in households, household composition-single member male and female, women headed households, livelihoods, unpaid care and domestic work responsibilities.
- Mapping households receiving state subsistence allowance assistance by analysing social assistance database across chosen regions.
- Carrying out group discussions and in-depth interviews with key stakeholders, ensuring an equal representation of women, persons over 65, disabled or family members of disabled, IDPs, community leaders and government officials for mapping hazards and risks, collection of existing community coping strategies, identifying local businesses and institutions able to contribute to DRM activities. To identify priority needs, responses to, separate coping mechanisms of women and men, elderly, disabled and poor.

Ensure collection of sex- and age-disaggregated data for project indicators

⁶¹ https://reliefweb.int/sites/reliefweb.int/files/resources/KHM_0601_ImpactonWomen_AA2014-01.pdf

- Gender disaggregated reporting will be further introduced beyond the project level at the national level through the multi-hazard disaster risk information and knowledge system to be designed by the project

Enhance gender considerations in disaster preparedness activities

- Disaster preparedness and response plans should be based on gender analysis and include gender considerations
- Men and women of different age groups, as well as those from vulnerable communities, should be involved in planning and implementation. Community consultancy groups with at least 40 percent representation of less represented sex should be established.
- Ensure that information on hazards, their character, probability of occurrence, threats to life, possible impact on livelihood, houses, crops, and livestock and on protecting measures is reaching both women and men through appropriately tailored channels.
- Increase preparedness of educational institutions. Provide information to teachers, pupils and students, academic and other staff on different types of disasters and on effective immediate response to them. Create emergency plans for kinder-gardens, schools, vocational institutions, and Universities. Inform parents on them. Ensure they are practicing periodically. Provide the staff with first aid training.
- Include Gender Mechanisms representative in disaster preparedness activities.

Warrant universal access to disaster warnings

- Tailor warning to the gender-differentiated needs and capabilities of specific population groups, such as children, senior citizens, IDPs, the sick and persons with disabilities.
- Set up community based early warning systems and make sure that they are tailored to effectively serve both women and men, based on prior gender responsive assessment of proposed systems.
- Use multiple methods for targeting messages for reaching broadest group of people, including TV, radio, Internet, sirens, flashing lights, registration-based alert systems sending messages to cell phones with information clearly stated orally and graphically. Use network of CSOs.
- Include pregnant women, IDPs, the elderly and disabled in emergency planning. Include CSOs.

Mainstream gender considerations and engagement of women in resilience building activities

- Target 40 percent representation of less represented sex in capacity building and training activities supported by the project.
- Make sure that women have equal access to livelihood generating activities facilitated by the project.

- Disseminate information and stimulate involvement of population of the hazard prone regions in insurance schemes, among them insurance of crops.
- Disseminate information on risk zones with recommendations for construction of dwellings, agricultural activities, and livestock husbandry.
- Support creation of employment opportunities for women and poor; ensure that women are part of employment-guarantee schemes planned in the framework of the community-based disaster risk management activities; provide social care services to redistribute burden of unpaid care work that falls on women. Provision of social care infrastructure can also generate jobs for women and men as part of disaster recovery.
- Strengthen social capital through creating community groups for planning for, safeguarding from and mitigating adverse effects of natural disasters with at least 30 percent less represented sex participations.

Conduct gender mainstreaming trainings and build capacity of national stakeholders targeted by project and responsible for various disaster risk reduction functions

- In 2016 UNDP Istanbul Regional Hub (IRH) developed and piloted a three-module training manual on Gender mainstreaming in disaster preparedness and response (GMiDPR). The tool is designed for UNDP staff and government officers working in disaster preparedness and disaster response in the Europe and CIS region. In 2017 the tool was piloted with the DRR project practitioners in the Western Balkans countries. But still there is a lack of knowledge within the governmental institutions, as well as CSOs organizations on GMiDPR. There is a need to organize a series of training and capacity building workshop on gender mainstreaming based on the UNDP training manual targeting various stakeholders and practitioners (government, CSOs, community leaders, youth groups, women's groups, farmers' groups, as well as Gender institutional mechanisms) at the national and local levels.

Engage women in decision-making

- Ensure at least 40 percent representation of less represented sex and their active participation in project stakeholder consultations, local and national decision-making bodies set up and/or facilitated by the project. This includes ensuring that women are properly represented in municipal investment planning process, formulation of community preparedness plans and that the of women are incorporated in all products are truly designed to promote and advance gender equality and social inclusion.
- Secure participation of the Gender Institutional Mechanism representatives from all the governmental level
- In line with the recent CO Engagement Strategy for Women CSOs and Grassroots, women's groups and women's advocacy organisations will be invited to contribute to advocacy and awareness raising as well as development of risk financing and transfer mechanisms products and tools to influence the offer of inclusive, responsible financial products. Women's associations know their real needs, barriers and main risks, which can support and guide the design of relevant products. Therefore, women's groups and gender equality advocates will be engaged with development and finance institutions, and governments, civil society and affected communities themselves – to develop mechanisms that bring transformative benefits to women and girls.

Affirmative Measures

- The recent UNDPs Intersectional Analysis of the COVID-19 and Inflation on Women confirms that 55.5% of women experienced a worsened financial situation while Report on Mapping the Gender Component in Data and Legislation in the area of Climate Change, Environment and Biodiversity discusses women's reduced access to credit as it requires land/asset ownership in order for commercial banks to approve credit funds. Therefore, affirmative measures will be provided for women led businesses and businesses that employ at least 50% of women and companies that have achieved parity in managerial positions-boards, by awarding additional scores to such companies by virtue of being embedded in the selection criteria.
- Furthermore, feasibility studies on micro-insurance, regional disaster risk insurance program and catastrophe bonds will take into account specific barriers women face (taking into account local dynamics and whether insurance products are sold through male-dominated groups or cooperatives, and whether men have better access to these products and could purchase them more easily.), inform targeting and identify women centered solutions (for example, feasibility of women-specific micro-level risk insurance schemes)

Requirements to the project staff

- Gender-responsive social vulnerability approach can be realized only by the gender and social vulnerability sensitive staff of the project. Therefore, project staff should be composed of both women and men represented at all levels of decision-making. Staff members should have a record of participation in trainings on gender mainstreaming and on social vulnerability approach. Also, it is suggested that staff members participate/complete a specific Gender mainstreaming in disaster preparedness and response trainings. Even employing new staff, the ToRs can consist of request of knowledge on gender mainstreaming.

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Table 1: Gender Action Plan

Gender Action Plan Template						
<i>Project-level outcome (e.g., Systems and capacity for sustainable land management established)</i>						
Gender-related activity	Indicator	Baseline	Targets, including sex-disaggregated targets	Budget	Timeline	Responsibility

Gender-related activity	Indicator	Baseline	Targets, including sex-disaggregated targets	Budget USD ⁶² 2,740,790	Timeline	Responsibility
Impact Statement: <i>Increase resilience and enhanced livelihoods of the most vulnerable people including women and girls, communities, and regions to the negative impacts of climate change.</i> Outcome Statement: <i>Gender-specific and gender-responsive climate risk and vulnerability information developed and used in decision-making; Reduced loss of life and assets of the most vulnerable including women, girls, disabled, old people through a strengthened country-wide gender-responsive flood forecasting and early warning system.</i> <i>Strengthened institutional and regulatory systems and capacity for gender sensitive climate responsive planning and decision-making; Increased climate risk financial resilience of the most vulnerable including women and female-headed households, through the development of a national flood and disaster risk insurance scheme and other risk financing mechanisms; Strengthened adaptive capacity and reduced exposure of the most vulnerable households, in particular female-headed households to climate risk through the implementation of structural, non-structural and EbA flood protection measures;</i> Output 1: Fully integrated impact-based Flood Forecasting and EWS facilitates timely preparation and response. Output 1 statement: <i>Gender-specific and gender responsive climate risk and vulnerability information developed and used in decision-making.</i> <i>Reduced loss of life and assets of the most vulnerable including women, girls, disabled, old people through a strengthened country-wide gender-responsive flood forecasting and early warning system.</i>						

⁶² A budget of \$100,640 has been allocated for the implementation of the Gender Action Plan and is included in the ESAR overall costs

Activity 0: Consider modifying Gender Analysis and Action Plan based on the outcome of stakeholder consultations during project implementation.	Indicators: Reviewed Gender Analysis and Action Plan	Gender Analysis and Action Plan are not revised	Targets: The Gender Analysis and Action Plan have been reviewed and adjusted to reflect the stakeholder's inputs following the consultation processes. This process will be done within the first 6 months of the project. <i>Note: The project is to ensure that the consultations process is participatory, consensus oriented, accountable, transparent, responsive, and inclusive. All potential changes to GAP will be communicated to the Secretariat through the APR.</i>	8,000	Year 1-4	UNDP project team, stakeholders
Activity 1.1: Upgrade and expand the coverage of the hydrometric network for enhanced monitoring of climate variables. Gender Mainstreaming Activities: Upgrade and expand the coverage of the hydrometric network with the specific focus on ways in which climate	Indicators: No of consultations on gender responsive hydrometric network expansion. No. of objections to new hydrometric network stations	Relevant consultations on hydrometric network expansion are not organized and women do not participate Women do not sufficiently benefit from work,	Targets: In consultations on hydrometric network expansions at least 5 consultations are held within the first 2 years of the project, 40% are women participate, ensuring that their specific needs and experiences are adequately taken into consideration.	30,000	Year 1 - 2	UNDP project team Water agencies in cooperation with local communities

variables affect and impact women. Assess options and propose ways in which women can benefit from hydrometric network expansion in terms of employment and income making opportunities. Ensure participation of both men and women in consultation process during the installation of the new hydrometric equipment at catchments and torrents where it requires community consultation	Number of women participating in consultations % of women that benefit from work and employment opportunities generated by implementing these measures	employment and income making opportunities generated in implementing these measures	No concerns or objections related to expansion of hydrometric equipment. Any complaints made are adequately addressed and resolved through the project complaints/grievance process. At least 30% of women participants have benefited from work, employment, and income making opportunities.			
Activity 1.2: Enhance climate-induced flood hazard, risk and vulnerability information for strategic risk management and sound decision making for climate induced flood management. Gender Mainstreaming Activities: Datasets for socio-economic risk and vulnerability assessment support gender disaggregated data for analyses and decision-making. Gender Advisor engaged to support gender mainstreaming activities.	Indicators: % of women in stakeholder consultations. % Of sex-aggregated socio-economic receptor vulnerability data collected for use in flood socio-economic risk and vulnerability modelling and mapping. % No. of datasets of gender-	17% women participated in stakeholders' consultations 16.6% Flood hazard, risk and vulnerability mapping is gender sensitive (in Vrbas basin only). 16.6% gender responsive climate risk	Targets: Gender expert, working as part of UNDP country office will be engaged immediately after project commencement and supported achievement of the following: At least 40% participants of consultations are women. 100% Flood hazard, risk and vulnerability mapping is gender responsible.	192,300 + 452,876 government co-financing	Year 1-3	UNDP project team Water agencies in cooperation with local communities

	responsive climate risk information produced and used.	information generated Lack of expert support in gender mainstreaming in socio-economic assessment, vulnerability, and risk mapping	100% of climate risk information generated used in at least 1 sectoral planning document.			
Activity 1.3: Develop an integrated impact-based centralized and community-based flood forecasting and early warning system. Gender Mainstreaming Activities: Secure equal participation and active role of women and men, in developing of FFEWS; Ensure that training and capacity building programme on EFEWS are engendered; Implement workshops on gender mainstreaming for FFEWS members and policy makers.	Indicators: No. of men and women trained in the development, operation, and maintenance of the integrated centralized and community based FFEWS.	0% participated in the development, operation, and maintenance of the integrated centralized and community based FFEWS.	Targets: At least 40% female participants trained in the development, operation, and maintenance of the integrated centralized and community based FFEWS. All the training and capacity building programmes in EFEWS are engendered (or gender expert participated in the development and implementation of the trainings and in the programmes.)	337,500	Year 1 - 2	UNDP project team Gender centers

Activity 1.4: Develop and implement protocols and SoPs on data generation, data management and communication for effective impact based FFEWS and flood risk management. Gender Mainstreaming Activities: Review National protocols and SOPs from gender point of view; Ensure that flood risk information and warnings are adapted to the needs and capabilities of women and other vulnerable groups (Roma women, children, elderly people, disabled, returnees); Develop communication products considering specific needs of man, women and other vulnerable groups, aiming to share information and raise knowledge on FFEWS; Use NGOs networks to reach women and the most vulnerable groups.	Indicators: No. of Gender responsive national protocols and SOPs; Flood risk information and warning messages meeting the needs of women and other vulnerable groups (Roma women, children, elderly people, disabled, returnees). No. Communication products engendered and adapted to the needs of all vulnerable groups. Number of gender-focused NGOs involved in development of SoPs and protocols on awareness raising,	Data sharing platform protocol developed for Vrbas and Una-Sana	Targets: Gender responsive national protocols and SOPs developed for data generation, data management and EWS communication. At least 3 different types of flood risk information and warning messages developed to meet the needs of women and other vulnerable groups (Roma women, children, elderly people, disabled, returnees). At least 2 gender-focused NGO groups engaged developing SoPs and protocols communication, awareness raising, knowledge and information sharing on FFEWS	176,000 _+ 100,000 governm ent co- financin g	Year 1 - 4	UNDP project team, water agencies, gender centers
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	knowledge and information sharing on FFEWS.					
Output 2: Non-structural flood risk reduction measures and nature-based solutions mainstreamed in sectoral policies and plans and effectively contribute to protection of people and livelihoods from climate-induced flood risk. Output 2 statement: <i>Strengthened institutional and regulatory systems and capacity for gender sensitive climate responsive planning and decision-making; Increased climate risk financial resilience of the most vulnerable including women and female-headed households, through the development of a national flood and disaster risk insurance scheme and other risk financing mechanisms;</i>						
Activity 2.1: Mainstream climate induced flood risk reduction into sectoral planning (agriculture, hydropower, critical infrastructure) and spatial planning. Gender mainstreaming Activities: Ensure participation of both men and women, but also representation of vulnerable groups into consultative processes organized to mainstream climate induced flood risk reduction into sectoral planning. Conduct gender mainstreaming trainings and build capacity of national stakeholders targeted by project and responsible for	Indicators: % of FRM decision makers and practitioners are trained on gender mainstreaming in FRM. No. of sectoral planning documents (reviewed by gender advisor) amended to include gender considerations. No. of women/men within the planning teams and consultation groups.	FRM decision makers and practitioners are not trained on gender mainstreaming in FRM Sectoral planning documents are not consulted with gender advisor	Targets: At least 80% of FRM decision makers and practitioners are trained on gender mainstreaming in FRM. At least 3 sectoral planning documents amended to include gender considerations. Minimum of 40% of less represented gender participated the planning teams and consultation groups.	88,000 + 90,000 government co-financing	Year 2 - 6	UNDP, sectoral ministries, gender centers

various disaster risk reduction functions						
Activity 2.2: Implement and mainstream new ecosystem-based flood risk reduction and climate change adaptation methods Gender mainstreaming Activities: Review of the community based EbA and EbA approaches piloted by the UNDP/SCCF Vrbas River project by gender expert to identify gender gaps, locate mainstreaming opportunities and uncover specific ways in which flood risk and climate change impact women, taking into account gender specific vulnerabilities and intersectional issues (single parenthood, age, income levels, disability, etc.); Mainstreaming of gender throughout all the new ecosystem-based flood risk reduction and climate change adaptation methods (e.g., afforestation, agroforestry) Conduct assessment and make	Indicators: % of community based EbA approaches that are gender responsive. Gender mainstreaming annex of new EbA methods and guidance documents to be applied in communities. % Of participants in local consultations related to EbA measures are women. % of women benefitting from work, employment and income making opportunities generated by	EBA FRM measures are not implemented Women insufficiently benefit from work, employment and income making opportunities generated in implementing these measures	Targets: 100 % of community based EbA approaches that are gender responsive. Gender mainstreaming annex of new EbA methods and guidance documents to be applied in communities. At least 40% participants in local consultations related to EbA measures are women. At least 30% of women participants have benefited from work, employment and income making opportunities.	8,000	Year 2 - 4	UNDP project team, gender centers, local communities

adequate provisions to ensure that women can equally benefit from the work, employment and income making opportunities associated with implementing these measures.	implementing these measures.					
Activity 2.3: Codify and mainstream EbA solutions into policies and regulations and promote non-structural measures. Gender mainstreaming Activities: Review of the existing policies and regulations by a gender advisor to identify gender gaps and mainstreaming opportunities; Include gender expert in the teams working on the preparation of new policies, plans and regulations and in promotion of new FRM concepts.	Indicators: No. of policies and regulations related to EbA solutions including gender equality considerations and mainstreaming	Lack of policies and regulations related to EbA solutions	Targets: Country-wide best-practice guidelines, technical specification, standards, and protocols for the application of eco-system based non-structural measures include gender equality considerations and mainstreaming	63,000 + 90,000 government co-financing	Year 3 - 6	UNDP project teams, gender agencies, water management ministries

Activity 2.4: Review and strengthen institutional capacity and develop long-term institutional capacity development plans for climate resilient FRM. Gender mainstreaming Activities: Gender sensitive review of institutional capacity. Participation of Gender expert in development of national training programmes in the technical and practical areas of flood risk management for practitioners, decision-makers, communities, emergency responders;	Indicators: <i>No. of institutions for which assessments of gender equality, and gender responsiveness of FRM is undertaken.</i> No. of gender-responsive long-term Capacity Development and training FRM plan. Min % female attendees to training in technical and practical areas of flood risk management for practitioners, decision-makers, communities, emergency responders	Assessments of gender equality, and gender responsiveness of FRM are not implemented for FRM institutions Lack of coherent institutional capacity development framework that embeds FRM and FFEWS capacities into relevant institutions	Targets: <i>Assessments of gender equality, and gender responsiveness of FRM is undertaken for all FRM institutions in both entities.</i> Gender-responsive long-term Capacity Development and training FRM plan developed for institutions in both entities. At least 40 % female attendance to 3-day training courses in technical and practical areas of flood risk management for practitioners, decision-makers, communities, emergency responders held every year for the first 3 years of project duration,	151,000 + 150,000 Government co-financing	Year 1 – 6	UNDP project team, gender centers, water management ministries
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Output 3 - Climate-proof flood protection measures scaled-up through new and improved national and local investment frameworks increasing resilience of the most vulnerable groups to climate induced flooding Output 3 statement: <i>Strengthened adaptive capacity and reduced exposure of the most vulnerable households, in particular female-headed households to climate risk through the implementation of structural, non-structural and EbA flood protection measures</i>						
Activity 3.1: Develop investment framework for climate induced flood risk reduction and management. Gender mainstreaming Activities: Representatives from the Gender mechanisms included in the teams for development of a national investment framework for climate induced floods risk reduction and management; Development of a gender responsive strategy and gender sensitive tools for private sector engagement in long-term climate-proof risk management. Development of risk financing and transfer mechanisms based on detailed gender sensitive assessment on socio-economic risk, damages, and losses.	Indicators: Gender responsive national investment framework for combating floods established and integrated into key development strategies/action plans. Ratio of women owner of private companies informed, consulted, and participating and benefitting from the development of the strategies and tools to the development of risk financing and transfer mechanism.	Insurance product developed for Vrbas basin applied gender responsive approach Gender mechanisms where not included in the teams for development of insurance products No other private sector investment frameworks in place	Targets: Two gender responsive national investment frameworks for combating floods established and integrated into key development strategies/action plans (one for each entity). At least 30% of private companies informed, consulted, and participating and benefitting from the development of the strategies and tools to the development of risk financing and transfer mechanism and women owned. Gender sensitive socio-economic risk, damages, and losses assessment for each basin, for the development of risk financing and transfer mechanisms.	193,400 + private sector co-financing 285,714		UNDP project team, gender institutions, ministries of finance

	Gender responsive socio-economic risk, damages, and losses assessment for the development of risk financing and transfer mechanisms.					
Activity 3.2: Formulate multi-year climate resilient municipal investment plans and gender responsive community preparedness plans in selected municipalities (10-12) and in one canton. Gender mainstreaming Activities: Both, men and women are adequately represented in municipal investment planning process; Men and women actively participate in formulation of community preparedness plans in selected municipalities; Needs of women and other vulnerable (elderly, disabled, children,	Indicators: No. of gender-responsive municipal investment plans developed. % Of women comprising the community planning teams and consultation groups.	Municipal investment plans are not developed;	Targets: 10-12 gender-responsive municipal investment plans developed. 40 % of women comprising the community planning teams and consultation groups	200,000	Year 3- 6	UNDP project team, selected municipalities, gender institutions

Roma) groups are incorporated in developed plans.						
Activity 3.3: Implement climate-proof structural flood risk reduction and anti-erosion interventions in Vrbas, Una-Sana, Bosna and Drina River basins Gender mainstreaming Activities: Both, women and men, and other vulnerable groups (elderly, disabled, children, Roma) are included in processes of stakeholders' consultation related to implementation of climate-proof flood risk measures; Ensure that specific ways in which flood risk and erosion impact women are analyzed and considered. Ensure that women and vulnerable group members can equally benefit from technical assistance ensured with GCF funds, to develop climate proofed designs based on new climate risk information and design methods. Assess and make provisions for women to equally benefit from the work, employment and income making opportunities associated with implementing these measures.	Indicators: % women and % men represented in the community consultation groups % Of women and men involved and benefited from GCF funds and technical assistance. % Of women that have benefited from the work, employment and income making opportunities associated with implementing these measures.	0%- climate-proof structural flood risk reduction and anti-erosion interventions are not implemented Women do not proportionately benefit from the work, employment and income making opportunities	Targets: at least 40% of less represented gender comprising the community consultation groups; At least 30% of less represented gender involved and benefited from GCF funds and technical assistance. At least 50% of women participants have benefited from the work, employment and income making opportunities associated with implementing these measures.	125,000	Year 2 - 6	UNDP project team, water agencies, local communities